

GTOS Detailed Volume - Volume 10

Data Database Reference

GTOS Enterprise Engineering Handbook - Final Edition 1.0

Source Chapters

- Data Architecture Introduction
- Conceptual Enterprise Data Model
- Domain Logical Data Models
- Database-per-Service Policy
- Schema Ownership
- Database and Schema Registry
- Table Family Documentation
- Identity and Keys
- Foreign Keys and Cross-Domain References
- Constraints and Invariants
- Indexes and Query Workloads
- Effective Time and Recorded Time
- Data Classification
- Privacy, Purpose and Residency
- Retention and Legal Hold
- Data Lineage
- Data Quality
- Correction and Retraction
- Materialized Views and Projections
- Partitioning and Archival
- Backup and Restore
- Disaster Recovery for Data
- Database Migration
- Data Contract Evolution
- Decommissioned Databases and Tables
- Database Users and Least Privilege

- Row-Level Security
- Encryption and Key Management
- Analytical Stores and Warehouse
- Vector Stores
- Search Indexes
- Caching and Invalidation
- CDC, Outbox and Replication
- Master and Reference Data
- Data Observability and Reconciliation
- Database Ownership and Data Lineage Matrix

Data Architecture Introduction

Metadata	Value
Volume	10
Chapter ID	V10-00
Subject	DataArchitectureIntroduction
Status	Generated First-Draft Architecture Skeleton - Domain-Specific Expansion Required
Content maturity	TEMPLATE_DERIVED_REFERENCE_SKELETON unless repository citations prove otherwise
Default implementation classification	REFERENCE_ARCHITECTURE / REQUIRES_REPOSITORY_REVIEW
Accountable owner	Data Architecture Council
Evidence rule	No implementation claim without inspected executable evidence
Repository	chinair88-prog/allinb2c

1. Executive Orientation

This chapter defines how GTOS shall **define enterprise and Domain data ownership, semantics, persistence, quality, lineage, correction, protection, operations, and verification for Data Architecture Introduction**.

The specification separates reality, observations, knowledge, Decisions, Intent, execution, and outcomes. A component name, database row, UI label, AI answer, or workflow state does not acquire authority merely because it is visible or technically convenient.

Reality or external outcome

- ≠ Observation
- ≠ Claim
- ≠ derived Perspective
- ≠ Prediction
- ≠ Recommendation
- ≠ authorized Decision
- ≠ execution Intent

1.1 Accountable ownership

The accountable owner is **Data Architecture Council**. Supporting applications, shared infrastructure, integration adapters, analytics, and AI coordinate or present the capability but shall not silently own its authoritative propositions.

1.2 Scope

- create and identify the subject
- validate evidence and policy
- make authorized Decisions
- execute typed Commands
- publish completed-fact Events
- query current and historical perspectives
- correct and reconcile outcomes

1.3 Non-goals

- treating a registered Gradle module, catalog record, README, frontend route, migration plan, or generated report as proof of end-to-end implementation;
- direct cross-Domain database writes;
- generic status values that combine lifecycle, approval, provider, document, payment, dispute, and correction state;

- silent replacement of history;
- AI-created authority or fabricated external outcomes.

2. Ubiquitous Language and Subject Model

Subject	Required interpretation
DataArchitectureIntroduction	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
DataArchitectureIntroductionVersion	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
DataArchitectureIntroductionDecision	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
DataArchitectureIntroductionEvidenceSet	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
DataArchitectureIntroductionException	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit

Reference aggregate:

```
DataArchitectureIntroduction {
  subjectId
  tenantId
  organizationContext
  sourceIdentityRefs[]
  sourceVersionRefs[]
  lifecycleState
  relationshipRefs[]
  decisionRefs[]
  intentRefs[]
  evidenceSetRef
  knowledgeCutoff
  policyRefs[]
  authorityRef
  jurisdictionRefs[]
  createdAt
  effectiveAt
  updatedAt
  version
  correctionOf?
  correlationId
}
```

2.1 Core invariants

- Stable identity precedes relationship, state, and cross-system correlation.
- Effective time and recorded time remain distinct where business truth changes over time.
- Recommendation, Decision, Intent, execution, and outcome are separate concepts.
- No Portal, report, search index, cache, workflow, or AI Agent becomes a shadow owner of Domain truth.
- Binding submitted, approved, issued, accepted, or signed versions are immutable.
- Corrections append governed facts and preserve the original record.
- External authority and provider outcomes retain their source and qualification.
- Tenant and represented-principal context come from trusted authentication and mandate evaluation.
- Retried writes are idempotent and concurrency guarded.
- Outcome Unknown is explicit and reconciled before risky duplicate execution.

2.2 Required standards

- stable identifiers
- typed contracts
- immutable binding versions
- transactional outbox
- idempotent Commands
- explicit Outcome Unknown
- correction lineage
- machine-verifiable conformance

3. Actors, Roles, and Authority

A conformant design distinguishes:

- the natural person acting;
- the service or AI identity invoking a Tool;
- the organization or principal represented;
- the role or relationship granting context;
- the mandate or delegation granting action authority;
- the Policy and jurisdiction limiting the action;
- the reviewer or approver responsible for a binding Decision.

Authority is evaluated at action time. Authentication alone does not grant business authority. Cached permissions and browser-supplied tenant headers are insufficient for high-impact actions.

4. Lifecycle and State Model

State	Semantics
PROPOSED	Distinct governed condition with entry, exit, timeout, correction and authority semantics
VALIDATION_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
APPROVAL_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
ACTIVE	Distinct governed condition with entry, exit, timeout, correction and authority semantics
SUSPENDED	Distinct governed condition with entry, exit, timeout, correction and authority semantics
CORRECTION_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
RETIRED	Distinct governed condition with entry, exit, timeout, correction and authority semantics

```
stateDiagram-v2
    [*] --> PROPOSED
    PROPOSED --> VALIDATION_PENDING
    VALIDATION_PENDING --> APPROVAL_PENDING
    APPROVAL_PENDING --> ACTIVE
    ACTIVE --> SUSPENDED
    SUSPENDED --> CORRECTION_PENDING
    CORRECTION_PENDING --> RETIRED
```

The diagram is a primary reading path. Real implementations may add parallel review, amendment, appeal, timeout, suspension, partial completion, reconciliation, correction, and terminal failure without collapsing their meaning.

4.1 Transition rules

- transition through a typed, authorized Command;
- require exact expected version where concurrent change matters;
- record actor, represented principal, mandate, Policy, reason, and Evidence;
- use an idempotency key for retried writes;
- publish a completed-fact Event after durable state change;
- expose Outcome Unknown for unresolved external completion;
- preserve original and corrected states in historical reconstruction.

5. Decision, Evidence, and Knowledge Cutoff

Reference Decision:

```
Decision {
    decisionId
    decisionType
    subjectRef
    sourceVersionRefs[]
    knowledgeCutoff
    evidenceSetRef
}
```

```

policyVersionRefs[]
recommendationRefs[]
principalId
representedOrganizationId
mandateRef
rationale
outcome
decidedAt
effectiveAt
correctionOf?
}

```

Evidence records identify issuer or observer, subject, version, observed time, received time, effective interval, integrity, confidence, qualification, contradiction, correction, retention, residency, and access class. A recommendation may assist the Decision but cannot promote itself to the Decision.

6. Commands, Events, Queries, and Contracts

Reference Commands:

- CreateDataArchitectureIntroduction
- ValidateDataArchitectureIntroduction
- ApproveDataArchitectureIntroduction
- ActivateDataArchitectureIntroduction
- SuspendDataArchitectureIntroduction
- CorrectDataArchitectureIntroduction
- RetireDataArchitectureIntroduction

Reference Events:

- DataArchitectureIntroductionCreated
- DataArchitectureIntroductionValidated
- DataArchitectureIntroductionApproved
- DataArchitectureIntroductionActivated
- DataArchitectureIntroductionSuspended
- DataArchitectureIntroductionCorrected
- DataArchitectureIntroductionRetired

Reference queries:

- GetDataArchitectureIntroduction
- ListDataArchitectureIntroductionRecords
- GetDataArchitectureIntroductionTimeline
- GetDataArchitectureIntroductionEvidence
- GetDataArchitectureIntroductionDecisionPackage
- GetDataArchitectureIntroductionExceptions
- GetDataArchitectureIntroductionAuditTrail

Reference API surface:

```

POST /api/v1/dataarchitectureintroduction/commands
GET  /api/v1/dataarchitectureintroduction/{id}
GET  /api/v1/dataarchitectureintroduction/{id}/timeline
GET  /api/v1/dataarchitectureintroduction/{id}/evidence
GET  /api/v1/dataarchitectureintroduction/{id}/decisions
POST /api/v1/dataarchitectureintroduction/{id}/exceptions
POST /api/v1/dataarchitectureintroduction/{id}/corrections

```

These endpoints are reference contracts, not claims that exact routes exist.

7. Workflow, Failure, and Compensation

flowchart TD

```

A[Validate identity version and authority] --> B[Collect required evidence]
B --> C{Policy and evidence gates pass?}
C -- No --> D[Open exception or request correction]
D --> B
C -- Yes --> E[Record Decision or execution Intent]
E --> F[Execute Domain command or external request]
F --> G{Outcome known?}
G -- Success --> H[Record fact and publish Event]
G -- Failure --> I[Record failure and compensation options]

```

```

G -- Unknown --> J[Enter Outcome Unknown]
J --> K[Query source or wait for verified callback]
K --> G
H --> L[Update projections and downstream workflow]

```

Representative failure modes:

- stale or wrong source version
- duplicate submission or replay
- external timeout after possible success
- contradictory source observations
- lost Event after committed state
- approval or mandate revoked during execution
- downstream action before a mandatory gate
- partial completion across multiple subjects
- evidence later retracted or corrected
- tenant, facility, channel or jurisdiction mismatch

Recovery shall query the authoritative source before duplicate execution, preserve partial success, use compensating Intent rather than destructive rewrite, publish corrections, quarantine high-impact ambiguity, and record affected downstream subjects.

8. Data and Persistence Architecture

Reference table families:

Table family	Purpose
dataarchitectureintroduction_aggregate	authoritative subject
dataarchitectureintroduction_version	immutable submitted, approved, issued, or signed versions
dataarchitectureintroduction_relationship	governed relationships
dataarchitectureintroduction_decision	binding Decisions and rationale
dataarchitectureintroduction_evidence_link	provenance and evidence relationships
dataarchitectureintroduction_event_outbox	transactional Event publication
dataarchitectureintroduction_idempotency	duplicate-write protection
dataarchitectureintroduction_exception	exception, claim, dispute, or hold
dataarchitectureintroduction_correction	correction and supersession lineage
dataarchitectureintroduction_audit	privileged action evidence

The owning service controls schema and migration. Cross-Domain references use stable identifiers or contracts. Analytics, reports, caches, search, and vector indexes remain projections. Backup, restore, retention, legal hold, privacy, residency, and decommissioning requirements are explicit.

9. Portal and Experience Requirements

- Primary participant Portal: initiate, review exact versions, supply evidence, and respond to required actions.
- Operator Portal: inspect exceptions, correlation, source responses, retries, compensation, and downstream impact.
- Executive or oversight view: consume governed metrics without becoming a source of Domain truth.

Every Portal displays exact version, owner, state, freshness, required action, and uncertainty. It supports loading, empty, partial, stale, denied, error, timeout, and Outcome Unknown states; accessibility; locale, currency, unit, date, and RTL/LTR correctness; evidence audit; safe retry; and correction history.

10. AI Participation and Tool Governance

Permitted AI tasks:

- assist schema and lineage analysis
- detect data-quality anomalies
- summarize migration and reconciliation risk
- never mutate authoritative data outside typed Domain Tools

AI shall not own Domain records, grant itself authority, suppress contradictions, silently edit approved versions, fabricate external outcomes, or execute high-impact actions outside typed Tools, policy, approval, idempotency, sandbox, and outcome observation.

Every Agent Run records source context, prompt/model/tool versions, structured output, confidence and limitations, Policy result, approval, Tool calls, provider responses, final outcome, and recovery actions.

11. Security, Privacy, and Trust Boundaries

- least privilege and separation of duties
- tenant, organization, facility, channel and jurisdiction isolation
- integrity protection for binding evidence
- verified external callbacks and anti-replay controls

Additional controls include secret isolation, callback signature verification, audit of privileged action, emergency-access review, sensitive-data minimization, purpose limitation, and threat models for impersonation, tampering, replay, confused deputy, fraud, prompt injection, and exfiltration.

12. Reliability, SLOs, and Operations

SLI	Reference objective
authoritative read availability	99.9% monthly unless a stricter tier applies
acknowledged Command durability	no acknowledged write may be silently lost
Event publication	99.9% within five minutes after committed fact
duplicate prevention	100% for same tenant, Command class, and idempotency key
binding Decision audit completeness	100%
stale-state detection	bounded by business deadline

Dashboards and runbooks cover backlog, error budget, dependency failure, retry budget, DLQ, workflow stall, provider outage, reconciliation, capacity, backup/restore, release/rollback, incident command, and post-incident correction.

13. Testing and Continuous Conformance

- aggregate invariant and transition tests
- authorization, delegation, separation-of-duties and tenant-isolation tests
- idempotency and optimistic-concurrency tests
- database migration, key, constraint and rollback tests
- contract, outbox/inbox, duplicate, replay and DLQ tests
- workflow restart, timeout and compensation tests
- Portal E2E tests for success, partial, stale, denied, error and Outcome Unknown states
- accessibility, locale, currency, unit and RTL/LTR tests
- security adversarial, fraud and data-exfiltration tests
- AI evaluation, prompt-injection, policy-bypass and unauthorized-tool tests
- historical reconstruction, correction and audit-completeness tests
- performance, capacity, recovery and operational exercise tests

Release evidence links each invariant and control to the test, environment, input data, result, owner, and artifact proving it. Generated test counts never replace semantic coverage.

14. Repository Review Boundary

- This chapter is a repository-review specification. Exact executable implementation has not been inspected for every target behavior.
- Registered modules, catalog entries, README claims, frontend contracts, or plans are not proof of complete implementation.

Area	Classification
target aggregate, API, tables, state and workflow in this chapter	REFERENCE_ARCHITECTURE
registered module or catalog claim	DOCUMENTATION_ONLY_CLAIM until source inspection
uninspected integration	REQUIRES_REPOSITORY_REVIEW

Area	Classification
behavior proven in another repository-grounded chapter contradictory ownership or path claims	retain that chapter's evidence and classification CONFLICT_REQUIRES_CANONICALIZATION

No statement upgrades implementation status merely because a file, directory, module name, frontend contract, database entry, plan, or report exists.

15. Worked Conformance Scenario

An authorized principal initiates a Command against an exact subject version. The service validates identity, tenant, mandate, Policy, Evidence, and idempotency. It records the state transition and outbox entry atomically. If an external dependency participates, timeout produces Outcome Unknown; reconciliation queries the source or waits for a verified callback instead of repeating the action blindly.

Portals show source, freshness, version, required action, and uncertainty. AI may extract, compare, summarize, detect anomalies, and recommend through typed Tools, but cannot own the subject or fabricate success. Later correction appends new facts, identifies affected projections and workflows, and preserves the historical Decision context.

16. Conformance Checklist

- ☐ stable identity and proposition ownership are explicit;
- ☐ authority and mandate are evaluated at action time;
- ☐ state, Recommendation, Decision, Intent, execution, and outcome remain distinct;
- ☐ Commands are typed, authorized, idempotent, and version guarded;
- ☐ Events describe completed facts;
- ☐ Evidence includes provenance, time, integrity, qualification, and correction;
- ☐ external outcomes retain their authority source;
- ☐ Portals show freshness and uncertainty;
- ☐ AI is bounded by typed Tools, Policy, approval, and outcome observation;
- ☐ security, tenant, facility, channel, and jurisdiction boundaries are enforced;
- ☐ Outcome Unknown and reconciliation exist where needed;
- ☐ corrections preserve history;
- ☐ tests prove invariants and recovery;
- ☐ implementation status is not upgraded without executable evidence.

17. Status Vocabulary

Status	Meaning
VERIFIED_IMPLEMENTED	Executable source and relevant behavior were inspected.
IMPLEMENTED_WITH_GAP	Executable behavior exists with a material governance or completeness gap.
PARTIAL_IMPLEMENTATION	Only part of the target capability is evidenced.
REFERENCE_ARCHITECTURE	Normative target behavior; no claim that it exists today.
PLANNED_NOT_IMPLEMENTED	Explicitly planned but not evidenced as executable.
DOCUMENTATION_ONLY_CLAIM	Documentation or client contract claims behavior without verified backend evidence.
CONFLICT_REQUIRES_CANONICALIZATION	Sources disagree and no final owner has been established.
REQUIRES_REPOSITORY_REVIEW	Necessary executable evidence has not yet been inspected.

Conceptual Enterprise Data Model

Metadata	Value
Volume	10
Chapter ID	V10-01
Subject	ConceptualEnterpriseDataModel
Status	Generated First-Draft Architecture Skeleton - Domain-Specific Expansion Required
Content maturity	TEMPLATE_DERIVED_REFERENCE_SKELETON unless repository citations prove otherwise
Default implementation classification	REFERENCE_ARCHITECTURE / REQUIRES_REPOSITORY_REVIEW
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1. Executive Orientation

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The specification separates reality, observations, knowledge, Decisions, Intent, execution, and outcomes. A component name, database row, UI label, AI answer, or workflow state does not acquire authority merely because it is visible or technically convenient.

Reality or external outcome

- ≠ Observation
- ≠ Claim
- ≠ derived Perspective
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- ≠ Recommendation
- ≠ authorized Decision
- ≠ execution Intent

1.1 Accountable ownership

The accountable owner is **Data Architecture Council**. Supporting applications, shared infrastructure, integration adapters, analytics, and AI coordinate or present the capability but shall not silently own its authoritative propositions.

1.2 Scope

- create and identify the subject
- validate evidence and policy
- make authorized Decisions
- execute typed Commands
- publish completed-fact Events
- query current and historical perspectives
- correct and reconcile outcomes

1.3 Non-goals

- treating a registered Gradle module, catalog record, README, frontend route, migration plan, or generated report as proof of end-to-end implementation;
- direct cross-Domain database writes;

- generic status values that combine lifecycle, approval, provider, document, payment, dispute, and correction state;
- silent replacement of history;
- AI-created authority or fabricated external outcomes.

2. Ubiquitous Language and Subject Model

Subject	Required interpretation
ConceptualEnterpriseDataModel	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
ConceptualEnterpriseDataModelVersion	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
ConceptualEnterpriseDataModelDecision	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
ConceptualEnterpriseDataModelEvidenceSet	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
ConceptualEnterpriseDataModelException	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit

Reference aggregate:

```
ConceptualEnterpriseDataModel {
  subjectId
  tenantId
  organizationContext
  sourceIdentityRefs[]
  sourceVersionRefs[]
  lifecycleState
  relationshipRefs[]
  decisionRefs[]
  intentRefs[]
  evidenceSetRef
  knowledgeCutoff
  policyRefs[]
  authorityRef
  jurisdictionRefs[]
  createdAt
  effectiveAt
  updatedAt
  version
  correctionOf?
  correlationId
}
```

2.1 Core invariants

- Stable identity precedes relationship, state, and cross-system correlation.
- Effective time and recorded time remain distinct where business truth changes over time.
- Recommendation, Decision, Intent, execution, and outcome are separate concepts.
- No Portal, report, search index, cache, workflow, or AI Agent becomes a shadow owner of Domain truth.
- Binding submitted, approved, issued, accepted, or signed versions are immutable.
- Corrections append governed facts and preserve the original record.
- External authority and provider outcomes retain their source and qualification.
- Tenant and represented-principal context come from trusted authentication and mandate evaluation.
- Retried writes are idempotent and concurrency guarded.
- Outcome Unknown is explicit and reconciled before risky duplicate execution.

2.2 Required standards

- stable identifiers
- typed contracts
- immutable binding versions
- transactional outbox
- idempotent Commands
- explicit Outcome Unknown
- correction lineage
- machine-verifiable conformance

3. Actors, Roles, and Authority

A conformant design distinguishes:

- the natural person acting;
- the service or AI identity invoking a Tool;
- the organization or principal represented;
- the role or relationship granting context;
- the mandate or delegation granting action authority;
- the Policy and jurisdiction limiting the action;
- the reviewer or approver responsible for a binding Decision.

Authority is evaluated at action time. Authentication alone does not grant business authority. Cached permissions and browser-supplied tenant headers are insufficient for high-impact actions.

4. Lifecycle and State Model

State	Semantics
PROPOSED	Distinct governed condition with entry, exit, timeout, correction and authority semantics
VALIDATION_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
APPROVAL_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
ACTIVE	Distinct governed condition with entry, exit, timeout, correction and authority semantics
SUSPENDED	Distinct governed condition with entry, exit, timeout, correction and authority semantics
CORRECTION_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
RETIRED	Distinct governed condition with entry, exit, timeout, correction and authority semantics

```
stateDiagram-v2
    [*] --> PROPOSED
    PROPOSED --> VALIDATION_PENDING
    VALIDATION_PENDING --> APPROVAL_PENDING
    APPROVAL_PENDING --> ACTIVE
    ACTIVE --> SUSPENDED
    SUSPENDED --> CORRECTION_PENDING
    CORRECTION_PENDING --> RETIRED
```

The diagram is a primary reading path. Real implementations may add parallel review, amendment, appeal, timeout, suspension, partial completion, reconciliation, correction, and terminal failure without collapsing their meaning.

4.1 Transition rules

- transition through a typed, authorized Command;
 - require exact expected version where concurrent change matters;
 - record actor, represented principal, mandate, Policy, reason, and Evidence;
 - use an idempotency key for retried writes;
 - publish a completed-fact Event after durable state change;
 - expose Outcome Unknown for unresolved external completion;
 - preserve original and corrected states in historical reconstruction.
-

5. Decision, Evidence, and Knowledge Cutoff

Reference Decision:

```
Decision {
    decisionId
    decisionType
    subjectRef
}
```

```

sourceVersionRefs[]
knowledgeCutoff
evidenceSetRef
policyVersionRefs[]
recommendationRefs[]
principalId
representedOrganizationId
mandateRef
rationale
outcome
decidedAt
effectiveAt
correctionOf?
}

```

Evidence records identify issuer or observer, subject, version, observed time, received time, effective interval, integrity, confidence, qualification, contradiction, correction, retention, residency, and access class. A recommendation may assist the Decision but cannot promote itself to the Decision.

6. Commands, Events, Queries, and Contracts

Reference Commands:

- CreateConceptualEnterpriseDataModel
- ValidateConceptualEnterpriseDataModel
- ApproveConceptualEnterpriseDataModel
- ActivateConceptualEnterpriseDataModel
- SuspendConceptualEnterpriseDataModel
- CorrectConceptualEnterpriseDataModel
- RetireConceptualEnterpriseDataModel

Reference Events:

- ConceptualEnterpriseDataModelCreated
- ConceptualEnterpriseDataModelValidated
- ConceptualEnterpriseDataModelApproved
- ConceptualEnterpriseDataModelActivated
- ConceptualEnterpriseDataModelSuspended
- ConceptualEnterpriseDataModelCorrected
- ConceptualEnterpriseDataModelRetired

Reference queries:

- GetConceptualEnterpriseDataModel
- ListConceptualEnterpriseDataModelRecords
- GetConceptualEnterpriseDataModelTimeline
- GetConceptualEnterpriseDataModelEvidence
- GetConceptualEnterpriseDataModelDecisionPackage
- GetConceptualEnterpriseDataModelExceptions
- GetConceptualEnterpriseDataModelAuditTrail

Reference API surface:

```

POST /api/v1/conceptualenterprisedatamodel/commands
GET  /api/v1/conceptualenterprisedatamodel/{id}
GET  /api/v1/conceptualenterprisedatamodel/{id}/timeline
GET  /api/v1/conceptualenterprisedatamodel/{id}/evidence
GET  /api/v1/conceptualenterprisedatamodel/{id}/decisions
POST /api/v1/conceptualenterprisedatamodel/{id}/exceptions
POST /api/v1/conceptualenterprisedatamodel/{id}/corrections

```

These endpoints are reference contracts, not claims that exact routes exist.

7. Workflow, Failure, and Compensation

flowchart TD

```

A[Validate identity version and authority] --> B[Collect required evidence]
B --> C{Policy and evidence gates pass?}
C -- No --> D[Open exception or request correction]
D --> B
C -- Yes --> E[Record Decision or execution Intent]
E --> F[Execute Domain command or external request]

```

```

F --> G{Outcome known?}
G -- Success --> H[Record fact and publish Event]
G -- Failure --> I[Record failure and compensation options]
G -- Unknown --> J[Enter Outcome Unknown]
J --> K[Query source or wait for verified callback]
K --> G
H --> L[Update projections and downstream workflow]

```

Representative failure modes:

- stale or wrong source version
- duplicate submission or replay
- external timeout after possible success
- contradictory source observations
- lost Event after committed state
- approval or mandate revoked during execution
- downstream action before a mandatory gate
- partial completion across multiple subjects
- evidence later retracted or corrected
- tenant, facility, channel or jurisdiction mismatch

Recovery shall query the authoritative source before duplicate execution, preserve partial success, use compensating Intent rather than destructive rewrite, publish corrections, quarantine high-impact ambiguity, and record affected downstream subjects.

8. Data and Persistence Architecture

Reference table families:

Table family	Purpose
conceptualenterprisedatamodel_aggregate	authoritative subject
conceptualenterprisedatamodel_version	immutable submitted, approved, issued, or signed versions
conceptualenterprisedatamodel_relationship	governed relationships
conceptualenterprisedatamodel_decision	binding Decisions and rationale
conceptualenterprisedatamodel_evidence_link	provenance and evidence relationships
conceptualenterprisedatamodel_event_outbox	transactional Event publication
conceptualenterprisedatamodel_idempotency	duplicate-write protection
conceptualenterprisedatamodel_exception	exception, claim, dispute, or hold
conceptualenterprisedatamodel_correction	correction and supersession lineage
conceptualenterprisedatamodel_audit	privileged action evidence

The owning service controls schema and migration. Cross-Domain references use stable identifiers or contracts. Analytics, reports, caches, search, and vector indexes remain projections. Backup, restore, retention, legal hold, privacy, residency, and decommissioning requirements are explicit.

9. Portal and Experience Requirements

- Primary participant Portal: initiate, review exact versions, supply evidence, and respond to required actions.
- Operator Portal: inspect exceptions, correlation, source responses, retries, compensation, and downstream impact.
- Executive or oversight view: consume governed metrics without becoming a source of Domain truth.

Every Portal displays exact version, owner, state, freshness, required action, and uncertainty. It supports loading, empty, partial, stale, denied, error, timeout, and Outcome Unknown states; accessibility; locale, currency, unit, date, and RTL/LTR correctness; evidence audit; safe retry; and correction history.

10. AI Participation and Tool Governance

Permitted AI tasks:

- assist schema and lineage analysis
- detect data-quality anomalies
- summarize migration and reconciliation risk
- never mutate authoritative data outside typed Domain Tools

AI shall not own Domain records, grant itself authority, suppress contradictions, silently edit approved versions, fabricate external outcomes, or execute high-impact actions outside typed Tools, policy, approval, idempotency, sandbox, and outcome observation.

Every Agent Run records source context, prompt/model/tool versions, structured output, confidence and limitations, Policy result, approval, Tool calls, provider responses, final outcome, and recovery actions.

11. Security, Privacy, and Trust Boundaries

- least privilege and separation of duties
- tenant, organization, facility, channel and jurisdiction isolation
- integrity protection for binding evidence
- verified external callbacks and anti-replay controls

Additional controls include secret isolation, callback signature verification, audit of privileged action, emergency-access review, sensitive-data minimization, purpose limitation, and threat models for impersonation, tampering, replay, confused deputy, fraud, prompt injection, and exfiltration.

12. Reliability, SLOs, and Operations

SLI	Reference objective
authoritative read availability	99.9% monthly unless a stricter tier applies
acknowledged Command durability	no acknowledged write may be silently lost
Event publication	99.9% within five minutes after committed fact
duplicate prevention	100% for same tenant, Command class, and idempotency key
binding Decision audit completeness	100%
state-state detection	bounded by business deadline

Dashboards and runbooks cover backlog, error budget, dependency failure, retry budget, DLQ, workflow stall, provider outage, reconciliation, capacity, backup/restore, release/rollback, incident command, and post-incident correction.

13. Testing and Continuous Conformance

- aggregate invariant and transition tests
- authorization, delegation, separation-of-duties and tenant-isolation tests
- idempotency and optimistic-concurrency tests
- database migration, key, constraint and rollback tests
- contract, outbox/inbox, duplicate, replay and DLQ tests
- workflow restart, timeout and compensation tests
- Portal E2E tests for success, partial, stale, denied, error and Outcome Unknown states
- accessibility, locale, currency, unit and RTL/LTR tests
- security adversarial, fraud and data-exfiltration tests
- AI evaluation, prompt-injection, policy-bypass and unauthorized-tool tests
- historical reconstruction, correction and audit-completeness tests
- performance, capacity, recovery and operational exercise tests

Release evidence links each invariant and control to the test, environment, input data, result, owner, and artifact proving it. Generated test counts never replace semantic coverage.

14. Repository Review Boundary

- This chapter is a repository-review specification. Exact executable implementation has not been inspected for every target behavior.
- Registered modules, catalog entries, README claims, frontend contracts, or plans are not proof of complete implementation.

Area	Classification
target aggregate, API, tables, state and workflow in this chapter	REFERENCE_ARCHITECTURE
registered module or catalog claim	DOCUMENTATION_ONLY_CLAIM until source inspection
uninspected integration	REQUIRES_REPOSITORY_REVIEW
behavior proven in another repository-grounded chapter	retain that chapter's evidence and classification
contradictory ownership or path claims	CONFLICT_REQUIRES_CANONICALIZATION

No statement upgrades implementation status merely because a file, directory, module name, frontend contract, database entry, plan, or report exists.

15. Worked Conformance Scenario

An authorized principal initiates a Command against an exact subject version. The service validates identity, tenant, mandate, Policy, Evidence, and idempotency. It records the state transition and outbox entry atomically. If an external dependency participates, timeout produces Outcome Unknown; reconciliation queries the source or waits for a verified callback instead of repeating the action blindly.

Portals show source, freshness, version, required action, and uncertainty. AI may extract, compare, summarize, detect anomalies, and recommend through typed Tools, but cannot own the subject or fabricate success. Later correction appends new facts, identifies affected projections and workflows, and preserves the historical Decision context.

16. Conformance Checklist

- ☐ stable identity and proposition ownership are explicit;
- ☐ authority and mandate are evaluated at action time;
- ☐ state, Recommendation, Decision, Intent, execution, and outcome remain distinct;
- ☐ Commands are typed, authorized, idempotent, and version guarded;
- ☐ Events describe completed facts;
- ☐ Evidence includes provenance, time, integrity, qualification, and correction;
- ☐ external outcomes retain their authority source;
- ☐ Portals show freshness and uncertainty;
- ☐ AI is bounded by typed Tools, Policy, approval, and outcome observation;
- ☐ security, tenant, facility, channel, and jurisdiction boundaries are enforced;
- ☐ Outcome Unknown and reconciliation exist where needed;
- ☐ corrections preserve history;
- ☐ tests prove invariants and recovery;
- ☐ implementation status is not upgraded without executable evidence.

17. Status Vocabulary

Status	Meaning
VERIFIED_IMPLEMENTED	Executable source and relevant behavior were inspected.
IMPLEMENTED_WITH_GAP	Executable behavior exists with a material governance or completeness gap.
PARTIAL_IMPLEMENTATION	Only part of the target capability is evidenced.
REFERENCE_ARCHITECTURE	Normative target behavior; no claim that it exists today.
PLANNED_NOT_IMPLEMENTED	Explicitly planned but not evidenced as executable.
DOCUMENTATION_ONLY_CLAIM	Documentation or client contract claims behavior without verified backend evidence.
CONFLICT_REQUIRES_CANONICALIZATION	Sources disagree and no final owner has been established.
REQUIRES_REPOSITORY_REVIEW	Necessary executable evidence has not yet been inspected.

Domain Logical Data Models

Metadata	Value
Volume	10
Chapter ID	V10-02
Subject	DomainLogicalDataModels
Status	Generated First-Draft Architecture Skeleton - Domain-Specific Expansion Required
Content maturity	TEMPLATE_DERIVED_REFERENCE_SKELETON unless repository citations prove otherwise
Default implementation classification	REFERENCE_ARCHITECTURE / REQUIRES_REPOSITORY_REVIEW
Accountable owner	Data Architecture Council
Evidence rule	No implementation claim without inspected executable evidence
Repository	chinair88-prog/allinb2c

1. Executive Orientation

This chapter defines how GTOS shall **define enterprise and Domain data ownership, semantics, persistence, quality, lineage, correction, protection, operations, and verification for Domain Logical Data Models.**

The specification separates reality, observations, knowledge, Decisions, Intent, execution, and outcomes. A component name, database row, UI label, AI answer, or workflow state does not acquire authority merely because it is visible or technically convenient.

Reality or external outcome

- ≠ Observation
- ≠ Claim
- ≠ derived Perspective
- ≠ Prediction
- ≠ Recommendation
- ≠ authorized Decision
- ≠ execution Intent

1.1 Accountable ownership

The accountable owner is **Data Architecture Council**. Supporting applications, shared infrastructure, integration adapters, analytics, and AI coordinate or present the capability but shall not silently own its authoritative propositions.

1.2 Scope

- create and identify the subject
- validate evidence and policy
- make authorized Decisions
- execute typed Commands
- publish completed-fact Events
- query current and historical perspectives
- correct and reconcile outcomes

1.3 Non-goals

- treating a registered Gradle module, catalog record, README, frontend route, migration plan, or generated report as proof of end-to-end implementation;
- direct cross-Domain database writes;

- generic status values that combine lifecycle, approval, provider, document, payment, dispute, and correction state;
- silent replacement of history;
- AI-created authority or fabricated external outcomes.

2. Ubiquitous Language and Subject Model

Subject	Required interpretation
DomainLogicalDataModels	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
DomainLogicalDataModelsVersion	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
DomainLogicalDataModelsDecision	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
DomainLogicalDataModelsEvidenceSet	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
DomainLogicalDataModelsException	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit

Reference aggregate:

```
DomainLogicalDataModels {
  subjectId
  tenantId
  organizationContext
  sourceIdentityRefs[]
  sourceVersionRefs[]
  lifecycleState
  relationshipRefs[]
  decisionRefs[]
  intentRefs[]
  evidenceSetRef
  knowledgeCutoff
  policyRefs[]
  authorityRef
  jurisdictionRefs[]
  createdAt
  effectiveAt
  updatedAt
  version
  correctionOf?
  correlationId
}
```

2.1 Core invariants

- Stable identity precedes relationship, state, and cross-system correlation.
- Effective time and recorded time remain distinct where business truth changes over time.
- Recommendation, Decision, Intent, execution, and outcome are separate concepts.
- No Portal, report, search index, cache, workflow, or AI Agent becomes a shadow owner of Domain truth.
- Binding submitted, approved, issued, accepted, or signed versions are immutable.
- Corrections append governed facts and preserve the original record.
- External authority and provider outcomes retain their source and qualification.
- Tenant and represented-principal context come from trusted authentication and mandate evaluation.
- Retried writes are idempotent and concurrency guarded.
- Outcome Unknown is explicit and reconciled before risky duplicate execution.

2.2 Required standards

- stable identifiers
- typed contracts
- immutable binding versions
- transactional outbox
- idempotent Commands
- explicit Outcome Unknown
- correction lineage
- machine-verifiable conformance

3. Actors, Roles, and Authority

A conformant design distinguishes:

- the natural person acting;
- the service or AI identity invoking a Tool;
- the organization or principal represented;
- the role or relationship granting context;
- the mandate or delegation granting action authority;
- the Policy and jurisdiction limiting the action;
- the reviewer or approver responsible for a binding Decision.

Authority is evaluated at action time. Authentication alone does not grant business authority. Cached permissions and browser-supplied tenant headers are insufficient for high-impact actions.

4. Lifecycle and State Model

State	Semantics
PROPOSED	Distinct governed condition with entry, exit, timeout, correction and authority semantics
VALIDATION_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
APPROVAL_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
ACTIVE	Distinct governed condition with entry, exit, timeout, correction and authority semantics
SUSPENDED	Distinct governed condition with entry, exit, timeout, correction and authority semantics
CORRECTION_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
RETIRED	Distinct governed condition with entry, exit, timeout, correction and authority semantics

```
stateDiagram-v2
    [*] --> PROPOSED
    PROPOSED --> VALIDATION_PENDING
    VALIDATION_PENDING --> APPROVAL_PENDING
    APPROVAL_PENDING --> ACTIVE
    ACTIVE --> SUSPENDED
    SUSPENDED --> CORRECTION_PENDING
    CORRECTION_PENDING --> RETIRED
```

The diagram is a primary reading path. Real implementations may add parallel review, amendment, appeal, timeout, suspension, partial completion, reconciliation, correction, and terminal failure without collapsing their meaning.

4.1 Transition rules

- transition through a typed, authorized Command;
 - require exact expected version where concurrent change matters;
 - record actor, represented principal, mandate, Policy, reason, and Evidence;
 - use an idempotency key for retried writes;
 - publish a completed-fact Event after durable state change;
 - expose Outcome Unknown for unresolved external completion;
 - preserve original and corrected states in historical reconstruction.
-

5. Decision, Evidence, and Knowledge Cutoff

Reference Decision:

```
Decision {
    decisionId
    decisionType
    subjectRef
}
```

```

sourceVersionRefs[]
knowledgeCutoff
evidenceSetRef
policyVersionRefs[]
recommendationRefs[]
principalId
representedOrganizationId
mandateRef
rationale
outcome
decidedAt
effectiveAt
correctionOf?
}

```

Evidence records identify issuer or observer, subject, version, observed time, received time, effective interval, integrity, confidence, qualification, contradiction, correction, retention, residency, and access class. A recommendation may assist the Decision but cannot promote itself to the Decision.

6. Commands, Events, Queries, and Contracts

Reference Commands:

- CreateDomainLogicalDataModels
- ValidateDomainLogicalDataModels
- ApproveDomainLogicalDataModels
- ActivateDomainLogicalDataModels
- SuspendDomainLogicalDataModels
- CorrectDomainLogicalDataModels
- RetireDomainLogicalDataModels

Reference Events:

- DomainLogicalDataModelsCreated
- DomainLogicalDataModelsValidated
- DomainLogicalDataModelsApproved
- DomainLogicalDataModelsActivated
- DomainLogicalDataModelsSuspended
- DomainLogicalDataModelsCorrected
- DomainLogicalDataModelsRetired

Reference queries:

- GetDomainLogicalDataModels
- ListDomainLogicalDataModelsRecords
- GetDomainLogicalDataModelsTimeline
- GetDomainLogicalDataModelsEvidence
- GetDomainLogicalDataModelsDecisionPackage
- GetDomainLogicalDataModelsExceptions
- GetDomainLogicalDataModelsAuditTrail

Reference API surface:

```

POST /api/v1/domainlogicaldatamodels/commands
GET  /api/v1/domainlogicaldatamodels/{id}
GET  /api/v1/domainlogicaldatamodels/{id}/timeline
GET  /api/v1/domainlogicaldatamodels/{id}/evidence
GET  /api/v1/domainlogicaldatamodels/{id}/decisions
POST /api/v1/domainlogicaldatamodels/{id}/exceptions
POST /api/v1/domainlogicaldatamodels/{id}/corrections

```

These endpoints are reference contracts, not claims that exact routes exist.

7. Workflow, Failure, and Compensation

flowchart TD

```

A[Validate identity version and authority] --> B[Collect required evidence]
B --> C{Policy and evidence gates pass?}
C -- No --> D[Open exception or request correction]
D --> B
C -- Yes --> E[Record Decision or execution Intent]
E --> F[Execute Domain command or external request]

```

```

F --> G{Outcome known?}
G -- Success --> H[Record fact and publish Event]
G -- Failure --> I[Record failure and compensation options]
G -- Unknown --> J[Enter Outcome Unknown]
J --> K[Query source or wait for verified callback]
K --> G
H --> L[Update projections and downstream workflow]

```

Representative failure modes:

- stale or wrong source version
- duplicate submission or replay
- external timeout after possible success
- contradictory source observations
- lost Event after committed state
- approval or mandate revoked during execution
- downstream action before a mandatory gate
- partial completion across multiple subjects
- evidence later retracted or corrected
- tenant, facility, channel or jurisdiction mismatch

Recovery shall query the authoritative source before duplicate execution, preserve partial success, use compensating Intent rather than destructive rewrite, publish corrections, quarantine high-impact ambiguity, and record affected downstream subjects.

8. Data and Persistence Architecture

Reference table families:

Table family	Purpose
domainlogicaldatamodels_aggregate	authoritative subject
domainlogicaldatamodels_version	immutable submitted, approved, issued, or signed versions
domainlogicaldatamodels_relationship	governed relationships
domainlogicaldatamodels_decision	binding Decisions and rationale
domainlogicaldatamodels_evidence_link	provenance and evidence relationships
domainlogicaldatamodels_event_outbox	transactional Event publication
domainlogicaldatamodels_idempotency	duplicate-write protection
domainlogicaldatamodels_exception	exception, claim, dispute, or hold
domainlogicaldatamodels_correction	correction and supersession lineage
domainlogicaldatamodels_audit	privileged action evidence

The owning service controls schema and migration. Cross-Domain references use stable identifiers or contracts. Analytics, reports, caches, search, and vector indexes remain projections. Backup, restore, retention, legal hold, privacy, residency, and decommissioning requirements are explicit.

9. Portal and Experience Requirements

- Primary participant Portal: initiate, review exact versions, supply evidence, and respond to required actions.
- Operator Portal: inspect exceptions, correlation, source responses, retries, compensation, and downstream impact.
- Executive or oversight view: consume governed metrics without becoming a source of Domain truth.

Every Portal displays exact version, owner, state, freshness, required action, and uncertainty. It supports loading, empty, partial, stale, denied, error, timeout, and Outcome Unknown states; accessibility; locale, currency, unit, date, and RTL/LTR correctness; evidence audit; safe retry; and correction history.

10. AI Participation and Tool Governance

Permitted AI tasks:

- assist schema and lineage analysis
- detect data-quality anomalies
- summarize migration and reconciliation risk
- never mutate authoritative data outside typed Domain Tools

AI shall not own Domain records, grant itself authority, suppress contradictions, silently edit approved versions, fabricate external outcomes, or execute high-impact actions outside typed Tools, policy, approval, idempotency, sandbox, and outcome observation.

Every Agent Run records source context, prompt/model/tool versions, structured output, confidence and limitations, Policy result, approval, Tool calls, provider responses, final outcome, and recovery actions.

11. Security, Privacy, and Trust Boundaries

- least privilege and separation of duties
- tenant, organization, facility, channel and jurisdiction isolation
- integrity protection for binding evidence
- verified external callbacks and anti-replay controls

Additional controls include secret isolation, callback signature verification, audit of privileged action, emergency-access review, sensitive-data minimization, purpose limitation, and threat models for impersonation, tampering, replay, confused deputy, fraud, prompt injection, and exfiltration.

12. Reliability, SLOs, and Operations

SLI	Reference objective
authoritative read availability	99.9% monthly unless a stricter tier applies
acknowledged Command durability	no acknowledged write may be silently lost
Event publication	99.9% within five minutes after committed fact
duplicate prevention	100% for same tenant, Command class, and idempotency key
binding Decision audit completeness	100%
state-state detection	bounded by business deadline

Dashboards and runbooks cover backlog, error budget, dependency failure, retry budget, DLQ, workflow stall, provider outage, reconciliation, capacity, backup/restore, release/rollback, incident command, and post-incident correction.

13. Testing and Continuous Conformance

- aggregate invariant and transition tests
- authorization, delegation, separation-of-duties and tenant-isolation tests
- idempotency and optimistic-concurrency tests
- database migration, key, constraint and rollback tests
- contract, outbox/inbox, duplicate, replay and DLQ tests
- workflow restart, timeout and compensation tests
- Portal E2E tests for success, partial, stale, denied, error and Outcome Unknown states
- accessibility, locale, currency, unit and RTL/LTR tests
- security adversarial, fraud and data-exfiltration tests
- AI evaluation, prompt-injection, policy-bypass and unauthorized-tool tests
- historical reconstruction, correction and audit-completeness tests
- performance, capacity, recovery and operational exercise tests

Release evidence links each invariant and control to the test, environment, input data, result, owner, and artifact proving it. Generated test counts never replace semantic coverage.

14. Repository Review Boundary

- This chapter is a repository-review specification. Exact executable implementation has not been inspected for every target behavior.
- Registered modules, catalog entries, README claims, frontend contracts, or plans are not proof of complete implementation.

Area	Classification
target aggregate, API, tables, state and workflow in this chapter	REFERENCE_ARCHITECTURE
registered module or catalog claim	DOCUMENTATION_ONLY_CLAIM until source inspection
uninspected integration	REQUIRES_REPOSITORY_REVIEW
behavior proven in another repository-grounded chapter	retain that chapter's evidence and classification
contradictory ownership or path claims	CONFLICT_REQUIRES_CANONICALIZATION

No statement upgrades implementation status merely because a file, directory, module name, frontend contract, database entry, plan, or report exists.

15. Worked Conformance Scenario

An authorized principal initiates a Command against an exact subject version. The service validates identity, tenant, mandate, Policy, Evidence, and idempotency. It records the state transition and outbox entry atomically. If an external dependency participates, timeout produces Outcome Unknown; reconciliation queries the source or waits for a verified callback instead of repeating the action blindly.

Portals show source, freshness, version, required action, and uncertainty. AI may extract, compare, summarize, detect anomalies, and recommend through typed Tools, but cannot own the subject or fabricate success. Later correction appends new facts, identifies affected projections and workflows, and preserves the historical Decision context.

16. Conformance Checklist

- ☐ stable identity and proposition ownership are explicit;
- ☐ authority and mandate are evaluated at action time;
- ☐ state, Recommendation, Decision, Intent, execution, and outcome remain distinct;
- ☐ Commands are typed, authorized, idempotent, and version guarded;
- ☐ Events describe completed facts;
- ☐ Evidence includes provenance, time, integrity, qualification, and correction;
- ☐ external outcomes retain their authority source;
- ☐ Portals show freshness and uncertainty;
- ☐ AI is bounded by typed Tools, Policy, approval, and outcome observation;
- ☐ security, tenant, facility, channel, and jurisdiction boundaries are enforced;
- ☐ Outcome Unknown and reconciliation exist where needed;
- ☐ corrections preserve history;
- ☐ tests prove invariants and recovery;
- ☐ implementation status is not upgraded without executable evidence.

17. Status Vocabulary

Status	Meaning
VERIFIED_IMPLEMENTED	Executable source and relevant behavior were inspected.
IMPLEMENTED_WITH_GAP	Executable behavior exists with a material governance or completeness gap.
PARTIAL_IMPLEMENTATION	Only part of the target capability is evidenced.
REFERENCE_ARCHITECTURE	Normative target behavior; no claim that it exists today.
PLANNED_NOT_IMPLEMENTED	Explicitly planned but not evidenced as executable.
DOCUMENTATION_ONLY_CLAIM	Documentation or client contract claims behavior without verified backend evidence.
CONFLICT_REQUIRES_CANONICALIZATION	Sources disagree and no final owner has been established.
REQUIRES_REPOSITORY_REVIEW	Necessary executable evidence has not yet been inspected.

Database-per-Service Policy

Metadata	Value
Volume	10
Chapter ID	V10-03
Subject	Database-per-ServicePolicy
Status	Generated First-Draft Architecture Skeleton - Domain-Specific Expansion Required
Content maturity	TEMPLATE_DERIVED_REFERENCE_SKELETON unless repository citations prove otherwise
Default implementation classification	REFERENCE_ARCHITECTURE / REQUIRES_REPOSITORY_REVIEW
Accountable owner	Data Architecture Council
Evidence rule	No implementation claim without inspected executable evidence
Repository	chinair88-prog/allinb2c

1. Executive Orientation

This chapter defines how GTOS shall **define enterprise and Domain data ownership, semantics, persistence, quality, lineage, correction, protection, operations, and verification for Database-per-Service Policy.**

The specification separates reality, observations, knowledge, Decisions, Intent, execution, and outcomes. A component name, database row, UI label, AI answer, or workflow state does not acquire authority merely because it is visible or technically convenient.

Reality or external outcome

- ≠ Observation
- ≠ Claim
- ≠ derived Perspective
- ≠ Prediction
- ≠ Recommendation
- ≠ authorized Decision
- ≠ execution Intent

1.1 Accountable ownership

The accountable owner is **Data Architecture Council**. Supporting applications, shared infrastructure, integration adapters, analytics, and AI coordinate or present the capability but shall not silently own its authoritative propositions.

1.2 Scope

- create and identify the subject
- validate evidence and policy
- make authorized Decisions
- execute typed Commands
- publish completed-fact Events
- query current and historical perspectives
- correct and reconcile outcomes

1.3 Non-goals

- treating a registered Gradle module, catalog record, README, frontend route, migration plan, or generated report as proof of end-to-end implementation;
- direct cross-Domain database writes;

- generic status values that combine lifecycle, approval, provider, document, payment, dispute, and correction state;
- silent replacement of history;
- AI-created authority or fabricated external outcomes.

2. Ubiquitous Language and Subject Model

Subject	Required interpretation
Database-per-ServicePolicy	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
Database-per-ServicePolicyVersion	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
Database-per-ServicePolicyDecision	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
Database-per-ServicePolicyEvidenceSet	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
Database-per-ServicePolicyException	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit

Reference aggregate:

```
Database-per-ServicePolicy {
  subjectId
  tenantId
  organizationContext
  sourceIdentityRefs[]
  sourceVersionRefs[]
  lifecycleState
  relationshipRefs[]
  decisionRefs[]
  intentRefs[]
  evidenceSetRef
  knowledgeCutoff
  policyRefs[]
  authorityRef
  jurisdictionRefs[]
  createdAt
  effectiveAt
  updatedAt
  version
  correctionOf?
  correlationId
}
```

2.1 Core invariants

- Stable identity precedes relationship, state, and cross-system correlation.
- Effective time and recorded time remain distinct where business truth changes over time.
- Recommendation, Decision, Intent, execution, and outcome are separate concepts.
- No Portal, report, search index, cache, workflow, or AI Agent becomes a shadow owner of Domain truth.
- Binding submitted, approved, issued, accepted, or signed versions are immutable.
- Corrections append governed facts and preserve the original record.
- External authority and provider outcomes retain their source and qualification.
- Tenant and represented-principal context come from trusted authentication and mandate evaluation.
- Retried writes are idempotent and concurrency guarded.
- Outcome Unknown is explicit and reconciled before risky duplicate execution.

2.2 Required standards

- stable identifiers
- typed contracts
- immutable binding versions
- transactional outbox
- idempotent Commands
- explicit Outcome Unknown
- correction lineage
- machine-verifiable conformance

3. Actors, Roles, and Authority

A conformant design distinguishes:

- the natural person acting;
- the service or AI identity invoking a Tool;
- the organization or principal represented;
- the role or relationship granting context;
- the mandate or delegation granting action authority;
- the Policy and jurisdiction limiting the action;
- the reviewer or approver responsible for a binding Decision.

Authority is evaluated at action time. Authentication alone does not grant business authority. Cached permissions and browser-supplied tenant headers are insufficient for high-impact actions.

4. Lifecycle and State Model

State	Semantics
PROPOSED	Distinct governed condition with entry, exit, timeout, correction and authority semantics
VALIDATION_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
APPROVAL_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
ACTIVE	Distinct governed condition with entry, exit, timeout, correction and authority semantics
SUSPENDED	Distinct governed condition with entry, exit, timeout, correction and authority semantics
CORRECTION_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
RETIRED	Distinct governed condition with entry, exit, timeout, correction and authority semantics

```
stateDiagram-v2
    [*] --> PROPOSED
    PROPOSED --> VALIDATION_PENDING
    VALIDATION_PENDING --> APPROVAL_PENDING
    APPROVAL_PENDING --> ACTIVE
    ACTIVE --> SUSPENDED
    SUSPENDED --> CORRECTION_PENDING
    CORRECTION_PENDING --> RETIRED
```

The diagram is a primary reading path. Real implementations may add parallel review, amendment, appeal, timeout, suspension, partial completion, reconciliation, correction, and terminal failure without collapsing their meaning.

4.1 Transition rules

- transition through a typed, authorized Command;
 - require exact expected version where concurrent change matters;
 - record actor, represented principal, mandate, Policy, reason, and Evidence;
 - use an idempotency key for retried writes;
 - publish a completed-fact Event after durable state change;
 - expose Outcome Unknown for unresolved external completion;
 - preserve original and corrected states in historical reconstruction.
-

5. Decision, Evidence, and Knowledge Cutoff

Reference Decision:

```
Decision {
    decisionId
    decisionType
    subjectRef
}
```

```

sourceVersionRefs[]
knowledgeCutoff
evidenceSetRef
policyVersionRefs[]
recommendationRefs[]
principalId
representedOrganizationId
mandateRef
rationale
outcome
decidedAt
effectiveAt
correctionOf?
}

```

Evidence records identify issuer or observer, subject, version, observed time, received time, effective interval, integrity, confidence, qualification, contradiction, correction, retention, residency, and access class. A recommendation may assist the Decision but cannot promote itself to the Decision.

6. Commands, Events, Queries, and Contracts

Reference Commands:

- CreateDatabase-per-ServicePolicy
- ValidateDatabase-per-ServicePolicy
- ApproveDatabase-per-ServicePolicy
- ActivateDatabase-per-ServicePolicy
- SuspendDatabase-per-ServicePolicy
- CorrectDatabase-per-ServicePolicy
- RetireDatabase-per-ServicePolicy

Reference Events:

- Database-per-ServicePolicyCreated
- Database-per-ServicePolicyValidated
- Database-per-ServicePolicyApproved
- Database-per-ServicePolicyActivated
- Database-per-ServicePolicySuspended
- Database-per-ServicePolicyCorrected
- Database-per-ServicePolicyRetired

Reference queries:

- GetDatabase-per-ServicePolicy
- ListDatabase-per-ServicePolicyRecords
- GetDatabase-per-ServicePolicyTimeline
- GetDatabase-per-ServicePolicyEvidence
- GetDatabase-per-ServicePolicyDecisionPackage
- GetDatabase-per-ServicePolicyExceptions
- GetDatabase-per-ServicePolicyAuditTrail

Reference API surface:

```

POST /api/v1/database_per_servicepolicy/commands
GET  /api/v1/database_per_servicepolicy/{id}
GET  /api/v1/database_per_servicepolicy/{id}/timeline
GET  /api/v1/database_per_servicepolicy/{id}/evidence
GET  /api/v1/database_per_servicepolicy/{id}/decisions
POST /api/v1/database_per_servicepolicy/{id}/exceptions
POST /api/v1/database_per_servicepolicy/{id}/corrections

```

These endpoints are reference contracts, not claims that exact routes exist.

7. Workflow, Failure, and Compensation

flowchart TD

```

A[Validate identity version and authority] --> B[Collect required evidence]
B --> C{Policy and evidence gates pass?}
C -- No --> D[Open exception or request correction]
D --> B
C -- Yes --> E[Record Decision or execution Intent]
E --> F[Execute Domain command or external request]

```

```

F --> G{Outcome known?}
G -- Success --> H[Record fact and publish Event]
G -- Failure --> I[Record failure and compensation options]
G -- Unknown --> J[Enter Outcome Unknown]
J --> K[Query source or wait for verified callback]
K --> G
H --> L[Update projections and downstream workflow]

```

Representative failure modes:

- stale or wrong source version
- duplicate submission or replay
- external timeout after possible success
- contradictory source observations
- lost Event after committed state
- approval or mandate revoked during execution
- downstream action before a mandatory gate
- partial completion across multiple subjects
- evidence later retracted or corrected
- tenant, facility, channel or jurisdiction mismatch

Recovery shall query the authoritative source before duplicate execution, preserve partial success, use compensating Intent rather than destructive rewrite, publish corrections, quarantine high-impact ambiguity, and record affected downstream subjects.

8. Data and Persistence Architecture

Reference table families:

Table family	Purpose
database_per_servicepolicy_aggregate	authoritative subject
database_per_servicepolicy_version	immutable submitted, approved, issued, or signed versions
database_per_servicepolicy_relationship	governed relationships
database_per_servicepolicy_decision	binding Decisions and rationale
database_per_servicepolicy_evidence_link	provenance and evidence relationships
database_per_servicepolicy_event_outbox	transactional Event publication
database_per_servicepolicy_idempotency	duplicate-write protection
database_per_servicepolicy_exception	exception, claim, dispute, or hold
database_per_servicepolicy_correction	correction and supersession lineage
database_per_servicepolicy_audit	privileged action evidence

The owning service controls schema and migration. Cross-Domain references use stable identifiers or contracts. Analytics, reports, caches, search, and vector indexes remain projections. Backup, restore, retention, legal hold, privacy, residency, and decommissioning requirements are explicit.

9. Portal and Experience Requirements

- Primary participant Portal: initiate, review exact versions, supply evidence, and respond to required actions.
- Operator Portal: inspect exceptions, correlation, source responses, retries, compensation, and downstream impact.
- Executive or oversight view: consume governed metrics without becoming a source of Domain truth.

Every Portal displays exact version, owner, state, freshness, required action, and uncertainty. It supports loading, empty, partial, stale, denied, error, timeout, and Outcome Unknown states; accessibility; locale, currency, unit, date, and RTL/LTR correctness; evidence audit; safe retry; and correction history.

10. AI Participation and Tool Governance

Permitted AI tasks:

- assist schema and lineage analysis
- detect data-quality anomalies
- summarize migration and reconciliation risk
- never mutate authoritative data outside typed Domain Tools

AI shall not own Domain records, grant itself authority, suppress contradictions, silently edit approved versions, fabricate external outcomes, or execute high-impact actions outside typed Tools, policy, approval, idempotency, sandbox, and outcome observation.

Every Agent Run records source context, prompt/model/tool versions, structured output, confidence and limitations, Policy result, approval, Tool calls, provider responses, final outcome, and recovery actions.

11. Security, Privacy, and Trust Boundaries

- least privilege and separation of duties
- tenant, organization, facility, channel and jurisdiction isolation
- integrity protection for binding evidence
- verified external callbacks and anti-replay controls

Additional controls include secret isolation, callback signature verification, audit of privileged action, emergency-access review, sensitive-data minimization, purpose limitation, and threat models for impersonation, tampering, replay, confused deputy, fraud, prompt injection, and exfiltration.

12. Reliability, SLOs, and Operations

SLI	Reference objective
authoritative read availability	99.9% monthly unless a stricter tier applies
acknowledged Command durability	no acknowledged write may be silently lost
Event publication	99.9% within five minutes after committed fact
duplicate prevention	100% for same tenant, Command class, and idempotency key
binding Decision audit completeness	100%
state-state detection	bounded by business deadline

Dashboards and runbooks cover backlog, error budget, dependency failure, retry budget, DLQ, workflow stall, provider outage, reconciliation, capacity, backup/restore, release/rollback, incident command, and post-incident correction.

13. Testing and Continuous Conformance

- aggregate invariant and transition tests
- authorization, delegation, separation-of-duties and tenant-isolation tests
- idempotency and optimistic-concurrency tests
- database migration, key, constraint and rollback tests
- contract, outbox/inbox, duplicate, replay and DLQ tests
- workflow restart, timeout and compensation tests
- Portal E2E tests for success, partial, stale, denied, error and Outcome Unknown states
- accessibility, locale, currency, unit and RTL/LTR tests
- security adversarial, fraud and data-exfiltration tests
- AI evaluation, prompt-injection, policy-bypass and unauthorized-tool tests
- historical reconstruction, correction and audit-completeness tests
- performance, capacity, recovery and operational exercise tests

Release evidence links each invariant and control to the test, environment, input data, result, owner, and artifact proving it. Generated test counts never replace semantic coverage.

14. Repository Review Boundary

- This chapter is a repository-review specification. Exact executable implementation has not been inspected for every target behavior.
- Registered modules, catalog entries, README claims, frontend contracts, or plans are not proof of complete implementation.

Area	Classification
target aggregate, API, tables, state and workflow in this chapter	REFERENCE_ARCHITECTURE
registered module or catalog claim	DOCUMENTATION_ONLY_CLAIM until source inspection
uninspected integration	REQUIRES_REPOSITORY_REVIEW
behavior proven in another repository-grounded chapter	retain that chapter's evidence and classification
contradictory ownership or path claims	CONFLICT_REQUIRES_CANONICALIZATION

No statement upgrades implementation status merely because a file, directory, module name, frontend contract, database entry, plan, or report exists.

15. Worked Conformance Scenario

An authorized principal initiates a Command against an exact subject version. The service validates identity, tenant, mandate, Policy, Evidence, and idempotency. It records the state transition and outbox entry atomically. If an external dependency participates, timeout produces Outcome Unknown; reconciliation queries the source or waits for a verified callback instead of repeating the action blindly.

Portals show source, freshness, version, required action, and uncertainty. AI may extract, compare, summarize, detect anomalies, and recommend through typed Tools, but cannot own the subject or fabricate success. Later correction appends new facts, identifies affected projections and workflows, and preserves the historical Decision context.

16. Conformance Checklist

- ☐ stable identity and proposition ownership are explicit;
- ☐ authority and mandate are evaluated at action time;
- ☐ state, Recommendation, Decision, Intent, execution, and outcome remain distinct;
- ☐ Commands are typed, authorized, idempotent, and version guarded;
- ☐ Events describe completed facts;
- ☐ Evidence includes provenance, time, integrity, qualification, and correction;
- ☐ external outcomes retain their authority source;
- ☐ Portals show freshness and uncertainty;
- ☐ AI is bounded by typed Tools, Policy, approval, and outcome observation;
- ☐ security, tenant, facility, channel, and jurisdiction boundaries are enforced;
- ☐ Outcome Unknown and reconciliation exist where needed;
- ☐ corrections preserve history;
- ☐ tests prove invariants and recovery;
- ☐ implementation status is not upgraded without executable evidence.

17. Status Vocabulary

Status	Meaning
VERIFIED_IMPLEMENTED	Executable source and relevant behavior were inspected.
IMPLEMENTED_WITH_GAP	Executable behavior exists with a material governance or completeness gap.
PARTIAL_IMPLEMENTATION	Only part of the target capability is evidenced.
REFERENCE_ARCHITECTURE	Normative target behavior; no claim that it exists today.
PLANNED_NOT_IMPLEMENTED	Explicitly planned but not evidenced as executable.
DOCUMENTATION_ONLY_CLAIM	Documentation or client contract claims behavior without verified backend evidence.
CONFLICT_REQUIRES_CANONICALIZATION	Sources disagree and no final owner has been established.
REQUIRES_REPOSITORY_REVIEW	Necessary executable evidence has not yet been inspected.

Schema Ownership

Metadata	Value
Volume	10
Chapter ID	V10-04
Subject	SchemaOwnership
Status	Generated First-Draft Architecture Skeleton - Domain-Specific Expansion Required
Content maturity	TEMPLATE_DERIVED_REFERENCE_SKELETON unless repository citations prove otherwise
Default implementation classification	REFERENCE_ARCHITECTURE / REQUIRES_REPOSITORY_REVIEW
Accountable owner	Data Architecture Council
Evidence rule	No implementation claim without inspected executable evidence
Repository	chinair88-prog/allinb2c

1. Executive Orientation

This chapter defines how GTOS shall **define enterprise and Domain data ownership, semantics, persistence, quality, lineage, correction, protection, operations, and verification for Schema Ownership**.

The specification separates reality, observations, knowledge, Decisions, Intent, execution, and outcomes. A component name, database row, UI label, AI answer, or workflow state does not acquire authority merely because it is visible or technically convenient.

Reality or external outcome

- ≠ Observation
- ≠ Claim
- ≠ derived Perspective
- ≠ Prediction
- ≠ Recommendation
- ≠ authorized Decision
- ≠ execution Intent

1.1 Accountable ownership

The accountable owner is **Data Architecture Council**. Supporting applications, shared infrastructure, integration adapters, analytics, and AI coordinate or present the capability but shall not silently own its authoritative propositions.

1.2 Scope

- create and identify the subject
- validate evidence and policy
- make authorized Decisions
- execute typed Commands
- publish completed-fact Events
- query current and historical perspectives
- correct and reconcile outcomes

1.3 Non-goals

- treating a registered Gradle module, catalog record, README, frontend route, migration plan, or generated report as proof of end-to-end implementation;
- direct cross-Domain database writes;

- generic status values that combine lifecycle, approval, provider, document, payment, dispute, and correction state;
- silent replacement of history;
- AI-created authority or fabricated external outcomes.

2. Ubiquitous Language and Subject Model

Subject	Required interpretation
SchemaOwnership	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
SchemaOwnershipVersion	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
SchemaOwnershipDecision	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
SchemaOwnershipEvidenceSet	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
SchemaOwnershipException	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit

Reference aggregate:

```
SchemaOwnership {
  subjectId
  tenantId
  organizationContext
  sourceIdentityRefs[]
  sourceVersionRefs[]
  lifecycleState
  relationshipRefs[]
  decisionRefs[]
  intentRefs[]
  evidenceSetRef
  knowledgeCutoff
  policyRefs[]
  authorityRef
  jurisdictionRefs[]
  createdAt
  effectiveAt
  updatedAt
  version
  correctionOf?
  correlationId
}
```

2.1 Core invariants

- Stable identity precedes relationship, state, and cross-system correlation.
- Effective time and recorded time remain distinct where business truth changes over time.
- Recommendation, Decision, Intent, execution, and outcome are separate concepts.
- No Portal, report, search index, cache, workflow, or AI Agent becomes a shadow owner of Domain truth.
- Binding submitted, approved, issued, accepted, or signed versions are immutable.
- Corrections append governed facts and preserve the original record.
- External authority and provider outcomes retain their source and qualification.
- Tenant and represented-principal context come from trusted authentication and mandate evaluation.
- Retried writes are idempotent and concurrency guarded.
- Outcome Unknown is explicit and reconciled before risky duplicate execution.

2.2 Required standards

- stable identifiers
- typed contracts
- immutable binding versions
- transactional outbox
- idempotent Commands
- explicit Outcome Unknown
- correction lineage
- machine-verifiable conformance

3. Actors, Roles, and Authority

A conformant design distinguishes:

- the natural person acting;
- the service or AI identity invoking a Tool;
- the organization or principal represented;
- the role or relationship granting context;
- the mandate or delegation granting action authority;
- the Policy and jurisdiction limiting the action;
- the reviewer or approver responsible for a binding Decision.

Authority is evaluated at action time. Authentication alone does not grant business authority. Cached permissions and browser-supplied tenant headers are insufficient for high-impact actions.

4. Lifecycle and State Model

State	Semantics
PROPOSED	Distinct governed condition with entry, exit, timeout, correction and authority semantics
VALIDATION_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
APPROVAL_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
ACTIVE	Distinct governed condition with entry, exit, timeout, correction and authority semantics
SUSPENDED	Distinct governed condition with entry, exit, timeout, correction and authority semantics
CORRECTION_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
RETIRED	Distinct governed condition with entry, exit, timeout, correction and authority semantics

```
stateDiagram-v2
    [*] --> PROPOSED
    PROPOSED --> VALIDATION_PENDING
    VALIDATION_PENDING --> APPROVAL_PENDING
    APPROVAL_PENDING --> ACTIVE
    ACTIVE --> SUSPENDED
    SUSPENDED --> CORRECTION_PENDING
    CORRECTION_PENDING --> RETIRED
```

The diagram is a primary reading path. Real implementations may add parallel review, amendment, appeal, timeout, suspension, partial completion, reconciliation, correction, and terminal failure without collapsing their meaning.

4.1 Transition rules

- transition through a typed, authorized Command;
 - require exact expected version where concurrent change matters;
 - record actor, represented principal, mandate, Policy, reason, and Evidence;
 - use an idempotency key for retried writes;
 - publish a completed-fact Event after durable state change;
 - expose Outcome Unknown for unresolved external completion;
 - preserve original and corrected states in historical reconstruction.
-

5. Decision, Evidence, and Knowledge Cutoff

Reference Decision:

```
Decision {
    decisionId
    decisionType
    subjectRef
}
```

```

sourceVersionRefs[]
knowledgeCutoff
evidenceSetRef
policyVersionRefs[]
recommendationRefs[]
principalId
representedOrganizationId
mandateRef
rationale
outcome
decidedAt
effectiveAt
correctionOf?
}

```

Evidence records identify issuer or observer, subject, version, observed time, received time, effective interval, integrity, confidence, qualification, contradiction, correction, retention, residency, and access class. A recommendation may assist the Decision but cannot promote itself to the Decision.

6. Commands, Events, Queries, and Contracts

Reference Commands:

- CreateSchemaOwnership
- ValidateSchemaOwnership
- ApproveSchemaOwnership
- ActivateSchemaOwnership
- SuspendSchemaOwnership
- CorrectSchemaOwnership
- RetireSchemaOwnership

Reference Events:

- SchemaOwnershipCreated
- SchemaOwnershipValidated
- SchemaOwnershipApproved
- SchemaOwnershipActivated
- SchemaOwnershipSuspended
- SchemaOwnershipCorrected
- SchemaOwnershipRetired

Reference queries:

- GetSchemaOwnership
- ListSchemaOwnershipRecords
- GetSchemaOwnershipTimeline
- GetSchemaOwnershipEvidence
- GetSchemaOwnershipDecisionPackage
- GetSchemaOwnershipExceptions
- GetSchemaOwnershipAuditTrail

Reference API surface:

```

POST /api/v1/schemaownership/commands
GET  /api/v1/schemaownership/{id}
GET  /api/v1/schemaownership/{id}/timeline
GET  /api/v1/schemaownership/{id}/evidence
GET  /api/v1/schemaownership/{id}/decisions
POST /api/v1/schemaownership/{id}/exceptions
POST /api/v1/schemaownership/{id}/corrections

```

These endpoints are reference contracts, not claims that exact routes exist.

7. Workflow, Failure, and Compensation

flowchart TD

```

A[Validate identity version and authority] --> B[Collect required evidence]
B --> C{Policy and evidence gates pass?}
C -- No --> D[Open exception or request correction]
D --> B
C -- Yes --> E[Record Decision or execution Intent]
E --> F[Execute Domain command or external request]

```

```

F --> G{Outcome known?}
G -- Success --> H[Record fact and publish Event]
G -- Failure --> I[Record failure and compensation options]
G -- Unknown --> J[Enter Outcome Unknown]
J --> K[Query source or wait for verified callback]
K --> G
H --> L[Update projections and downstream workflow]

```

Representative failure modes:

- stale or wrong source version
- duplicate submission or replay
- external timeout after possible success
- contradictory source observations
- lost Event after committed state
- approval or mandate revoked during execution
- downstream action before a mandatory gate
- partial completion across multiple subjects
- evidence later retracted or corrected
- tenant, facility, channel or jurisdiction mismatch

Recovery shall query the authoritative source before duplicate execution, preserve partial success, use compensating Intent rather than destructive rewrite, publish corrections, quarantine high-impact ambiguity, and record affected downstream subjects.

8. Data and Persistence Architecture

Reference table families:

Table family	Purpose
schemaownership_aggregate	authoritative subject
schemaownership_version	immutable submitted, approved, issued, or signed versions
schemaownership_relationship	governed relationships
schemaownership_decision	binding Decisions and rationale
schemaownership_evidence_link	provenance and evidence relationships
schemaownership_event_outbox	transactional Event publication
schemaownership_idempotency	duplicate-write protection
schemaownership_exception	exception, claim, dispute, or hold
schemaownership_correction	correction and supersession lineage
schemaownership_audit	privileged action evidence

The owning service controls schema and migration. Cross-Domain references use stable identifiers or contracts. Analytics, reports, caches, search, and vector indexes remain projections. Backup, restore, retention, legal hold, privacy, residency, and decommissioning requirements are explicit.

9. Portal and Experience Requirements

- Primary participant Portal: initiate, review exact versions, supply evidence, and respond to required actions.
- Operator Portal: inspect exceptions, correlation, source responses, retries, compensation, and downstream impact.
- Executive or oversight view: consume governed metrics without becoming a source of Domain truth.

Every Portal displays exact version, owner, state, freshness, required action, and uncertainty. It supports loading, empty, partial, stale, denied, error, timeout, and Outcome Unknown states; accessibility; locale, currency, unit, date, and RTL/LTR correctness; evidence audit; safe retry; and correction history.

10. AI Participation and Tool Governance

Permitted AI tasks:

- assist schema and lineage analysis
- detect data-quality anomalies
- summarize migration and reconciliation risk
- never mutate authoritative data outside typed Domain Tools

AI shall not own Domain records, grant itself authority, suppress contradictions, silently edit approved versions, fabricate external outcomes, or execute high-impact actions outside typed Tools, policy, approval, idempotency, sandbox, and outcome observation.

Every Agent Run records source context, prompt/model/tool versions, structured output, confidence and limitations, Policy result, approval, Tool calls, provider responses, final outcome, and recovery actions.

11. Security, Privacy, and Trust Boundaries

- least privilege and separation of duties
- tenant, organization, facility, channel and jurisdiction isolation
- integrity protection for binding evidence
- verified external callbacks and anti-replay controls

Additional controls include secret isolation, callback signature verification, audit of privileged action, emergency-access review, sensitive-data minimization, purpose limitation, and threat models for impersonation, tampering, replay, confused deputy, fraud, prompt injection, and exfiltration.

12. Reliability, SLOs, and Operations

SLI	Reference objective
authoritative read availability	99.9% monthly unless a stricter tier applies
acknowledged Command durability	no acknowledged write may be silently lost
Event publication	99.9% within five minutes after committed fact
duplicate prevention	100% for same tenant, Command class, and idempotency key
binding Decision audit completeness	100%
stale-state detection	bounded by business deadline

Dashboards and runbooks cover backlog, error budget, dependency failure, retry budget, DLQ, workflow stall, provider outage, reconciliation, capacity, backup/restore, release/rollback, incident command, and post-incident correction.

13. Testing and Continuous Conformance

- aggregate invariant and transition tests
- authorization, delegation, separation-of-duties and tenant-isolation tests
- idempotency and optimistic-concurrency tests
- database migration, key, constraint and rollback tests
- contract, outbox/inbox, duplicate, replay and DLQ tests
- workflow restart, timeout and compensation tests
- Portal E2E tests for success, partial, stale, denied, error and Outcome Unknown states
- accessibility, locale, currency, unit and RTL/LTR tests
- security adversarial, fraud and data-exfiltration tests
- AI evaluation, prompt-injection, policy-bypass and unauthorized-tool tests
- historical reconstruction, correction and audit-completeness tests
- performance, capacity, recovery and operational exercise tests

Release evidence links each invariant and control to the test, environment, input data, result, owner, and artifact proving it. Generated test counts never replace semantic coverage.

14. Repository Review Boundary

- This chapter is a repository-review specification. Exact executable implementation has not been inspected for every target behavior.
- Registered modules, catalog entries, README claims, frontend contracts, or plans are not proof of complete implementation.

Area	Classification
target aggregate, API, tables, state and workflow in this chapter	REFERENCE_ARCHITECTURE
registered module or catalog claim	DOCUMENTATION_ONLY_CLAIM until source inspection
uninspected integration	REQUIRES_REPOSITORY_REVIEW
behavior proven in another repository-grounded chapter	retain that chapter's evidence and classification
contradictory ownership or path claims	CONFLICT_REQUIRES_CANONICALIZATION

No statement upgrades implementation status merely because a file, directory, module name, frontend contract, database entry, plan, or report exists.

15. Worked Conformance Scenario

An authorized principal initiates a Command against an exact subject version. The service validates identity, tenant, mandate, Policy, Evidence, and idempotency. It records the state transition and outbox entry atomically. If an external dependency participates, timeout produces Outcome Unknown; reconciliation queries the source or waits for a verified callback instead of repeating the action blindly.

Portals show source, freshness, version, required action, and uncertainty. AI may extract, compare, summarize, detect anomalies, and recommend through typed Tools, but cannot own the subject or fabricate success. Later correction appends new facts, identifies affected projections and workflows, and preserves the historical Decision context.

16. Conformance Checklist

- ☐ stable identity and proposition ownership are explicit;
- ☐ authority and mandate are evaluated at action time;
- ☐ state, Recommendation, Decision, Intent, execution, and outcome remain distinct;
- ☐ Commands are typed, authorized, idempotent, and version guarded;
- ☐ Events describe completed facts;
- ☐ Evidence includes provenance, time, integrity, qualification, and correction;
- ☐ external outcomes retain their authority source;
- ☐ Portals show freshness and uncertainty;
- ☐ AI is bounded by typed Tools, Policy, approval, and outcome observation;
- ☐ security, tenant, facility, channel, and jurisdiction boundaries are enforced;
- ☐ Outcome Unknown and reconciliation exist where needed;
- ☐ corrections preserve history;
- ☐ tests prove invariants and recovery;
- ☐ implementation status is not upgraded without executable evidence.

17. Status Vocabulary

Status	Meaning
VERIFIED_IMPLEMENTED	Executable source and relevant behavior were inspected.
IMPLEMENTED_WITH_GAP	Executable behavior exists with a material governance or completeness gap.
PARTIAL_IMPLEMENTATION	Only part of the target capability is evidenced.
REFERENCE_ARCHITECTURE	Normative target behavior; no claim that it exists today.
PLANNED_NOT_IMPLEMENTED	Explicitly planned but not evidenced as executable.
DOCUMENTATION_ONLY_CLAIM	Documentation or client contract claims behavior without verified backend evidence.
CONFLICT_REQUIRES_CANONICALIZATION	Sources disagree and no final owner has been established.
REQUIRES_REPOSITORY_REVIEW	Necessary executable evidence has not yet been inspected.

Database and Schema Registry

Metadata	Value
Volume	10
Chapter ID	V10-05
Subject	DatabaseandSchemaRegistry
Status	Generated First-Draft Architecture Skeleton - Domain-Specific Expansion Required
Content maturity	TEMPLATE_DERIVED_REFERENCE_SKELETON unless repository citations prove otherwise
Default implementation classification	REFERENCE_ARCHITECTURE / REQUIRES_REPOSITORY_REVIEW
Accountable owner	Data Architecture Council
Evidence rule	No implementation claim without inspected executable evidence
Repository	chinair88-prog/allinb2c

1. Executive Orientation

This chapter defines how GTOS shall **define enterprise and Domain data ownership, semantics, persistence, quality, lineage, correction, protection, operations, and verification for Database and Schema Registry.**

The specification separates reality, observations, knowledge, Decisions, Intent, execution, and outcomes. A component name, database row, UI label, AI answer, or workflow state does not acquire authority merely because it is visible or technically convenient.

Reality or external outcome

- ≠ Observation
- ≠ Claim
- ≠ derived Perspective
- ≠ Prediction
- ≠ Recommendation
- ≠ authorized Decision
- ≠ execution Intent

1.1 Accountable ownership

The accountable owner is **Data Architecture Council**. Supporting applications, shared infrastructure, integration adapters, analytics, and AI coordinate or present the capability but shall not silently own its authoritative propositions.

1.2 Scope

- create and identify the subject
- validate evidence and policy
- make authorized Decisions
- execute typed Commands
- publish completed-fact Events
- query current and historical perspectives
- correct and reconcile outcomes

1.3 Non-goals

- treating a registered Gradle module, catalog record, README, frontend route, migration plan, or generated report as proof of end-to-end implementation;
- direct cross-Domain database writes;

- generic status values that combine lifecycle, approval, provider, document, payment, dispute, and correction state;
- silent replacement of history;
- AI-created authority or fabricated external outcomes.

2. Ubiquitous Language and Subject Model

Subject	Required interpretation
DatabaseandSchemaRegistry	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
DatabaseandSchemaRegistryVersion	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
DatabaseandSchemaRegistryDecision	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
DatabaseandSchemaRegistryEvidenceSet	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
DatabaseandSchemaRegistryException	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit

Reference aggregate:

```
DatabaseandSchemaRegistry {
  subjectId
  tenantId
  organizationContext
  sourceIdentityRefs[]
  sourceVersionRefs[]
  lifecycleState
  relationshipRefs[]
  decisionRefs[]
  intentRefs[]
  evidenceSetRef
  knowledgeCutoff
  policyRefs[]
  authorityRef
  jurisdictionRefs[]
  createdAt
  effectiveAt
  updatedAt
  version
  correctionOf?
  correlationId
}
```

2.1 Core invariants

- Stable identity precedes relationship, state, and cross-system correlation.
- Effective time and recorded time remain distinct where business truth changes over time.
- Recommendation, Decision, Intent, execution, and outcome are separate concepts.
- No Portal, report, search index, cache, workflow, or AI Agent becomes a shadow owner of Domain truth.
- Binding submitted, approved, issued, accepted, or signed versions are immutable.
- Corrections append governed facts and preserve the original record.
- External authority and provider outcomes retain their source and qualification.
- Tenant and represented-principal context come from trusted authentication and mandate evaluation.
- Retried writes are idempotent and concurrency guarded.
- Outcome Unknown is explicit and reconciled before risky duplicate execution.

2.2 Required standards

- stable identifiers
- typed contracts
- immutable binding versions
- transactional outbox
- idempotent Commands
- explicit Outcome Unknown
- correction lineage
- machine-verifiable conformance

3. Actors, Roles, and Authority

A conformant design distinguishes:

- the natural person acting;
- the service or AI identity invoking a Tool;
- the organization or principal represented;
- the role or relationship granting context;
- the mandate or delegation granting action authority;
- the Policy and jurisdiction limiting the action;
- the reviewer or approver responsible for a binding Decision.

Authority is evaluated at action time. Authentication alone does not grant business authority. Cached permissions and browser-supplied tenant headers are insufficient for high-impact actions.

4. Lifecycle and State Model

State	Semantics
PROPOSED	Distinct governed condition with entry, exit, timeout, correction and authority semantics
VALIDATION_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
APPROVAL_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
ACTIVE	Distinct governed condition with entry, exit, timeout, correction and authority semantics
SUSPENDED	Distinct governed condition with entry, exit, timeout, correction and authority semantics
CORRECTION_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
RETIRED	Distinct governed condition with entry, exit, timeout, correction and authority semantics

```
stateDiagram-v2
    [*] --> PROPOSED
    PROPOSED --> VALIDATION_PENDING
    VALIDATION_PENDING --> APPROVAL_PENDING
    APPROVAL_PENDING --> ACTIVE
    ACTIVE --> SUSPENDED
    SUSPENDED --> CORRECTION_PENDING
    CORRECTION_PENDING --> RETIRED
```

The diagram is a primary reading path. Real implementations may add parallel review, amendment, appeal, timeout, suspension, partial completion, reconciliation, correction, and terminal failure without collapsing their meaning.

4.1 Transition rules

- transition through a typed, authorized Command;
 - require exact expected version where concurrent change matters;
 - record actor, represented principal, mandate, Policy, reason, and Evidence;
 - use an idempotency key for retried writes;
 - publish a completed-fact Event after durable state change;
 - expose Outcome Unknown for unresolved external completion;
 - preserve original and corrected states in historical reconstruction.
-

5. Decision, Evidence, and Knowledge Cutoff

Reference Decision:

```
Decision {
    decisionId
    decisionType
    subjectRef
}
```

```

sourceVersionRefs[]
knowledgeCutoff
evidenceSetRef
policyVersionRefs[]
recommendationRefs[]
principalId
representedOrganizationId
mandateRef
rationale
outcome
decidedAt
effectiveAt
correctionOf?
}

```

Evidence records identify issuer or observer, subject, version, observed time, received time, effective interval, integrity, confidence, qualification, contradiction, correction, retention, residency, and access class. A recommendation may assist the Decision but cannot promote itself to the Decision.

6. Commands, Events, Queries, and Contracts

Reference Commands:

- CreateDatabaseandSchemaRegistry
- ValidateDatabaseandSchemaRegistry
- ApproveDatabaseandSchemaRegistry
- ActivateDatabaseandSchemaRegistry
- SuspendDatabaseandSchemaRegistry
- CorrectDatabaseandSchemaRegistry
- RetireDatabaseandSchemaRegistry

Reference Events:

- DatabaseandSchemaRegistryCreated
- DatabaseandSchemaRegistryValidated
- DatabaseandSchemaRegistryApproved
- DatabaseandSchemaRegistryActivated
- DatabaseandSchemaRegistrySuspended
- DatabaseandSchemaRegistryCorrected
- DatabaseandSchemaRegistryRetired

Reference queries:

- GetDatabaseandSchemaRegistry
- ListDatabaseandSchemaRegistryRecords
- GetDatabaseandSchemaRegistryTimeline
- GetDatabaseandSchemaRegistryEvidence
- GetDatabaseandSchemaRegistryDecisionPackage
- GetDatabaseandSchemaRegistryExceptions
- GetDatabaseandSchemaRegistryAuditTrail

Reference API surface:

```

POST /api/v1/databaseandschemaregistry/commands
GET  /api/v1/databaseandschemaregistry/{id}
GET  /api/v1/databaseandschemaregistry/{id}/timeline
GET  /api/v1/databaseandschemaregistry/{id}/evidence
GET  /api/v1/databaseandschemaregistry/{id}/decisions
POST /api/v1/databaseandschemaregistry/{id}/exceptions
POST /api/v1/databaseandschemaregistry/{id}/corrections

```

These endpoints are reference contracts, not claims that exact routes exist.

7. Workflow, Failure, and Compensation

flowchart TD

```

A[Validate identity version and authority] --> B[Collect required evidence]
B --> C{Policy and evidence gates pass?}
C -- No --> D[Open exception or request correction]
D --> B
C -- Yes --> E[Record Decision or execution Intent]
E --> F[Execute Domain command or external request]

```

```

F --> G{Outcome known?}
G -- Success --> H[Record fact and publish Event]
G -- Failure --> I[Record failure and compensation options]
G -- Unknown --> J[Enter Outcome Unknown]
J --> K[Query source or wait for verified callback]
K --> G
H --> L[Update projections and downstream workflow]

```

Representative failure modes:

- stale or wrong source version
- duplicate submission or replay
- external timeout after possible success
- contradictory source observations
- lost Event after committed state
- approval or mandate revoked during execution
- downstream action before a mandatory gate
- partial completion across multiple subjects
- evidence later retracted or corrected
- tenant, facility, channel or jurisdiction mismatch

Recovery shall query the authoritative source before duplicate execution, preserve partial success, use compensating Intent rather than destructive rewrite, publish corrections, quarantine high-impact ambiguity, and record affected downstream subjects.

8. Data and Persistence Architecture

Reference table families:

Table family	Purpose
databaseandschemaregistry_aggregate	authoritative subject
databaseandschemaregistry_version	immutable submitted, approved, issued, or signed versions
databaseandschemaregistry_relationship	governed relationships
databaseandschemaregistry_decision	binding Decisions and rationale
databaseandschemaregistry_evidence_link	provenance and evidence relationships
databaseandschemaregistry_event_outbox	transactional Event publication
databaseandschemaregistry_idempotency	duplicate-write protection
databaseandschemaregistry_exception	exception, claim, dispute, or hold
databaseandschemaregistry_correction	correction and supersession lineage
databaseandschemaregistry_audit	privileged action evidence

The owning service controls schema and migration. Cross-Domain references use stable identifiers or contracts. Analytics, reports, caches, search, and vector indexes remain projections. Backup, restore, retention, legal hold, privacy, residency, and decommissioning requirements are explicit.

9. Portal and Experience Requirements

- Primary participant Portal: initiate, review exact versions, supply evidence, and respond to required actions.
- Operator Portal: inspect exceptions, correlation, source responses, retries, compensation, and downstream impact.
- Executive or oversight view: consume governed metrics without becoming a source of Domain truth.

Every Portal displays exact version, owner, state, freshness, required action, and uncertainty. It supports loading, empty, partial, stale, denied, error, timeout, and Outcome Unknown states; accessibility; locale, currency, unit, date, and RTL/LTR correctness; evidence audit; safe retry; and correction history.

10. AI Participation and Tool Governance

Permitted AI tasks:

- assist schema and lineage analysis
- detect data-quality anomalies
- summarize migration and reconciliation risk
- never mutate authoritative data outside typed Domain Tools

AI shall not own Domain records, grant itself authority, suppress contradictions, silently edit approved versions, fabricate external outcomes, or execute high-impact actions outside typed Tools, policy, approval, idempotency, sandbox, and outcome observation.

Every Agent Run records source context, prompt/model/tool versions, structured output, confidence and limitations, Policy result, approval, Tool calls, provider responses, final outcome, and recovery actions.

11. Security, Privacy, and Trust Boundaries

- least privilege and separation of duties
- tenant, organization, facility, channel and jurisdiction isolation
- integrity protection for binding evidence
- verified external callbacks and anti-replay controls

Additional controls include secret isolation, callback signature verification, audit of privileged action, emergency-access review, sensitive-data minimization, purpose limitation, and threat models for impersonation, tampering, replay, confused deputy, fraud, prompt injection, and exfiltration.

12. Reliability, SLOs, and Operations

SLI	Reference objective
authoritative read availability	99.9% monthly unless a stricter tier applies
acknowledged Command durability	no acknowledged write may be silently lost
Event publication	99.9% within five minutes after committed fact
duplicate prevention	100% for same tenant, Command class, and idempotency key
binding Decision audit completeness	100%
state-state detection	bounded by business deadline

Dashboards and runbooks cover backlog, error budget, dependency failure, retry budget, DLQ, workflow stall, provider outage, reconciliation, capacity, backup/restore, release/rollback, incident command, and post-incident correction.

13. Testing and Continuous Conformance

- aggregate invariant and transition tests
- authorization, delegation, separation-of-duties and tenant-isolation tests
- idempotency and optimistic-concurrency tests
- database migration, key, constraint and rollback tests
- contract, outbox/inbox, duplicate, replay and DLQ tests
- workflow restart, timeout and compensation tests
- Portal E2E tests for success, partial, stale, denied, error and Outcome Unknown states
- accessibility, locale, currency, unit and RTL/LTR tests
- security adversarial, fraud and data-exfiltration tests
- AI evaluation, prompt-injection, policy-bypass and unauthorized-tool tests
- historical reconstruction, correction and audit-completeness tests
- performance, capacity, recovery and operational exercise tests

Release evidence links each invariant and control to the test, environment, input data, result, owner, and artifact proving it. Generated test counts never replace semantic coverage.

14. Repository Review Boundary

- This chapter is a repository-review specification. Exact executable implementation has not been inspected for every target behavior.
- Registered modules, catalog entries, README claims, frontend contracts, or plans are not proof of complete implementation.

Area	Classification
target aggregate, API, tables, state and workflow in this chapter	REFERENCE_ARCHITECTURE
registered module or catalog claim	DOCUMENTATION_ONLY_CLAIM until source inspection
uninspected integration	REQUIRES_REPOSITORY_REVIEW
behavior proven in another repository-grounded chapter	retain that chapter's evidence and classification
contradictory ownership or path claims	CONFLICT_REQUIRES_CANONICALIZATION

No statement upgrades implementation status merely because a file, directory, module name, frontend contract, database entry, plan, or report exists.

15. Worked Conformance Scenario

An authorized principal initiates a Command against an exact subject version. The service validates identity, tenant, mandate, Policy, Evidence, and idempotency. It records the state transition and outbox entry atomically. If an external dependency participates, timeout produces Outcome Unknown; reconciliation queries the source or waits for a verified callback instead of repeating the action blindly.

Portals show source, freshness, version, required action, and uncertainty. AI may extract, compare, summarize, detect anomalies, and recommend through typed Tools, but cannot own the subject or fabricate success. Later correction appends new facts, identifies affected projections and workflows, and preserves the historical Decision context.

16. Conformance Checklist

- ☐ stable identity and proposition ownership are explicit;
- ☐ authority and mandate are evaluated at action time;
- ☐ state, Recommendation, Decision, Intent, execution, and outcome remain distinct;
- ☐ Commands are typed, authorized, idempotent, and version guarded;
- ☐ Events describe completed facts;
- ☐ Evidence includes provenance, time, integrity, qualification, and correction;
- ☐ external outcomes retain their authority source;
- ☐ Portals show freshness and uncertainty;
- ☐ AI is bounded by typed Tools, Policy, approval, and outcome observation;
- ☐ security, tenant, facility, channel, and jurisdiction boundaries are enforced;
- ☐ Outcome Unknown and reconciliation exist where needed;
- ☐ corrections preserve history;
- ☐ tests prove invariants and recovery;
- ☐ implementation status is not upgraded without executable evidence.

17. Status Vocabulary

Status	Meaning
VERIFIED_IMPLEMENTED	Executable source and relevant behavior were inspected.
IMPLEMENTED_WITH_GAP	Executable behavior exists with a material governance or completeness gap.
PARTIAL_IMPLEMENTATION	Only part of the target capability is evidenced.
REFERENCE_ARCHITECTURE	Normative target behavior; no claim that it exists today.
PLANNED_NOT_IMPLEMENTED	Explicitly planned but not evidenced as executable.
DOCUMENTATION_ONLY_CLAIM	Documentation or client contract claims behavior without verified backend evidence.
CONFLICT_REQUIRES_CANONICALIZATION	Sources disagree and no final owner has been established.
REQUIRES_REPOSITORY_REVIEW	Necessary executable evidence has not yet been inspected.

Table Family Documentation

Metadata	Value
Volume	10
Chapter ID	V10-06
Subject	TableFamilyDocumentation
Status	Generated First-Draft Architecture Skeleton - Domain-Specific Expansion Required
Content maturity	TEMPLATE_DERIVED_REFERENCE_SKELETON unless repository citations prove otherwise
Default implementation classification	REFERENCE_ARCHITECTURE / REQUIRES_REPOSITORY_REVIEW
Accountable owner	Data Architecture Council
Evidence rule	No implementation claim without inspected executable evidence
Repository	chinair88-prog/allinb2c

1. Executive Orientation

This chapter defines how GTOS shall **define enterprise and Domain data ownership, semantics, persistence, quality, lineage, correction, protection, operations, and verification for Table Family Documentation.**

The specification separates reality, observations, knowledge, Decisions, Intent, execution, and outcomes. A component name, database row, UI label, AI answer, or workflow state does not acquire authority merely because it is visible or technically convenient.

Reality or external outcome

- ≠ Observation
- ≠ Claim
- ≠ derived Perspective
- ≠ Prediction
- ≠ Recommendation
- ≠ authorized Decision
- ≠ execution Intent

1.1 Accountable ownership

The accountable owner is **Data Architecture Council**. Supporting applications, shared infrastructure, integration adapters, analytics, and AI coordinate or present the capability but shall not silently own its authoritative propositions.

1.2 Scope

- create and identify the subject
- validate evidence and policy
- make authorized Decisions
- execute typed Commands
- publish completed-fact Events
- query current and historical perspectives
- correct and reconcile outcomes

1.3 Non-goals

- treating a registered Gradle module, catalog record, README, frontend route, migration plan, or generated report as proof of end-to-end implementation;
- direct cross-Domain database writes;

- generic status values that combine lifecycle, approval, provider, document, payment, dispute, and correction state;
- silent replacement of history;
- AI-created authority or fabricated external outcomes.

2. Ubiquitous Language and Subject Model

Subject	Required interpretation
TableFamilyDocumentation	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
TableFamilyDocumentationVersion	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
TableFamilyDocumentationDecision	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
TableFamilyDocumentationEvidenceSet	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
TableFamilyDocumentationException	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit

Reference aggregate:

```
TableFamilyDocumentation {
  subjectId
  tenantId
  organizationContext
  sourceIdentityRefs[]
  sourceVersionRefs[]
  lifecycleState
  relationshipRefs[]
  decisionRefs[]
  intentRefs[]
  evidenceSetRef
  knowledgeCutoff
  policyRefs[]
  authorityRef
  jurisdictionRefs[]
  createdAt
  effectiveAt
  updatedAt
  version
  correctionOf?
  correlationId
}
```

2.1 Core invariants

- Stable identity precedes relationship, state, and cross-system correlation.
- Effective time and recorded time remain distinct where business truth changes over time.
- Recommendation, Decision, Intent, execution, and outcome are separate concepts.
- No Portal, report, search index, cache, workflow, or AI Agent becomes a shadow owner of Domain truth.
- Binding submitted, approved, issued, accepted, or signed versions are immutable.
- Corrections append governed facts and preserve the original record.
- External authority and provider outcomes retain their source and qualification.
- Tenant and represented-principal context come from trusted authentication and mandate evaluation.
- Retried writes are idempotent and concurrency guarded.
- Outcome Unknown is explicit and reconciled before risky duplicate execution.

2.2 Required standards

- stable identifiers
- typed contracts
- immutable binding versions
- transactional outbox
- idempotent Commands
- explicit Outcome Unknown
- correction lineage
- machine-verifiable conformance

3. Actors, Roles, and Authority

A conformant design distinguishes:

- the natural person acting;
- the service or AI identity invoking a Tool;
- the organization or principal represented;
- the role or relationship granting context;
- the mandate or delegation granting action authority;
- the Policy and jurisdiction limiting the action;
- the reviewer or approver responsible for a binding Decision.

Authority is evaluated at action time. Authentication alone does not grant business authority. Cached permissions and browser-supplied tenant headers are insufficient for high-impact actions.

4. Lifecycle and State Model

State	Semantics
PROPOSED	Distinct governed condition with entry, exit, timeout, correction and authority semantics
VALIDATION_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
APPROVAL_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
ACTIVE	Distinct governed condition with entry, exit, timeout, correction and authority semantics
SUSPENDED	Distinct governed condition with entry, exit, timeout, correction and authority semantics
CORRECTION_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
RETIRED	Distinct governed condition with entry, exit, timeout, correction and authority semantics

```
stateDiagram-v2
    [*] --> PROPOSED
    PROPOSED --> VALIDATION_PENDING
    VALIDATION_PENDING --> APPROVAL_PENDING
    APPROVAL_PENDING --> ACTIVE
    ACTIVE --> SUSPENDED
    SUSPENDED --> CORRECTION_PENDING
    CORRECTION_PENDING --> RETIRED
```

The diagram is a primary reading path. Real implementations may add parallel review, amendment, appeal, timeout, suspension, partial completion, reconciliation, correction, and terminal failure without collapsing their meaning.

4.1 Transition rules

- transition through a typed, authorized Command;
 - require exact expected version where concurrent change matters;
 - record actor, represented principal, mandate, Policy, reason, and Evidence;
 - use an idempotency key for retried writes;
 - publish a completed-fact Event after durable state change;
 - expose Outcome Unknown for unresolved external completion;
 - preserve original and corrected states in historical reconstruction.
-

5. Decision, Evidence, and Knowledge Cutoff

Reference Decision:

```
Decision {
    decisionId
    decisionType
    subjectRef
}
```

```

sourceVersionRefs[]
knowledgeCutoff
evidenceSetRef
policyVersionRefs[]
recommendationRefs[]
principalId
representedOrganizationId
mandateRef
rationale
outcome
decidedAt
effectiveAt
correctionOf?
}

```

Evidence records identify issuer or observer, subject, version, observed time, received time, effective interval, integrity, confidence, qualification, contradiction, correction, retention, residency, and access class. A recommendation may assist the Decision but cannot promote itself to the Decision.

6. Commands, Events, Queries, and Contracts

Reference Commands:

- CreateTableFamilyDocumentation
- ValidateTableFamilyDocumentation
- ApproveTableFamilyDocumentation
- ActivateTableFamilyDocumentation
- SuspendTableFamilyDocumentation
- CorrectTableFamilyDocumentation
- RetireTableFamilyDocumentation

Reference Events:

- TableFamilyDocumentationCreated
- TableFamilyDocumentationValidated
- TableFamilyDocumentationApproved
- TableFamilyDocumentationActivated
- TableFamilyDocumentationSuspended
- TableFamilyDocumentationCorrected
- TableFamilyDocumentationRetired

Reference queries:

- GetTableFamilyDocumentation
- ListTableFamilyDocumentationRecords
- GetTableFamilyDocumentationTimeline
- GetTableFamilyDocumentationEvidence
- GetTableFamilyDocumentationDecisionPackage
- GetTableFamilyDocumentationExceptions
- GetTableFamilyDocumentationAuditTrail

Reference API surface:

```

POST /api/v1/tablefamilydocumentation/commands
GET  /api/v1/tablefamilydocumentation/{id}
GET  /api/v1/tablefamilydocumentation/{id}/timeline
GET  /api/v1/tablefamilydocumentation/{id}/evidence
GET  /api/v1/tablefamilydocumentation/{id}/decisions
POST /api/v1/tablefamilydocumentation/{id}/exceptions
POST /api/v1/tablefamilydocumentation/{id}/corrections

```

These endpoints are reference contracts, not claims that exact routes exist.

7. Workflow, Failure, and Compensation

flowchart TD

```

A[Validate identity version and authority] --> B[Collect required evidence]
B --> C{Policy and evidence gates pass?}
C -- No --> D[Open exception or request correction]
D --> B
C -- Yes --> E[Record Decision or execution Intent]
E --> F[Execute Domain command or external request]

```

```

F --> G{Outcome known?}
G -- Success --> H[Record fact and publish Event]
G -- Failure --> I[Record failure and compensation options]
G -- Unknown --> J[Enter Outcome Unknown]
J --> K[Query source or wait for verified callback]
K --> G
H --> L[Update projections and downstream workflow]

```

Representative failure modes:

- stale or wrong source version
- duplicate submission or replay
- external timeout after possible success
- contradictory source observations
- lost Event after committed state
- approval or mandate revoked during execution
- downstream action before a mandatory gate
- partial completion across multiple subjects
- evidence later retracted or corrected
- tenant, facility, channel or jurisdiction mismatch

Recovery shall query the authoritative source before duplicate execution, preserve partial success, use compensating Intent rather than destructive rewrite, publish corrections, quarantine high-impact ambiguity, and record affected downstream subjects.

8. Data and Persistence Architecture

Reference table families:

Table family	Purpose
tablefamilydocumentation_aggregate	authoritative subject
tablefamilydocumentation_version	immutable submitted, approved, issued, or signed versions
tablefamilydocumentation_relationship	governed relationships
tablefamilydocumentation_decision	binding Decisions and rationale
tablefamilydocumentation_evidence_link	provenance and evidence relationships
tablefamilydocumentation_event_outbox	transactional Event publication
tablefamilydocumentation_idempotency	duplicate-write protection
tablefamilydocumentation_exception	exception, claim, dispute, or hold
tablefamilydocumentation_correction	correction and supersession lineage
tablefamilydocumentation_audit	privileged action evidence

The owning service controls schema and migration. Cross-Domain references use stable identifiers or contracts. Analytics, reports, caches, search, and vector indexes remain projections. Backup, restore, retention, legal hold, privacy, residency, and decommissioning requirements are explicit.

9. Portal and Experience Requirements

- Primary participant Portal: initiate, review exact versions, supply evidence, and respond to required actions.
- Operator Portal: inspect exceptions, correlation, source responses, retries, compensation, and downstream impact.
- Executive or oversight view: consume governed metrics without becoming a source of Domain truth.

Every Portal displays exact version, owner, state, freshness, required action, and uncertainty. It supports loading, empty, partial, stale, denied, error, timeout, and Outcome Unknown states; accessibility; locale, currency, unit, date, and RTL/LTR correctness; evidence audit; safe retry; and correction history.

10. AI Participation and Tool Governance

Permitted AI tasks:

- assist schema and lineage analysis
- detect data-quality anomalies
- summarize migration and reconciliation risk
- never mutate authoritative data outside typed Domain Tools

AI shall not own Domain records, grant itself authority, suppress contradictions, silently edit approved versions, fabricate external outcomes, or execute high-impact actions outside typed Tools, policy, approval, idempotency, sandbox, and outcome observation.

Every Agent Run records source context, prompt/model/tool versions, structured output, confidence and limitations, Policy result, approval, Tool calls, provider responses, final outcome, and recovery actions.

11. Security, Privacy, and Trust Boundaries

- least privilege and separation of duties
- tenant, organization, facility, channel and jurisdiction isolation
- integrity protection for binding evidence
- verified external callbacks and anti-replay controls

Additional controls include secret isolation, callback signature verification, audit of privileged action, emergency-access review, sensitive-data minimization, purpose limitation, and threat models for impersonation, tampering, replay, confused deputy, fraud, prompt injection, and exfiltration.

12. Reliability, SLOs, and Operations

SLI	Reference objective
authoritative read availability	99.9% monthly unless a stricter tier applies
acknowledged Command durability	no acknowledged write may be silently lost
Event publication	99.9% within five minutes after committed fact
duplicate prevention	100% for same tenant, Command class, and idempotency key
binding Decision audit completeness	100%
stale-state detection	bounded by business deadline

Dashboards and runbooks cover backlog, error budget, dependency failure, retry budget, DLQ, workflow stall, provider outage, reconciliation, capacity, backup/restore, release/rollback, incident command, and post-incident correction.

13. Testing and Continuous Conformance

- aggregate invariant and transition tests
- authorization, delegation, separation-of-duties and tenant-isolation tests
- idempotency and optimistic-concurrency tests
- database migration, key, constraint and rollback tests
- contract, outbox/inbox, duplicate, replay and DLQ tests
- workflow restart, timeout and compensation tests
- Portal E2E tests for success, partial, stale, denied, error and Outcome Unknown states
- accessibility, locale, currency, unit and RTL/LTR tests
- security adversarial, fraud and data-exfiltration tests
- AI evaluation, prompt-injection, policy-bypass and unauthorized-tool tests
- historical reconstruction, correction and audit-completeness tests
- performance, capacity, recovery and operational exercise tests

Release evidence links each invariant and control to the test, environment, input data, result, owner, and artifact proving it. Generated test counts never replace semantic coverage.

14. Repository Review Boundary

- This chapter is a repository-review specification. Exact executable implementation has not been inspected for every target behavior.
- Registered modules, catalog entries, README claims, frontend contracts, or plans are not proof of complete implementation.

Area	Classification
target aggregate, API, tables, state and workflow in this chapter	REFERENCE_ARCHITECTURE
registered module or catalog claim	DOCUMENTATION_ONLY_CLAIM until source inspection
uninspected integration	REQUIRES_REPOSITORY_REVIEW
behavior proven in another repository-grounded chapter	retain that chapter's evidence and classification
contradictory ownership or path claims	CONFLICT_REQUIRES_CANONICALIZATION

No statement upgrades implementation status merely because a file, directory, module name, frontend contract, database entry, plan, or report exists.

15. Worked Conformance Scenario

An authorized principal initiates a Command against an exact subject version. The service validates identity, tenant, mandate, Policy, Evidence, and idempotency. It records the state transition and outbox entry atomically. If an external dependency participates, timeout produces Outcome Unknown; reconciliation queries the source or waits for a verified callback instead of repeating the action blindly.

Portals show source, freshness, version, required action, and uncertainty. AI may extract, compare, summarize, detect anomalies, and recommend through typed Tools, but cannot own the subject or fabricate success. Later correction appends new facts, identifies affected projections and workflows, and preserves the historical Decision context.

16. Conformance Checklist

- ☐ stable identity and proposition ownership are explicit;
- ☐ authority and mandate are evaluated at action time;
- ☐ state, Recommendation, Decision, Intent, execution, and outcome remain distinct;
- ☐ Commands are typed, authorized, idempotent, and version guarded;
- ☐ Events describe completed facts;
- ☐ Evidence includes provenance, time, integrity, qualification, and correction;
- ☐ external outcomes retain their authority source;
- ☐ Portals show freshness and uncertainty;
- ☐ AI is bounded by typed Tools, Policy, approval, and outcome observation;
- ☐ security, tenant, facility, channel, and jurisdiction boundaries are enforced;
- ☐ Outcome Unknown and reconciliation exist where needed;
- ☐ corrections preserve history;
- ☐ tests prove invariants and recovery;
- ☐ implementation status is not upgraded without executable evidence.

17. Status Vocabulary

Status	Meaning
VERIFIED_IMPLEMENTED	Executable source and relevant behavior were inspected.
IMPLEMENTED_WITH_GAP	Executable behavior exists with a material governance or completeness gap.
PARTIAL_IMPLEMENTATION	Only part of the target capability is evidenced.
REFERENCE_ARCHITECTURE	Normative target behavior; no claim that it exists today.
PLANNED_NOT_IMPLEMENTED	Explicitly planned but not evidenced as executable.
DOCUMENTATION_ONLY_CLAIM	Documentation or client contract claims behavior without verified backend evidence.
CONFLICT_REQUIRES_CANONICALIZATION	Sources disagree and no final owner has been established.
REQUIRES_REPOSITORY_REVIEW	Necessary executable evidence has not yet been inspected.

Identity and Keys

Metadata	Value
Volume	10
Chapter ID	V10-07
Subject	IdentityandKeys
Status	Generated First-Draft Architecture Skeleton - Domain-Specific Expansion Required
Content maturity	TEMPLATE_DERIVED_REFERENCE_SKELETON unless repository citations prove otherwise
Default implementation classification	REFERENCE_ARCHITECTURE / REQUIRES_REPOSITORY_REVIEW
Accountable owner	Data Architecture Council
Evidence rule	No implementation claim without inspected executable evidence
Repository	chinair88-prog/allinb2c

1. Executive Orientation

This chapter defines how GTOS shall **define enterprise and Domain data ownership, semantics, persistence, quality, lineage, correction, protection, operations, and verification for Identity and Keys.**

The specification separates reality, observations, knowledge, Decisions, Intent, execution, and outcomes. A component name, database row, UI label, AI answer, or workflow state does not acquire authority merely because it is visible or technically convenient.

Reality or external outcome

- ≠ Observation
- ≠ Claim
- ≠ derived Perspective
- ≠ Prediction
- ≠ Recommendation
- ≠ authorized Decision
- ≠ execution Intent

1.1 Accountable ownership

The accountable owner is **Data Architecture Council**. Supporting applications, shared infrastructure, integration adapters, analytics, and AI coordinate or present the capability but shall not silently own its authoritative propositions.

1.2 Scope

- create and identify the subject
- validate evidence and policy
- make authorized Decisions
- execute typed Commands
- publish completed-fact Events
- query current and historical perspectives
- correct and reconcile outcomes

1.3 Non-goals

- treating a registered Gradle module, catalog record, README, frontend route, migration plan, or generated report as proof of end-to-end implementation;
- direct cross-Domain database writes;

- generic status values that combine lifecycle, approval, provider, document, payment, dispute, and correction state;
- silent replacement of history;
- AI-created authority or fabricated external outcomes.

2. Ubiquitous Language and Subject Model

Subject	Required interpretation
IdentityandKeys	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
IdentityandKeysVersion	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
IdentityandKeysDecision	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
IdentityandKeysEvidenceSet	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
IdentityandKeysException	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit

Reference aggregate:

```
IdentityandKeys {
  subjectId
  tenantId
  organizationContext
  sourceIdentityRefs[]
  sourceVersionRefs[]
  lifecycleState
  relationshipRefs[]
  decisionRefs[]
  intentRefs[]
  evidenceSetRef
  knowledgeCutoff
  policyRefs[]
  authorityRef
  jurisdictionRefs[]
  createdAt
  effectiveAt
  updatedAt
  version
  correctionOf?
  correlationId
}
```

2.1 Core invariants

- Stable identity precedes relationship, state, and cross-system correlation.
- Effective time and recorded time remain distinct where business truth changes over time.
- Recommendation, Decision, Intent, execution, and outcome are separate concepts.
- No Portal, report, search index, cache, workflow, or AI Agent becomes a shadow owner of Domain truth.
- Binding submitted, approved, issued, accepted, or signed versions are immutable.
- Corrections append governed facts and preserve the original record.
- External authority and provider outcomes retain their source and qualification.
- Tenant and represented-principal context come from trusted authentication and mandate evaluation.
- Retried writes are idempotent and concurrency guarded.
- Outcome Unknown is explicit and reconciled before risky duplicate execution.

2.2 Required standards

- stable identifiers
- typed contracts
- immutable binding versions
- transactional outbox
- idempotent Commands
- explicit Outcome Unknown
- correction lineage
- machine-verifiable conformance

3. Actors, Roles, and Authority

A conformant design distinguishes:

- the natural person acting;
- the service or AI identity invoking a Tool;
- the organization or principal represented;
- the role or relationship granting context;
- the mandate or delegation granting action authority;
- the Policy and jurisdiction limiting the action;
- the reviewer or approver responsible for a binding Decision.

Authority is evaluated at action time. Authentication alone does not grant business authority. Cached permissions and browser-supplied tenant headers are insufficient for high-impact actions.

4. Lifecycle and State Model

State	Semantics
PROPOSED	Distinct governed condition with entry, exit, timeout, correction and authority semantics
VALIDATION_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
APPROVAL_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
ACTIVE	Distinct governed condition with entry, exit, timeout, correction and authority semantics
SUSPENDED	Distinct governed condition with entry, exit, timeout, correction and authority semantics
CORRECTION_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
RETIRED	Distinct governed condition with entry, exit, timeout, correction and authority semantics

```
stateDiagram-v2
    [*] --> PROPOSED
    PROPOSED --> VALIDATION_PENDING
    VALIDATION_PENDING --> APPROVAL_PENDING
    APPROVAL_PENDING --> ACTIVE
    ACTIVE --> SUSPENDED
    SUSPENDED --> CORRECTION_PENDING
    CORRECTION_PENDING --> RETIRED
```

The diagram is a primary reading path. Real implementations may add parallel review, amendment, appeal, timeout, suspension, partial completion, reconciliation, correction, and terminal failure without collapsing their meaning.

4.1 Transition rules

- transition through a typed, authorized Command;
 - require exact expected version where concurrent change matters;
 - record actor, represented principal, mandate, Policy, reason, and Evidence;
 - use an idempotency key for retried writes;
 - publish a completed-fact Event after durable state change;
 - expose Outcome Unknown for unresolved external completion;
 - preserve original and corrected states in historical reconstruction.
-

5. Decision, Evidence, and Knowledge Cutoff

Reference Decision:

```
Decision {
    decisionId
    decisionType
    subjectRef
}
```

```

sourceVersionRefs[]
knowledgeCutoff
evidenceSetRef
policyVersionRefs[]
recommendationRefs[]
principalId
representedOrganizationId
mandateRef
rationale
outcome
decidedAt
effectiveAt
correctionOf?
}

```

Evidence records identify issuer or observer, subject, version, observed time, received time, effective interval, integrity, confidence, qualification, contradiction, correction, retention, residency, and access class. A recommendation may assist the Decision but cannot promote itself to the Decision.

6. Commands, Events, Queries, and Contracts

Reference Commands:

- CreateIdentityandKeys
- ValidateIdentityandKeys
- ApproveIdentityandKeys
- ActivateIdentityandKeys
- SuspendIdentityandKeys
- CorrectIdentityandKeys
- RetireIdentityandKeys

Reference Events:

- IdentityandKeysCreated
- IdentityandKeysValidated
- IdentityandKeysApproved
- IdentityandKeysActivated
- IdentityandKeysSuspended
- IdentityandKeysCorrected
- IdentityandKeysRetired

Reference queries:

- GetIdentityandKeys
- ListIdentityandKeysRecords
- GetIdentityandKeysTimeline
- GetIdentityandKeysEvidence
- GetIdentityandKeysDecisionPackage
- GetIdentityandKeysExceptions
- GetIdentityandKeysAuditTrail

Reference API surface:

```

POST /api/v1/identityandkeys/commands
GET  /api/v1/identityandkeys/{id}
GET  /api/v1/identityandkeys/{id}/timeline
GET  /api/v1/identityandkeys/{id}/evidence
GET  /api/v1/identityandkeys/{id}/decisions
POST /api/v1/identityandkeys/{id}/exceptions
POST /api/v1/identityandkeys/{id}/corrections

```

These endpoints are reference contracts, not claims that exact routes exist.

7. Workflow, Failure, and Compensation

flowchart TD

```

A[Validate identity version and authority] --> B[Collect required evidence]
B --> C{Policy and evidence gates pass?}
C -- No --> D[Open exception or request correction]
D --> B
C -- Yes --> E[Record Decision or execution Intent]
E --> F[Execute Domain command or external request]

```

```

F --> G{Outcome known?}
G -- Success --> H[Record fact and publish Event]
G -- Failure --> I[Record failure and compensation options]
G -- Unknown --> J[Enter Outcome Unknown]
J --> K[Query source or wait for verified callback]
K --> G
H --> L[Update projections and downstream workflow]

```

Representative failure modes:

- stale or wrong source version
- duplicate submission or replay
- external timeout after possible success
- contradictory source observations
- lost Event after committed state
- approval or mandate revoked during execution
- downstream action before a mandatory gate
- partial completion across multiple subjects
- evidence later retracted or corrected
- tenant, facility, channel or jurisdiction mismatch

Recovery shall query the authoritative source before duplicate execution, preserve partial success, use compensating Intent rather than destructive rewrite, publish corrections, quarantine high-impact ambiguity, and record affected downstream subjects.

8. Data and Persistence Architecture

Reference table families:

Table family	Purpose
identityandkeys_aggregate	authoritative subject
identityandkeys_version	immutable submitted, approved, issued, or signed versions
identityandkeys_relationship	governed relationships
identityandkeys_decision	binding Decisions and rationale
identityandkeys_evidence_link	provenance and evidence relationships
identityandkeys_event_outbox	transactional Event publication
identityandkeys_idempotency	duplicate-write protection
identityandkeys_exception	exception, claim, dispute, or hold
identityandkeys_correction	correction and supersession lineage
identityandkeys_audit	privileged action evidence

The owning service controls schema and migration. Cross-Domain references use stable identifiers or contracts. Analytics, reports, caches, search, and vector indexes remain projections. Backup, restore, retention, legal hold, privacy, residency, and decommissioning requirements are explicit.

9. Portal and Experience Requirements

- Primary participant Portal: initiate, review exact versions, supply evidence, and respond to required actions.
- Operator Portal: inspect exceptions, correlation, source responses, retries, compensation, and downstream impact.
- Executive or oversight view: consume governed metrics without becoming a source of Domain truth.

Every Portal displays exact version, owner, state, freshness, required action, and uncertainty. It supports loading, empty, partial, stale, denied, error, timeout, and Outcome Unknown states; accessibility; locale, currency, unit, date, and RTL/LTR correctness; evidence audit; safe retry; and correction history.

10. AI Participation and Tool Governance

Permitted AI tasks:

- assist schema and lineage analysis
- detect data-quality anomalies
- summarize migration and reconciliation risk
- never mutate authoritative data outside typed Domain Tools

AI shall not own Domain records, grant itself authority, suppress contradictions, silently edit approved versions, fabricate external outcomes, or execute high-impact actions outside typed Tools, policy, approval, idempotency, sandbox, and outcome observation.

Every Agent Run records source context, prompt/model/tool versions, structured output, confidence and limitations, Policy result, approval, Tool calls, provider responses, final outcome, and recovery actions.

11. Security, Privacy, and Trust Boundaries

- least privilege and separation of duties
- tenant, organization, facility, channel and jurisdiction isolation
- integrity protection for binding evidence
- verified external callbacks and anti-replay controls

Additional controls include secret isolation, callback signature verification, audit of privileged action, emergency-access review, sensitive-data minimization, purpose limitation, and threat models for impersonation, tampering, replay, confused deputy, fraud, prompt injection, and exfiltration.

12. Reliability, SLOs, and Operations

SLI	Reference objective
authoritative read availability	99.9% monthly unless a stricter tier applies
acknowledged Command durability	no acknowledged write may be silently lost
Event publication	99.9% within five minutes after committed fact
duplicate prevention	100% for same tenant, Command class, and idempotency key
binding Decision audit completeness	100%
stale-state detection	bounded by business deadline

Dashboards and runbooks cover backlog, error budget, dependency failure, retry budget, DLQ, workflow stall, provider outage, reconciliation, capacity, backup/restore, release/rollback, incident command, and post-incident correction.

13. Testing and Continuous Conformance

- aggregate invariant and transition tests
- authorization, delegation, separation-of-duties and tenant-isolation tests
- idempotency and optimistic-concurrency tests
- database migration, key, constraint and rollback tests
- contract, outbox/inbox, duplicate, replay and DLQ tests
- workflow restart, timeout and compensation tests
- Portal E2E tests for success, partial, stale, denied, error and Outcome Unknown states
- accessibility, locale, currency, unit and RTL/LTR tests
- security adversarial, fraud and data-exfiltration tests
- AI evaluation, prompt-injection, policy-bypass and unauthorized-tool tests
- historical reconstruction, correction and audit-completeness tests
- performance, capacity, recovery and operational exercise tests

Release evidence links each invariant and control to the test, environment, input data, result, owner, and artifact proving it. Generated test counts never replace semantic coverage.

14. Repository Review Boundary

- This chapter is a repository-review specification. Exact executable implementation has not been inspected for every target behavior.
- Registered modules, catalog entries, README claims, frontend contracts, or plans are not proof of complete implementation.

Area	Classification
target aggregate, API, tables, state and workflow in this chapter	REFERENCE_ARCHITECTURE
registered module or catalog claim	DOCUMENTATION_ONLY_CLAIM until source inspection
uninspected integration	REQUIRES_REPOSITORY_REVIEW
behavior proven in another repository-grounded chapter	retain that chapter's evidence and classification
contradictory ownership or path claims	CONFLICT_REQUIRES_CANONICALIZATION

No statement upgrades implementation status merely because a file, directory, module name, frontend contract, database entry, plan, or report exists.

15. Worked Conformance Scenario

An authorized principal initiates a Command against an exact subject version. The service validates identity, tenant, mandate, Policy, Evidence, and idempotency. It records the state transition and outbox entry atomically. If an external dependency participates, timeout produces Outcome Unknown; reconciliation queries the source or waits for a verified callback instead of repeating the action blindly.

Portals show source, freshness, version, required action, and uncertainty. AI may extract, compare, summarize, detect anomalies, and recommend through typed Tools, but cannot own the subject or fabricate success. Later correction appends new facts, identifies affected projections and workflows, and preserves the historical Decision context.

16. Conformance Checklist

- ☐ stable identity and proposition ownership are explicit;
- ☐ authority and mandate are evaluated at action time;
- ☐ state, Recommendation, Decision, Intent, execution, and outcome remain distinct;
- ☐ Commands are typed, authorized, idempotent, and version guarded;
- ☐ Events describe completed facts;
- ☐ Evidence includes provenance, time, integrity, qualification, and correction;
- ☐ external outcomes retain their authority source;
- ☐ Portals show freshness and uncertainty;
- ☐ AI is bounded by typed Tools, Policy, approval, and outcome observation;
- ☐ security, tenant, facility, channel, and jurisdiction boundaries are enforced;
- ☐ Outcome Unknown and reconciliation exist where needed;
- ☐ corrections preserve history;
- ☐ tests prove invariants and recovery;
- ☐ implementation status is not upgraded without executable evidence.

17. Status Vocabulary

Status	Meaning
VERIFIED_IMPLEMENTED	Executable source and relevant behavior were inspected.
IMPLEMENTED_WITH_GAP	Executable behavior exists with a material governance or completeness gap.
PARTIAL_IMPLEMENTATION	Only part of the target capability is evidenced.
REFERENCE_ARCHITECTURE	Normative target behavior; no claim that it exists today.
PLANNED_NOT_IMPLEMENTED	Explicitly planned but not evidenced as executable.
DOCUMENTATION_ONLY_CLAIM	Documentation or client contract claims behavior without verified backend evidence.
CONFLICT_REQUIRES_CANONICALIZATION	Sources disagree and no final owner has been established.
REQUIRES_REPOSITORY_REVIEW	Necessary executable evidence has not yet been inspected.

Foreign Keys and Cross-Domain References

Metadata	Value
Volume	10
Chapter ID	V10-08
Subject	ForeignKeysandCross-DomainReferences
Status	Generated First-Draft Architecture Skeleton - Domain-Specific Expansion Required
Content maturity	TEMPLATE_DERIVED_REFERENCE_SKELETON unless repository citations prove otherwise
Default implementation classification	REFERENCE_ARCHITECTURE / REQUIRES_REPOSITORY_REVIEW
Accountable owner	Data Architecture Council
Evidence rule	No implementation claim without inspected executable evidence
Repository	chinair88-prog/allinb2c

1. Executive Orientation

This chapter defines how GTOS shall **define enterprise and Domain data ownership, semantics, persistence, quality, lineage, correction, protection, operations, and verification for Foreign Keys and Cross-Domain References**.

The specification separates reality, observations, knowledge, Decisions, Intent, execution, and outcomes. A component name, database row, UI label, AI answer, or workflow state does not acquire authority merely because it is visible or technically convenient.

Reality or external outcome
≠ Observation
≠ Claim
≠ derived Perspective
≠ Prediction
≠ Recommendation
≠ authorized Decision
≠ execution Intent

1.1 Accountable ownership

The accountable owner is **Data Architecture Council**. Supporting applications, shared infrastructure, integration adapters, analytics, and AI coordinate or present the capability but shall not silently own its authoritative propositions.

1.2 Scope

- create and identify the subject
- validate evidence and policy
- make authorized Decisions
- execute typed Commands
- publish completed-fact Events
- query current and historical perspectives
- correct and reconcile outcomes

1.3 Non-goals

- treating a registered Gradle module, catalog record, README, frontend route, migration plan, or generated report as proof of end-to-end implementation;

- direct cross-Domain database writes;
- generic status values that combine lifecycle, approval, provider, document, payment, dispute, and correction state;
- silent replacement of history;
- AI-created authority or fabricated external outcomes.

2. Ubiquitous Language and Subject Model

Subject	Required interpretation
ForeignKeysandCross-DomainReferences	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
ForeignKeysandCross-DomainReferencesVersion	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
ForeignKeysandCross-DomainReferencesDecision	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
ForeignKeysandCross-DomainReferencesEvidenceSet	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
ForeignKeysandCross-DomainReferencesException	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit

Reference aggregate:

```
ForeignKeysandCross-DomainReferences {
  subjectId
  tenantId
  organizationContext
  sourceIdentityRefs[]
  sourceVersionRefs[]
  lifecycleState
  relationshipRefs[]
  decisionRefs[]
  intentRefs[]
  evidenceSetRef
  knowledgeCutoff
  policyRefs[]
  authorityRef
  jurisdictionRefs[]
  createdAt
  effectiveAt
  updatedAt
  version
  correctionOf?
  correlationId
}
```

2.1 Core invariants

- Stable identity precedes relationship, state, and cross-system correlation.
- Effective time and recorded time remain distinct where business truth changes over time.
- Recommendation, Decision, Intent, execution, and outcome are separate concepts.
- No Portal, report, search index, cache, workflow, or AI Agent becomes a shadow owner of Domain truth.
- Binding submitted, approved, issued, accepted, or signed versions are immutable.
- Corrections append governed facts and preserve the original record.
- External authority and provider outcomes retain their source and qualification.
- Tenant and represented-principal context come from trusted authentication and mandate evaluation.
- Retried writes are idempotent and concurrency guarded.
- Outcome Unknown is explicit and reconciled before risky duplicate execution.

2.2 Required standards

- stable identifiers
- typed contracts
- immutable binding versions
- transactional outbox
- idempotent Commands
- explicit Outcome Unknown
- correction lineage

- machine-verifiable conformance

3. Actors, Roles, and Authority

A conformant design distinguishes:

- the natural person acting;
- the service or AI identity invoking a Tool;
- the organization or principal represented;
- the role or relationship granting context;
- the mandate or delegation granting action authority;
- the Policy and jurisdiction limiting the action;
- the reviewer or approver responsible for a binding Decision.

Authority is evaluated at action time. Authentication alone does not grant business authority. Cached permissions and browser-supplied tenant headers are insufficient for high-impact actions.

4. Lifecycle and State Model

State	Semantics
PROPOSED	Distinct governed condition with entry, exit, timeout, correction and authority semantics
VALIDATION_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
APPROVAL_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
ACTIVE	Distinct governed condition with entry, exit, timeout, correction and authority semantics
SUSPENDED	Distinct governed condition with entry, exit, timeout, correction and authority semantics
CORRECTION_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
RETIRED	Distinct governed condition with entry, exit, timeout, correction and authority semantics

```
stateDiagram-v2
    [*] --> PROPOSED
    PROPOSED --> VALIDATION_PENDING
    VALIDATION_PENDING --> APPROVAL_PENDING
    APPROVAL_PENDING --> ACTIVE
    ACTIVE --> SUSPENDED
    SUSPENDED --> CORRECTION_PENDING
    CORRECTION_PENDING --> RETIRED
```

The diagram is a primary reading path. Real implementations may add parallel review, amendment, appeal, timeout, suspension, partial completion, reconciliation, correction, and terminal failure without collapsing their meaning.

4.1 Transition rules

- transition through a typed, authorized Command;
- require exact expected version where concurrent change matters;
- record actor, represented principal, mandate, Policy, reason, and Evidence;
- use an idempotency key for retried writes;
- publish a completed-fact Event after durable state change;
- expose Outcome Unknown for unresolved external completion;
- preserve original and corrected states in historical reconstruction.

5. Decision, Evidence, and Knowledge Cutoff

Reference Decision:

```
Decision {
    decisionId
```

```

decisionType
subjectRef
sourceVersionRefs[]
knowledgeCutoff
evidenceSetRef
policyVersionRefs[]
recommendationRefs[]
principalId
representedOrganizationId
mandateRef
rationale
outcome
decidedAt
effectiveAt
correctionOf?
}

```

Evidence records identify issuer or observer, subject, version, observed time, received time, effective interval, integrity, confidence, qualification, contradiction, correction, retention, residency, and access class. A recommendation may assist the Decision but cannot promote itself to the Decision.

6. Commands, Events, Queries, and Contracts

Reference Commands:

- CreateForeignKeysandCross-DomainReferences
- ValidateForeignKeysandCross-DomainReferences
- ApproveForeignKeysandCross-DomainReferences
- ActivateForeignKeysandCross-DomainReferences
- SuspendForeignKeysandCross-DomainReferences
- CorrectForeignKeysandCross-DomainReferences
- RetireForeignKeysandCross-DomainReferences

Reference Events:

- ForeignKeysandCross-DomainReferencesCreated
- ForeignKeysandCross-DomainReferencesValidated
- ForeignKeysandCross-DomainReferencesApproved
- ForeignKeysandCross-DomainReferencesActivated
- ForeignKeysandCross-DomainReferencesSuspended
- ForeignKeysandCross-DomainReferencesCorrected
- ForeignKeysandCross-DomainReferencesRetired

Reference queries:

- GetForeignKeysandCross-DomainReferences
- ListForeignKeysandCross-DomainReferencesRecords
- GetForeignKeysandCross-DomainReferencesTimeline
- GetForeignKeysandCross-DomainReferencesEvidence
- GetForeignKeysandCross-DomainReferencesDecisionPackage
- GetForeignKeysandCross-DomainReferencesExceptions
- GetForeignKeysandCross-DomainReferencesAuditTrail

Reference API surface:

```

POST /api/v1/foreignkeysandcross_domainreferences/commands
GET  /api/v1/foreignkeysandcross_domainreferences/{id}
GET  /api/v1/foreignkeysandcross_domainreferences/{id}/timeline
GET  /api/v1/foreignkeysandcross_domainreferences/{id}/evidence
GET  /api/v1/foreignkeysandcross_domainreferences/{id}/decisions
POST /api/v1/foreignkeysandcross_domainreferences/{id}/exceptions
POST /api/v1/foreignkeysandcross_domainreferences/{id}/corrections

```

These endpoints are reference contracts, not claims that exact routes exist.

7. Workflow, Failure, and Compensation

flowchart TD

```

A[Validate identity version and authority] --> B[Collect required evidence]
B --> C[Policy and evidence gates pass?]
C -- No --> D[Open exception or request correction]
D --> B

```

```

C -- Yes --> E[Record Decision or execution Intent]
E --> F[Execute Domain command or external request]
F --> G{Outcome known?}
G -- Success --> H[Record fact and publish Event]
G -- Failure --> I[Record failure and compensation options]
G -- Unknown --> J[Enter Outcome Unknown]
J --> K[Query source or wait for verified callback]
K --> G
H --> L[Update projections and downstream workflow]

```

Representative failure modes:

- stale or wrong source version
- duplicate submission or replay
- external timeout after possible success
- contradictory source observations
- lost Event after committed state
- approval or mandate revoked during execution
- downstream action before a mandatory gate
- partial completion across multiple subjects
- evidence later retracted or corrected
- tenant, facility, channel or jurisdiction mismatch

Recovery shall query the authoritative source before duplicate execution, preserve partial success, use compensating Intent rather than destructive rewrite, publish corrections, quarantine high-impact ambiguity, and record affected downstream subjects.

8. Data and Persistence Architecture

Reference table families:

Table family	Purpose
foreignkeysandcross_domainreferences_aggregate	authoritative subject
foreignkeysandcross_domainreferences_version	immutable submitted, approved, issued, or signed versions
foreignkeysandcross_domainreferences_relationship	governed relationships
foreignkeysandcross_domainreferences_decision	binding Decisions and rationale
foreignkeysandcross_domainreferences_evidence_link	provenance and evidence relationships
foreignkeysandcross_domainreferences_event_outbox	transactional Event publication
foreignkeysandcross_domainreferences_idempotency	duplicate-write protection
foreignkeysandcross_domainreferences_exception	exception, claim, dispute, or hold
foreignkeysandcross_domainreferences_correction	correction and supersession lineage
foreignkeysandcross_domainreferences_audit	privileged action evidence

The owning service controls schema and migration. Cross-Domain references use stable identifiers or contracts. Analytics, reports, caches, search, and vector indexes remain projections. Backup, restore, retention, legal hold, privacy, residency, and decommissioning requirements are explicit.

9. Portal and Experience Requirements

- Primary participant Portal: initiate, review exact versions, supply evidence, and respond to required actions.
- Operator Portal: inspect exceptions, correlation, source responses, retries, compensation, and downstream impact.
- Executive or oversight view: consume governed metrics without becoming a source of Domain truth.

Every Portal displays exact version, owner, state, freshness, required action, and uncertainty. It supports loading, empty, partial, stale, denied, error, timeout, and Outcome Unknown states; accessibility; locale, currency, unit, date, and RTL/LTR correctness; evidence audit; safe retry; and correction history.

10. AI Participation and Tool Governance

Permitted AI tasks:

- assist schema and lineage analysis
- detect data-quality anomalies

- summarize migration and reconciliation risk
- never mutate authoritative data outside typed Domain Tools

AI shall not own Domain records, grant itself authority, suppress contradictions, silently edit approved versions, fabricate external outcomes, or execute high-impact actions outside typed Tools, policy, approval, idempotency, sandbox, and outcome observation.

Every Agent Run records source context, prompt/model/tool versions, structured output, confidence and limitations, Policy result, approval, Tool calls, provider responses, final outcome, and recovery actions.

11. Security, Privacy, and Trust Boundaries

- least privilege and separation of duties
- tenant, organization, facility, channel and jurisdiction isolation
- integrity protection for binding evidence
- verified external callbacks and anti-replay controls

Additional controls include secret isolation, callback signature verification, audit of privileged action, emergency-access review, sensitive-data minimization, purpose limitation, and threat models for impersonation, tampering, replay, confused deputy, fraud, prompt injection, and exfiltration.

12. Reliability, SLOs, and Operations

SLI	Reference objective
authoritative read availability	99.9% monthly unless a stricter tier applies
acknowledged Command durability	no acknowledged write may be silently lost
Event publication	99.9% within five minutes after committed fact
duplicate prevention	100% for same tenant, Command class, and idempotency key
binding Decision audit completeness	100%
state-state detection	bounded by business deadline

Dashboards and runbooks cover backlog, error budget, dependency failure, retry budget, DLQ, workflow stall, provider outage, reconciliation, capacity, backup/restore, release/rollback, incident command, and post-incident correction.

13. Testing and Continuous Conformance

- aggregate invariant and transition tests
- authorization, delegation, separation-of-duties and tenant-isolation tests
- idempotency and optimistic-concurrency tests
- database migration, key, constraint and rollback tests
- contract, outbox/inbox, duplicate, replay and DLQ tests
- workflow restart, timeout and compensation tests
- Portal E2E tests for success, partial, stale, denied, error and Outcome Unknown states
- accessibility, locale, currency, unit and RTL/LTR tests
- security adversarial, fraud and data-exfiltration tests
- AI evaluation, prompt-injection, policy-bypass and unauthorized-tool tests
- historical reconstruction, correction and audit-completeness tests
- performance, capacity, recovery and operational exercise tests

Release evidence links each invariant and control to the test, environment, input data, result, owner, and artifact proving it. Generated test counts never replace semantic coverage.

14. Repository Review Boundary

- This chapter is a repository-review specification. Exact executable implementation has not been inspected for every target behavior.
- Registered modules, catalog entries, README claims, frontend contracts, or plans are not proof of complete implementation.

Area	Classification
target aggregate, API, tables, state and workflow in this chapter	REFERENCE_ARCHITECTURE
registered module or catalog claim	DOCUMENTATION_ONLY_CLAIM until source inspection
uninspected integration	REQUIRES_REPOSITORY_REVIEW
behavior proven in another repository-grounded chapter	retain that chapter's evidence and classification
contradictory ownership or path claims	CONFLICT_REQUIRES_CANONICALIZATION

No statement upgrades implementation status merely because a file, directory, module name, frontend contract, database entry, plan, or report exists.

15. Worked Conformance Scenario

An authorized principal initiates a Command against an exact subject version. The service validates identity, tenant, mandate, Policy, Evidence, and idempotency. It records the state transition and outbox entry atomically. If an external dependency participates, timeout produces Outcome Unknown; reconciliation queries the source or waits for a verified callback instead of repeating the action blindly.

Portals show source, freshness, version, required action, and uncertainty. AI may extract, compare, summarize, detect anomalies, and recommend through typed Tools, but cannot own the subject or fabricate success. Later correction appends new facts, identifies affected projections and workflows, and preserves the historical Decision context.

16. Conformance Checklist

- ☐ stable identity and proposition ownership are explicit;
- ☐ authority and mandate are evaluated at action time;
- ☐ state, Recommendation, Decision, Intent, execution, and outcome remain distinct;
- ☐ Commands are typed, authorized, idempotent, and version guarded;
- ☐ Events describe completed facts;
- ☐ Evidence includes provenance, time, integrity, qualification, and correction;
- ☐ external outcomes retain their authority source;
- ☐ Portals show freshness and uncertainty;
- ☐ AI is bounded by typed Tools, Policy, approval, and outcome observation;
- ☐ security, tenant, facility, channel, and jurisdiction boundaries are enforced;
- ☐ Outcome Unknown and reconciliation exist where needed;
- ☐ corrections preserve history;
- ☐ tests prove invariants and recovery;
- ☐ implementation status is not upgraded without executable evidence.

17. Status Vocabulary

Status	Meaning
VERIFIED_IMPLEMENTED	Executable source and relevant behavior were inspected.
IMPLEMENTED_WITH_GAP	Executable behavior exists with a material governance or completeness gap.
PARTIAL_IMPLEMENTATION	Only part of the target capability is evidenced.
REFERENCE_ARCHITECTURE	Normative target behavior; no claim that it exists today.
PLANNED_NOT_IMPLEMENTED	Explicitly planned but not evidenced as executable.
DOCUMENTATION_ONLY_CLAIM	Documentation or client contract claims behavior without verified backend evidence.
CONFLICT_REQUIRES_CANONICALIZATION	Sources disagree and no final owner has been established.
REQUIRES_REPOSITORY_REVIEW	Necessary executable evidence has not yet been inspected.

Constraints and Invariants

Metadata	Value
Volume	10
Chapter ID	V10-09
Subject	ConstraintsandInvariants
Status	Generated First-Draft Architecture Skeleton - Domain-Specific Expansion Required
Content maturity	TEMPLATE_DERIVED_REFERENCE_SKELETON unless repository citations prove otherwise
Default implementation classification	REFERENCE_ARCHITECTURE / REQUIRES_REPOSITORY_REVIEW
Accountable owner	Data Architecture Council
Evidence rule	No implementation claim without inspected executable evidence
Repository	chinair88-prog/allinb2c

1. Executive Orientation

This chapter defines how GTOS shall **define enterprise and Domain data ownership, semantics, persistence, quality, lineage, correction, protection, operations, and verification for Constraints and Invariants.**

The specification separates reality, observations, knowledge, Decisions, Intent, execution, and outcomes. A component name, database row, UI label, AI answer, or workflow state does not acquire authority merely because it is visible or technically convenient.

Reality or external outcome

- ≠ Observation
- ≠ Claim
- ≠ derived Perspective
- ≠ Prediction
- ≠ Recommendation
- ≠ authorized Decision
- ≠ execution Intent

1.1 Accountable ownership

The accountable owner is **Data Architecture Council**. Supporting applications, shared infrastructure, integration adapters, analytics, and AI coordinate or present the capability but shall not silently own its authoritative propositions.

1.2 Scope

- create and identify the subject
- validate evidence and policy
- make authorized Decisions
- execute typed Commands
- publish completed-fact Events
- query current and historical perspectives
- correct and reconcile outcomes

1.3 Non-goals

- treating a registered Gradle module, catalog record, README, frontend route, migration plan, or generated report as proof of end-to-end implementation;
- direct cross-Domain database writes;
- generic status values that combine lifecycle, approval, provider, document, payment, dispute, and correction state;

- silent replacement of history;
- AI-created authority or fabricated external outcomes.

2. Ubiquitous Language and Subject Model

Subject	Required interpretation
ConstraintsandInvariants	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
ConstraintsandInvariantsVersion	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
ConstraintsandInvariantsDecision	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
ConstraintsandInvariantsEvidenceSet	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
ConstraintsandInvariantsException	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit

Reference aggregate:

```
ConstraintsandInvariants {
  subjectId
  tenantId
  organizationContext
  sourceIdentityRefs[]
  sourceVersionRefs[]
  lifecycleState
  relationshipRefs[]
  decisionRefs[]
  intentRefs[]
  evidenceSetRef
  knowledgeCutoff
  policyRefs[]
  authorityRef
  jurisdictionRefs[]
  createdAt
  effectiveAt
  updatedAt
  version
  correctionOf?
  correlationId
}
```

2.1 Core invariants

- Stable identity precedes relationship, state, and cross-system correlation.
- Effective time and recorded time remain distinct where business truth changes over time.
- Recommendation, Decision, Intent, execution, and outcome are separate concepts.
- No Portal, report, search index, cache, workflow, or AI Agent becomes a shadow owner of Domain truth.
- Binding submitted, approved, issued, accepted, or signed versions are immutable.
- Corrections append governed facts and preserve the original record.
- External authority and provider outcomes retain their source and qualification.
- Tenant and represented-principal context come from trusted authentication and mandate evaluation.
- Retried writes are idempotent and concurrency guarded.
- Outcome Unknown is explicit and reconciled before risky duplicate execution.

2.2 Required standards

- stable identifiers
- typed contracts
- immutable binding versions
- transactional outbox
- idempotent Commands
- explicit Outcome Unknown
- correction lineage
- machine-verifiable conformance

3. Actors, Roles, and Authority

A conformant design distinguishes:

- the natural person acting;
- the service or AI identity invoking a Tool;
- the organization or principal represented;
- the role or relationship granting context;
- the mandate or delegation granting action authority;
- the Policy and jurisdiction limiting the action;
- the reviewer or approver responsible for a binding Decision.

Authority is evaluated at action time. Authentication alone does not grant business authority. Cached permissions and browser-supplied tenant headers are insufficient for high-impact actions.

4. Lifecycle and State Model

State	Semantics
PROPOSED	Distinct governed condition with entry, exit, timeout, correction and authority semantics
VALIDATION_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
APPROVAL_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
ACTIVE	Distinct governed condition with entry, exit, timeout, correction and authority semantics
SUSPENDED	Distinct governed condition with entry, exit, timeout, correction and authority semantics
CORRECTION_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
RETIRED	Distinct governed condition with entry, exit, timeout, correction and authority semantics

```
stateDiagram-v2
    [*] --> PROPOSED
    PROPOSED --> VALIDATION_PENDING
    VALIDATION_PENDING --> APPROVAL_PENDING
    APPROVAL_PENDING --> ACTIVE
    ACTIVE --> SUSPENDED
    SUSPENDED --> CORRECTION_PENDING
    CORRECTION_PENDING --> RETIRED
```

The diagram is a primary reading path. Real implementations may add parallel review, amendment, appeal, timeout, suspension, partial completion, reconciliation, correction, and terminal failure without collapsing their meaning.

4.1 Transition rules

- transition through a typed, authorized Command;
- require exact expected version where concurrent change matters;
- record actor, represented principal, mandate, Policy, reason, and Evidence;
- use an idempotency key for retried writes;
- publish a completed-fact Event after durable state change;
- expose Outcome Unknown for unresolved external completion;
- preserve original and corrected states in historical reconstruction.

5. Decision, Evidence, and Knowledge Cutoff

Reference Decision:

```
Decision {
    decisionId
    decisionType
    subjectRef
    sourceVersionRefs[]
    knowledgeCutoff
    evidenceSetRef
}
```

```

policyVersionRefs[]
recommendationRefs[]
principalId
representedOrganizationId
mandateRef
rationale
outcome
decidedAt
effectiveAt
correctionOf?
}

```

Evidence records identify issuer or observer, subject, version, observed time, received time, effective interval, integrity, confidence, qualification, contradiction, correction, retention, residency, and access class. A recommendation may assist the Decision but cannot promote itself to the Decision.

6. Commands, Events, Queries, and Contracts

Reference Commands:

- CreateConstraintsandInvariants
- ValidateConstraintsandInvariants
- ApproveConstraintsandInvariants
- ActivateConstraintsandInvariants
- SuspendConstraintsandInvariants
- CorrectConstraintsandInvariants
- RetireConstraintsandInvariants

Reference Events:

- ConstraintsandInvariantsCreated
- ConstraintsandInvariantsValidated
- ConstraintsandInvariantsApproved
- ConstraintsandInvariantsActivated
- ConstraintsandInvariantsSuspended
- ConstraintsandInvariantsCorrected
- ConstraintsandInvariantsRetired

Reference queries:

- GetConstraintsandInvariants
- ListConstraintsandInvariantsRecords
- GetConstraintsandInvariantsTimeline
- GetConstraintsandInvariantsEvidence
- GetConstraintsandInvariantsDecisionPackage
- GetConstraintsandInvariantsExceptions
- GetConstraintsandInvariantsAuditTrail

Reference API surface:

```

POST /api/v1/constraintsandinvariants/commands
GET  /api/v1/constraintsandinvariants/{id}
GET  /api/v1/constraintsandinvariants/{id}/timeline
GET  /api/v1/constraintsandinvariants/{id}/evidence
GET  /api/v1/constraintsandinvariants/{id}/decisions
POST /api/v1/constraintsandinvariants/{id}/exceptions
POST /api/v1/constraintsandinvariants/{id}/corrections

```

These endpoints are reference contracts, not claims that exact routes exist.

7. Workflow, Failure, and Compensation

flowchart TD

```

A[Validate identity version and authority] --> B[Collect required evidence]
B --> C{Policy and evidence gates pass?}
C -- No --> D[Open exception or request correction]
D --> B
C -- Yes --> E[Record Decision or execution Intent]
E --> F[Execute Domain command or external request]
F --> G{Outcome known?}
G -- Success --> H[Record fact and publish Event]
G -- Failure --> I[Record failure and compensation options]

```

```

G -- Unknown --> J[Enter Outcome Unknown]
J --> K[Query source or wait for verified callback]
K --> G
H --> L[Update projections and downstream workflow]

```

Representative failure modes:

- stale or wrong source version
- duplicate submission or replay
- external timeout after possible success
- contradictory source observations
- lost Event after committed state
- approval or mandate revoked during execution
- downstream action before a mandatory gate
- partial completion across multiple subjects
- evidence later retracted or corrected
- tenant, facility, channel or jurisdiction mismatch

Recovery shall query the authoritative source before duplicate execution, preserve partial success, use compensating Intent rather than destructive rewrite, publish corrections, quarantine high-impact ambiguity, and record affected downstream subjects.

8. Data and Persistence Architecture

Reference table families:

Table family	Purpose
constraintsandinvariants_aggregate	authoritative subject
constraintsandinvariants_version	immutable submitted, approved, issued, or signed versions
constraintsandinvariants_relationship	governed relationships
constraintsandinvariants_decision	binding Decisions and rationale
constraintsandinvariants_evidence_link	provenance and evidence relationships
constraintsandinvariants_event_outbox	transactional Event publication
constraintsandinvariants_idempotency	duplicate-write protection
constraintsandinvariants_exception	exception, claim, dispute, or hold
constraintsandinvariants_correction	correction and supersession lineage
constraintsandinvariants_audit	privileged action evidence

The owning service controls schema and migration. Cross-Domain references use stable identifiers or contracts. Analytics, reports, caches, search, and vector indexes remain projections. Backup, restore, retention, legal hold, privacy, residency, and decommissioning requirements are explicit.

9. Portal and Experience Requirements

- Primary participant Portal: initiate, review exact versions, supply evidence, and respond to required actions.
- Operator Portal: inspect exceptions, correlation, source responses, retries, compensation, and downstream impact.
- Executive or oversight view: consume governed metrics without becoming a source of Domain truth.

Every Portal displays exact version, owner, state, freshness, required action, and uncertainty. It supports loading, empty, partial, stale, denied, error, timeout, and Outcome Unknown states; accessibility; locale, currency, unit, date, and RTL/LTR correctness; evidence audit; safe retry; and correction history.

10. AI Participation and Tool Governance

Permitted AI tasks:

- assist schema and lineage analysis
- detect data-quality anomalies
- summarize migration and reconciliation risk
- never mutate authoritative data outside typed Domain Tools

AI shall not own Domain records, grant itself authority, suppress contradictions, silently edit approved versions, fabricate external outcomes, or execute high-impact actions outside typed Tools, policy, approval, idempotency, sandbox, and outcome observation.

Every Agent Run records source context, prompt/model/tool versions, structured output, confidence and limitations, Policy result, approval, Tool calls, provider responses, final outcome, and recovery actions.

11. Security, Privacy, and Trust Boundaries

- least privilege and separation of duties
- tenant, organization, facility, channel and jurisdiction isolation
- integrity protection for binding evidence
- verified external callbacks and anti-replay controls

Additional controls include secret isolation, callback signature verification, audit of privileged action, emergency-access review, sensitive-data minimization, purpose limitation, and threat models for impersonation, tampering, replay, confused deputy, fraud, prompt injection, and exfiltration.

12. Reliability, SLOs, and Operations

SLI	Reference objective
authoritative read availability	99.9% monthly unless a stricter tier applies
acknowledged Command durability	no acknowledged write may be silently lost
Event publication	99.9% within five minutes after committed fact
duplicate prevention	100% for same tenant, Command class, and idempotency key
binding Decision audit completeness	100%
stale-state detection	bounded by business deadline

Dashboards and runbooks cover backlog, error budget, dependency failure, retry budget, DLQ, workflow stall, provider outage, reconciliation, capacity, backup/restore, release/rollback, incident command, and post-incident correction.

13. Testing and Continuous Conformance

- aggregate invariant and transition tests
- authorization, delegation, separation-of-duties and tenant-isolation tests
- idempotency and optimistic-concurrency tests
- database migration, key, constraint and rollback tests
- contract, outbox/inbox, duplicate, replay and DLQ tests
- workflow restart, timeout and compensation tests
- Portal E2E tests for success, partial, stale, denied, error and Outcome Unknown states
- accessibility, locale, currency, unit and RTL/LTR tests
- security adversarial, fraud and data-exfiltration tests
- AI evaluation, prompt-injection, policy-bypass and unauthorized-tool tests
- historical reconstruction, correction and audit-completeness tests
- performance, capacity, recovery and operational exercise tests

Release evidence links each invariant and control to the test, environment, input data, result, owner, and artifact proving it. Generated test counts never replace semantic coverage.

14. Repository Review Boundary

- This chapter is a repository-review specification. Exact executable implementation has not been inspected for every target behavior.
- Registered modules, catalog entries, README claims, frontend contracts, or plans are not proof of complete implementation.

Area	Classification
target aggregate, API, tables, state and workflow in this chapter	REFERENCE_ARCHITECTURE
registered module or catalog claim	DOCUMENTATION_ONLY_CLAIM until source inspection
uninspected integration	REQUIRES_REPOSITORY_REVIEW

Area	Classification
behavior proven in another repository-grounded chapter contradictory ownership or path claims	retain that chapter's evidence and classification CONFLICT_REQUIRES_CANONICALIZATION

No statement upgrades implementation status merely because a file, directory, module name, frontend contract, database entry, plan, or report exists.

15. Worked Conformance Scenario

An authorized principal initiates a Command against an exact subject version. The service validates identity, tenant, mandate, Policy, Evidence, and idempotency. It records the state transition and outbox entry atomically. If an external dependency participates, timeout produces Outcome Unknown; reconciliation queries the source or waits for a verified callback instead of repeating the action blindly.

Portals show source, freshness, version, required action, and uncertainty. AI may extract, compare, summarize, detect anomalies, and recommend through typed Tools, but cannot own the subject or fabricate success. Later correction appends new facts, identifies affected projections and workflows, and preserves the historical Decision context.

16. Conformance Checklist

- ☐ stable identity and proposition ownership are explicit;
- ☐ authority and mandate are evaluated at action time;
- ☐ state, Recommendation, Decision, Intent, execution, and outcome remain distinct;
- ☐ Commands are typed, authorized, idempotent, and version guarded;
- ☐ Events describe completed facts;
- ☐ Evidence includes provenance, time, integrity, qualification, and correction;
- ☐ external outcomes retain their authority source;
- ☐ Portals show freshness and uncertainty;
- ☐ AI is bounded by typed Tools, Policy, approval, and outcome observation;
- ☐ security, tenant, facility, channel, and jurisdiction boundaries are enforced;
- ☐ Outcome Unknown and reconciliation exist where needed;
- ☐ corrections preserve history;
- ☐ tests prove invariants and recovery;
- ☐ implementation status is not upgraded without executable evidence.

17. Status Vocabulary

Status	Meaning
VERIFIED_IMPLEMENTED	Executable source and relevant behavior were inspected.
IMPLEMENTED_WITH_GAP	Executable behavior exists with a material governance or completeness gap.
PARTIAL_IMPLEMENTATION	Only part of the target capability is evidenced.
REFERENCE_ARCHITECTURE	Normative target behavior; no claim that it exists today.
PLANNED_NOT_IMPLEMENTED	Explicitly planned but not evidenced as executable.
DOCUMENTATION_ONLY_CLAIM	Documentation or client contract claims behavior without verified backend evidence.
CONFLICT_REQUIRES_CANONICALIZATION	Sources disagree and no final owner has been established.
REQUIRES_REPOSITORY_REVIEW	Necessary executable evidence has not yet been inspected.

Indexes and Query Workloads

Metadata	Value
Volume	10
Chapter ID	V10-10
Subject	IndexesandQueryWorkloads
Status	Generated First-Draft Architecture Skeleton - Domain-Specific Expansion Required
Content maturity	TEMPLATE_DERIVED_REFERENCE_SKELETON unless repository citations prove otherwise
Default implementation classification	REFERENCE_ARCHITECTURE / REQUIRES_REPOSITORY_REVIEW
Accountable owner	Data Architecture Council
Evidence rule	No implementation claim without inspected executable evidence
Repository	chinair88-prog/allinb2c

1. Executive Orientation

This chapter defines how GTOS shall **define enterprise and Domain data ownership, semantics, persistence, quality, lineage, correction, protection, operations, and verification for Indexes and Query Workloads.**

The specification separates reality, observations, knowledge, Decisions, Intent, execution, and outcomes. A component name, database row, UI label, AI answer, or workflow state does not acquire authority merely because it is visible or technically convenient.

Reality or external outcome

- ≠ Observation
- ≠ Claim
- ≠ derived Perspective
- ≠ Prediction
- ≠ Recommendation
- ≠ authorized Decision
- ≠ execution Intent

1.1 Accountable ownership

The accountable owner is **Data Architecture Council**. Supporting applications, shared infrastructure, integration adapters, analytics, and AI coordinate or present the capability but shall not silently own its authoritative propositions.

1.2 Scope

- create and identify the subject
- validate evidence and policy
- make authorized Decisions
- execute typed Commands
- publish completed-fact Events
- query current and historical perspectives
- correct and reconcile outcomes

1.3 Non-goals

- treating a registered Gradle module, catalog record, README, frontend route, migration plan, or generated report as proof of end-to-end implementation;
- direct cross-Domain database writes;

- generic status values that combine lifecycle, approval, provider, document, payment, dispute, and correction state;
- silent replacement of history;
- AI-created authority or fabricated external outcomes.

2. Ubiquitous Language and Subject Model

Subject	Required interpretation
IndexesandQueryWorkloads	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
IndexesandQueryWorkloadsVersion	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
IndexesandQueryWorkloadsDecision	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
IndexesandQueryWorkloadsEvidenceSet	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
IndexesandQueryWorkloadsException	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit

Reference aggregate:

```
IndexesandQueryWorkloads {
  subjectId
  tenantId
  organizationContext
  sourceIdentityRefs[]
  sourceVersionRefs[]
  lifecycleState
  relationshipRefs[]
  decisionRefs[]
  intentRefs[]
  evidenceSetRef
  knowledgeCutoff
  policyRefs[]
  authorityRef
  jurisdictionRefs[]
  createdAt
  effectiveAt
  updatedAt
  version
  correctionOf?
  correlationId
}
```

2.1 Core invariants

- Stable identity precedes relationship, state, and cross-system correlation.
- Effective time and recorded time remain distinct where business truth changes over time.
- Recommendation, Decision, Intent, execution, and outcome are separate concepts.
- No Portal, report, search index, cache, workflow, or AI Agent becomes a shadow owner of Domain truth.
- Binding submitted, approved, issued, accepted, or signed versions are immutable.
- Corrections append governed facts and preserve the original record.
- External authority and provider outcomes retain their source and qualification.
- Tenant and represented-principal context come from trusted authentication and mandate evaluation.
- Retried writes are idempotent and concurrency guarded.
- Outcome Unknown is explicit and reconciled before risky duplicate execution.

2.2 Required standards

- stable identifiers
- typed contracts
- immutable binding versions
- transactional outbox
- idempotent Commands
- explicit Outcome Unknown
- correction lineage
- machine-verifiable conformance

3. Actors, Roles, and Authority

A conformant design distinguishes:

- the natural person acting;
- the service or AI identity invoking a Tool;
- the organization or principal represented;
- the role or relationship granting context;
- the mandate or delegation granting action authority;
- the Policy and jurisdiction limiting the action;
- the reviewer or approver responsible for a binding Decision.

Authority is evaluated at action time. Authentication alone does not grant business authority. Cached permissions and browser-supplied tenant headers are insufficient for high-impact actions.

4. Lifecycle and State Model

State	Semantics
PROPOSED	Distinct governed condition with entry, exit, timeout, correction and authority semantics
VALIDATION_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
APPROVAL_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
ACTIVE	Distinct governed condition with entry, exit, timeout, correction and authority semantics
SUSPENDED	Distinct governed condition with entry, exit, timeout, correction and authority semantics
CORRECTION_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
RETIRED	Distinct governed condition with entry, exit, timeout, correction and authority semantics

```
stateDiagram-v2
    [*] --> PROPOSED
    PROPOSED --> VALIDATION_PENDING
    VALIDATION_PENDING --> APPROVAL_PENDING
    APPROVAL_PENDING --> ACTIVE
    ACTIVE --> SUSPENDED
    SUSPENDED --> CORRECTION_PENDING
    CORRECTION_PENDING --> RETIRED
```

The diagram is a primary reading path. Real implementations may add parallel review, amendment, appeal, timeout, suspension, partial completion, reconciliation, correction, and terminal failure without collapsing their meaning.

4.1 Transition rules

- transition through a typed, authorized Command;
 - require exact expected version where concurrent change matters;
 - record actor, represented principal, mandate, Policy, reason, and Evidence;
 - use an idempotency key for retried writes;
 - publish a completed-fact Event after durable state change;
 - expose Outcome Unknown for unresolved external completion;
 - preserve original and corrected states in historical reconstruction.
-

5. Decision, Evidence, and Knowledge Cutoff

Reference Decision:

```
Decision {
    decisionId
    decisionType
    subjectRef
}
```

```

sourceVersionRefs[]
knowledgeCutoff
evidenceSetRef
policyVersionRefs[]
recommendationRefs[]
principalId
representedOrganizationId
mandateRef
rationale
outcome
decidedAt
effectiveAt
correctionOf?
}

```

Evidence records identify issuer or observer, subject, version, observed time, received time, effective interval, integrity, confidence, qualification, contradiction, correction, retention, residency, and access class. A recommendation may assist the Decision but cannot promote itself to the Decision.

6. Commands, Events, Queries, and Contracts

Reference Commands:

- CreateIndexesandQueryWorkloads
- ValidateIndexesandQueryWorkloads
- ApproveIndexesandQueryWorkloads
- ActivateIndexesandQueryWorkloads
- SuspendIndexesandQueryWorkloads
- CorrectIndexesandQueryWorkloads
- RetireIndexesandQueryWorkloads

Reference Events:

- IndexesandQueryWorkloadsCreated
- IndexesandQueryWorkloadsValidated
- IndexesandQueryWorkloadsApproved
- IndexesandQueryWorkloadsActivated
- IndexesandQueryWorkloadsSuspended
- IndexesandQueryWorkloadsCorrected
- IndexesandQueryWorkloadsRetired

Reference queries:

- GetIndexesandQueryWorkloads
- ListIndexesandQueryWorkloadsRecords
- GetIndexesandQueryWorkloadsTimeline
- GetIndexesandQueryWorkloadsEvidence
- GetIndexesandQueryWorkloadsDecisionPackage
- GetIndexesandQueryWorkloadsExceptions
- GetIndexesandQueryWorkloadsAuditTrail

Reference API surface:

```

POST /api/v1/indexesandqueryworkloads/commands
GET  /api/v1/indexesandqueryworkloads/{id}
GET  /api/v1/indexesandqueryworkloads/{id}/timeline
GET  /api/v1/indexesandqueryworkloads/{id}/evidence
GET  /api/v1/indexesandqueryworkloads/{id}/decisions
POST /api/v1/indexesandqueryworkloads/{id}/exceptions
POST /api/v1/indexesandqueryworkloads/{id}/corrections

```

These endpoints are reference contracts, not claims that exact routes exist.

7. Workflow, Failure, and Compensation

flowchart TD

```

A[Validate identity version and authority] --> B[Collect required evidence]
B --> C{Policy and evidence gates pass?}
C -- No --> D[Open exception or request correction]
D --> B
C -- Yes --> E[Record Decision or execution Intent]
E --> F[Execute Domain command or external request]

```

```

F --> G{Outcome known?}
G -- Success --> H[Record fact and publish Event]
G -- Failure --> I[Record failure and compensation options]
G -- Unknown --> J[Enter Outcome Unknown]
J --> K[Query source or wait for verified callback]
K --> G
H --> L[Update projections and downstream workflow]

```

Representative failure modes:

- stale or wrong source version
- duplicate submission or replay
- external timeout after possible success
- contradictory source observations
- lost Event after committed state
- approval or mandate revoked during execution
- downstream action before a mandatory gate
- partial completion across multiple subjects
- evidence later retracted or corrected
- tenant, facility, channel or jurisdiction mismatch

Recovery shall query the authoritative source before duplicate execution, preserve partial success, use compensating Intent rather than destructive rewrite, publish corrections, quarantine high-impact ambiguity, and record affected downstream subjects.

8. Data and Persistence Architecture

Reference table families:

Table family	Purpose
indexesandqueryworkloads_aggregate	authoritative subject
indexesandqueryworkloads_version	immutable submitted, approved, issued, or signed versions
indexesandqueryworkloads_relationship	governed relationships
indexesandqueryworkloads_decision	binding Decisions and rationale
indexesandqueryworkloads_evidence_link	provenance and evidence relationships
indexesandqueryworkloads_event_outbox	transactional Event publication
indexesandqueryworkloads_idempotency	duplicate-write protection
indexesandqueryworkloads_exception	exception, claim, dispute, or hold
indexesandqueryworkloads_correction	correction and supersession lineage
indexesandqueryworkloads_audit	privileged action evidence

The owning service controls schema and migration. Cross-Domain references use stable identifiers or contracts. Analytics, reports, caches, search, and vector indexes remain projections. Backup, restore, retention, legal hold, privacy, residency, and decommissioning requirements are explicit.

9. Portal and Experience Requirements

- Primary participant Portal: initiate, review exact versions, supply evidence, and respond to required actions.
- Operator Portal: inspect exceptions, correlation, source responses, retries, compensation, and downstream impact.
- Executive or oversight view: consume governed metrics without becoming a source of Domain truth.

Every Portal displays exact version, owner, state, freshness, required action, and uncertainty. It supports loading, empty, partial, stale, denied, error, timeout, and Outcome Unknown states; accessibility; locale, currency, unit, date, and RTL/LTR correctness; evidence audit; safe retry; and correction history.

10. AI Participation and Tool Governance

Permitted AI tasks:

- assist schema and lineage analysis
- detect data-quality anomalies
- summarize migration and reconciliation risk
- never mutate authoritative data outside typed Domain Tools

AI shall not own Domain records, grant itself authority, suppress contradictions, silently edit approved versions, fabricate external outcomes, or execute high-impact actions outside typed Tools, policy, approval, idempotency, sandbox, and outcome observation.

Every Agent Run records source context, prompt/model/tool versions, structured output, confidence and limitations, Policy result, approval, Tool calls, provider responses, final outcome, and recovery actions.

11. Security, Privacy, and Trust Boundaries

- least privilege and separation of duties
- tenant, organization, facility, channel and jurisdiction isolation
- integrity protection for binding evidence
- verified external callbacks and anti-replay controls

Additional controls include secret isolation, callback signature verification, audit of privileged action, emergency-access review, sensitive-data minimization, purpose limitation, and threat models for impersonation, tampering, replay, confused deputy, fraud, prompt injection, and exfiltration.

12. Reliability, SLOs, and Operations

SLI	Reference objective
authoritative read availability	99.9% monthly unless a stricter tier applies
acknowledged Command durability	no acknowledged write may be silently lost
Event publication	99.9% within five minutes after committed fact
duplicate prevention	100% for same tenant, Command class, and idempotency key
binding Decision audit completeness	100%
stale-state detection	bounded by business deadline

Dashboards and runbooks cover backlog, error budget, dependency failure, retry budget, DLQ, workflow stall, provider outage, reconciliation, capacity, backup/restore, release/rollback, incident command, and post-incident correction.

13. Testing and Continuous Conformance

- aggregate invariant and transition tests
- authorization, delegation, separation-of-duties and tenant-isolation tests
- idempotency and optimistic-concurrency tests
- database migration, key, constraint and rollback tests
- contract, outbox/inbox, duplicate, replay and DLQ tests
- workflow restart, timeout and compensation tests
- Portal E2E tests for success, partial, stale, denied, error and Outcome Unknown states
- accessibility, locale, currency, unit and RTL/LTR tests
- security adversarial, fraud and data-exfiltration tests
- AI evaluation, prompt-injection, policy-bypass and unauthorized-tool tests
- historical reconstruction, correction and audit-completeness tests
- performance, capacity, recovery and operational exercise tests

Release evidence links each invariant and control to the test, environment, input data, result, owner, and artifact proving it. Generated test counts never replace semantic coverage.

14. Repository Review Boundary

- This chapter is a repository-review specification. Exact executable implementation has not been inspected for every target behavior.
- Registered modules, catalog entries, README claims, frontend contracts, or plans are not proof of complete implementation.

Area	Classification
target aggregate, API, tables, state and workflow in this chapter	REFERENCE_ARCHITECTURE
registered module or catalog claim	DOCUMENTATION_ONLY_CLAIM until source inspection
uninspected integration	REQUIRES_REPOSITORY_REVIEW
behavior proven in another repository-grounded chapter	retain that chapter's evidence and classification
contradictory ownership or path claims	CONFLICT_REQUIRES_CANONICALIZATION

No statement upgrades implementation status merely because a file, directory, module name, frontend contract, database entry, plan, or report exists.

15. Worked Conformance Scenario

An authorized principal initiates a Command against an exact subject version. The service validates identity, tenant, mandate, Policy, Evidence, and idempotency. It records the state transition and outbox entry atomically. If an external dependency participates, timeout produces Outcome Unknown; reconciliation queries the source or waits for a verified callback instead of repeating the action blindly.

Portals show source, freshness, version, required action, and uncertainty. AI may extract, compare, summarize, detect anomalies, and recommend through typed Tools, but cannot own the subject or fabricate success. Later correction appends new facts, identifies affected projections and workflows, and preserves the historical Decision context.

16. Conformance Checklist

- ☐ stable identity and proposition ownership are explicit;
- ☐ authority and mandate are evaluated at action time;
- ☐ state, Recommendation, Decision, Intent, execution, and outcome remain distinct;
- ☐ Commands are typed, authorized, idempotent, and version guarded;
- ☐ Events describe completed facts;
- ☐ Evidence includes provenance, time, integrity, qualification, and correction;
- ☐ external outcomes retain their authority source;
- ☐ Portals show freshness and uncertainty;
- ☐ AI is bounded by typed Tools, Policy, approval, and outcome observation;
- ☐ security, tenant, facility, channel, and jurisdiction boundaries are enforced;
- ☐ Outcome Unknown and reconciliation exist where needed;
- ☐ corrections preserve history;
- ☐ tests prove invariants and recovery;
- ☐ implementation status is not upgraded without executable evidence.

17. Status Vocabulary

Status	Meaning
VERIFIED_IMPLEMENTED	Executable source and relevant behavior were inspected.
IMPLEMENTED_WITH_GAP	Executable behavior exists with a material governance or completeness gap.
PARTIAL_IMPLEMENTATION	Only part of the target capability is evidenced.
REFERENCE_ARCHITECTURE	Normative target behavior; no claim that it exists today.
PLANNED_NOT_IMPLEMENTED	Explicitly planned but not evidenced as executable.
DOCUMENTATION_ONLY_CLAIM	Documentation or client contract claims behavior without verified backend evidence.
CONFLICT_REQUIRES_CANONICALIZATION	Sources disagree and no final owner has been established.
REQUIRES_REPOSITORY_REVIEW	Necessary executable evidence has not yet been inspected.

Effective Time and Recorded Time

Metadata	Value
Volume	10
Chapter ID	V10-11
Subject	EffectiveTimeandRecordedTime
Status	Generated First-Draft Architecture Skeleton - Domain-Specific Expansion Required
Content maturity	TEMPLATE_DERIVED_REFERENCE_SKELETON unless repository citations prove otherwise
Default implementation classification	REFERENCE_ARCHITECTURE / REQUIRES_REPOSITORY_REVIEW
Accountable owner	Data Architecture Council
Evidence rule	No implementation claim without inspected executable evidence
Repository	chinair88-prog/allinb2c

1. Executive Orientation

This chapter defines how GTOS shall **define enterprise and Domain data ownership, semantics, persistence, quality, lineage, correction, protection, operations, and verification for Effective Time and Recorded Time.**

The specification separates reality, observations, knowledge, Decisions, Intent, execution, and outcomes. A component name, database row, UI label, AI answer, or workflow state does not acquire authority merely because it is visible or technically convenient.

Reality or external outcome

- ≠ Observation
- ≠ Claim
- ≠ derived Perspective
- ≠ Prediction
- ≠ Recommendation
- ≠ authorized Decision
- ≠ execution Intent

1.1 Accountable ownership

The accountable owner is **Data Architecture Council**. Supporting applications, shared infrastructure, integration adapters, analytics, and AI coordinate or present the capability but shall not silently own its authoritative propositions.

1.2 Scope

- create and identify the subject
- validate evidence and policy
- make authorized Decisions
- execute typed Commands
- publish completed-fact Events
- query current and historical perspectives
- correct and reconcile outcomes

1.3 Non-goals

- treating a registered Gradle module, catalog record, README, frontend route, migration plan, or generated report as proof of end-to-end implementation;
- direct cross-Domain database writes;
- generic status values that combine lifecycle, approval, provider, document, payment, dispute, and correction state;

- silent replacement of history;
- AI-created authority or fabricated external outcomes.

2. Ubiquitous Language and Subject Model

Subject	Required interpretation
EffectiveTimeandRecordedTime	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
EffectiveTimeandRecordedTimeVersion	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
EffectiveTimeandRecordedTimeDecision	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
EffectiveTimeandRecordedTimeEvidenceSet	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
EffectiveTimeandRecordedTimeException	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit

Reference aggregate:

```
EffectiveTimeandRecordedTime {
  subjectId
  tenantId
  organizationContext
  sourceIdentityRefs[]
  sourceVersionRefs[]
  lifecycleState
  relationshipRefs[]
  decisionRefs[]
  intentRefs[]
  evidenceSetRef
  knowledgeCutoff
  policyRefs[]
  authorityRef
  jurisdictionRefs[]
  createdAt
  effectiveAt
  updatedAt
  version
  correctionOf?
  correlationId
}
```

2.1 Core invariants

- Stable identity precedes relationship, state, and cross-system correlation.
- Effective time and recorded time remain distinct where business truth changes over time.
- Recommendation, Decision, Intent, execution, and outcome are separate concepts.
- No Portal, report, search index, cache, workflow, or AI Agent becomes a shadow owner of Domain truth.
- Binding submitted, approved, issued, accepted, or signed versions are immutable.
- Corrections append governed facts and preserve the original record.
- External authority and provider outcomes retain their source and qualification.
- Tenant and represented-principal context come from trusted authentication and mandate evaluation.
- Retried writes are idempotent and concurrency guarded.
- Outcome Unknown is explicit and reconciled before risky duplicate execution.

2.2 Required standards

- stable identifiers
- typed contracts
- immutable binding versions
- transactional outbox
- idempotent Commands
- explicit Outcome Unknown
- correction lineage
- machine-verifiable conformance

3. Actors, Roles, and Authority

A conformant design distinguishes:

- the natural person acting;
- the service or AI identity invoking a Tool;
- the organization or principal represented;
- the role or relationship granting context;
- the mandate or delegation granting action authority;
- the Policy and jurisdiction limiting the action;
- the reviewer or approver responsible for a binding Decision.

Authority is evaluated at action time. Authentication alone does not grant business authority. Cached permissions and browser-supplied tenant headers are insufficient for high-impact actions.

4. Lifecycle and State Model

State	Semantics
PROPOSED	Distinct governed condition with entry, exit, timeout, correction and authority semantics
VALIDATION_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
APPROVAL_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
ACTIVE	Distinct governed condition with entry, exit, timeout, correction and authority semantics
SUSPENDED	Distinct governed condition with entry, exit, timeout, correction and authority semantics
CORRECTION_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
RETIRED	Distinct governed condition with entry, exit, timeout, correction and authority semantics

```
stateDiagram-v2
    [*] --> PROPOSED
    PROPOSED --> VALIDATION_PENDING
    VALIDATION_PENDING --> APPROVAL_PENDING
    APPROVAL_PENDING --> ACTIVE
    ACTIVE --> SUSPENDED
    SUSPENDED --> CORRECTION_PENDING
    CORRECTION_PENDING --> RETIRED
```

The diagram is a primary reading path. Real implementations may add parallel review, amendment, appeal, timeout, suspension, partial completion, reconciliation, correction, and terminal failure without collapsing their meaning.

4.1 Transition rules

- transition through a typed, authorized Command;
- require exact expected version where concurrent change matters;
- record actor, represented principal, mandate, Policy, reason, and Evidence;
- use an idempotency key for retried writes;
- publish a completed-fact Event after durable state change;
- expose Outcome Unknown for unresolved external completion;
- preserve original and corrected states in historical reconstruction.

5. Decision, Evidence, and Knowledge Cutoff

Reference Decision:

```
Decision {
    decisionId
    decisionType
    subjectRef
    sourceVersionRefs[]
    knowledgeCutoff
    evidenceSetRef
}
```

```

policyVersionRefs[]
recommendationRefs[]
principalId
representedOrganizationId
mandateRef
rationale
outcome
decidedAt
effectiveAt
correctionOf?
}

```

Evidence records identify issuer or observer, subject, version, observed time, received time, effective interval, integrity, confidence, qualification, contradiction, correction, retention, residency, and access class. A recommendation may assist the Decision but cannot promote itself to the Decision.

6. Commands, Events, Queries, and Contracts

Reference Commands:

- CreateEffectiveTimeandRecordedTime
- ValidateEffectiveTimeandRecordedTime
- ApproveEffectiveTimeandRecordedTime
- ActivateEffectiveTimeandRecordedTime
- SuspendEffectiveTimeandRecordedTime
- CorrectEffectiveTimeandRecordedTime
- RetireEffectiveTimeandRecordedTime

Reference Events:

- EffectiveTimeandRecordedTimeCreated
- EffectiveTimeandRecordedTimeValidated
- EffectiveTimeandRecordedTimeApproved
- EffectiveTimeandRecordedTimeActivated
- EffectiveTimeandRecordedTimeSuspended
- EffectiveTimeandRecordedTimeCorrected
- EffectiveTimeandRecordedTimeRetired

Reference queries:

- GetEffectiveTimeandRecordedTime
- ListEffectiveTimeandRecordedTimeRecords
- GetEffectiveTimeandRecordedTimeTimeline
- GetEffectiveTimeandRecordedTimeEvidence
- GetEffectiveTimeandRecordedTimeDecisionPackage
- GetEffectiveTimeandRecordedTimeExceptions
- GetEffectiveTimeandRecordedTimeAuditTrail

Reference API surface:

```

POST /api/v1/effectivetimeandrecordedtime/commands
GET  /api/v1/effectivetimeandrecordedtime/{id}
GET  /api/v1/effectivetimeandrecordedtime/{id}/timeline
GET  /api/v1/effectivetimeandrecordedtime/{id}/evidence
GET  /api/v1/effectivetimeandrecordedtime/{id}/decisions
POST /api/v1/effectivetimeandrecordedtime/{id}/exceptions
POST /api/v1/effectivetimeandrecordedtime/{id}/corrections

```

These endpoints are reference contracts, not claims that exact routes exist.

7. Workflow, Failure, and Compensation

flowchart TD

```

A[Validate identity version and authority] --> B[Collect required evidence]
B --> C{Policy and evidence gates pass?}
C -- No --> D[Open exception or request correction]
D --> B
C -- Yes --> E[Record Decision or execution Intent]
E --> F[Execute Domain command or external request]
F --> G{Outcome known?}
G -- Success --> H[Record fact and publish Event]
G -- Failure --> I[Record failure and compensation options]

```

```

G -- Unknown --> J[Enter Outcome Unknown]
J --> K[Query source or wait for verified callback]
K --> G
H --> L[Update projections and downstream workflow]

```

Representative failure modes:

- stale or wrong source version
- duplicate submission or replay
- external timeout after possible success
- contradictory source observations
- lost Event after committed state
- approval or mandate revoked during execution
- downstream action before a mandatory gate
- partial completion across multiple subjects
- evidence later retracted or corrected
- tenant, facility, channel or jurisdiction mismatch

Recovery shall query the authoritative source before duplicate execution, preserve partial success, use compensating Intent rather than destructive rewrite, publish corrections, quarantine high-impact ambiguity, and record affected downstream subjects.

8. Data and Persistence Architecture

Reference table families:

Table family	Purpose
effectivetimeandrecordedtime_aggregate	authoritative subject
effectivetimeandrecordedtime_version	immutable submitted, approved, issued, or signed versions
effectivetimeandrecordedtime_relationship	governed relationships
effectivetimeandrecordedtime_decision	binding Decisions and rationale
effectivetimeandrecordedtime_evidence_link	provenance and evidence relationships
effectivetimeandrecordedtime_event_outbox	transactional Event publication
effectivetimeandrecordedtime_idempotency	duplicate-write protection
effectivetimeandrecordedtime_exception	exception, claim, dispute, or hold
effectivetimeandrecordedtime_correction	correction and supersession lineage
effectivetimeandrecordedtime_audit	privileged action evidence

The owning service controls schema and migration. Cross-Domain references use stable identifiers or contracts. Analytics, reports, caches, search, and vector indexes remain projections. Backup, restore, retention, legal hold, privacy, residency, and decommissioning requirements are explicit.

9. Portal and Experience Requirements

- Primary participant Portal: initiate, review exact versions, supply evidence, and respond to required actions.
- Operator Portal: inspect exceptions, correlation, source responses, retries, compensation, and downstream impact.
- Executive or oversight view: consume governed metrics without becoming a source of Domain truth.

Every Portal displays exact version, owner, state, freshness, required action, and uncertainty. It supports loading, empty, partial, stale, denied, error, timeout, and Outcome Unknown states; accessibility; locale, currency, unit, date, and RTL/LTR correctness; evidence audit; safe retry; and correction history.

10. AI Participation and Tool Governance

Permitted AI tasks:

- assist schema and lineage analysis
- detect data-quality anomalies
- summarize migration and reconciliation risk
- never mutate authoritative data outside typed Domain Tools

AI shall not own Domain records, grant itself authority, suppress contradictions, silently edit approved versions, fabricate external outcomes, or execute high-impact actions outside typed Tools, policy, approval, idempotency, sandbox, and outcome observation.

Every Agent Run records source context, prompt/model/tool versions, structured output, confidence and limitations, Policy result, approval, Tool calls, provider responses, final outcome, and recovery actions.

11. Security, Privacy, and Trust Boundaries

- least privilege and separation of duties
- tenant, organization, facility, channel and jurisdiction isolation
- integrity protection for binding evidence
- verified external callbacks and anti-replay controls

Additional controls include secret isolation, callback signature verification, audit of privileged action, emergency-access review, sensitive-data minimization, purpose limitation, and threat models for impersonation, tampering, replay, confused deputy, fraud, prompt injection, and exfiltration.

12. Reliability, SLOs, and Operations

SLI	Reference objective
authoritative read availability	99.9% monthly unless a stricter tier applies
acknowledged Command durability	no acknowledged write may be silently lost
Event publication	99.9% within five minutes after committed fact
duplicate prevention	100% for same tenant, Command class, and idempotency key
binding Decision audit completeness	100%
stale-state detection	bounded by business deadline

Dashboards and runbooks cover backlog, error budget, dependency failure, retry budget, DLQ, workflow stall, provider outage, reconciliation, capacity, backup/restore, release/rollback, incident command, and post-incident correction.

13. Testing and Continuous Conformance

- aggregate invariant and transition tests
- authorization, delegation, separation-of-duties and tenant-isolation tests
- idempotency and optimistic-concurrency tests
- database migration, key, constraint and rollback tests
- contract, outbox/inbox, duplicate, replay and DLQ tests
- workflow restart, timeout and compensation tests
- Portal E2E tests for success, partial, stale, denied, error and Outcome Unknown states
- accessibility, locale, currency, unit and RTL/LTR tests
- security adversarial, fraud and data-exfiltration tests
- AI evaluation, prompt-injection, policy-bypass and unauthorized-tool tests
- historical reconstruction, correction and audit-completeness tests
- performance, capacity, recovery and operational exercise tests

Release evidence links each invariant and control to the test, environment, input data, result, owner, and artifact proving it. Generated test counts never replace semantic coverage.

14. Repository Review Boundary

- This chapter is a repository-review specification. Exact executable implementation has not been inspected for every target behavior.
- Registered modules, catalog entries, README claims, frontend contracts, or plans are not proof of complete implementation.

Area	Classification
target aggregate, API, tables, state and workflow in this chapter	REFERENCE_ARCHITECTURE
registered module or catalog claim	DOCUMENTATION_ONLY_CLAIM until source inspection
uninspected integration	REQUIRES_REPOSITORY_REVIEW

Area	Classification
behavior proven in another repository-grounded chapter contradictory ownership or path claims	retain that chapter's evidence and classification CONFLICT_REQUIRES_CANONICALIZATION

No statement upgrades implementation status merely because a file, directory, module name, frontend contract, database entry, plan, or report exists.

15. Worked Conformance Scenario

An authorized principal initiates a Command against an exact subject version. The service validates identity, tenant, mandate, Policy, Evidence, and idempotency. It records the state transition and outbox entry atomically. If an external dependency participates, timeout produces Outcome Unknown; reconciliation queries the source or waits for a verified callback instead of repeating the action blindly.

Portals show source, freshness, version, required action, and uncertainty. AI may extract, compare, summarize, detect anomalies, and recommend through typed Tools, but cannot own the subject or fabricate success. Later correction appends new facts, identifies affected projections and workflows, and preserves the historical Decision context.

16. Conformance Checklist

- ☐ stable identity and proposition ownership are explicit;
- ☐ authority and mandate are evaluated at action time;
- ☐ state, Recommendation, Decision, Intent, execution, and outcome remain distinct;
- ☐ Commands are typed, authorized, idempotent, and version guarded;
- ☐ Events describe completed facts;
- ☐ Evidence includes provenance, time, integrity, qualification, and correction;
- ☐ external outcomes retain their authority source;
- ☐ Portals show freshness and uncertainty;
- ☐ AI is bounded by typed Tools, Policy, approval, and outcome observation;
- ☐ security, tenant, facility, channel, and jurisdiction boundaries are enforced;
- ☐ Outcome Unknown and reconciliation exist where needed;
- ☐ corrections preserve history;
- ☐ tests prove invariants and recovery;
- ☐ implementation status is not upgraded without executable evidence.

17. Status Vocabulary

Status	Meaning
VERIFIED_IMPLEMENTED IMPLEMENTED_WITH_GAP	Executable source and relevant behavior were inspected. Executable behavior exists with a material governance or completeness gap.
PARTIAL_IMPLEMENTATION REFERENCE_ARCHITECTURE PLANNED_NOT_IMPLEMENTED DOCUMENTATION_ONLY_CLAIM	Only part of the target capability is evidenced. Normative target behavior; no claim that it exists today. Explicitly planned but not evidenced as executable. Documentation or client contract claims behavior without verified backend evidence.
CONFLICT_REQUIRES_CANONICALIZATION REQUIRES_REPOSITORY_REVIEW	Sources disagree and no final owner has been established. Necessary executable evidence has not yet been inspected.

Data Classification

Metadata	Value
Volume	10
Chapter ID	V10-12
Subject	DataClassification
Status	Generated First-Draft Architecture Skeleton - Domain-Specific Expansion Required
Content maturity	TEMPLATE_DERIVED_REFERENCE_SKELETON unless repository citations prove otherwise
Default implementation classification	REFERENCE_ARCHITECTURE / REQUIRES_REPOSITORY_REVIEW
Accountable owner	Data Architecture Council
Evidence rule	No implementation claim without inspected executable evidence
Repository	chinair88-prog/allinb2c

1. Executive Orientation

This chapter defines how GTOS shall **define enterprise and Domain data ownership, semantics, persistence, quality, lineage, correction, protection, operations, and verification for Data Classification.**

The specification separates reality, observations, knowledge, Decisions, Intent, execution, and outcomes. A component name, database row, UI label, AI answer, or workflow state does not acquire authority merely because it is visible or technically convenient.

Reality or external outcome

- ≠ Observation
- ≠ Claim
- ≠ derived Perspective
- ≠ Prediction
- ≠ Recommendation
- ≠ authorized Decision
- ≠ execution Intent

1.1 Accountable ownership

The accountable owner is **Data Architecture Council**. Supporting applications, shared infrastructure, integration adapters, analytics, and AI coordinate or present the capability but shall not silently own its authoritative propositions.

1.2 Scope

- create and identify the subject
- validate evidence and policy
- make authorized Decisions
- execute typed Commands
- publish completed-fact Events
- query current and historical perspectives
- correct and reconcile outcomes

1.3 Non-goals

- treating a registered Gradle module, catalog record, README, frontend route, migration plan, or generated report as proof of end-to-end implementation;
- direct cross-Domain database writes;
- generic status values that combine lifecycle, approval, provider, document, payment, dispute, and correction state;

- silent replacement of history;
- AI-created authority or fabricated external outcomes.

2. Ubiquitous Language and Subject Model

Subject	Required interpretation
DataClassification	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
DataClassificationVersion	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
DataClassificationDecision	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
DataClassificationEvidenceSet	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
DataClassificationException	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit

Reference aggregate:

```
DataClassification {
  subjectId
  tenantId
  organizationContext
  sourceIdentityRefs[]
  sourceVersionRefs[]
  lifecycleState
  relationshipRefs[]
  decisionRefs[]
  intentRefs[]
  evidenceSetRef
  knowledgeCutoff
  policyRefs[]
  authorityRef
  jurisdictionRefs[]
  createdAt
  effectiveAt
  updatedAt
  version
  correctionOf?
  correlationId
}
```

2.1 Core invariants

- Stable identity precedes relationship, state, and cross-system correlation.
- Effective time and recorded time remain distinct where business truth changes over time.
- Recommendation, Decision, Intent, execution, and outcome are separate concepts.
- No Portal, report, search index, cache, workflow, or AI Agent becomes a shadow owner of Domain truth.
- Binding submitted, approved, issued, accepted, or signed versions are immutable.
- Corrections append governed facts and preserve the original record.
- External authority and provider outcomes retain their source and qualification.
- Tenant and represented-principal context come from trusted authentication and mandate evaluation.
- Retried writes are idempotent and concurrency guarded.
- Outcome Unknown is explicit and reconciled before risky duplicate execution.

2.2 Required standards

- stable identifiers
- typed contracts
- immutable binding versions
- transactional outbox
- idempotent Commands
- explicit Outcome Unknown
- correction lineage
- machine-verifiable conformance

3. Actors, Roles, and Authority

A conformant design distinguishes:

- the natural person acting;
- the service or AI identity invoking a Tool;
- the organization or principal represented;
- the role or relationship granting context;
- the mandate or delegation granting action authority;
- the Policy and jurisdiction limiting the action;
- the reviewer or approver responsible for a binding Decision.

Authority is evaluated at action time. Authentication alone does not grant business authority. Cached permissions and browser-supplied tenant headers are insufficient for high-impact actions.

4. Lifecycle and State Model

State	Semantics
PROPOSED	Distinct governed condition with entry, exit, timeout, correction and authority semantics
VALIDATION_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
APPROVAL_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
ACTIVE	Distinct governed condition with entry, exit, timeout, correction and authority semantics
SUSPENDED	Distinct governed condition with entry, exit, timeout, correction and authority semantics
CORRECTION_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
RETIRED	Distinct governed condition with entry, exit, timeout, correction and authority semantics

```
stateDiagram-v2
    [*] --> PROPOSED
    PROPOSED --> VALIDATION_PENDING
    VALIDATION_PENDING --> APPROVAL_PENDING
    APPROVAL_PENDING --> ACTIVE
    ACTIVE --> SUSPENDED
    SUSPENDED --> CORRECTION_PENDING
    CORRECTION_PENDING --> RETIRED
```

The diagram is a primary reading path. Real implementations may add parallel review, amendment, appeal, timeout, suspension, partial completion, reconciliation, correction, and terminal failure without collapsing their meaning.

4.1 Transition rules

- transition through a typed, authorized Command;
- require exact expected version where concurrent change matters;
- record actor, represented principal, mandate, Policy, reason, and Evidence;
- use an idempotency key for retried writes;
- publish a completed-fact Event after durable state change;
- expose Outcome Unknown for unresolved external completion;
- preserve original and corrected states in historical reconstruction.

5. Decision, Evidence, and Knowledge Cutoff

Reference Decision:

```
Decision {
    decisionId
    decisionType
    subjectRef
    sourceVersionRefs[]
    knowledgeCutoff
    evidenceSetRef
}
```

```

policyVersionRefs[]
recommendationRefs[]
principalId
representedOrganizationId
mandateRef
rationale
outcome
decidedAt
effectiveAt
correctionOf?
}

```

Evidence records identify issuer or observer, subject, version, observed time, received time, effective interval, integrity, confidence, qualification, contradiction, correction, retention, residency, and access class. A recommendation may assist the Decision but cannot promote itself to the Decision.

6. Commands, Events, Queries, and Contracts

Reference Commands:

- CreateDataClassification
- ValidateDataClassification
- ApproveDataClassification
- ActivateDataClassification
- SuspendDataClassification
- CorrectDataClassification
- RetireDataClassification

Reference Events:

- DataClassificationCreated
- DataClassificationValidated
- DataClassificationApproved
- DataClassificationActivated
- DataClassificationSuspended
- DataClassificationCorrected
- DataClassificationRetired

Reference queries:

- GetDataClassification
- ListDataClassificationRecords
- GetDataClassificationTimeline
- GetDataClassificationEvidence
- GetDataClassificationDecisionPackage
- GetDataClassificationExceptions
- GetDataClassificationAuditTrail

Reference API surface:

```

POST /api/v1/dataclassification/commands
GET  /api/v1/dataclassification/{id}
GET  /api/v1/dataclassification/{id}/timeline
GET  /api/v1/dataclassification/{id}/evidence
GET  /api/v1/dataclassification/{id}/decisions
POST /api/v1/dataclassification/{id}/exceptions
POST /api/v1/dataclassification/{id}/corrections

```

These endpoints are reference contracts, not claims that exact routes exist.

7. Workflow, Failure, and Compensation

flowchart TD

```

A[Validate identity version and authority] --> B[Collect required evidence]
B --> C{Policy and evidence gates pass?}
C -- No --> D[Open exception or request correction]
D --> B
C -- Yes --> E[Record Decision or execution Intent]
E --> F[Execute Domain command or external request]
F --> G{Outcome known?}
G -- Success --> H[Record fact and publish Event]
G -- Failure --> I[Record failure and compensation options]

```

```

G -- Unknown --> J[Enter Outcome Unknown]
J --> K[Query source or wait for verified callback]
K --> G
H --> L[Update projections and downstream workflow]

```

Representative failure modes:

- stale or wrong source version
- duplicate submission or replay
- external timeout after possible success
- contradictory source observations
- lost Event after committed state
- approval or mandate revoked during execution
- downstream action before a mandatory gate
- partial completion across multiple subjects
- evidence later retracted or corrected
- tenant, facility, channel or jurisdiction mismatch

Recovery shall query the authoritative source before duplicate execution, preserve partial success, use compensating Intent rather than destructive rewrite, publish corrections, quarantine high-impact ambiguity, and record affected downstream subjects.

8. Data and Persistence Architecture

Reference table families:

Table family	Purpose
dataclassification_aggregate	authoritative subject
dataclassification_version	immutable submitted, approved, issued, or signed versions
dataclassification_relationship	governed relationships
dataclassification_decision	binding Decisions and rationale
dataclassification_evidence_link	provenance and evidence relationships
dataclassification_event_outbox	transactional Event publication
dataclassification_idempotency	duplicate-write protection
dataclassification_exception	exception, claim, dispute, or hold
dataclassification_correction	correction and supersession lineage
dataclassification_audit	privileged action evidence

The owning service controls schema and migration. Cross-Domain references use stable identifiers or contracts. Analytics, reports, caches, search, and vector indexes remain projections. Backup, restore, retention, legal hold, privacy, residency, and decommissioning requirements are explicit.

9. Portal and Experience Requirements

- Primary participant Portal: initiate, review exact versions, supply evidence, and respond to required actions.
- Operator Portal: inspect exceptions, correlation, source responses, retries, compensation, and downstream impact.
- Executive or oversight view: consume governed metrics without becoming a source of Domain truth.

Every Portal displays exact version, owner, state, freshness, required action, and uncertainty. It supports loading, empty, partial, stale, denied, error, timeout, and Outcome Unknown states; accessibility; locale, currency, unit, date, and RTL/LTR correctness; evidence audit; safe retry; and correction history.

10. AI Participation and Tool Governance

Permitted AI tasks:

- assist schema and lineage analysis
- detect data-quality anomalies
- summarize migration and reconciliation risk
- never mutate authoritative data outside typed Domain Tools

AI shall not own Domain records, grant itself authority, suppress contradictions, silently edit approved versions, fabricate external outcomes, or execute high-impact actions outside typed Tools, policy, approval, idempotency, sandbox, and outcome observation.

Every Agent Run records source context, prompt/model/tool versions, structured output, confidence and limitations, Policy result, approval, Tool calls, provider responses, final outcome, and recovery actions.

11. Security, Privacy, and Trust Boundaries

- least privilege and separation of duties
- tenant, organization, facility, channel and jurisdiction isolation
- integrity protection for binding evidence
- verified external callbacks and anti-replay controls

Additional controls include secret isolation, callback signature verification, audit of privileged action, emergency-access review, sensitive-data minimization, purpose limitation, and threat models for impersonation, tampering, replay, confused deputy, fraud, prompt injection, and exfiltration.

12. Reliability, SLOs, and Operations

SLI	Reference objective
authoritative read availability	99.9% monthly unless a stricter tier applies
acknowledged Command durability	no acknowledged write may be silently lost
Event publication	99.9% within five minutes after committed fact
duplicate prevention	100% for same tenant, Command class, and idempotency key
binding Decision audit completeness	100%
stale-state detection	bounded by business deadline

Dashboards and runbooks cover backlog, error budget, dependency failure, retry budget, DLQ, workflow stall, provider outage, reconciliation, capacity, backup/restore, release/rollback, incident command, and post-incident correction.

13. Testing and Continuous Conformance

- aggregate invariant and transition tests
- authorization, delegation, separation-of-duties and tenant-isolation tests
- idempotency and optimistic-concurrency tests
- database migration, key, constraint and rollback tests
- contract, outbox/inbox, duplicate, replay and DLQ tests
- workflow restart, timeout and compensation tests
- Portal E2E tests for success, partial, stale, denied, error and Outcome Unknown states
- accessibility, locale, currency, unit and RTL/LTR tests
- security adversarial, fraud and data-exfiltration tests
- AI evaluation, prompt-injection, policy-bypass and unauthorized-tool tests
- historical reconstruction, correction and audit-completeness tests
- performance, capacity, recovery and operational exercise tests

Release evidence links each invariant and control to the test, environment, input data, result, owner, and artifact proving it. Generated test counts never replace semantic coverage.

14. Repository Review Boundary

- This chapter is a repository-review specification. Exact executable implementation has not been inspected for every target behavior.
- Registered modules, catalog entries, README claims, frontend contracts, or plans are not proof of complete implementation.

Area	Classification
target aggregate, API, tables, state and workflow in this chapter	REFERENCE_ARCHITECTURE
registered module or catalog claim	DOCUMENTATION_ONLY_CLAIM until source inspection
uninspected integration	REQUIRES_REPOSITORY_REVIEW

Area	Classification
behavior proven in another repository-grounded chapter contradictory ownership or path claims	retain that chapter's evidence and classification CONFLICT_REQUIRES_CANONICALIZATION

No statement upgrades implementation status merely because a file, directory, module name, frontend contract, database entry, plan, or report exists.

15. Worked Conformance Scenario

An authorized principal initiates a Command against an exact subject version. The service validates identity, tenant, mandate, Policy, Evidence, and idempotency. It records the state transition and outbox entry atomically. If an external dependency participates, timeout produces Outcome Unknown; reconciliation queries the source or waits for a verified callback instead of repeating the action blindly.

Portals show source, freshness, version, required action, and uncertainty. AI may extract, compare, summarize, detect anomalies, and recommend through typed Tools, but cannot own the subject or fabricate success. Later correction appends new facts, identifies affected projections and workflows, and preserves the historical Decision context.

16. Conformance Checklist

- ☐ stable identity and proposition ownership are explicit;
- ☐ authority and mandate are evaluated at action time;
- ☐ state, Recommendation, Decision, Intent, execution, and outcome remain distinct;
- ☐ Commands are typed, authorized, idempotent, and version guarded;
- ☐ Events describe completed facts;
- ☐ Evidence includes provenance, time, integrity, qualification, and correction;
- ☐ external outcomes retain their authority source;
- ☐ Portals show freshness and uncertainty;
- ☐ AI is bounded by typed Tools, Policy, approval, and outcome observation;
- ☐ security, tenant, facility, channel, and jurisdiction boundaries are enforced;
- ☐ Outcome Unknown and reconciliation exist where needed;
- ☐ corrections preserve history;
- ☐ tests prove invariants and recovery;
- ☐ implementation status is not upgraded without executable evidence.

17. Status Vocabulary

Status	Meaning
VERIFIED_IMPLEMENTED	Executable source and relevant behavior were inspected.
IMPLEMENTED_WITH_GAP	Executable behavior exists with a material governance or completeness gap.
PARTIAL_IMPLEMENTATION	Only part of the target capability is evidenced.
REFERENCE_ARCHITECTURE	Normative target behavior; no claim that it exists today.
PLANNED_NOT_IMPLEMENTED	Explicitly planned but not evidenced as executable.
DOCUMENTATION_ONLY_CLAIM	Documentation or client contract claims behavior without verified backend evidence.
CONFLICT_REQUIRES_CANONICALIZATION	Sources disagree and no final owner has been established.
REQUIRES_REPOSITORY_REVIEW	Necessary executable evidence has not yet been inspected.

Privacy, Purpose and Residency

Metadata	Value
Volume	10
Chapter ID	V10-13
Subject	Privacy, Purpose and Residency
Status	Generated First-Draft Architecture Skeleton - Domain-Specific Expansion Required
Content maturity	TEMPLATE_DERIVED_REFERENCE_SKELETON unless repository citations prove otherwise
Default implementation classification	REFERENCE_ARCHITECTURE / REQUIRES_REPOSITORY_REVIEW
Accountable owner	Data Architecture Council
Evidence rule	No implementation claim without inspected executable evidence
Repository	chinair88-prog/allinb2c

1. Executive Orientation

This chapter defines how GTOS shall **define enterprise and Domain data ownership, semantics, persistence, quality, lineage, correction, protection, operations, and verification for Privacy, Purpose and Residency.**

The specification separates reality, observations, knowledge, Decisions, Intent, execution, and outcomes. A component name, database row, UI label, AI answer, or workflow state does not acquire authority merely because it is visible or technically convenient.

Reality or external outcome

- ≠ Observation
- ≠ Claim
- ≠ derived Perspective
- ≠ Prediction
- ≠ Recommendation
- ≠ authorized Decision
- ≠ execution Intent

1.1 Accountable ownership

The accountable owner is **Data Architecture Council**. Supporting applications, shared infrastructure, integration adapters, analytics, and AI coordinate or present the capability but shall not silently own its authoritative propositions.

1.2 Scope

- create and identify the subject
- validate evidence and policy
- make authorized Decisions
- execute typed Commands
- publish completed-fact Events
- query current and historical perspectives
- correct and reconcile outcomes

1.3 Non-goals

- treating a registered Gradle module, catalog record, README, frontend route, migration plan, or generated report as proof of end-to-end implementation;
- direct cross-Domain database writes;

- generic status values that combine lifecycle, approval, provider, document, payment, dispute, and correction state;
- silent replacement of history;
- AI-created authority or fabricated external outcomes.

2. Ubiquitous Language and Subject Model

Subject	Required interpretation
Privacy,PurposeandResidency	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
Privacy,PurposeandResidencyVersion	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
Privacy,PurposeandResidencyDecision	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
Privacy,PurposeandResidencyEvidenceSet	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
Privacy,PurposeandResidencyException	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit

Reference aggregate:

```
Privacy,PurposeandResidency {
  subjectId
  tenantId
  organizationContext
  sourceIdentityRefs[]
  sourceVersionRefs[]
  lifecycleState
  relationshipRefs[]
  decisionRefs[]
  intentRefs[]
  evidenceSetRef
  knowledgeCutoff
  policyRefs[]
  authorityRef
  jurisdictionRefs[]
  createdAt
  effectiveAt
  updatedAt
  version
  correctionOf?
  correlationId
}
```

2.1 Core invariants

- Stable identity precedes relationship, state, and cross-system correlation.
- Effective time and recorded time remain distinct where business truth changes over time.
- Recommendation, Decision, Intent, execution, and outcome are separate concepts.
- No Portal, report, search index, cache, workflow, or AI Agent becomes a shadow owner of Domain truth.
- Binding submitted, approved, issued, accepted, or signed versions are immutable.
- Corrections append governed facts and preserve the original record.
- External authority and provider outcomes retain their source and qualification.
- Tenant and represented-principal context come from trusted authentication and mandate evaluation.
- Retried writes are idempotent and concurrency guarded.
- Outcome Unknown is explicit and reconciled before risky duplicate execution.

2.2 Required standards

- stable identifiers
- typed contracts
- immutable binding versions
- transactional outbox
- idempotent Commands
- explicit Outcome Unknown
- correction lineage
- machine-verifiable conformance

3. Actors, Roles, and Authority

A conformant design distinguishes:

- the natural person acting;
- the service or AI identity invoking a Tool;
- the organization or principal represented;
- the role or relationship granting context;
- the mandate or delegation granting action authority;
- the Policy and jurisdiction limiting the action;
- the reviewer or approver responsible for a binding Decision.

Authority is evaluated at action time. Authentication alone does not grant business authority. Cached permissions and browser-supplied tenant headers are insufficient for high-impact actions.

4. Lifecycle and State Model

State	Semantics
PROPOSED	Distinct governed condition with entry, exit, timeout, correction and authority semantics
VALIDATION_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
APPROVAL_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
ACTIVE	Distinct governed condition with entry, exit, timeout, correction and authority semantics
SUSPENDED	Distinct governed condition with entry, exit, timeout, correction and authority semantics
CORRECTION_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
RETIRED	Distinct governed condition with entry, exit, timeout, correction and authority semantics

```
stateDiagram-v2
    [*] --> PROPOSED
    PROPOSED --> VALIDATION_PENDING
    VALIDATION_PENDING --> APPROVAL_PENDING
    APPROVAL_PENDING --> ACTIVE
    ACTIVE --> SUSPENDED
    SUSPENDED --> CORRECTION_PENDING
    CORRECTION_PENDING --> RETIRED
```

The diagram is a primary reading path. Real implementations may add parallel review, amendment, appeal, timeout, suspension, partial completion, reconciliation, correction, and terminal failure without collapsing their meaning.

4.1 Transition rules

- transition through a typed, authorized Command;
 - require exact expected version where concurrent change matters;
 - record actor, represented principal, mandate, Policy, reason, and Evidence;
 - use an idempotency key for retried writes;
 - publish a completed-fact Event after durable state change;
 - expose Outcome Unknown for unresolved external completion;
 - preserve original and corrected states in historical reconstruction.
-

5. Decision, Evidence, and Knowledge Cutoff

Reference Decision:

```
Decision {
    decisionId
    decisionType
    subjectRef
}
```

```

sourceVersionRefs[]
knowledgeCutoff
evidenceSetRef
policyVersionRefs[]
recommendationRefs[]
principalId
representedOrganizationId
mandateRef
rationale
outcome
decidedAt
effectiveAt
correctionOf?
}

```

Evidence records identify issuer or observer, subject, version, observed time, received time, effective interval, integrity, confidence, qualification, contradiction, correction, retention, residency, and access class. A recommendation may assist the Decision but cannot promote itself to the Decision.

6. Commands, Events, Queries, and Contracts

Reference Commands:

- CreatePrivacy, PurposeandResidency
- ValidatePrivacy, PurposeandResidency
- ApprovePrivacy, PurposeandResidency
- ActivatePrivacy, PurposeandResidency
- SuspendPrivacy, PurposeandResidency
- CorrectPrivacy, PurposeandResidency
- RetirePrivacy, PurposeandResidency

Reference Events:

- Privacy, PurposeandResidencyCreated
- Privacy, PurposeandResidencyValidated
- Privacy, PurposeandResidencyApproved
- Privacy, PurposeandResidencyActivated
- Privacy, PurposeandResidencySuspended
- Privacy, PurposeandResidencyCorrected
- Privacy, PurposeandResidencyRetired

Reference queries:

- GetPrivacy, PurposeandResidency
- ListPrivacy, PurposeandResidencyRecords
- GetPrivacy, PurposeandResidencyTimeline
- GetPrivacy, PurposeandResidencyEvidence
- GetPrivacy, PurposeandResidencyDecisionPackage
- GetPrivacy, PurposeandResidencyExceptions
- GetPrivacy, PurposeandResidencyAuditTrail

Reference API surface:

```

POST /api/v1/privacy_purposeandresidency/commands
GET  /api/v1/privacy_purposeandresidency/{id}
GET  /api/v1/privacy_purposeandresidency/{id}/timeline
GET  /api/v1/privacy_purposeandresidency/{id}/evidence
GET  /api/v1/privacy_purposeandresidency/{id}/decisions
POST /api/v1/privacy_purposeandresidency/{id}/exceptions
POST /api/v1/privacy_purposeandresidency/{id}/corrections

```

These endpoints are reference contracts, not claims that exact routes exist.

7. Workflow, Failure, and Compensation

flowchart TD

```

A[Validate identity version and authority] --> B[Collect required evidence]
B --> C{Policy and evidence gates pass?}
C -- No --> D[Open exception or request correction]
D --> B
C -- Yes --> E[Record Decision or execution Intent]
E --> F[Execute Domain command or external request]

```

```

F --> G{Outcome known?}
G -- Success --> H[Record fact and publish Event]
G -- Failure --> I[Record failure and compensation options]
G -- Unknown --> J[Enter Outcome Unknown]
J --> K[Query source or wait for verified callback]
K --> G
H --> L[Update projections and downstream workflow]

```

Representative failure modes:

- stale or wrong source version
- duplicate submission or replay
- external timeout after possible success
- contradictory source observations
- lost Event after committed state
- approval or mandate revoked during execution
- downstream action before a mandatory gate
- partial completion across multiple subjects
- evidence later retracted or corrected
- tenant, facility, channel or jurisdiction mismatch

Recovery shall query the authoritative source before duplicate execution, preserve partial success, use compensating Intent rather than destructive rewrite, publish corrections, quarantine high-impact ambiguity, and record affected downstream subjects.

8. Data and Persistence Architecture

Reference table families:

Table family	Purpose
privacy_purposeandresidency_aggregate	authoritative subject
privacy_purposeandresidency_version	immutable submitted, approved, issued, or signed versions
privacy_purposeandresidency_relationship	governed relationships
privacy_purposeandresidency_decision	binding Decisions and rationale
privacy_purposeandresidency_evidence_link	provenance and evidence relationships
privacy_purposeandresidency_event_outbox	transactional Event publication
privacy_purposeandresidency_idempotency	duplicate-write protection
privacy_purposeandresidency_exception	exception, claim, dispute, or hold
privacy_purposeandresidency_correction	correction and supersession lineage
privacy_purposeandresidency_audit	privileged action evidence

The owning service controls schema and migration. Cross-Domain references use stable identifiers or contracts. Analytics, reports, caches, search, and vector indexes remain projections. Backup, restore, retention, legal hold, privacy, residency, and decommissioning requirements are explicit.

9. Portal and Experience Requirements

- Primary participant Portal: initiate, review exact versions, supply evidence, and respond to required actions.
- Operator Portal: inspect exceptions, correlation, source responses, retries, compensation, and downstream impact.
- Executive or oversight view: consume governed metrics without becoming a source of Domain truth.

Every Portal displays exact version, owner, state, freshness, required action, and uncertainty. It supports loading, empty, partial, stale, denied, error, timeout, and Outcome Unknown states; accessibility; locale, currency, unit, date, and RTL/LTR correctness; evidence audit; safe retry; and correction history.

10. AI Participation and Tool Governance

Permitted AI tasks:

- assist schema and lineage analysis
- detect data-quality anomalies
- summarize migration and reconciliation risk
- never mutate authoritative data outside typed Domain Tools

AI shall not own Domain records, grant itself authority, suppress contradictions, silently edit approved versions, fabricate external outcomes, or execute high-impact actions outside typed Tools, policy, approval, idempotency, sandbox, and outcome observation.

Every Agent Run records source context, prompt/model/tool versions, structured output, confidence and limitations, Policy result, approval, Tool calls, provider responses, final outcome, and recovery actions.

11. Security, Privacy, and Trust Boundaries

- least privilege and separation of duties
- tenant, organization, facility, channel and jurisdiction isolation
- integrity protection for binding evidence
- verified external callbacks and anti-replay controls

Additional controls include secret isolation, callback signature verification, audit of privileged action, emergency-access review, sensitive-data minimization, purpose limitation, and threat models for impersonation, tampering, replay, confused deputy, fraud, prompt injection, and exfiltration.

12. Reliability, SLOs, and Operations

SLI	Reference objective
authoritative read availability	99.9% monthly unless a stricter tier applies
acknowledged Command durability	no acknowledged write may be silently lost
Event publication	99.9% within five minutes after committed fact
duplicate prevention	100% for same tenant, Command class, and idempotency key
binding Decision audit completeness	100%
state-state detection	bounded by business deadline

Dashboards and runbooks cover backlog, error budget, dependency failure, retry budget, DLQ, workflow stall, provider outage, reconciliation, capacity, backup/restore, release/rollback, incident command, and post-incident correction.

13. Testing and Continuous Conformance

- aggregate invariant and transition tests
- authorization, delegation, separation-of-duties and tenant-isolation tests
- idempotency and optimistic-concurrency tests
- database migration, key, constraint and rollback tests
- contract, outbox/inbox, duplicate, replay and DLQ tests
- workflow restart, timeout and compensation tests
- Portal E2E tests for success, partial, stale, denied, error and Outcome Unknown states
- accessibility, locale, currency, unit and RTL/LTR tests
- security adversarial, fraud and data-exfiltration tests
- AI evaluation, prompt-injection, policy-bypass and unauthorized-tool tests
- historical reconstruction, correction and audit-completeness tests
- performance, capacity, recovery and operational exercise tests

Release evidence links each invariant and control to the test, environment, input data, result, owner, and artifact proving it. Generated test counts never replace semantic coverage.

14. Repository Review Boundary

- This chapter is a repository-review specification. Exact executable implementation has not been inspected for every target behavior.
- Registered modules, catalog entries, README claims, frontend contracts, or plans are not proof of complete implementation.

Area	Classification
target aggregate, API, tables, state and workflow in this chapter	REFERENCE_ARCHITECTURE
registered module or catalog claim	DOCUMENTATION_ONLY_CLAIM until source inspection
uninspected integration	REQUIRES_REPOSITORY_REVIEW
behavior proven in another repository-grounded chapter	retain that chapter's evidence and classification
contradictory ownership or path claims	CONFLICT_REQUIRES_CANONICALIZATION

No statement upgrades implementation status merely because a file, directory, module name, frontend contract, database entry, plan, or report exists.

15. Worked Conformance Scenario

An authorized principal initiates a Command against an exact subject version. The service validates identity, tenant, mandate, Policy, Evidence, and idempotency. It records the state transition and outbox entry atomically. If an external dependency participates, timeout produces Outcome Unknown; reconciliation queries the source or waits for a verified callback instead of repeating the action blindly.

Portals show source, freshness, version, required action, and uncertainty. AI may extract, compare, summarize, detect anomalies, and recommend through typed Tools, but cannot own the subject or fabricate success. Later correction appends new facts, identifies affected projections and workflows, and preserves the historical Decision context.

16. Conformance Checklist

- ☐ stable identity and proposition ownership are explicit;
- ☐ authority and mandate are evaluated at action time;
- ☐ state, Recommendation, Decision, Intent, execution, and outcome remain distinct;
- ☐ Commands are typed, authorized, idempotent, and version guarded;
- ☐ Events describe completed facts;
- ☐ Evidence includes provenance, time, integrity, qualification, and correction;
- ☐ external outcomes retain their authority source;
- ☐ Portals show freshness and uncertainty;
- ☐ AI is bounded by typed Tools, Policy, approval, and outcome observation;
- ☐ security, tenant, facility, channel, and jurisdiction boundaries are enforced;
- ☐ Outcome Unknown and reconciliation exist where needed;
- ☐ corrections preserve history;
- ☐ tests prove invariants and recovery;
- ☐ implementation status is not upgraded without executable evidence.

17. Status Vocabulary

Status	Meaning
VERIFIED_IMPLEMENTED	Executable source and relevant behavior were inspected.
IMPLEMENTED_WITH_GAP	Executable behavior exists with a material governance or completeness gap.
PARTIAL_IMPLEMENTATION	Only part of the target capability is evidenced.
REFERENCE_ARCHITECTURE	Normative target behavior; no claim that it exists today.
PLANNED_NOT_IMPLEMENTED	Explicitly planned but not evidenced as executable.
DOCUMENTATION_ONLY_CLAIM	Documentation or client contract claims behavior without verified backend evidence.
CONFLICT_REQUIRES_CANONICALIZATION	Sources disagree and no final owner has been established.
REQUIRES_REPOSITORY_REVIEW	Necessary executable evidence has not yet been inspected.

Retention and Legal Hold

Metadata	Value
Volume	10
Chapter ID	V10-14
Subject	RetentionandLegalHold
Status	Generated First-Draft Architecture Skeleton - Domain-Specific Expansion Required
Content maturity	TEMPLATE_DERIVED_REFERENCE_SKELETON unless repository citations prove otherwise
Default implementation classification	REFERENCE_ARCHITECTURE / REQUIRES_REPOSITORY_REVIEW
Accountable owner	Data Architecture Council
Evidence rule	No implementation claim without inspected executable evidence
Repository	chinair88-prog/allinb2c

1. Executive Orientation

This chapter defines how GTOS shall **define enterprise and Domain data ownership, semantics, persistence, quality, lineage, correction, protection, operations, and verification for Retention and Legal Hold.**

The specification separates reality, observations, knowledge, Decisions, Intent, execution, and outcomes. A component name, database row, UI label, AI answer, or workflow state does not acquire authority merely because it is visible or technically convenient.

Reality or external outcome

- ≠ Observation
- ≠ Claim
- ≠ derived Perspective
- ≠ Prediction
- ≠ Recommendation
- ≠ authorized Decision
- ≠ execution Intent

1.1 Accountable ownership

The accountable owner is **Data Architecture Council**. Supporting applications, shared infrastructure, integration adapters, analytics, and AI coordinate or present the capability but shall not silently own its authoritative propositions.

1.2 Scope

- create and identify the subject
- validate evidence and policy
- make authorized Decisions
- execute typed Commands
- publish completed-fact Events
- query current and historical perspectives
- correct and reconcile outcomes

1.3 Non-goals

- treating a registered Gradle module, catalog record, README, frontend route, migration plan, or generated report as proof of end-to-end implementation;
- direct cross-Domain database writes;

- generic status values that combine lifecycle, approval, provider, document, payment, dispute, and correction state;
- silent replacement of history;
- AI-created authority or fabricated external outcomes.

2. Ubiquitous Language and Subject Model

Subject	Required interpretation
RetentionandLegalHold	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
RetentionandLegalHoldVersion	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
RetentionandLegalHoldDecision	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
RetentionandLegalHoldEvidenceSet	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
RetentionandLegalHoldException	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit

Reference aggregate:

```
RetentionandLegalHold {
  subjectId
  tenantId
  organizationContext
  sourceIdentityRefs[]
  sourceVersionRefs[]
  lifecycleState
  relationshipRefs[]
  decisionRefs[]
  intentRefs[]
  evidenceSetRef
  knowledgeCutoff
  policyRefs[]
  authorityRef
  jurisdictionRefs[]
  createdAt
  effectiveAt
  updatedAt
  version
  correctionOf?
  correlationId
}
```

2.1 Core invariants

- Stable identity precedes relationship, state, and cross-system correlation.
- Effective time and recorded time remain distinct where business truth changes over time.
- Recommendation, Decision, Intent, execution, and outcome are separate concepts.
- No Portal, report, search index, cache, workflow, or AI Agent becomes a shadow owner of Domain truth.
- Binding submitted, approved, issued, accepted, or signed versions are immutable.
- Corrections append governed facts and preserve the original record.
- External authority and provider outcomes retain their source and qualification.
- Tenant and represented-principal context come from trusted authentication and mandate evaluation.
- Retried writes are idempotent and concurrency guarded.
- Outcome Unknown is explicit and reconciled before risky duplicate execution.

2.2 Required standards

- stable identifiers
- typed contracts
- immutable binding versions
- transactional outbox
- idempotent Commands
- explicit Outcome Unknown
- correction lineage
- machine-verifiable conformance

3. Actors, Roles, and Authority

A conformant design distinguishes:

- the natural person acting;
- the service or AI identity invoking a Tool;
- the organization or principal represented;
- the role or relationship granting context;
- the mandate or delegation granting action authority;
- the Policy and jurisdiction limiting the action;
- the reviewer or approver responsible for a binding Decision.

Authority is evaluated at action time. Authentication alone does not grant business authority. Cached permissions and browser-supplied tenant headers are insufficient for high-impact actions.

4. Lifecycle and State Model

State	Semantics
PROPOSED	Distinct governed condition with entry, exit, timeout, correction and authority semantics
VALIDATION_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
APPROVAL_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
ACTIVE	Distinct governed condition with entry, exit, timeout, correction and authority semantics
SUSPENDED	Distinct governed condition with entry, exit, timeout, correction and authority semantics
CORRECTION_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
RETIRED	Distinct governed condition with entry, exit, timeout, correction and authority semantics

```
stateDiagram-v2
    [*] --> PROPOSED
    PROPOSED --> VALIDATION_PENDING
    VALIDATION_PENDING --> APPROVAL_PENDING
    APPROVAL_PENDING --> ACTIVE
    ACTIVE --> SUSPENDED
    SUSPENDED --> CORRECTION_PENDING
    CORRECTION_PENDING --> RETIRED
```

The diagram is a primary reading path. Real implementations may add parallel review, amendment, appeal, timeout, suspension, partial completion, reconciliation, correction, and terminal failure without collapsing their meaning.

4.1 Transition rules

- transition through a typed, authorized Command;
 - require exact expected version where concurrent change matters;
 - record actor, represented principal, mandate, Policy, reason, and Evidence;
 - use an idempotency key for retried writes;
 - publish a completed-fact Event after durable state change;
 - expose Outcome Unknown for unresolved external completion;
 - preserve original and corrected states in historical reconstruction.
-

5. Decision, Evidence, and Knowledge Cutoff

Reference Decision:

```
Decision {
    decisionId
    decisionType
    subjectRef
}
```

```

sourceVersionRefs[]
knowledgeCutoff
evidenceSetRef
policyVersionRefs[]
recommendationRefs[]
principalId
representedOrganizationId
mandateRef
rationale
outcome
decidedAt
effectiveAt
correctionOf?
}

```

Evidence records identify issuer or observer, subject, version, observed time, received time, effective interval, integrity, confidence, qualification, contradiction, correction, retention, residency, and access class. A recommendation may assist the Decision but cannot promote itself to the Decision.

6. Commands, Events, Queries, and Contracts

Reference Commands:

- CreateRetentionandLegalHold
- ValidateRetentionandLegalHold
- ApproveRetentionandLegalHold
- ActivateRetentionandLegalHold
- SuspendRetentionandLegalHold
- CorrectRetentionandLegalHold
- RetireRetentionandLegalHold

Reference Events:

- RetentionandLegalHoldCreated
- RetentionandLegalHoldValidated
- RetentionandLegalHoldApproved
- RetentionandLegalHoldActivated
- RetentionandLegalHoldSuspended
- RetentionandLegalHoldCorrected
- RetentionandLegalHoldRetired

Reference queries:

- GetRetentionandLegalHold
- ListRetentionandLegalHoldRecords
- GetRetentionandLegalHoldTimeline
- GetRetentionandLegalHoldEvidence
- GetRetentionandLegalHoldDecisionPackage
- GetRetentionandLegalHoldExceptions
- GetRetentionandLegalHoldAuditTrail

Reference API surface:

```

POST /api/v1/retentionandlegalhold/commands
GET  /api/v1/retentionandlegalhold/{id}
GET  /api/v1/retentionandlegalhold/{id}/timeline
GET  /api/v1/retentionandlegalhold/{id}/evidence
GET  /api/v1/retentionandlegalhold/{id}/decisions
POST /api/v1/retentionandlegalhold/{id}/exceptions
POST /api/v1/retentionandlegalhold/{id}/corrections

```

These endpoints are reference contracts, not claims that exact routes exist.

7. Workflow, Failure, and Compensation

flowchart TD

```

A[Validate identity version and authority] --> B[Collect required evidence]
B --> C{Policy and evidence gates pass?}
C -- No --> D[Open exception or request correction]
D --> B
C -- Yes --> E[Record Decision or execution Intent]
E --> F[Execute Domain command or external request]

```

```

F --> G{Outcome known?}
G -- Success --> H[Record fact and publish Event]
G -- Failure --> I[Record failure and compensation options]
G -- Unknown --> J[Enter Outcome Unknown]
J --> K[Query source or wait for verified callback]
K --> G
H --> L[Update projections and downstream workflow]

```

Representative failure modes:

- stale or wrong source version
- duplicate submission or replay
- external timeout after possible success
- contradictory source observations
- lost Event after committed state
- approval or mandate revoked during execution
- downstream action before a mandatory gate
- partial completion across multiple subjects
- evidence later retracted or corrected
- tenant, facility, channel or jurisdiction mismatch

Recovery shall query the authoritative source before duplicate execution, preserve partial success, use compensating Intent rather than destructive rewrite, publish corrections, quarantine high-impact ambiguity, and record affected downstream subjects.

8. Data and Persistence Architecture

Reference table families:

Table family	Purpose
retentionandlegalhold_aggregate	authoritative subject
retentionandlegalhold_version	immutable submitted, approved, issued, or signed versions
retentionandlegalhold_relationship	governed relationships
retentionandlegalhold_decision	binding Decisions and rationale
retentionandlegalhold_evidence_link	provenance and evidence relationships
retentionandlegalhold_event_outbox	transactional Event publication
retentionandlegalhold_idempotency	duplicate-write protection
retentionandlegalhold_exception	exception, claim, dispute, or hold
retentionandlegalhold_correction	correction and supersession lineage
retentionandlegalhold_audit	privileged action evidence

The owning service controls schema and migration. Cross-Domain references use stable identifiers or contracts. Analytics, reports, caches, search, and vector indexes remain projections. Backup, restore, retention, legal hold, privacy, residency, and decommissioning requirements are explicit.

9. Portal and Experience Requirements

- Primary participant Portal: initiate, review exact versions, supply evidence, and respond to required actions.
- Operator Portal: inspect exceptions, correlation, source responses, retries, compensation, and downstream impact.
- Executive or oversight view: consume governed metrics without becoming a source of Domain truth.

Every Portal displays exact version, owner, state, freshness, required action, and uncertainty. It supports loading, empty, partial, stale, denied, error, timeout, and Outcome Unknown states; accessibility; locale, currency, unit, date, and RTL/LTR correctness; evidence audit; safe retry; and correction history.

10. AI Participation and Tool Governance

Permitted AI tasks:

- assist schema and lineage analysis
- detect data-quality anomalies
- summarize migration and reconciliation risk
- never mutate authoritative data outside typed Domain Tools

AI shall not own Domain records, grant itself authority, suppress contradictions, silently edit approved versions, fabricate external outcomes, or execute high-impact actions outside typed Tools, policy, approval, idempotency, sandbox, and outcome observation.

Every Agent Run records source context, prompt/model/tool versions, structured output, confidence and limitations, Policy result, approval, Tool calls, provider responses, final outcome, and recovery actions.

11. Security, Privacy, and Trust Boundaries

- least privilege and separation of duties
- tenant, organization, facility, channel and jurisdiction isolation
- integrity protection for binding evidence
- verified external callbacks and anti-replay controls

Additional controls include secret isolation, callback signature verification, audit of privileged action, emergency-access review, sensitive-data minimization, purpose limitation, and threat models for impersonation, tampering, replay, confused deputy, fraud, prompt injection, and exfiltration.

12. Reliability, SLOs, and Operations

SLI	Reference objective
authoritative read availability	99.9% monthly unless a stricter tier applies
acknowledged Command durability	no acknowledged write may be silently lost
Event publication	99.9% within five minutes after committed fact
duplicate prevention	100% for same tenant, Command class, and idempotency key
binding Decision audit completeness	100%
stale-state detection	bounded by business deadline

Dashboards and runbooks cover backlog, error budget, dependency failure, retry budget, DLQ, workflow stall, provider outage, reconciliation, capacity, backup/restore, release/rollback, incident command, and post-incident correction.

13. Testing and Continuous Conformance

- aggregate invariant and transition tests
- authorization, delegation, separation-of-duties and tenant-isolation tests
- idempotency and optimistic-concurrency tests
- database migration, key, constraint and rollback tests
- contract, outbox/inbox, duplicate, replay and DLQ tests
- workflow restart, timeout and compensation tests
- Portal E2E tests for success, partial, stale, denied, error and Outcome Unknown states
- accessibility, locale, currency, unit and RTL/LTR tests
- security adversarial, fraud and data-exfiltration tests
- AI evaluation, prompt-injection, policy-bypass and unauthorized-tool tests
- historical reconstruction, correction and audit-completeness tests
- performance, capacity, recovery and operational exercise tests

Release evidence links each invariant and control to the test, environment, input data, result, owner, and artifact proving it. Generated test counts never replace semantic coverage.

14. Repository Review Boundary

- This chapter is a repository-review specification. Exact executable implementation has not been inspected for every target behavior.
- Registered modules, catalog entries, README claims, frontend contracts, or plans are not proof of complete implementation.

Area	Classification
target aggregate, API, tables, state and workflow in this chapter	REFERENCE_ARCHITECTURE
registered module or catalog claim	DOCUMENTATION_ONLY_CLAIM until source inspection
uninspected integration	REQUIRES_REPOSITORY_REVIEW
behavior proven in another repository-grounded chapter	retain that chapter's evidence and classification
contradictory ownership or path claims	CONFLICT_REQUIRES_CANONICALIZATION

No statement upgrades implementation status merely because a file, directory, module name, frontend contract, database entry, plan, or report exists.

15. Worked Conformance Scenario

An authorized principal initiates a Command against an exact subject version. The service validates identity, tenant, mandate, Policy, Evidence, and idempotency. It records the state transition and outbox entry atomically. If an external dependency participates, timeout produces Outcome Unknown; reconciliation queries the source or waits for a verified callback instead of repeating the action blindly.

Portals show source, freshness, version, required action, and uncertainty. AI may extract, compare, summarize, detect anomalies, and recommend through typed Tools, but cannot own the subject or fabricate success. Later correction appends new facts, identifies affected projections and workflows, and preserves the historical Decision context.

16. Conformance Checklist

- ☐ stable identity and proposition ownership are explicit;
- ☐ authority and mandate are evaluated at action time;
- ☐ state, Recommendation, Decision, Intent, execution, and outcome remain distinct;
- ☐ Commands are typed, authorized, idempotent, and version guarded;
- ☐ Events describe completed facts;
- ☐ Evidence includes provenance, time, integrity, qualification, and correction;
- ☐ external outcomes retain their authority source;
- ☐ Portals show freshness and uncertainty;
- ☐ AI is bounded by typed Tools, Policy, approval, and outcome observation;
- ☐ security, tenant, facility, channel, and jurisdiction boundaries are enforced;
- ☐ Outcome Unknown and reconciliation exist where needed;
- ☐ corrections preserve history;
- ☐ tests prove invariants and recovery;
- ☐ implementation status is not upgraded without executable evidence.

17. Status Vocabulary

Status	Meaning
VERIFIED_IMPLEMENTED	Executable source and relevant behavior were inspected.
IMPLEMENTED_WITH_GAP	Executable behavior exists with a material governance or completeness gap.
PARTIAL_IMPLEMENTATION	Only part of the target capability is evidenced.
REFERENCE_ARCHITECTURE	Normative target behavior; no claim that it exists today.
PLANNED_NOT_IMPLEMENTED	Explicitly planned but not evidenced as executable.
DOCUMENTATION_ONLY_CLAIM	Documentation or client contract claims behavior without verified backend evidence.
CONFLICT_REQUIRES_CANONICALIZATION	Sources disagree and no final owner has been established.
REQUIRES_REPOSITORY_REVIEW	Necessary executable evidence has not yet been inspected.

Data Lineage

Metadata	Value
Volume	10
Chapter ID	V10-15
Subject	DataLineage
Status	Generated First-Draft Architecture Skeleton - Domain-Specific Expansion Required
Content maturity	TEMPLATE_DERIVED_REFERENCE_SKELETON unless repository citations prove otherwise
Default implementation classification	REFERENCE_ARCHITECTURE / REQUIRES_REPOSITORY_REVIEW
Accountable owner	Data Architecture Council
Evidence rule	No implementation claim without inspected executable evidence
Repository	chinair88-prog/allinb2c

1. Executive Orientation

This chapter defines how GTOS shall **define enterprise and Domain data ownership, semantics, persistence, quality, lineage, correction, protection, operations, and verification for Data Lineage.**

The specification separates reality, observations, knowledge, Decisions, Intent, execution, and outcomes. A component name, database row, UI label, AI answer, or workflow state does not acquire authority merely because it is visible or technically convenient.

Reality or external outcome

- ≠ Observation
- ≠ Claim
- ≠ derived Perspective
- ≠ Prediction
- ≠ Recommendation
- ≠ authorized Decision
- ≠ execution Intent

1.1 Accountable ownership

The accountable owner is **Data Architecture Council**. Supporting applications, shared infrastructure, integration adapters, analytics, and AI coordinate or present the capability but shall not silently own its authoritative propositions.

1.2 Scope

- create and identify the subject
- validate evidence and policy
- make authorized Decisions
- execute typed Commands
- publish completed-fact Events
- query current and historical perspectives
- correct and reconcile outcomes

1.3 Non-goals

- treating a registered Gradle module, catalog record, README, frontend route, migration plan, or generated report as proof of end-to-end implementation;
- direct cross-Domain database writes;

- generic status values that combine lifecycle, approval, provider, document, payment, dispute, and correction state;
- silent replacement of history;
- AI-created authority or fabricated external outcomes.

2. Ubiquitous Language and Subject Model

Subject	Required interpretation
DataLineage	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
DataLineageVersion	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
DataLineageDecision	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
DataLineageEvidenceSet	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
DataLineageException	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit

Reference aggregate:

```
DataLineage {
  subjectId
  tenantId
  organizationContext
  sourceIdentityRefs[]
  sourceVersionRefs[]
  lifecycleState
  relationshipRefs[]
  decisionRefs[]
  intentRefs[]
  evidenceSetRef
  knowledgeCutoff
  policyRefs[]
  authorityRef
  jurisdictionRefs[]
  createdAt
  effectiveAt
  updatedAt
  version
  correctionOf?
  correlationId
}
```

2.1 Core invariants

- Stable identity precedes relationship, state, and cross-system correlation.
- Effective time and recorded time remain distinct where business truth changes over time.
- Recommendation, Decision, Intent, execution, and outcome are separate concepts.
- No Portal, report, search index, cache, workflow, or AI Agent becomes a shadow owner of Domain truth.
- Binding submitted, approved, issued, accepted, or signed versions are immutable.
- Corrections append governed facts and preserve the original record.
- External authority and provider outcomes retain their source and qualification.
- Tenant and represented-principal context come from trusted authentication and mandate evaluation.
- Retried writes are idempotent and concurrency guarded.
- Outcome Unknown is explicit and reconciled before risky duplicate execution.

2.2 Required standards

- stable identifiers
- typed contracts
- immutable binding versions
- transactional outbox
- idempotent Commands
- explicit Outcome Unknown
- correction lineage
- machine-verifiable conformance

3. Actors, Roles, and Authority

A conformant design distinguishes:

- the natural person acting;
- the service or AI identity invoking a Tool;
- the organization or principal represented;
- the role or relationship granting context;
- the mandate or delegation granting action authority;
- the Policy and jurisdiction limiting the action;
- the reviewer or approver responsible for a binding Decision.

Authority is evaluated at action time. Authentication alone does not grant business authority. Cached permissions and browser-supplied tenant headers are insufficient for high-impact actions.

4. Lifecycle and State Model

State	Semantics
PROPOSED	Distinct governed condition with entry, exit, timeout, correction and authority semantics
VALIDATION_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
APPROVAL_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
ACTIVE	Distinct governed condition with entry, exit, timeout, correction and authority semantics
SUSPENDED	Distinct governed condition with entry, exit, timeout, correction and authority semantics
CORRECTION_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
RETIRED	Distinct governed condition with entry, exit, timeout, correction and authority semantics

```
stateDiagram-v2
    [*] --> PROPOSED
    PROPOSED --> VALIDATION_PENDING
    VALIDATION_PENDING --> APPROVAL_PENDING
    APPROVAL_PENDING --> ACTIVE
    ACTIVE --> SUSPENDED
    SUSPENDED --> CORRECTION_PENDING
    CORRECTION_PENDING --> RETIRED
```

The diagram is a primary reading path. Real implementations may add parallel review, amendment, appeal, timeout, suspension, partial completion, reconciliation, correction, and terminal failure without collapsing their meaning.

4.1 Transition rules

- transition through a typed, authorized Command;
 - require exact expected version where concurrent change matters;
 - record actor, represented principal, mandate, Policy, reason, and Evidence;
 - use an idempotency key for retried writes;
 - publish a completed-fact Event after durable state change;
 - expose Outcome Unknown for unresolved external completion;
 - preserve original and corrected states in historical reconstruction.
-

5. Decision, Evidence, and Knowledge Cutoff

Reference Decision:

```
Decision {
    decisionId
    decisionType
    subjectRef
}
```

```

sourceVersionRefs[]
knowledgeCutoff
evidenceSetRef
policyVersionRefs[]
recommendationRefs[]
principalId
representedOrganizationId
mandateRef
rationale
outcome
decidedAt
effectiveAt
correctionOf?
}

```

Evidence records identify issuer or observer, subject, version, observed time, received time, effective interval, integrity, confidence, qualification, contradiction, correction, retention, residency, and access class. A recommendation may assist the Decision but cannot promote itself to the Decision.

6. Commands, Events, Queries, and Contracts

Reference Commands:

- CreateDataLineage
- ValidateDataLineage
- ApproveDataLineage
- ActivateDataLineage
- SuspendDataLineage
- CorrectDataLineage
- RetireDataLineage

Reference Events:

- DataLineageCreated
- DataLineageValidated
- DataLineageApproved
- DataLineageActivated
- DataLineageSuspended
- DataLineageCorrected
- DataLineageRetired

Reference queries:

- GetDataLineage
- ListDataLineageRecords
- GetDataLineageTimeline
- GetDataLineageEvidence
- GetDataLineageDecisionPackage
- GetDataLineageExceptions
- GetDataLineageAuditTrail

Reference API surface:

```

POST /api/v1/datalineage/commands
GET  /api/v1/datalineage/{id}
GET  /api/v1/datalineage/{id}/timeline
GET  /api/v1/datalineage/{id}/evidence
GET  /api/v1/datalineage/{id}/decisions
POST /api/v1/datalineage/{id}/exceptions
POST /api/v1/datalineage/{id}/corrections

```

These endpoints are reference contracts, not claims that exact routes exist.

7. Workflow, Failure, and Compensation

flowchart TD

```

A[Validate identity version and authority] --> B[Collect required evidence]
B --> C{Policy and evidence gates pass?}
C -- No --> D[Open exception or request correction]
D --> B
C -- Yes --> E[Record Decision or execution Intent]
E --> F[Execute Domain command or external request]

```

```

F --> G{Outcome known?}
G -- Success --> H[Record fact and publish Event]
G -- Failure --> I[Record failure and compensation options]
G -- Unknown --> J[Enter Outcome Unknown]
J --> K[Query source or wait for verified callback]
K --> G
H --> L[Update projections and downstream workflow]

```

Representative failure modes:

- stale or wrong source version
- duplicate submission or replay
- external timeout after possible success
- contradictory source observations
- lost Event after committed state
- approval or mandate revoked during execution
- downstream action before a mandatory gate
- partial completion across multiple subjects
- evidence later retracted or corrected
- tenant, facility, channel or jurisdiction mismatch

Recovery shall query the authoritative source before duplicate execution, preserve partial success, use compensating Intent rather than destructive rewrite, publish corrections, quarantine high-impact ambiguity, and record affected downstream subjects.

8. Data and Persistence Architecture

Reference table families:

Table family	Purpose
<code>datalineage_aggregate</code>	authoritative subject
<code>datalineage_version</code>	immutable submitted, approved, issued, or signed versions
<code>datalineage_relationship</code>	governed relationships
<code>datalineage_decision</code>	binding Decisions and rationale
<code>datalineage_evidence_link</code>	provenance and evidence relationships
<code>datalineage_event_outbox</code>	transactional Event publication
<code>datalineage_idempotency</code>	duplicate-write protection
<code>datalineage_exception</code>	exception, claim, dispute, or hold
<code>datalineage_correction</code>	correction and supersession lineage
<code>datalineage_audit</code>	privileged action evidence

The owning service controls schema and migration. Cross-Domain references use stable identifiers or contracts. Analytics, reports, caches, search, and vector indexes remain projections. Backup, restore, retention, legal hold, privacy, residency, and decommissioning requirements are explicit.

9. Portal and Experience Requirements

- Primary participant Portal: initiate, review exact versions, supply evidence, and respond to required actions.
- Operator Portal: inspect exceptions, correlation, source responses, retries, compensation, and downstream impact.
- Executive or oversight view: consume governed metrics without becoming a source of Domain truth.

Every Portal displays exact version, owner, state, freshness, required action, and uncertainty. It supports loading, empty, partial, stale, denied, error, timeout, and Outcome Unknown states; accessibility; locale, currency, unit, date, and RTL/LTR correctness; evidence audit; safe retry; and correction history.

10. AI Participation and Tool Governance

Permitted AI tasks:

- assist schema and lineage analysis
- detect data-quality anomalies
- summarize migration and reconciliation risk
- never mutate authoritative data outside typed Domain Tools

AI shall not own Domain records, grant itself authority, suppress contradictions, silently edit approved versions, fabricate external outcomes, or execute high-impact actions outside typed Tools, policy, approval, idempotency, sandbox, and outcome observation.

Every Agent Run records source context, prompt/model/tool versions, structured output, confidence and limitations, Policy result, approval, Tool calls, provider responses, final outcome, and recovery actions.

11. Security, Privacy, and Trust Boundaries

- least privilege and separation of duties
- tenant, organization, facility, channel and jurisdiction isolation
- integrity protection for binding evidence
- verified external callbacks and anti-replay controls

Additional controls include secret isolation, callback signature verification, audit of privileged action, emergency-access review, sensitive-data minimization, purpose limitation, and threat models for impersonation, tampering, replay, confused deputy, fraud, prompt injection, and exfiltration.

12. Reliability, SLOs, and Operations

SLI	Reference objective
authoritative read availability	99.9% monthly unless a stricter tier applies
acknowledged Command durability	no acknowledged write may be silently lost
Event publication	99.9% within five minutes after committed fact
duplicate prevention	100% for same tenant, Command class, and idempotency key
binding Decision audit completeness	100%
stale-state detection	bounded by business deadline

Dashboards and runbooks cover backlog, error budget, dependency failure, retry budget, DLQ, workflow stall, provider outage, reconciliation, capacity, backup/restore, release/rollback, incident command, and post-incident correction.

13. Testing and Continuous Conformance

- aggregate invariant and transition tests
- authorization, delegation, separation-of-duties and tenant-isolation tests
- idempotency and optimistic-concurrency tests
- database migration, key, constraint and rollback tests
- contract, outbox/inbox, duplicate, replay and DLQ tests
- workflow restart, timeout and compensation tests
- Portal E2E tests for success, partial, stale, denied, error and Outcome Unknown states
- accessibility, locale, currency, unit and RTL/LTR tests
- security adversarial, fraud and data-exfiltration tests
- AI evaluation, prompt-injection, policy-bypass and unauthorized-tool tests
- historical reconstruction, correction and audit-completeness tests
- performance, capacity, recovery and operational exercise tests

Release evidence links each invariant and control to the test, environment, input data, result, owner, and artifact proving it. Generated test counts never replace semantic coverage.

14. Repository Review Boundary

- This chapter is a repository-review specification. Exact executable implementation has not been inspected for every target behavior.
- Registered modules, catalog entries, README claims, frontend contracts, or plans are not proof of complete implementation.

Area	Classification
target aggregate, API, tables, state and workflow in this chapter	REFERENCE_ARCHITECTURE
registered module or catalog claim	DOCUMENTATION_ONLY_CLAIM until source inspection
uninspected integration	REQUIRES_REPOSITORY_REVIEW
behavior proven in another repository-grounded chapter	retain that chapter's evidence and classification
contradictory ownership or path claims	CONFLICT_REQUIRES_CANONICALIZATION

No statement upgrades implementation status merely because a file, directory, module name, frontend contract, database entry, plan, or report exists.

15. Worked Conformance Scenario

An authorized principal initiates a Command against an exact subject version. The service validates identity, tenant, mandate, Policy, Evidence, and idempotency. It records the state transition and outbox entry atomically. If an external dependency participates, timeout produces Outcome Unknown; reconciliation queries the source or waits for a verified callback instead of repeating the action blindly.

Portals show source, freshness, version, required action, and uncertainty. AI may extract, compare, summarize, detect anomalies, and recommend through typed Tools, but cannot own the subject or fabricate success. Later correction appends new facts, identifies affected projections and workflows, and preserves the historical Decision context.

16. Conformance Checklist

- ☐ stable identity and proposition ownership are explicit;
- ☐ authority and mandate are evaluated at action time;
- ☐ state, Recommendation, Decision, Intent, execution, and outcome remain distinct;
- ☐ Commands are typed, authorized, idempotent, and version guarded;
- ☐ Events describe completed facts;
- ☐ Evidence includes provenance, time, integrity, qualification, and correction;
- ☐ external outcomes retain their authority source;
- ☐ Portals show freshness and uncertainty;
- ☐ AI is bounded by typed Tools, Policy, approval, and outcome observation;
- ☐ security, tenant, facility, channel, and jurisdiction boundaries are enforced;
- ☐ Outcome Unknown and reconciliation exist where needed;
- ☐ corrections preserve history;
- ☐ tests prove invariants and recovery;
- ☐ implementation status is not upgraded without executable evidence.

17. Status Vocabulary

Status	Meaning
VERIFIED_IMPLEMENTED	Executable source and relevant behavior were inspected.
IMPLEMENTED_WITH_GAP	Executable behavior exists with a material governance or completeness gap.
PARTIAL_IMPLEMENTATION	Only part of the target capability is evidenced.
REFERENCE_ARCHITECTURE	Normative target behavior; no claim that it exists today.
PLANNED_NOT_IMPLEMENTED	Explicitly planned but not evidenced as executable.
DOCUMENTATION_ONLY_CLAIM	Documentation or client contract claims behavior without verified backend evidence.
CONFLICT_REQUIRES_CANONICALIZATION	Sources disagree and no final owner has been established.
REQUIRES_REPOSITORY_REVIEW	Necessary executable evidence has not yet been inspected.

Data Quality

Metadata	Value
Volume	10
Chapter ID	V10-16
Subject	DataQuality
Status	Generated First-Draft Architecture Skeleton - Domain-Specific Expansion Required
Content maturity	TEMPLATE_DERIVED_REFERENCE_SKELETON unless repository citations prove otherwise
Default implementation classification	REFERENCE_ARCHITECTURE / REQUIRES_REPOSITORY_REVIEW
Accountable owner	Data Architecture Council
Evidence rule	No implementation claim without inspected executable evidence
Repository	chinair88-prog/allinb2c

1. Executive Orientation

This chapter defines how GTOS shall **define enterprise and Domain data ownership, semantics, persistence, quality, lineage, correction, protection, operations, and verification for Data Quality.**

The specification separates reality, observations, knowledge, Decisions, Intent, execution, and outcomes. A component name, database row, UI label, AI answer, or workflow state does not acquire authority merely because it is visible or technically convenient.

Reality or external outcome

- ≠ Observation
- ≠ Claim
- ≠ derived Perspective
- ≠ Prediction
- ≠ Recommendation
- ≠ authorized Decision
- ≠ execution Intent

1.1 Accountable ownership

The accountable owner is **Data Architecture Council**. Supporting applications, shared infrastructure, integration adapters, analytics, and AI coordinate or present the capability but shall not silently own its authoritative propositions.

1.2 Scope

- create and identify the subject
- validate evidence and policy
- make authorized Decisions
- execute typed Commands
- publish completed-fact Events
- query current and historical perspectives
- correct and reconcile outcomes

1.3 Non-goals

- treating a registered Gradle module, catalog record, README, frontend route, migration plan, or generated report as proof of end-to-end implementation;
- direct cross-Domain database writes;

- generic status values that combine lifecycle, approval, provider, document, payment, dispute, and correction state;
- silent replacement of history;
- AI-created authority or fabricated external outcomes.

2. Ubiquitous Language and Subject Model

Subject	Required interpretation
DataQuality	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
DataQualityVersion	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
DataQualityDecision	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
DataQualityEvidenceSet	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
DataQualityException	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit

Reference aggregate:

```
DataQuality {
  subjectId
  tenantId
  organizationContext
  sourceIdentityRefs[]
  sourceVersionRefs[]
  lifecycleState
  relationshipRefs[]
  decisionRefs[]
  intentRefs[]
  evidenceSetRef
  knowledgeCutoff
  policyRefs[]
  authorityRef
  jurisdictionRefs[]
  createdAt
  effectiveAt
  updatedAt
  version
  correctionOf?
  correlationId
}
```

2.1 Core invariants

- Stable identity precedes relationship, state, and cross-system correlation.
- Effective time and recorded time remain distinct where business truth changes over time.
- Recommendation, Decision, Intent, execution, and outcome are separate concepts.
- No Portal, report, search index, cache, workflow, or AI Agent becomes a shadow owner of Domain truth.
- Binding submitted, approved, issued, accepted, or signed versions are immutable.
- Corrections append governed facts and preserve the original record.
- External authority and provider outcomes retain their source and qualification.
- Tenant and represented-principal context come from trusted authentication and mandate evaluation.
- Retried writes are idempotent and concurrency guarded.
- Outcome Unknown is explicit and reconciled before risky duplicate execution.

2.2 Required standards

- stable identifiers
- typed contracts
- immutable binding versions
- transactional outbox
- idempotent Commands
- explicit Outcome Unknown
- correction lineage
- machine-verifiable conformance

3. Actors, Roles, and Authority

A conformant design distinguishes:

- the natural person acting;
- the service or AI identity invoking a Tool;
- the organization or principal represented;
- the role or relationship granting context;
- the mandate or delegation granting action authority;
- the Policy and jurisdiction limiting the action;
- the reviewer or approver responsible for a binding Decision.

Authority is evaluated at action time. Authentication alone does not grant business authority. Cached permissions and browser-supplied tenant headers are insufficient for high-impact actions.

4. Lifecycle and State Model

State	Semantics
PROPOSED	Distinct governed condition with entry, exit, timeout, correction and authority semantics
VALIDATION_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
APPROVAL_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
ACTIVE	Distinct governed condition with entry, exit, timeout, correction and authority semantics
SUSPENDED	Distinct governed condition with entry, exit, timeout, correction and authority semantics
CORRECTION_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
RETIRED	Distinct governed condition with entry, exit, timeout, correction and authority semantics

```
stateDiagram-v2
    [*] --> PROPOSED
    PROPOSED --> VALIDATION_PENDING
    VALIDATION_PENDING --> APPROVAL_PENDING
    APPROVAL_PENDING --> ACTIVE
    ACTIVE --> SUSPENDED
    SUSPENDED --> CORRECTION_PENDING
    CORRECTION_PENDING --> RETIRED
```

The diagram is a primary reading path. Real implementations may add parallel review, amendment, appeal, timeout, suspension, partial completion, reconciliation, correction, and terminal failure without collapsing their meaning.

4.1 Transition rules

- transition through a typed, authorized Command;
 - require exact expected version where concurrent change matters;
 - record actor, represented principal, mandate, Policy, reason, and Evidence;
 - use an idempotency key for retried writes;
 - publish a completed-fact Event after durable state change;
 - expose Outcome Unknown for unresolved external completion;
 - preserve original and corrected states in historical reconstruction.
-

5. Decision, Evidence, and Knowledge Cutoff

Reference Decision:

```
Decision {
    decisionId
    decisionType
    subjectRef
}
```

```

sourceVersionRefs[]
knowledgeCutoff
evidenceSetRef
policyVersionRefs[]
recommendationRefs[]
principalId
representedOrganizationId
mandateRef
rationale
outcome
decidedAt
effectiveAt
correctionOf?
}

```

Evidence records identify issuer or observer, subject, version, observed time, received time, effective interval, integrity, confidence, qualification, contradiction, correction, retention, residency, and access class. A recommendation may assist the Decision but cannot promote itself to the Decision.

6. Commands, Events, Queries, and Contracts

Reference Commands:

- CreateDataQuality
- ValidateDataQuality
- ApproveDataQuality
- ActivateDataQuality
- SuspendDataQuality
- CorrectDataQuality
- RetireDataQuality

Reference Events:

- DataQualityCreated
- DataQualityValidated
- DataQualityApproved
- DataQualityActivated
- DataQualitySuspended
- DataQualityCorrected
- DataQualityRetired

Reference queries:

- GetDataQuality
- ListDataQualityRecords
- GetDataQualityTimeline
- GetDataQualityEvidence
- GetDataQualityDecisionPackage
- GetDataQualityExceptions
- GetDataQualityAuditTrail

Reference API surface:

```

POST /api/v1/dataquality/commands
GET  /api/v1/dataquality/{id}
GET  /api/v1/dataquality/{id}/timeline
GET  /api/v1/dataquality/{id}/evidence
GET  /api/v1/dataquality/{id}/decisions
POST /api/v1/dataquality/{id}/exceptions
POST /api/v1/dataquality/{id}/corrections

```

These endpoints are reference contracts, not claims that exact routes exist.

7. Workflow, Failure, and Compensation

flowchart TD

```

A[Validate identity version and authority] --> B[Collect required evidence]
B --> C{Policy and evidence gates pass?}
C -- No --> D[Open exception or request correction]
D --> B
C -- Yes --> E[Record Decision or execution Intent]
E --> F[Execute Domain command or external request]

```

```

F --> G{Outcome known?}
G -- Success --> H[Record fact and publish Event]
G -- Failure --> I[Record failure and compensation options]
G -- Unknown --> J[Enter Outcome Unknown]
J --> K[Query source or wait for verified callback]
K --> G
H --> L[Update projections and downstream workflow]

```

Representative failure modes:

- stale or wrong source version
- duplicate submission or replay
- external timeout after possible success
- contradictory source observations
- lost Event after committed state
- approval or mandate revoked during execution
- downstream action before a mandatory gate
- partial completion across multiple subjects
- evidence later retracted or corrected
- tenant, facility, channel or jurisdiction mismatch

Recovery shall query the authoritative source before duplicate execution, preserve partial success, use compensating Intent rather than destructive rewrite, publish corrections, quarantine high-impact ambiguity, and record affected downstream subjects.

8. Data and Persistence Architecture

Reference table families:

Table family	Purpose
dataquality_aggregate	authoritative subject
dataquality_version	immutable submitted, approved, issued, or signed versions
dataquality_relationship	governed relationships
dataquality_decision	binding Decisions and rationale
dataquality_evidence_link	provenance and evidence relationships
dataquality_event_outbox	transactional Event publication
dataquality_idempotency	duplicate-write protection
dataquality_exception	exception, claim, dispute, or hold
dataquality_correction	correction and supersession lineage
dataquality_audit	privileged action evidence

The owning service controls schema and migration. Cross-Domain references use stable identifiers or contracts. Analytics, reports, caches, search, and vector indexes remain projections. Backup, restore, retention, legal hold, privacy, residency, and decommissioning requirements are explicit.

9. Portal and Experience Requirements

- Primary participant Portal: initiate, review exact versions, supply evidence, and respond to required actions.
- Operator Portal: inspect exceptions, correlation, source responses, retries, compensation, and downstream impact.
- Executive or oversight view: consume governed metrics without becoming a source of Domain truth.

Every Portal displays exact version, owner, state, freshness, required action, and uncertainty. It supports loading, empty, partial, stale, denied, error, timeout, and Outcome Unknown states; accessibility; locale, currency, unit, date, and RTL/LTR correctness; evidence audit; safe retry; and correction history.

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- assist schema and lineage analysis
- detect data-quality anomalies
- summarize migration and reconciliation risk
- never mutate authoritative data outside typed Domain Tools

AI shall not own Domain records, grant itself authority, suppress contradictions, silently edit approved versions, fabricate external outcomes, or execute high-impact actions outside typed Tools, policy, approval, idempotency, sandbox, and outcome observation.

Every Agent Run records source context, prompt/model/tool versions, structured output, confidence and limitations, Policy result, approval, Tool calls, provider responses, final outcome, and recovery actions.

11. Security, Privacy, and Trust Boundaries

- least privilege and separation of duties
- tenant, organization, facility, channel and jurisdiction isolation
- integrity protection for binding evidence
- verified external callbacks and anti-replay controls

Additional controls include secret isolation, callback signature verification, audit of privileged action, emergency-access review, sensitive-data minimization, purpose limitation, and threat models for impersonation, tampering, replay, confused deputy, fraud, prompt injection, and exfiltration.

12. Reliability, SLOs, and Operations

SLI	Reference objective
authoritative read availability	99.9% monthly unless a stricter tier applies
acknowledged Command durability	no acknowledged write may be silently lost
Event publication	99.9% within five minutes after committed fact
duplicate prevention	100% for same tenant, Command class, and idempotency key
binding Decision audit completeness	100%
state-state detection	bounded by business deadline

Dashboards and runbooks cover backlog, error budget, dependency failure, retry budget, DLQ, workflow stall, provider outage, reconciliation, capacity, backup/restore, release/rollback, incident command, and post-incident correction.

13. Testing and Continuous Conformance

- aggregate invariant and transition tests
- authorization, delegation, separation-of-duties and tenant-isolation tests
- idempotency and optimistic-concurrency tests
- database migration, key, constraint and rollback tests
- contract, outbox/inbox, duplicate, replay and DLQ tests
- workflow restart, timeout and compensation tests
- Portal E2E tests for success, partial, stale, denied, error and Outcome Unknown states
- accessibility, locale, currency, unit and RTL/LTR tests
- security adversarial, fraud and data-exfiltration tests
- AI evaluation, prompt-injection, policy-bypass and unauthorized-tool tests
- historical reconstruction, correction and audit-completeness tests
- performance, capacity, recovery and operational exercise tests

Release evidence links each invariant and control to the test, environment, input data, result, owner, and artifact proving it. Generated test counts never replace semantic coverage.

14. Repository Review Boundary

- This chapter is a repository-review specification. Exact executable implementation has not been inspected for every target behavior.
- Registered modules, catalog entries, README claims, frontend contracts, or plans are not proof of complete implementation.

Area	Classification
target aggregate, API, tables, state and workflow in this chapter	REFERENCE_ARCHITECTURE
registered module or catalog claim	DOCUMENTATION_ONLY_CLAIM until source inspection
uninspected integration	REQUIRES_REPOSITORY_REVIEW
behavior proven in another repository-grounded chapter	retain that chapter's evidence and classification
contradictory ownership or path claims	CONFLICT_REQUIRES_CANONICALIZATION

No statement upgrades implementation status merely because a file, directory, module name, frontend contract, database entry, plan, or report exists.

15. Worked Conformance Scenario

An authorized principal initiates a Command against an exact subject version. The service validates identity, tenant, mandate, Policy, Evidence, and idempotency. It records the state transition and outbox entry atomically. If an external dependency participates, timeout produces Outcome Unknown; reconciliation queries the source or waits for a verified callback instead of repeating the action blindly.

Portals show source, freshness, version, required action, and uncertainty. AI may extract, compare, summarize, detect anomalies, and recommend through typed Tools, but cannot own the subject or fabricate success. Later correction appends new facts, identifies affected projections and workflows, and preserves the historical Decision context.

16. Conformance Checklist

- ☐ stable identity and proposition ownership are explicit;
- ☐ authority and mandate are evaluated at action time;
- ☐ state, Recommendation, Decision, Intent, execution, and outcome remain distinct;
- ☐ Commands are typed, authorized, idempotent, and version guarded;
- ☐ Events describe completed facts;
- ☐ Evidence includes provenance, time, integrity, qualification, and correction;
- ☐ external outcomes retain their authority source;
- ☐ Portals show freshness and uncertainty;
- ☐ AI is bounded by typed Tools, Policy, approval, and outcome observation;
- ☐ security, tenant, facility, channel, and jurisdiction boundaries are enforced;
- ☐ Outcome Unknown and reconciliation exist where needed;
- ☐ corrections preserve history;
- ☐ tests prove invariants and recovery;
- ☐ implementation status is not upgraded without executable evidence.

17. Status Vocabulary

Status	Meaning
VERIFIED_IMPLEMENTED	Executable source and relevant behavior were inspected.
IMPLEMENTED_WITH_GAP	Executable behavior exists with a material governance or completeness gap.
PARTIAL_IMPLEMENTATION	Only part of the target capability is evidenced.
REFERENCE_ARCHITECTURE	Normative target behavior; no claim that it exists today.
PLANNED_NOT_IMPLEMENTED	Explicitly planned but not evidenced as executable.
DOCUMENTATION_ONLY_CLAIM	Documentation or client contract claims behavior without verified backend evidence.
CONFLICT_REQUIRES_CANONICALIZATION	Sources disagree and no final owner has been established.
REQUIRES_REPOSITORY_REVIEW	Necessary executable evidence has not yet been inspected.

Correction and Retraction

Metadata	Value
Volume	10
Chapter ID	V10-17
Subject	CorrectionandRetraction
Status	Generated First-Draft Architecture Skeleton - Domain-Specific Expansion Required
Content maturity	TEMPLATE_DERIVED_REFERENCE_SKELETON unless repository citations prove otherwise
Default implementation classification	REFERENCE_ARCHITECTURE / REQUIRES_REPOSITORY_REVIEW
Accountable owner	Data Architecture Council
Evidence rule	No implementation claim without inspected executable evidence
Repository	chinair88-prog/allinb2c

1. Executive Orientation

This chapter defines how GTOS shall **define enterprise and Domain data ownership, semantics, persistence, quality, lineage, correction, protection, operations, and verification for Correction and Retraction.**

The specification separates reality, observations, knowledge, Decisions, Intent, execution, and outcomes. A component name, database row, UI label, AI answer, or workflow state does not acquire authority merely because it is visible or technically convenient.

Reality or external outcome

- ≠ Observation
- ≠ Claim
- ≠ derived Perspective
- ≠ Prediction
- ≠ Recommendation
- ≠ authorized Decision
- ≠ execution Intent

1.1 Accountable ownership

The accountable owner is **Data Architecture Council**. Supporting applications, shared infrastructure, integration adapters, analytics, and AI coordinate or present the capability but shall not silently own its authoritative propositions.

1.2 Scope

- create and identify the subject
- validate evidence and policy
- make authorized Decisions
- execute typed Commands
- publish completed-fact Events
- query current and historical perspectives
- correct and reconcile outcomes

1.3 Non-goals

- treating a registered Gradle module, catalog record, README, frontend route, migration plan, or generated report as proof of end-to-end implementation;
- direct cross-Domain database writes;
- generic status values that combine lifecycle, approval, provider, document, payment, dispute, and correction state;

- silent replacement of history;
- AI-created authority or fabricated external outcomes.

2. Ubiquitous Language and Subject Model

Subject	Required interpretation
CorrectionandRetraction	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
CorrectionandRetractionVersion	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
CorrectionandRetractionDecision	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
CorrectionandRetractionEvidenceSet	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
CorrectionandRetractionException	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit

Reference aggregate:

```
CorrectionandRetraction {
  subjectId
  tenantId
  organizationContext
  sourceIdentityRefs[]
  sourceVersionRefs[]
  lifecycleState
  relationshipRefs[]
  decisionRefs[]
  intentRefs[]
  evidenceSetRef
  knowledgeCutoff
  policyRefs[]
  authorityRef
  jurisdictionRefs[]
  createdAt
  effectiveAt
  updatedAt
  version
  correctionOf?
  correlationId
}
```

2.1 Core invariants

- Stable identity precedes relationship, state, and cross-system correlation.
- Effective time and recorded time remain distinct where business truth changes over time.
- Recommendation, Decision, Intent, execution, and outcome are separate concepts.
- No Portal, report, search index, cache, workflow, or AI Agent becomes a shadow owner of Domain truth.
- Binding submitted, approved, issued, accepted, or signed versions are immutable.
- Corrections append governed facts and preserve the original record.
- External authority and provider outcomes retain their source and qualification.
- Tenant and represented-principal context come from trusted authentication and mandate evaluation.
- Retried writes are idempotent and concurrency guarded.
- Outcome Unknown is explicit and reconciled before risky duplicate execution.

2.2 Required standards

- stable identifiers
- typed contracts
- immutable binding versions
- transactional outbox
- idempotent Commands
- explicit Outcome Unknown
- correction lineage
- machine-verifiable conformance

3. Actors, Roles, and Authority

A conformant design distinguishes:

- the natural person acting;
- the service or AI identity invoking a Tool;
- the organization or principal represented;
- the role or relationship granting context;
- the mandate or delegation granting action authority;
- the Policy and jurisdiction limiting the action;
- the reviewer or approver responsible for a binding Decision.

Authority is evaluated at action time. Authentication alone does not grant business authority. Cached permissions and browser-supplied tenant headers are insufficient for high-impact actions.

4. Lifecycle and State Model

State	Semantics
PROPOSED	Distinct governed condition with entry, exit, timeout, correction and authority semantics
VALIDATION_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
APPROVAL_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
ACTIVE	Distinct governed condition with entry, exit, timeout, correction and authority semantics
SUSPENDED	Distinct governed condition with entry, exit, timeout, correction and authority semantics
CORRECTION_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
RETIRED	Distinct governed condition with entry, exit, timeout, correction and authority semantics

```
stateDiagram-v2
    [*] --> PROPOSED
    PROPOSED --> VALIDATION_PENDING
    VALIDATION_PENDING --> APPROVAL_PENDING
    APPROVAL_PENDING --> ACTIVE
    ACTIVE --> SUSPENDED
    SUSPENDED --> CORRECTION_PENDING
    CORRECTION_PENDING --> RETIRED
```

The diagram is a primary reading path. Real implementations may add parallel review, amendment, appeal, timeout, suspension, partial completion, reconciliation, correction, and terminal failure without collapsing their meaning.

4.1 Transition rules

- transition through a typed, authorized Command;
- require exact expected version where concurrent change matters;
- record actor, represented principal, mandate, Policy, reason, and Evidence;
- use an idempotency key for retried writes;
- publish a completed-fact Event after durable state change;
- expose Outcome Unknown for unresolved external completion;
- preserve original and corrected states in historical reconstruction.

5. Decision, Evidence, and Knowledge Cutoff

Reference Decision:

```
Decision {
    decisionId
    decisionType
    subjectRef
    sourceVersionRefs[]
    knowledgeCutoff
    evidenceSetRef
}
```

```

policyVersionRefs[]
recommendationRefs[]
principalId
representedOrganizationId
mandateRef
rationale
outcome
decidedAt
effectiveAt
correctionOf?
}

```

Evidence records identify issuer or observer, subject, version, observed time, received time, effective interval, integrity, confidence, qualification, contradiction, correction, retention, residency, and access class. A recommendation may assist the Decision but cannot promote itself to the Decision.

6. Commands, Events, Queries, and Contracts

Reference Commands:

- CreateCorrectionandRetraction
- ValidateCorrectionandRetraction
- ApproveCorrectionandRetraction
- ActivateCorrectionandRetraction
- SuspendCorrectionandRetraction
- CorrectCorrectionandRetraction
- RetireCorrectionandRetraction

Reference Events:

- CorrectionandRetractionCreated
- CorrectionandRetractionValidated
- CorrectionandRetractionApproved
- CorrectionandRetractionActivated
- CorrectionandRetractionSuspended
- CorrectionandRetractionCorrected
- CorrectionandRetractionRetired

Reference queries:

- GetCorrectionandRetraction
- ListCorrectionandRetractionRecords
- GetCorrectionandRetractionTimeline
- GetCorrectionandRetractionEvidence
- GetCorrectionandRetractionDecisionPackage
- GetCorrectionandRetractionExceptions
- GetCorrectionandRetractionAuditTrail

Reference API surface:

```

POST /api/v1/correctionandretraction/commands
GET  /api/v1/correctionandretraction/{id}
GET  /api/v1/correctionandretraction/{id}/timeline
GET  /api/v1/correctionandretraction/{id}/evidence
GET  /api/v1/correctionandretraction/{id}/decisions
POST /api/v1/correctionandretraction/{id}/exceptions
POST /api/v1/correctionandretraction/{id}/corrections

```

These endpoints are reference contracts, not claims that exact routes exist.

7. Workflow, Failure, and Compensation

flowchart TD

```

A[Validate identity version and authority] --> B[Collect required evidence]
B --> C{Policy and evidence gates pass?}
C -- No --> D[Open exception or request correction]
D --> B
C -- Yes --> E[Record Decision or execution Intent]
E --> F[Execute Domain command or external request]
F --> G{Outcome known?}
G -- Success --> H[Record fact and publish Event]
G -- Failure --> I[Record failure and compensation options]

```

```

G -- Unknown --> J[Enter Outcome Unknown]
J --> K[Query source or wait for verified callback]
K --> G
H --> L[Update projections and downstream workflow]

```

Representative failure modes:

- stale or wrong source version
- duplicate submission or replay
- external timeout after possible success
- contradictory source observations
- lost Event after committed state
- approval or mandate revoked during execution
- downstream action before a mandatory gate
- partial completion across multiple subjects
- evidence later retracted or corrected
- tenant, facility, channel or jurisdiction mismatch

Recovery shall query the authoritative source before duplicate execution, preserve partial success, use compensating Intent rather than destructive rewrite, publish corrections, quarantine high-impact ambiguity, and record affected downstream subjects.

8. Data and Persistence Architecture

Reference table families:

Table family	Purpose
correctionandretraction_aggregate	authoritative subject
correctionandretraction_version	immutable submitted, approved, issued, or signed versions
correctionandretraction_relationship	governed relationships
correctionandretraction_decision	binding Decisions and rationale
correctionandretraction_evidence_link	provenance and evidence relationships
correctionandretraction_event_outbox	transactional Event publication
correctionandretraction_idempotency	duplicate-write protection
correctionandretraction_exception	exception, claim, dispute, or hold
correctionandretraction_correction	correction and supersession lineage
correctionandretraction_audit	privileged action evidence

The owning service controls schema and migration. Cross-Domain references use stable identifiers or contracts. Analytics, reports, caches, search, and vector indexes remain projections. Backup, restore, retention, legal hold, privacy, residency, and decommissioning requirements are explicit.

9. Portal and Experience Requirements

- Primary participant Portal: initiate, review exact versions, supply evidence, and respond to required actions.
- Operator Portal: inspect exceptions, correlation, source responses, retries, compensation, and downstream impact.
- Executive or oversight view: consume governed metrics without becoming a source of Domain truth.

Every Portal displays exact version, owner, state, freshness, required action, and uncertainty. It supports loading, empty, partial, stale, denied, error, timeout, and Outcome Unknown states; accessibility; locale, currency, unit, date, and RTL/LTR correctness; evidence audit; safe retry; and correction history.

10. AI Participation and Tool Governance

Permitted AI tasks:

- assist schema and lineage analysis
- detect data-quality anomalies
- summarize migration and reconciliation risk
- never mutate authoritative data outside typed Domain Tools

AI shall not own Domain records, grant itself authority, suppress contradictions, silently edit approved versions, fabricate external outcomes, or execute high-impact actions outside typed Tools, policy, approval, idempotency, sandbox, and outcome observation.

Every Agent Run records source context, prompt/model/tool versions, structured output, confidence and limitations, Policy result, approval, Tool calls, provider responses, final outcome, and recovery actions.

11. Security, Privacy, and Trust Boundaries

- least privilege and separation of duties
- tenant, organization, facility, channel and jurisdiction isolation
- integrity protection for binding evidence
- verified external callbacks and anti-replay controls

Additional controls include secret isolation, callback signature verification, audit of privileged action, emergency-access review, sensitive-data minimization, purpose limitation, and threat models for impersonation, tampering, replay, confused deputy, fraud, prompt injection, and exfiltration.

12. Reliability, SLOs, and Operations

SLI	Reference objective
authoritative read availability	99.9% monthly unless a stricter tier applies
acknowledged Command durability	no acknowledged write may be silently lost
Event publication	99.9% within five minutes after committed fact
duplicate prevention	100% for same tenant, Command class, and idempotency key
binding Decision audit completeness	100%
stale-state detection	bounded by business deadline

Dashboards and runbooks cover backlog, error budget, dependency failure, retry budget, DLQ, workflow stall, provider outage, reconciliation, capacity, backup/restore, release/rollback, incident command, and post-incident correction.

13. Testing and Continuous Conformance

- aggregate invariant and transition tests
- authorization, delegation, separation-of-duties and tenant-isolation tests
- idempotency and optimistic-concurrency tests
- database migration, key, constraint and rollback tests
- contract, outbox/inbox, duplicate, replay and DLQ tests
- workflow restart, timeout and compensation tests
- Portal E2E tests for success, partial, stale, denied, error and Outcome Unknown states
- accessibility, locale, currency, unit and RTL/LTR tests
- security adversarial, fraud and data-exfiltration tests
- AI evaluation, prompt-injection, policy-bypass and unauthorized-tool tests
- historical reconstruction, correction and audit-completeness tests
- performance, capacity, recovery and operational exercise tests

Release evidence links each invariant and control to the test, environment, input data, result, owner, and artifact proving it. Generated test counts never replace semantic coverage.

14. Repository Review Boundary

- This chapter is a repository-review specification. Exact executable implementation has not been inspected for every target behavior.
- Registered modules, catalog entries, README claims, frontend contracts, or plans are not proof of complete implementation.

Area	Classification
target aggregate, API, tables, state and workflow in this chapter	REFERENCE_ARCHITECTURE
registered module or catalog claim	DOCUMENTATION_ONLY_CLAIM until source inspection
uninspected integration	REQUIRES_REPOSITORY_REVIEW

Area	Classification
behavior proven in another repository-grounded chapter contradictory ownership or path claims	retain that chapter's evidence and classification CONFLICT_REQUIRES_CANONICALIZATION

No statement upgrades implementation status merely because a file, directory, module name, frontend contract, database entry, plan, or report exists.

15. Worked Conformance Scenario

An authorized principal initiates a Command against an exact subject version. The service validates identity, tenant, mandate, Policy, Evidence, and idempotency. It records the state transition and outbox entry atomically. If an external dependency participates, timeout produces Outcome Unknown; reconciliation queries the source or waits for a verified callback instead of repeating the action blindly.

Portals show source, freshness, version, required action, and uncertainty. AI may extract, compare, summarize, detect anomalies, and recommend through typed Tools, but cannot own the subject or fabricate success. Later correction appends new facts, identifies affected projections and workflows, and preserves the historical Decision context.

16. Conformance Checklist

- ☐ stable identity and proposition ownership are explicit;
- ☐ authority and mandate are evaluated at action time;
- ☐ state, Recommendation, Decision, Intent, execution, and outcome remain distinct;
- ☐ Commands are typed, authorized, idempotent, and version guarded;
- ☐ Events describe completed facts;
- ☐ Evidence includes provenance, time, integrity, qualification, and correction;
- ☐ external outcomes retain their authority source;
- ☐ Portals show freshness and uncertainty;
- ☐ AI is bounded by typed Tools, Policy, approval, and outcome observation;
- ☐ security, tenant, facility, channel, and jurisdiction boundaries are enforced;
- ☐ Outcome Unknown and reconciliation exist where needed;
- ☐ corrections preserve history;
- ☐ tests prove invariants and recovery;
- ☐ implementation status is not upgraded without executable evidence.

17. Status Vocabulary

Status	Meaning
VERIFIED_IMPLEMENTED	Executable source and relevant behavior were inspected.
IMPLEMENTED_WITH_GAP	Executable behavior exists with a material governance or completeness gap.
PARTIAL_IMPLEMENTATION	Only part of the target capability is evidenced.
REFERENCE_ARCHITECTURE	Normative target behavior; no claim that it exists today.
PLANNED_NOT_IMPLEMENTED	Explicitly planned but not evidenced as executable.
DOCUMENTATION_ONLY_CLAIM	Documentation or client contract claims behavior without verified backend evidence.
CONFLICT_REQUIRES_CANONICALIZATION	Sources disagree and no final owner has been established.
REQUIRES_REPOSITORY_REVIEW	Necessary executable evidence has not yet been inspected.

Materialized Views and Projections

Metadata	Value
Volume	10
Chapter ID	V10-18
Subject	MaterializedViewsandProjections
Status	Generated First-Draft Architecture Skeleton - Domain-Specific Expansion Required
Content maturity	TEMPLATE_DERIVED_REFERENCE_SKELETON unless repository citations prove otherwise
Default implementation classification	REFERENCE_ARCHITECTURE / REQUIRES_REPOSITORY_REVIEW
Accountable owner	Data Architecture Council
Evidence rule	No implementation claim without inspected executable evidence
Repository	chinair88-prog/allinb2c

1. Executive Orientation

This chapter defines how GTOS shall **define enterprise and Domain data ownership, semantics, persistence, quality, lineage, correction, protection, operations, and verification for Materialized Views and Projections.**

The specification separates reality, observations, knowledge, Decisions, Intent, execution, and outcomes. A component name, database row, UI label, AI answer, or workflow state does not acquire authority merely because it is visible or technically convenient.

Reality or external outcome

- ≠ Observation
- ≠ Claim
- ≠ derived Perspective
- ≠ Prediction
- ≠ Recommendation
- ≠ authorized Decision
- ≠ execution Intent

1.1 Accountable ownership

The accountable owner is **Data Architecture Council**. Supporting applications, shared infrastructure, integration adapters, analytics, and AI coordinate or present the capability but shall not silently own its authoritative propositions.

1.2 Scope

- create and identify the subject
- validate evidence and policy
- make authorized Decisions
- execute typed Commands
- publish completed-fact Events
- query current and historical perspectives
- correct and reconcile outcomes

1.3 Non-goals

- treating a registered Gradle module, catalog record, README, frontend route, migration plan, or generated report as proof of end-to-end implementation;
- direct cross-Domain database writes;

- generic status values that combine lifecycle, approval, provider, document, payment, dispute, and correction state;
- silent replacement of history;
- AI-created authority or fabricated external outcomes.

2. Ubiquitous Language and Subject Model

Subject	Required interpretation
MaterializedViewsandProjections	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
MaterializedViewsandProjectionsVersion	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
MaterializedViewsandProjectionsDecision	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
MaterializedViewsandProjectionsEvidenceSet	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
MaterializedViewsandProjectionsException	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit

Reference aggregate:

```
MaterializedViewsandProjections {
  subjectId
  tenantId
  organizationContext
  sourceIdentityRefs[]
  sourceVersionRefs[]
  lifecycleState
  relationshipRefs[]
  decisionRefs[]
  intentRefs[]
  evidenceSetRef
  knowledgeCutoff
  policyRefs[]
  authorityRef
  jurisdictionRefs[]
  createdAt
  effectiveAt
  updatedAt
  version
  correctionOf?
  correlationId
}
```

2.1 Core invariants

- Stable identity precedes relationship, state, and cross-system correlation.
- Effective time and recorded time remain distinct where business truth changes over time.
- Recommendation, Decision, Intent, execution, and outcome are separate concepts.
- No Portal, report, search index, cache, workflow, or AI Agent becomes a shadow owner of Domain truth.
- Binding submitted, approved, issued, accepted, or signed versions are immutable.
- Corrections append governed facts and preserve the original record.
- External authority and provider outcomes retain their source and qualification.
- Tenant and represented-principal context come from trusted authentication and mandate evaluation.
- Retried writes are idempotent and concurrency guarded.
- Outcome Unknown is explicit and reconciled before risky duplicate execution.

2.2 Required standards

- stable identifiers
- typed contracts
- immutable binding versions
- transactional outbox
- idempotent Commands
- explicit Outcome Unknown
- correction lineage
- machine-verifiable conformance

3. Actors, Roles, and Authority

A conformant design distinguishes:

- the natural person acting;
- the service or AI identity invoking a Tool;
- the organization or principal represented;
- the role or relationship granting context;
- the mandate or delegation granting action authority;
- the Policy and jurisdiction limiting the action;
- the reviewer or approver responsible for a binding Decision.

Authority is evaluated at action time. Authentication alone does not grant business authority. Cached permissions and browser-supplied tenant headers are insufficient for high-impact actions.

4. Lifecycle and State Model

State	Semantics
PROPOSED	Distinct governed condition with entry, exit, timeout, correction and authority semantics
VALIDATION_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
APPROVAL_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
ACTIVE	Distinct governed condition with entry, exit, timeout, correction and authority semantics
SUSPENDED	Distinct governed condition with entry, exit, timeout, correction and authority semantics
CORRECTION_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
RETIRED	Distinct governed condition with entry, exit, timeout, correction and authority semantics

```
stateDiagram-v2
    [*] --> PROPOSED
    PROPOSED --> VALIDATION_PENDING
    VALIDATION_PENDING --> APPROVAL_PENDING
    APPROVAL_PENDING --> ACTIVE
    ACTIVE --> SUSPENDED
    SUSPENDED --> CORRECTION_PENDING
    CORRECTION_PENDING --> RETIRED
```

The diagram is a primary reading path. Real implementations may add parallel review, amendment, appeal, timeout, suspension, partial completion, reconciliation, correction, and terminal failure without collapsing their meaning.

4.1 Transition rules

- transition through a typed, authorized Command;
 - require exact expected version where concurrent change matters;
 - record actor, represented principal, mandate, Policy, reason, and Evidence;
 - use an idempotency key for retried writes;
 - publish a completed-fact Event after durable state change;
 - expose Outcome Unknown for unresolved external completion;
 - preserve original and corrected states in historical reconstruction.
-

5. Decision, Evidence, and Knowledge Cutoff

Reference Decision:

```
Decision {
    decisionId
    decisionType
    subjectRef
}
```

```

sourceVersionRefs[]
knowledgeCutoff
evidenceSetRef
policyVersionRefs[]
recommendationRefs[]
principalId
representedOrganizationId
mandateRef
rationale
outcome
decidedAt
effectiveAt
correctionOf?
}

```

Evidence records identify issuer or observer, subject, version, observed time, received time, effective interval, integrity, confidence, qualification, contradiction, correction, retention, residency, and access class. A recommendation may assist the Decision but cannot promote itself to the Decision.

6. Commands, Events, Queries, and Contracts

Reference Commands:

- CreateMaterializedViewsandProjections
- ValidateMaterializedViewsandProjections
- ApproveMaterializedViewsandProjections
- ActivateMaterializedViewsandProjections
- SuspendMaterializedViewsandProjections
- CorrectMaterializedViewsandProjections
- RetireMaterializedViewsandProjections

Reference Events:

- MaterializedViewsandProjectionsCreated
- MaterializedViewsandProjectionsValidated
- MaterializedViewsandProjectionsApproved
- MaterializedViewsandProjectionsActivated
- MaterializedViewsandProjectionsSuspended
- MaterializedViewsandProjectionsCorrected
- MaterializedViewsandProjectionsRetired

Reference queries:

- GetMaterializedViewsandProjections
- ListMaterializedViewsandProjectionsRecords
- GetMaterializedViewsandProjectionsTimeline
- GetMaterializedViewsandProjectionsEvidence
- GetMaterializedViewsandProjectionsDecisionPackage
- GetMaterializedViewsandProjectionsExceptions
- GetMaterializedViewsandProjectionsAuditTrail

Reference API surface:

```

POST /api/v1/materializedviewsandprojections/commands
GET  /api/v1/materializedviewsandprojections/{id}
GET  /api/v1/materializedviewsandprojections/{id}/timeline
GET  /api/v1/materializedviewsandprojections/{id}/evidence
GET  /api/v1/materializedviewsandprojections/{id}/decisions
POST /api/v1/materializedviewsandprojections/{id}/exceptions
POST /api/v1/materializedviewsandprojections/{id}/corrections

```

These endpoints are reference contracts, not claims that exact routes exist.

7. Workflow, Failure, and Compensation

flowchart TD

```

A[Validate identity version and authority] --> B[Collect required evidence]
B --> C{Policy and evidence gates pass?}
C -- No --> D[Open exception or request correction]
D --> B
C -- Yes --> E[Record Decision or execution Intent]
E --> F[Execute Domain command or external request]

```

```

F --> G{Outcome known?}
G -- Success --> H[Record fact and publish Event]
G -- Failure --> I[Record failure and compensation options]
G -- Unknown --> J[Enter Outcome Unknown]
J --> K[Query source or wait for verified callback]
K --> G
H --> L[Update projections and downstream workflow]

```

Representative failure modes:

- stale or wrong source version
- duplicate submission or replay
- external timeout after possible success
- contradictory source observations
- lost Event after committed state
- approval or mandate revoked during execution
- downstream action before a mandatory gate
- partial completion across multiple subjects
- evidence later retracted or corrected
- tenant, facility, channel or jurisdiction mismatch

Recovery shall query the authoritative source before duplicate execution, preserve partial success, use compensating Intent rather than destructive rewrite, publish corrections, quarantine high-impact ambiguity, and record affected downstream subjects.

8. Data and Persistence Architecture

Reference table families:

Table family	Purpose
materializedviewsandprojections_aggregate	authoritative subject
materializedviewsandprojections_version	immutable submitted, approved, issued, or signed versions
materializedviewsandprojections_relationship	governed relationships
materializedviewsandprojections_decision	binding Decisions and rationale
materializedviewsandprojections_evidence_link	provenance and evidence relationships
materializedviewsandprojections_event_outbox	transactional Event publication
materializedviewsandprojections_idempotency	duplicate-write protection
materializedviewsandprojections_exception	exception, claim, dispute, or hold
materializedviewsandprojections_correction	correction and supersession lineage
materializedviewsandprojections_audit	privileged action evidence

The owning service controls schema and migration. Cross-Domain references use stable identifiers or contracts. Analytics, reports, caches, search, and vector indexes remain projections. Backup, restore, retention, legal hold, privacy, residency, and decommissioning requirements are explicit.

9. Portal and Experience Requirements

- Primary participant Portal: initiate, review exact versions, supply evidence, and respond to required actions.
- Operator Portal: inspect exceptions, correlation, source responses, retries, compensation, and downstream impact.
- Executive or oversight view: consume governed metrics without becoming a source of Domain truth.

Every Portal displays exact version, owner, state, freshness, required action, and uncertainty. It supports loading, empty, partial, stale, denied, error, timeout, and Outcome Unknown states; accessibility; locale, currency, unit, date, and RTL/LTR correctness; evidence audit; safe retry; and correction history.

10. AI Participation and Tool Governance

Permitted AI tasks:

- assist schema and lineage analysis
- detect data-quality anomalies
- summarize migration and reconciliation risk
- never mutate authoritative data outside typed Domain Tools

AI shall not own Domain records, grant itself authority, suppress contradictions, silently edit approved versions, fabricate external outcomes, or execute high-impact actions outside typed Tools, policy, approval, idempotency, sandbox, and outcome observation.

Every Agent Run records source context, prompt/model/tool versions, structured output, confidence and limitations, Policy result, approval, Tool calls, provider responses, final outcome, and recovery actions.

11. Security, Privacy, and Trust Boundaries

- least privilege and separation of duties
- tenant, organization, facility, channel and jurisdiction isolation
- integrity protection for binding evidence
- verified external callbacks and anti-replay controls

Additional controls include secret isolation, callback signature verification, audit of privileged action, emergency-access review, sensitive-data minimization, purpose limitation, and threat models for impersonation, tampering, replay, confused deputy, fraud, prompt injection, and exfiltration.

12. Reliability, SLOs, and Operations

SLI	Reference objective
authoritative read availability	99.9% monthly unless a stricter tier applies
acknowledged Command durability	no acknowledged write may be silently lost
Event publication	99.9% within five minutes after committed fact
duplicate prevention	100% for same tenant, Command class, and idempotency key
binding Decision audit completeness	100%
state-state detection	bounded by business deadline

Dashboards and runbooks cover backlog, error budget, dependency failure, retry budget, DLQ, workflow stall, provider outage, reconciliation, capacity, backup/restore, release/rollback, incident command, and post-incident correction.

13. Testing and Continuous Conformance

- aggregate invariant and transition tests
- authorization, delegation, separation-of-duties and tenant-isolation tests
- idempotency and optimistic-concurrency tests
- database migration, key, constraint and rollback tests
- contract, outbox/inbox, duplicate, replay and DLQ tests
- workflow restart, timeout and compensation tests
- Portal E2E tests for success, partial, stale, denied, error and Outcome Unknown states
- accessibility, locale, currency, unit and RTL/LTR tests
- security adversarial, fraud and data-exfiltration tests
- AI evaluation, prompt-injection, policy-bypass and unauthorized-tool tests
- historical reconstruction, correction and audit-completeness tests
- performance, capacity, recovery and operational exercise tests

Release evidence links each invariant and control to the test, environment, input data, result, owner, and artifact proving it. Generated test counts never replace semantic coverage.

14. Repository Review Boundary

- This chapter is a repository-review specification. Exact executable implementation has not been inspected for every target behavior.
- Registered modules, catalog entries, README claims, frontend contracts, or plans are not proof of complete implementation.

Area	Classification
target aggregate, API, tables, state and workflow in this chapter	REFERENCE_ARCHITECTURE
registered module or catalog claim	DOCUMENTATION_ONLY_CLAIM until source inspection
uninspected integration	REQUIRES_REPOSITORY_REVIEW
behavior proven in another repository-grounded chapter	retain that chapter's evidence and classification
contradictory ownership or path claims	CONFLICT_REQUIRES_CANONICALIZATION

No statement upgrades implementation status merely because a file, directory, module name, frontend contract, database entry, plan, or report exists.

15. Worked Conformance Scenario

An authorized principal initiates a Command against an exact subject version. The service validates identity, tenant, mandate, Policy, Evidence, and idempotency. It records the state transition and outbox entry atomically. If an external dependency participates, timeout produces Outcome Unknown; reconciliation queries the source or waits for a verified callback instead of repeating the action blindly.

Portals show source, freshness, version, required action, and uncertainty. AI may extract, compare, summarize, detect anomalies, and recommend through typed Tools, but cannot own the subject or fabricate success. Later correction appends new facts, identifies affected projections and workflows, and preserves the historical Decision context.

16. Conformance Checklist

- ☐ stable identity and proposition ownership are explicit;
- ☐ authority and mandate are evaluated at action time;
- ☐ state, Recommendation, Decision, Intent, execution, and outcome remain distinct;
- ☐ Commands are typed, authorized, idempotent, and version guarded;
- ☐ Events describe completed facts;
- ☐ Evidence includes provenance, time, integrity, qualification, and correction;
- ☐ external outcomes retain their authority source;
- ☐ Portals show freshness and uncertainty;
- ☐ AI is bounded by typed Tools, Policy, approval, and outcome observation;
- ☐ security, tenant, facility, channel, and jurisdiction boundaries are enforced;
- ☐ Outcome Unknown and reconciliation exist where needed;
- ☐ corrections preserve history;
- ☐ tests prove invariants and recovery;
- ☐ implementation status is not upgraded without executable evidence.

17. Status Vocabulary

Status	Meaning
VERIFIED_IMPLEMENTED	Executable source and relevant behavior were inspected.
IMPLEMENTED_WITH_GAP	Executable behavior exists with a material governance or completeness gap.
PARTIAL_IMPLEMENTATION	Only part of the target capability is evidenced.
REFERENCE_ARCHITECTURE	Normative target behavior; no claim that it exists today.
PLANNED_NOT_IMPLEMENTED	Explicitly planned but not evidenced as executable.
DOCUMENTATION_ONLY_CLAIM	Documentation or client contract claims behavior without verified backend evidence.
CONFLICT_REQUIRES_CANONICALIZATION	Sources disagree and no final owner has been established.
REQUIRES_REPOSITORY_REVIEW	Necessary executable evidence has not yet been inspected.

Partitioning and Archival

Metadata	Value
Volume	10
Chapter ID	V10-19
Subject	PartitioningandArchival
Status	Generated First-Draft Architecture Skeleton - Domain-Specific Expansion Required
Content maturity	TEMPLATE_DERIVED_REFERENCE_SKELETON unless repository citations prove otherwise
Default implementation classification	REFERENCE_ARCHITECTURE / REQUIRES_REPOSITORY_REVIEW
Accountable owner	Data Architecture Council
Evidence rule	No implementation claim without inspected executable evidence
Repository	chinair88-prog/allinb2c

1. Executive Orientation

This chapter defines how GTOS shall **define enterprise and Domain data ownership, semantics, persistence, quality, lineage, correction, protection, operations, and verification for Partitioning and Archival.**

The specification separates reality, observations, knowledge, Decisions, Intent, execution, and outcomes. A component name, database row, UI label, AI answer, or workflow state does not acquire authority merely because it is visible or technically convenient.

Reality or external outcome

- ≠ Observation
- ≠ Claim
- ≠ derived Perspective
- ≠ Prediction
- ≠ Recommendation
- ≠ authorized Decision
- ≠ execution Intent

1.1 Accountable ownership

The accountable owner is **Data Architecture Council**. Supporting applications, shared infrastructure, integration adapters, analytics, and AI coordinate or present the capability but shall not silently own its authoritative propositions.

1.2 Scope

- create and identify the subject
- validate evidence and policy
- make authorized Decisions
- execute typed Commands
- publish completed-fact Events
- query current and historical perspectives
- correct and reconcile outcomes

1.3 Non-goals

- treating a registered Gradle module, catalog record, README, frontend route, migration plan, or generated report as proof of end-to-end implementation;
- direct cross-Domain database writes;

- generic status values that combine lifecycle, approval, provider, document, payment, dispute, and correction state;
- silent replacement of history;
- AI-created authority or fabricated external outcomes.

2. Ubiquitous Language and Subject Model

Subject	Required interpretation
PartitioningandArchival	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
PartitioningandArchivalVersion	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
PartitioningandArchivalDecision	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
PartitioningandArchivalEvidenceSet	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
PartitioningandArchivalException	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit

Reference aggregate:

```
PartitioningandArchival {
  subjectId
  tenantId
  organizationContext
  sourceIdentityRefs[]
  sourceVersionRefs[]
  lifecycleState
  relationshipRefs[]
  decisionRefs[]
  intentRefs[]
  evidenceSetRef
  knowledgeCutoff
  policyRefs[]
  authorityRef
  jurisdictionRefs[]
  createdAt
  effectiveAt
  updatedAt
  version
  correctionOf?
  correlationId
}
```

2.1 Core invariants

- Stable identity precedes relationship, state, and cross-system correlation.
- Effective time and recorded time remain distinct where business truth changes over time.
- Recommendation, Decision, Intent, execution, and outcome are separate concepts.
- No Portal, report, search index, cache, workflow, or AI Agent becomes a shadow owner of Domain truth.
- Binding submitted, approved, issued, accepted, or signed versions are immutable.
- Corrections append governed facts and preserve the original record.
- External authority and provider outcomes retain their source and qualification.
- Tenant and represented-principal context come from trusted authentication and mandate evaluation.
- Retried writes are idempotent and concurrency guarded.
- Outcome Unknown is explicit and reconciled before risky duplicate execution.

2.2 Required standards

- stable identifiers
- typed contracts
- immutable binding versions
- transactional outbox
- idempotent Commands
- explicit Outcome Unknown
- correction lineage
- machine-verifiable conformance

3. Actors, Roles, and Authority

A conformant design distinguishes:

- the natural person acting;
- the service or AI identity invoking a Tool;
- the organization or principal represented;
- the role or relationship granting context;
- the mandate or delegation granting action authority;
- the Policy and jurisdiction limiting the action;
- the reviewer or approver responsible for a binding Decision.

Authority is evaluated at action time. Authentication alone does not grant business authority. Cached permissions and browser-supplied tenant headers are insufficient for high-impact actions.

4. Lifecycle and State Model

State	Semantics
PROPOSED	Distinct governed condition with entry, exit, timeout, correction and authority semantics
VALIDATION_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
APPROVAL_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
ACTIVE	Distinct governed condition with entry, exit, timeout, correction and authority semantics
SUSPENDED	Distinct governed condition with entry, exit, timeout, correction and authority semantics
CORRECTION_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
RETIRED	Distinct governed condition with entry, exit, timeout, correction and authority semantics

```
stateDiagram-v2
    [*] --> PROPOSED
    PROPOSED --> VALIDATION_PENDING
    VALIDATION_PENDING --> APPROVAL_PENDING
    APPROVAL_PENDING --> ACTIVE
    ACTIVE --> SUSPENDED
    SUSPENDED --> CORRECTION_PENDING
    CORRECTION_PENDING --> RETIRED
```

The diagram is a primary reading path. Real implementations may add parallel review, amendment, appeal, timeout, suspension, partial completion, reconciliation, correction, and terminal failure without collapsing their meaning.

4.1 Transition rules

- transition through a typed, authorized Command;
 - require exact expected version where concurrent change matters;
 - record actor, represented principal, mandate, Policy, reason, and Evidence;
 - use an idempotency key for retried writes;
 - publish a completed-fact Event after durable state change;
 - expose Outcome Unknown for unresolved external completion;
 - preserve original and corrected states in historical reconstruction.
-

5. Decision, Evidence, and Knowledge Cutoff

Reference Decision:

```
Decision {
    decisionId
    decisionType
    subjectRef
}
```

```

sourceVersionRefs[]
knowledgeCutoff
evidenceSetRef
policyVersionRefs[]
recommendationRefs[]
principalId
representedOrganizationId
mandateRef
rationale
outcome
decidedAt
effectiveAt
correctionOf?
}

```

Evidence records identify issuer or observer, subject, version, observed time, received time, effective interval, integrity, confidence, qualification, contradiction, correction, retention, residency, and access class. A recommendation may assist the Decision but cannot promote itself to the Decision.

6. Commands, Events, Queries, and Contracts

Reference Commands:

- CreatePartitioningandArchival
- ValidatePartitioningandArchival
- ApprovePartitioningandArchival
- ActivatePartitioningandArchival
- SuspendPartitioningandArchival
- CorrectPartitioningandArchival
- RetirePartitioningandArchival

Reference Events:

- PartitioningandArchivalCreated
- PartitioningandArchivalValidated
- PartitioningandArchivalApproved
- PartitioningandArchivalActivated
- PartitioningandArchivalSuspended
- PartitioningandArchivalCorrected
- PartitioningandArchivalRetired

Reference queries:

- GetPartitioningandArchival
- ListPartitioningandArchivalRecords
- GetPartitioningandArchivalTimeline
- GetPartitioningandArchivalEvidence
- GetPartitioningandArchivalDecisionPackage
- GetPartitioningandArchivalExceptions
- GetPartitioningandArchivalAuditTrail

Reference API surface:

```

POST /api/v1/partitioningandarchival/commands
GET  /api/v1/partitioningandarchival/{id}
GET  /api/v1/partitioningandarchival/{id}/timeline
GET  /api/v1/partitioningandarchival/{id}/evidence
GET  /api/v1/partitioningandarchival/{id}/decisions
POST /api/v1/partitioningandarchival/{id}/exceptions
POST /api/v1/partitioningandarchival/{id}/corrections

```

These endpoints are reference contracts, not claims that exact routes exist.

7. Workflow, Failure, and Compensation

flowchart TD

```

A[Validate identity version and authority] --> B[Collect required evidence]
B --> C{Policy and evidence gates pass?}
C -- No --> D[Open exception or request correction]
D --> B
C -- Yes --> E[Record Decision or execution Intent]
E --> F[Execute Domain command or external request]

```

```

F --> G{Outcome known?}
G -- Success --> H[Record fact and publish Event]
G -- Failure --> I[Record failure and compensation options]
G -- Unknown --> J[Enter Outcome Unknown]
J --> K[Query source or wait for verified callback]
K --> G
H --> L[Update projections and downstream workflow]

```

Representative failure modes:

- stale or wrong source version
- duplicate submission or replay
- external timeout after possible success
- contradictory source observations
- lost Event after committed state
- approval or mandate revoked during execution
- downstream action before a mandatory gate
- partial completion across multiple subjects
- evidence later retracted or corrected
- tenant, facility, channel or jurisdiction mismatch

Recovery shall query the authoritative source before duplicate execution, preserve partial success, use compensating Intent rather than destructive rewrite, publish corrections, quarantine high-impact ambiguity, and record affected downstream subjects.

8. Data and Persistence Architecture

Reference table families:

Table family	Purpose
partitioningandarchival_aggregate	authoritative subject
partitioningandarchival_version	immutable submitted, approved, issued, or signed versions
partitioningandarchival_relationship	governed relationships
partitioningandarchival_decision	binding Decisions and rationale
partitioningandarchival_evidence_link	provenance and evidence relationships
partitioningandarchival_event_outbox	transactional Event publication
partitioningandarchival_idempotency	duplicate-write protection
partitioningandarchival_exception	exception, claim, dispute, or hold
partitioningandarchival_correction	correction and supersession lineage
partitioningandarchival_audit	privileged action evidence

The owning service controls schema and migration. Cross-Domain references use stable identifiers or contracts. Analytics, reports, caches, search, and vector indexes remain projections. Backup, restore, retention, legal hold, privacy, residency, and decommissioning requirements are explicit.

9. Portal and Experience Requirements

- Primary participant Portal: initiate, review exact versions, supply evidence, and respond to required actions.
- Operator Portal: inspect exceptions, correlation, source responses, retries, compensation, and downstream impact.
- Executive or oversight view: consume governed metrics without becoming a source of Domain truth.

Every Portal displays exact version, owner, state, freshness, required action, and uncertainty. It supports loading, empty, partial, stale, denied, error, timeout, and Outcome Unknown states; accessibility; locale, currency, unit, date, and RTL/LTR correctness; evidence audit; safe retry; and correction history.

10. AI Participation and Tool Governance

Permitted AI tasks:

- assist schema and lineage analysis
- detect data-quality anomalies
- summarize migration and reconciliation risk
- never mutate authoritative data outside typed Domain Tools

AI shall not own Domain records, grant itself authority, suppress contradictions, silently edit approved versions, fabricate external outcomes, or execute high-impact actions outside typed Tools, policy, approval, idempotency, sandbox, and outcome observation.

Every Agent Run records source context, prompt/model/tool versions, structured output, confidence and limitations, Policy result, approval, Tool calls, provider responses, final outcome, and recovery actions.

11. Security, Privacy, and Trust Boundaries

- least privilege and separation of duties
- tenant, organization, facility, channel and jurisdiction isolation
- integrity protection for binding evidence
- verified external callbacks and anti-replay controls

Additional controls include secret isolation, callback signature verification, audit of privileged action, emergency-access review, sensitive-data minimization, purpose limitation, and threat models for impersonation, tampering, replay, confused deputy, fraud, prompt injection, and exfiltration.

12. Reliability, SLOs, and Operations

SLI	Reference objective
authoritative read availability	99.9% monthly unless a stricter tier applies
acknowledged Command durability	no acknowledged write may be silently lost
Event publication	99.9% within five minutes after committed fact
duplicate prevention	100% for same tenant, Command class, and idempotency key
binding Decision audit completeness	100%
stale-state detection	bounded by business deadline

Dashboards and runbooks cover backlog, error budget, dependency failure, retry budget, DLQ, workflow stall, provider outage, reconciliation, capacity, backup/restore, release/rollback, incident command, and post-incident correction.

13. Testing and Continuous Conformance

- aggregate invariant and transition tests
- authorization, delegation, separation-of-duties and tenant-isolation tests
- idempotency and optimistic-concurrency tests
- database migration, key, constraint and rollback tests
- contract, outbox/inbox, duplicate, replay and DLQ tests
- workflow restart, timeout and compensation tests
- Portal E2E tests for success, partial, stale, denied, error and Outcome Unknown states
- accessibility, locale, currency, unit and RTL/LTR tests
- security adversarial, fraud and data-exfiltration tests
- AI evaluation, prompt-injection, policy-bypass and unauthorized-tool tests
- historical reconstruction, correction and audit-completeness tests
- performance, capacity, recovery and operational exercise tests

Release evidence links each invariant and control to the test, environment, input data, result, owner, and artifact proving it. Generated test counts never replace semantic coverage.

14. Repository Review Boundary

- This chapter is a repository-review specification. Exact executable implementation has not been inspected for every target behavior.
- Registered modules, catalog entries, README claims, frontend contracts, or plans are not proof of complete implementation.

Area	Classification
target aggregate, API, tables, state and workflow in this chapter	REFERENCE_ARCHITECTURE
registered module or catalog claim	DOCUMENTATION_ONLY_CLAIM until source inspection
uninspected integration	REQUIRES_REPOSITORY_REVIEW
behavior proven in another repository-grounded chapter	retain that chapter's evidence and classification
contradictory ownership or path claims	CONFLICT_REQUIRES_CANONICALIZATION

No statement upgrades implementation status merely because a file, directory, module name, frontend contract, database entry, plan, or report exists.

15. Worked Conformance Scenario

An authorized principal initiates a Command against an exact subject version. The service validates identity, tenant, mandate, Policy, Evidence, and idempotency. It records the state transition and outbox entry atomically. If an external dependency participates, timeout produces Outcome Unknown; reconciliation queries the source or waits for a verified callback instead of repeating the action blindly.

Portals show source, freshness, version, required action, and uncertainty. AI may extract, compare, summarize, detect anomalies, and recommend through typed Tools, but cannot own the subject or fabricate success. Later correction appends new facts, identifies affected projections and workflows, and preserves the historical Decision context.

16. Conformance Checklist

- ☐ stable identity and proposition ownership are explicit;
- ☐ authority and mandate are evaluated at action time;
- ☐ state, Recommendation, Decision, Intent, execution, and outcome remain distinct;
- ☐ Commands are typed, authorized, idempotent, and version guarded;
- ☐ Events describe completed facts;
- ☐ Evidence includes provenance, time, integrity, qualification, and correction;
- ☐ external outcomes retain their authority source;
- ☐ Portals show freshness and uncertainty;
- ☐ AI is bounded by typed Tools, Policy, approval, and outcome observation;
- ☐ security, tenant, facility, channel, and jurisdiction boundaries are enforced;
- ☐ Outcome Unknown and reconciliation exist where needed;
- ☐ corrections preserve history;
- ☐ tests prove invariants and recovery;
- ☐ implementation status is not upgraded without executable evidence.

17. Status Vocabulary

Status	Meaning
VERIFIED_IMPLEMENTED	Executable source and relevant behavior were inspected.
IMPLEMENTED_WITH_GAP	Executable behavior exists with a material governance or completeness gap.
PARTIAL_IMPLEMENTATION	Only part of the target capability is evidenced.
REFERENCE_ARCHITECTURE	Normative target behavior; no claim that it exists today.
PLANNED_NOT_IMPLEMENTED	Explicitly planned but not evidenced as executable.
DOCUMENTATION_ONLY_CLAIM	Documentation or client contract claims behavior without verified backend evidence.
CONFLICT_REQUIRES_CANONICALIZATION	Sources disagree and no final owner has been established.
REQUIRES_REPOSITORY_REVIEW	Necessary executable evidence has not yet been inspected.

Backup and Restore

Metadata	Value
Volume	10
Chapter ID	V10-20
Subject	BackupandRestore
Status	Generated First-Draft Architecture Skeleton - Domain-Specific Expansion Required
Content maturity	TEMPLATE_DERIVED_REFERENCE_SKELETON unless repository citations prove otherwise
Default implementation classification	REFERENCE_ARCHITECTURE / REQUIRES_REPOSITORY_REVIEW
Accountable owner	Data Architecture Council
Evidence rule	No implementation claim without inspected executable evidence
Repository	chinair88-prog/allinb2c

1. Executive Orientation

This chapter defines how GTOS shall **define enterprise and Domain data ownership, semantics, persistence, quality, lineage, correction, protection, operations, and verification for Backup and Restore.**

The specification separates reality, observations, knowledge, Decisions, Intent, execution, and outcomes. A component name, database row, UI label, AI answer, or workflow state does not acquire authority merely because it is visible or technically convenient.

Reality or external outcome

- ≠ Observation
- ≠ Claim
- ≠ derived Perspective
- ≠ Prediction
- ≠ Recommendation
- ≠ authorized Decision
- ≠ execution Intent

1.1 Accountable ownership

The accountable owner is **Data Architecture Council**. Supporting applications, shared infrastructure, integration adapters, analytics, and AI coordinate or present the capability but shall not silently own its authoritative propositions.

1.2 Scope

- create and identify the subject
- validate evidence and policy
- make authorized Decisions
- execute typed Commands
- publish completed-fact Events
- query current and historical perspectives
- correct and reconcile outcomes

1.3 Non-goals

- treating a registered Gradle module, catalog record, README, frontend route, migration plan, or generated report as proof of end-to-end implementation;
- direct cross-Domain database writes;

- generic status values that combine lifecycle, approval, provider, document, payment, dispute, and correction state;
- silent replacement of history;
- AI-created authority or fabricated external outcomes.

2. Ubiquitous Language and Subject Model

Subject	Required interpretation
BackupandRestore	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
BackupandRestoreVersion	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
BackupandRestoreDecision	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
BackupandRestoreEvidenceSet	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
BackupandRestoreException	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit

Reference aggregate:

```
BackupandRestore {
  subjectId
  tenantId
  organizationContext
  sourceIdentityRefs[]
  sourceVersionRefs[]
  lifecycleState
  relationshipRefs[]
  decisionRefs[]
  intentRefs[]
  evidenceSetRef
  knowledgeCutoff
  policyRefs[]
  authorityRef
  jurisdictionRefs[]
  createdAt
  effectiveAt
  updatedAt
  version
  correctionOf?
  correlationId
}
```

2.1 Core invariants

- Stable identity precedes relationship, state, and cross-system correlation.
- Effective time and recorded time remain distinct where business truth changes over time.
- Recommendation, Decision, Intent, execution, and outcome are separate concepts.
- No Portal, report, search index, cache, workflow, or AI Agent becomes a shadow owner of Domain truth.
- Binding submitted, approved, issued, accepted, or signed versions are immutable.
- Corrections append governed facts and preserve the original record.
- External authority and provider outcomes retain their source and qualification.
- Tenant and represented-principal context come from trusted authentication and mandate evaluation.
- Retried writes are idempotent and concurrency guarded.
- Outcome Unknown is explicit and reconciled before risky duplicate execution.

2.2 Required standards

- stable identifiers
- typed contracts
- immutable binding versions
- transactional outbox
- idempotent Commands
- explicit Outcome Unknown
- correction lineage
- machine-verifiable conformance

3. Actors, Roles, and Authority

A conformant design distinguishes:

- the natural person acting;
- the service or AI identity invoking a Tool;
- the organization or principal represented;
- the role or relationship granting context;
- the mandate or delegation granting action authority;
- the Policy and jurisdiction limiting the action;
- the reviewer or approver responsible for a binding Decision.

Authority is evaluated at action time. Authentication alone does not grant business authority. Cached permissions and browser-supplied tenant headers are insufficient for high-impact actions.

4. Lifecycle and State Model

State	Semantics
PROPOSED	Distinct governed condition with entry, exit, timeout, correction and authority semantics
VALIDATION_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
APPROVAL_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
ACTIVE	Distinct governed condition with entry, exit, timeout, correction and authority semantics
SUSPENDED	Distinct governed condition with entry, exit, timeout, correction and authority semantics
CORRECTION_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
RETIRED	Distinct governed condition with entry, exit, timeout, correction and authority semantics

```
stateDiagram-v2
    [*] --> PROPOSED
    PROPOSED --> VALIDATION_PENDING
    VALIDATION_PENDING --> APPROVAL_PENDING
    APPROVAL_PENDING --> ACTIVE
    ACTIVE --> SUSPENDED
    SUSPENDED --> CORRECTION_PENDING
    CORRECTION_PENDING --> RETIRED
```

The diagram is a primary reading path. Real implementations may add parallel review, amendment, appeal, timeout, suspension, partial completion, reconciliation, correction, and terminal failure without collapsing their meaning.

4.1 Transition rules

- transition through a typed, authorized Command;
 - require exact expected version where concurrent change matters;
 - record actor, represented principal, mandate, Policy, reason, and Evidence;
 - use an idempotency key for retried writes;
 - publish a completed-fact Event after durable state change;
 - expose Outcome Unknown for unresolved external completion;
 - preserve original and corrected states in historical reconstruction.
-

5. Decision, Evidence, and Knowledge Cutoff

Reference Decision:

```
Decision {
    decisionId
    decisionType
    subjectRef
}
```

```

sourceVersionRefs[]
knowledgeCutoff
evidenceSetRef
policyVersionRefs[]
recommendationRefs[]
principalId
representedOrganizationId
mandateRef
rationale
outcome
decidedAt
effectiveAt
correctionOf?
}

```

Evidence records identify issuer or observer, subject, version, observed time, received time, effective interval, integrity, confidence, qualification, contradiction, correction, retention, residency, and access class. A recommendation may assist the Decision but cannot promote itself to the Decision.

6. Commands, Events, Queries, and Contracts

Reference Commands:

- CreateBackupandRestore
- ValidateBackupandRestore
- ApproveBackupandRestore
- ActivateBackupandRestore
- SuspendBackupandRestore
- CorrectBackupandRestore
- RetireBackupandRestore

Reference Events:

- BackupandRestoreCreated
- BackupandRestoreValidated
- BackupandRestoreApproved
- BackupandRestoreActivated
- BackupandRestoreSuspended
- BackupandRestoreCorrected
- BackupandRestoreRetired

Reference queries:

- GetBackupandRestore
- ListBackupandRestoreRecords
- GetBackupandRestoreTimeline
- GetBackupandRestoreEvidence
- GetBackupandRestoreDecisionPackage
- GetBackupandRestoreExceptions
- GetBackupandRestoreAuditTrail

Reference API surface:

```

POST /api/v1/backupandrestore/commands
GET  /api/v1/backupandrestore/{id}
GET  /api/v1/backupandrestore/{id}/timeline
GET  /api/v1/backupandrestore/{id}/evidence
GET  /api/v1/backupandrestore/{id}/decisions
POST /api/v1/backupandrestore/{id}/exceptions
POST /api/v1/backupandrestore/{id}/corrections

```

These endpoints are reference contracts, not claims that exact routes exist.

7. Workflow, Failure, and Compensation

flowchart TD

```

A[Validate identity version and authority] --> B[Collect required evidence]
B --> C{Policy and evidence gates pass?}
C -- No --> D[Open exception or request correction]
D --> B
C -- Yes --> E[Record Decision or execution Intent]
E --> F[Execute Domain command or external request]

```

```

F --> G{Outcome known?}
G -- Success --> H[Record fact and publish Event]
G -- Failure --> I[Record failure and compensation options]
G -- Unknown --> J[Enter Outcome Unknown]
J --> K[Query source or wait for verified callback]
K --> G
H --> L[Update projections and downstream workflow]

```

Representative failure modes:

- stale or wrong source version
- duplicate submission or replay
- external timeout after possible success
- contradictory source observations
- lost Event after committed state
- approval or mandate revoked during execution
- downstream action before a mandatory gate
- partial completion across multiple subjects
- evidence later retracted or corrected
- tenant, facility, channel or jurisdiction mismatch

Recovery shall query the authoritative source before duplicate execution, preserve partial success, use compensating Intent rather than destructive rewrite, publish corrections, quarantine high-impact ambiguity, and record affected downstream subjects.

8. Data and Persistence Architecture

Reference table families:

Table family	Purpose
backupandrestore_aggregate	authoritative subject
backupandrestore_version	immutable submitted, approved, issued, or signed versions
backupandrestore_relationship	governed relationships
backupandrestore_decision	binding Decisions and rationale
backupandrestore_evidence_link	provenance and evidence relationships
backupandrestore_event_outbox	transactional Event publication
backupandrestore_idempotency	duplicate-write protection
backupandrestore_exception	exception, claim, dispute, or hold
backupandrestore_correction	correction and supersession lineage
backupandrestore_audit	privileged action evidence

The owning service controls schema and migration. Cross-Domain references use stable identifiers or contracts. Analytics, reports, caches, search, and vector indexes remain projections. Backup, restore, retention, legal hold, privacy, residency, and decommissioning requirements are explicit.

9. Portal and Experience Requirements

- Primary participant Portal: initiate, review exact versions, supply evidence, and respond to required actions.
- Operator Portal: inspect exceptions, correlation, source responses, retries, compensation, and downstream impact.
- Executive or oversight view: consume governed metrics without becoming a source of Domain truth.

Every Portal displays exact version, owner, state, freshness, required action, and uncertainty. It supports loading, empty, partial, stale, denied, error, timeout, and Outcome Unknown states; accessibility; locale, currency, unit, date, and RTL/LTR correctness; evidence audit; safe retry; and correction history.

10. AI Participation and Tool Governance

Permitted AI tasks:

- assist schema and lineage analysis
- detect data-quality anomalies
- summarize migration and reconciliation risk
- never mutate authoritative data outside typed Domain Tools

AI shall not own Domain records, grant itself authority, suppress contradictions, silently edit approved versions, fabricate external outcomes, or execute high-impact actions outside typed Tools, policy, approval, idempotency, sandbox, and outcome observation.

Every Agent Run records source context, prompt/model/tool versions, structured output, confidence and limitations, Policy result, approval, Tool calls, provider responses, final outcome, and recovery actions.

11. Security, Privacy, and Trust Boundaries

- least privilege and separation of duties
- tenant, organization, facility, channel and jurisdiction isolation
- integrity protection for binding evidence
- verified external callbacks and anti-replay controls

Additional controls include secret isolation, callback signature verification, audit of privileged action, emergency-access review, sensitive-data minimization, purpose limitation, and threat models for impersonation, tampering, replay, confused deputy, fraud, prompt injection, and exfiltration.

12. Reliability, SLOs, and Operations

SLI	Reference objective
authoritative read availability	99.9% monthly unless a stricter tier applies
acknowledged Command durability	no acknowledged write may be silently lost
Event publication	99.9% within five minutes after committed fact
duplicate prevention	100% for same tenant, Command class, and idempotency key
binding Decision audit completeness	100%
stale-state detection	bounded by business deadline

Dashboards and runbooks cover backlog, error budget, dependency failure, retry budget, DLQ, workflow stall, provider outage, reconciliation, capacity, backup/restore, release/rollback, incident command, and post-incident correction.

13. Testing and Continuous Conformance

- aggregate invariant and transition tests
- authorization, delegation, separation-of-duties and tenant-isolation tests
- idempotency and optimistic-concurrency tests
- database migration, key, constraint and rollback tests
- contract, outbox/inbox, duplicate, replay and DLQ tests
- workflow restart, timeout and compensation tests
- Portal E2E tests for success, partial, stale, denied, error and Outcome Unknown states
- accessibility, locale, currency, unit and RTL/LTR tests
- security adversarial, fraud and data-exfiltration tests
- AI evaluation, prompt-injection, policy-bypass and unauthorized-tool tests
- historical reconstruction, correction and audit-completeness tests
- performance, capacity, recovery and operational exercise tests

Release evidence links each invariant and control to the test, environment, input data, result, owner, and artifact proving it. Generated test counts never replace semantic coverage.

14. Repository Review Boundary

- This chapter is a repository-review specification. Exact executable implementation has not been inspected for every target behavior.
- Registered modules, catalog entries, README claims, frontend contracts, or plans are not proof of complete implementation.

Area	Classification
target aggregate, API, tables, state and workflow in this chapter	REFERENCE_ARCHITECTURE
registered module or catalog claim	DOCUMENTATION_ONLY_CLAIM until source inspection
uninspected integration	REQUIRES_REPOSITORY_REVIEW
behavior proven in another repository-grounded chapter	retain that chapter's evidence and classification
contradictory ownership or path claims	CONFLICT_REQUIRES_CANONICALIZATION

No statement upgrades implementation status merely because a file, directory, module name, frontend contract, database entry, plan, or report exists.

15. Worked Conformance Scenario

An authorized principal initiates a Command against an exact subject version. The service validates identity, tenant, mandate, Policy, Evidence, and idempotency. It records the state transition and outbox entry atomically. If an external dependency participates, timeout produces Outcome Unknown; reconciliation queries the source or waits for a verified callback instead of repeating the action blindly.

Portals show source, freshness, version, required action, and uncertainty. AI may extract, compare, summarize, detect anomalies, and recommend through typed Tools, but cannot own the subject or fabricate success. Later correction appends new facts, identifies affected projections and workflows, and preserves the historical Decision context.

16. Conformance Checklist

- ☐ stable identity and proposition ownership are explicit;
- ☐ authority and mandate are evaluated at action time;
- ☐ state, Recommendation, Decision, Intent, execution, and outcome remain distinct;
- ☐ Commands are typed, authorized, idempotent, and version guarded;
- ☐ Events describe completed facts;
- ☐ Evidence includes provenance, time, integrity, qualification, and correction;
- ☐ external outcomes retain their authority source;
- ☐ Portals show freshness and uncertainty;
- ☐ AI is bounded by typed Tools, Policy, approval, and outcome observation;
- ☐ security, tenant, facility, channel, and jurisdiction boundaries are enforced;
- ☐ Outcome Unknown and reconciliation exist where needed;
- ☐ corrections preserve history;
- ☐ tests prove invariants and recovery;
- ☐ implementation status is not upgraded without executable evidence.

17. Status Vocabulary

Status	Meaning
VERIFIED_IMPLEMENTED	Executable source and relevant behavior were inspected.
IMPLEMENTED_WITH_GAP	Executable behavior exists with a material governance or completeness gap.
PARTIAL_IMPLEMENTATION	Only part of the target capability is evidenced.
REFERENCE_ARCHITECTURE	Normative target behavior; no claim that it exists today.
PLANNED_NOT_IMPLEMENTED	Explicitly planned but not evidenced as executable.
DOCUMENTATION_ONLY_CLAIM	Documentation or client contract claims behavior without verified backend evidence.
CONFLICT_REQUIRES_CANONICALIZATION	Sources disagree and no final owner has been established.
REQUIRES_REPOSITORY_REVIEW	Necessary executable evidence has not yet been inspected.

Disaster Recovery for Data

Metadata	Value
Volume	10
Chapter ID	V10-21
Subject	DisasterRecoveryforData
Status	Generated First-Draft Architecture Skeleton - Domain-Specific Expansion Required
Content maturity	TEMPLATE_DERIVED_REFERENCE_SKELETON unless repository citations prove otherwise
Default implementation classification	REFERENCE_ARCHITECTURE / REQUIRES_REPOSITORY_REVIEW
Accountable owner	Data Architecture Council
Evidence rule	No implementation claim without inspected executable evidence
Repository	chinair88-prog/allinb2c

1. Executive Orientation

This chapter defines how GTOS shall **define enterprise and Domain data ownership, semantics, persistence, quality, lineage, correction, protection, operations, and verification for Disaster Recovery for Data.**

The specification separates reality, observations, knowledge, Decisions, Intent, execution, and outcomes. A component name, database row, UI label, AI answer, or workflow state does not acquire authority merely because it is visible or technically convenient.

Reality or external outcome

- ≠ Observation
- ≠ Claim
- ≠ derived Perspective
- ≠ Prediction
- ≠ Recommendation
- ≠ authorized Decision
- ≠ execution Intent

1.1 Accountable ownership

The accountable owner is **Data Architecture Council**. Supporting applications, shared infrastructure, integration adapters, analytics, and AI coordinate or present the capability but shall not silently own its authoritative propositions.

1.2 Scope

- create and identify the subject
- validate evidence and policy
- make authorized Decisions
- execute typed Commands
- publish completed-fact Events
- query current and historical perspectives
- correct and reconcile outcomes

1.3 Non-goals

- treating a registered Gradle module, catalog record, README, frontend route, migration plan, or generated report as proof of end-to-end implementation;
- direct cross-Domain database writes;

- generic status values that combine lifecycle, approval, provider, document, payment, dispute, and correction state;
- silent replacement of history;
- AI-created authority or fabricated external outcomes.

2. Ubiquitous Language and Subject Model

Subject	Required interpretation
DisasterRecoveryforData	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
DisasterRecoveryforDataVersion	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
DisasterRecoveryforDataDecision	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
DisasterRecoveryforDataEvidenceSet	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
DisasterRecoveryforDataException	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit

Reference aggregate:

```
DisasterRecoveryforData {
  subjectId
  tenantId
  organizationContext
  sourceIdentityRefs[]
  sourceVersionRefs[]
  lifecycleState
  relationshipRefs[]
  decisionRefs[]
  intentRefs[]
  evidenceSetRef
  knowledgeCutoff
  policyRefs[]
  authorityRef
  jurisdictionRefs[]
  createdAt
  effectiveAt
  updatedAt
  version
  correctionOf?
  correlationId
}
```

2.1 Core invariants

- Stable identity precedes relationship, state, and cross-system correlation.
- Effective time and recorded time remain distinct where business truth changes over time.
- Recommendation, Decision, Intent, execution, and outcome are separate concepts.
- No Portal, report, search index, cache, workflow, or AI Agent becomes a shadow owner of Domain truth.
- Binding submitted, approved, issued, accepted, or signed versions are immutable.
- Corrections append governed facts and preserve the original record.
- External authority and provider outcomes retain their source and qualification.
- Tenant and represented-principal context come from trusted authentication and mandate evaluation.
- Retried writes are idempotent and concurrency guarded.
- Outcome Unknown is explicit and reconciled before risky duplicate execution.

2.2 Required standards

- stable identifiers
- typed contracts
- immutable binding versions
- transactional outbox
- idempotent Commands
- explicit Outcome Unknown
- correction lineage
- machine-verifiable conformance

3. Actors, Roles, and Authority

A conformant design distinguishes:

- the natural person acting;
- the service or AI identity invoking a Tool;
- the organization or principal represented;
- the role or relationship granting context;
- the mandate or delegation granting action authority;
- the Policy and jurisdiction limiting the action;
- the reviewer or approver responsible for a binding Decision.

Authority is evaluated at action time. Authentication alone does not grant business authority. Cached permissions and browser-supplied tenant headers are insufficient for high-impact actions.

4. Lifecycle and State Model

State	Semantics
PROPOSED	Distinct governed condition with entry, exit, timeout, correction and authority semantics
VALIDATION_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
APPROVAL_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
ACTIVE	Distinct governed condition with entry, exit, timeout, correction and authority semantics
SUSPENDED	Distinct governed condition with entry, exit, timeout, correction and authority semantics
CORRECTION_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
RETIRED	Distinct governed condition with entry, exit, timeout, correction and authority semantics

```
stateDiagram-v2
    [*] --> PROPOSED
    PROPOSED --> VALIDATION_PENDING
    VALIDATION_PENDING --> APPROVAL_PENDING
    APPROVAL_PENDING --> ACTIVE
    ACTIVE --> SUSPENDED
    SUSPENDED --> CORRECTION_PENDING
    CORRECTION_PENDING --> RETIRED
```

The diagram is a primary reading path. Real implementations may add parallel review, amendment, appeal, timeout, suspension, partial completion, reconciliation, correction, and terminal failure without collapsing their meaning.

4.1 Transition rules

- transition through a typed, authorized Command;
 - require exact expected version where concurrent change matters;
 - record actor, represented principal, mandate, Policy, reason, and Evidence;
 - use an idempotency key for retried writes;
 - publish a completed-fact Event after durable state change;
 - expose Outcome Unknown for unresolved external completion;
 - preserve original and corrected states in historical reconstruction.
-

5. Decision, Evidence, and Knowledge Cutoff

Reference Decision:

```
Decision {
    decisionId
    decisionType
    subjectRef
}
```

```

sourceVersionRefs[]
knowledgeCutoff
evidenceSetRef
policyVersionRefs[]
recommendationRefs[]
principalId
representedOrganizationId
mandateRef
rationale
outcome
decidedAt
effectiveAt
correctionOf?
}

```

Evidence records identify issuer or observer, subject, version, observed time, received time, effective interval, integrity, confidence, qualification, contradiction, correction, retention, residency, and access class. A recommendation may assist the Decision but cannot promote itself to the Decision.

6. Commands, Events, Queries, and Contracts

Reference Commands:

- CreateDisasterRecoveryforData
- ValidateDisasterRecoveryforData
- ApproveDisasterRecoveryforData
- ActivateDisasterRecoveryforData
- SuspendDisasterRecoveryforData
- CorrectDisasterRecoveryforData
- RetireDisasterRecoveryforData

Reference Events:

- DisasterRecoveryforDataCreated
- DisasterRecoveryforDataValidated
- DisasterRecoveryforDataApproved
- DisasterRecoveryforDataActivated
- DisasterRecoveryforDataSuspended
- DisasterRecoveryforDataCorrected
- DisasterRecoveryforDataRetired

Reference queries:

- GetDisasterRecoveryforData
- ListDisasterRecoveryforDataRecords
- GetDisasterRecoveryforDataTimeline
- GetDisasterRecoveryforDataEvidence
- GetDisasterRecoveryforDataDecisionPackage
- GetDisasterRecoveryforDataExceptions
- GetDisasterRecoveryforDataAuditTrail

Reference API surface:

```

POST /api/v1/disasterrecoveryfordata/commands
GET  /api/v1/disasterrecoveryfordata/{id}
GET  /api/v1/disasterrecoveryfordata/{id}/timeline
GET  /api/v1/disasterrecoveryfordata/{id}/evidence
GET  /api/v1/disasterrecoveryfordata/{id}/decisions
POST /api/v1/disasterrecoveryfordata/{id}/exceptions
POST /api/v1/disasterrecoveryfordata/{id}/corrections

```

These endpoints are reference contracts, not claims that exact routes exist.

7. Workflow, Failure, and Compensation

flowchart TD

```

A[Validate identity version and authority] --> B[Collect required evidence]
B --> C{Policy and evidence gates pass?}
C -- No --> D[Open exception or request correction]
D --> B
C -- Yes --> E[Record Decision or execution Intent]
E --> F[Execute Domain command or external request]

```

```

F --> G{Outcome known?}
G -- Success --> H[Record fact and publish Event]
G -- Failure --> I[Record failure and compensation options]
G -- Unknown --> J[Enter Outcome Unknown]
J --> K[Query source or wait for verified callback]
K --> G
H --> L[Update projections and downstream workflow]

```

Representative failure modes:

- stale or wrong source version
- duplicate submission or replay
- external timeout after possible success
- contradictory source observations
- lost Event after committed state
- approval or mandate revoked during execution
- downstream action before a mandatory gate
- partial completion across multiple subjects
- evidence later retracted or corrected
- tenant, facility, channel or jurisdiction mismatch

Recovery shall query the authoritative source before duplicate execution, preserve partial success, use compensating Intent rather than destructive rewrite, publish corrections, quarantine high-impact ambiguity, and record affected downstream subjects.

8. Data and Persistence Architecture

Reference table families:

Table family	Purpose
disasterrecoveryfordata_aggregate	authoritative subject
disasterrecoveryfordata_version	immutable submitted, approved, issued, or signed versions
disasterrecoveryfordata_relationship	governed relationships
disasterrecoveryfordata_decision	binding Decisions and rationale
disasterrecoveryfordata_evidence_link	provenance and evidence relationships
disasterrecoveryfordata_event_outbox	transactional Event publication
disasterrecoveryfordata_idempotency	duplicate-write protection
disasterrecoveryfordata_exception	exception, claim, dispute, or hold
disasterrecoveryfordata_correction	correction and supersession lineage
disasterrecoveryfordata_audit	privileged action evidence

The owning service controls schema and migration. Cross-Domain references use stable identifiers or contracts. Analytics, reports, caches, search, and vector indexes remain projections. Backup, restore, retention, legal hold, privacy, residency, and decommissioning requirements are explicit.

9. Portal and Experience Requirements

- Primary participant Portal: initiate, review exact versions, supply evidence, and respond to required actions.
- Operator Portal: inspect exceptions, correlation, source responses, retries, compensation, and downstream impact.
- Executive or oversight view: consume governed metrics without becoming a source of Domain truth.

Every Portal displays exact version, owner, state, freshness, required action, and uncertainty. It supports loading, empty, partial, stale, denied, error, timeout, and Outcome Unknown states; accessibility; locale, currency, unit, date, and RTL/LTR correctness; evidence audit; safe retry; and correction history.

10. AI Participation and Tool Governance

Permitted AI tasks:

- assist schema and lineage analysis
- detect data-quality anomalies
- summarize migration and reconciliation risk
- never mutate authoritative data outside typed Domain Tools

AI shall not own Domain records, grant itself authority, suppress contradictions, silently edit approved versions, fabricate external outcomes, or execute high-impact actions outside typed Tools, policy, approval, idempotency, sandbox, and outcome observation.

Every Agent Run records source context, prompt/model/tool versions, structured output, confidence and limitations, Policy result, approval, Tool calls, provider responses, final outcome, and recovery actions.

11. Security, Privacy, and Trust Boundaries

- least privilege and separation of duties
- tenant, organization, facility, channel and jurisdiction isolation
- integrity protection for binding evidence
- verified external callbacks and anti-replay controls

Additional controls include secret isolation, callback signature verification, audit of privileged action, emergency-access review, sensitive-data minimization, purpose limitation, and threat models for impersonation, tampering, replay, confused deputy, fraud, prompt injection, and exfiltration.

12. Reliability, SLOs, and Operations

SLI	Reference objective
authoritative read availability	99.9% monthly unless a stricter tier applies
acknowledged Command durability	no acknowledged write may be silently lost
Event publication	99.9% within five minutes after committed fact
duplicate prevention	100% for same tenant, Command class, and idempotency key
binding Decision audit completeness	100%
state-state detection	bounded by business deadline

Dashboards and runbooks cover backlog, error budget, dependency failure, retry budget, DLQ, workflow stall, provider outage, reconciliation, capacity, backup/restore, release/rollback, incident command, and post-incident correction.

13. Testing and Continuous Conformance

- aggregate invariant and transition tests
- authorization, delegation, separation-of-duties and tenant-isolation tests
- idempotency and optimistic-concurrency tests
- database migration, key, constraint and rollback tests
- contract, outbox/inbox, duplicate, replay and DLQ tests
- workflow restart, timeout and compensation tests
- Portal E2E tests for success, partial, stale, denied, error and Outcome Unknown states
- accessibility, locale, currency, unit and RTL/LTR tests
- security adversarial, fraud and data-exfiltration tests
- AI evaluation, prompt-injection, policy-bypass and unauthorized-tool tests
- historical reconstruction, correction and audit-completeness tests
- performance, capacity, recovery and operational exercise tests

Release evidence links each invariant and control to the test, environment, input data, result, owner, and artifact proving it. Generated test counts never replace semantic coverage.

14. Repository Review Boundary

- This chapter is a repository-review specification. Exact executable implementation has not been inspected for every target behavior.
- Registered modules, catalog entries, README claims, frontend contracts, or plans are not proof of complete implementation.

Area	Classification
target aggregate, API, tables, state and workflow in this chapter	REFERENCE_ARCHITECTURE
registered module or catalog claim	DOCUMENTATION_ONLY_CLAIM until source inspection
uninspected integration	REQUIRES_REPOSITORY_REVIEW
behavior proven in another repository-grounded chapter	retain that chapter's evidence and classification
contradictory ownership or path claims	CONFLICT_REQUIRES_CANONICALIZATION

No statement upgrades implementation status merely because a file, directory, module name, frontend contract, database entry, plan, or report exists.

15. Worked Conformance Scenario

An authorized principal initiates a Command against an exact subject version. The service validates identity, tenant, mandate, Policy, Evidence, and idempotency. It records the state transition and outbox entry atomically. If an external dependency participates, timeout produces Outcome Unknown; reconciliation queries the source or waits for a verified callback instead of repeating the action blindly.

Portals show source, freshness, version, required action, and uncertainty. AI may extract, compare, summarize, detect anomalies, and recommend through typed Tools, but cannot own the subject or fabricate success. Later correction appends new facts, identifies affected projections and workflows, and preserves the historical Decision context.

16. Conformance Checklist

- ☐ stable identity and proposition ownership are explicit;
- ☐ authority and mandate are evaluated at action time;
- ☐ state, Recommendation, Decision, Intent, execution, and outcome remain distinct;
- ☐ Commands are typed, authorized, idempotent, and version guarded;
- ☐ Events describe completed facts;
- ☐ Evidence includes provenance, time, integrity, qualification, and correction;
- ☐ external outcomes retain their authority source;
- ☐ Portals show freshness and uncertainty;
- ☐ AI is bounded by typed Tools, Policy, approval, and outcome observation;
- ☐ security, tenant, facility, channel, and jurisdiction boundaries are enforced;
- ☐ Outcome Unknown and reconciliation exist where needed;
- ☐ corrections preserve history;
- ☐ tests prove invariants and recovery;
- ☐ implementation status is not upgraded without executable evidence.

17. Status Vocabulary

Status	Meaning
VERIFIED_IMPLEMENTED	Executable source and relevant behavior were inspected.
IMPLEMENTED_WITH_GAP	Executable behavior exists with a material governance or completeness gap.
PARTIAL_IMPLEMENTATION	Only part of the target capability is evidenced.
REFERENCE_ARCHITECTURE	Normative target behavior; no claim that it exists today.
PLANNED_NOT_IMPLEMENTED	Explicitly planned but not evidenced as executable.
DOCUMENTATION_ONLY_CLAIM	Documentation or client contract claims behavior without verified backend evidence.
CONFLICT_REQUIRES_CANONICALIZATION	Sources disagree and no final owner has been established.
REQUIRES_REPOSITORY_REVIEW	Necessary executable evidence has not yet been inspected.

Database Migration

Metadata	Value
Volume	10
Chapter ID	V10-22
Subject	DatabaseMigration
Status	Generated First-Draft Architecture Skeleton - Domain-Specific Expansion Required
Content maturity	TEMPLATE_DERIVED_REFERENCE_SKELETON unless repository citations prove otherwise
Default implementation classification	REFERENCE_ARCHITECTURE / REQUIRES_REPOSITORY_REVIEW
Accountable owner	Data Architecture Council
Evidence rule	No implementation claim without inspected executable evidence
Repository	chinair88-prog/allinb2c

1. Executive Orientation

This chapter defines how GTOS shall **define enterprise and Domain data ownership, semantics, persistence, quality, lineage, correction, protection, operations, and verification for Database Migration.**

The specification separates reality, observations, knowledge, Decisions, Intent, execution, and outcomes. A component name, database row, UI label, AI answer, or workflow state does not acquire authority merely because it is visible or technically convenient.

Reality or external outcome

- ≠ Observation
- ≠ Claim
- ≠ derived Perspective
- ≠ Prediction
- ≠ Recommendation
- ≠ authorized Decision
- ≠ execution Intent

1.1 Accountable ownership

The accountable owner is **Data Architecture Council**. Supporting applications, shared infrastructure, integration adapters, analytics, and AI coordinate or present the capability but shall not silently own its authoritative propositions.

1.2 Scope

- create and identify the subject
- validate evidence and policy
- make authorized Decisions
- execute typed Commands
- publish completed-fact Events
- query current and historical perspectives
- correct and reconcile outcomes

1.3 Non-goals

- treating a registered Gradle module, catalog record, README, frontend route, migration plan, or generated report as proof of end-to-end implementation;
- direct cross-Domain database writes;

- generic status values that combine lifecycle, approval, provider, document, payment, dispute, and correction state;
- silent replacement of history;
- AI-created authority or fabricated external outcomes.

2. Ubiquitous Language and Subject Model

Subject	Required interpretation
DatabaseMigration	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
DatabaseMigrationVersion	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
DatabaseMigrationDecision	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
DatabaseMigrationEvidenceSet	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
DatabaseMigrationException	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit

Reference aggregate:

```
DatabaseMigration {
  subjectId
  tenantId
  organizationContext
  sourceIdentityRefs[]
  sourceVersionRefs[]
  lifecycleState
  relationshipRefs[]
  decisionRefs[]
  intentRefs[]
  evidenceSetRef
  knowledgeCutoff
  policyRefs[]
  authorityRef
  jurisdictionRefs[]
  createdAt
  effectiveAt
  updatedAt
  version
  correctionOf?
  correlationId
}
```

2.1 Core invariants

- Stable identity precedes relationship, state, and cross-system correlation.
- Effective time and recorded time remain distinct where business truth changes over time.
- Recommendation, Decision, Intent, execution, and outcome are separate concepts.
- No Portal, report, search index, cache, workflow, or AI Agent becomes a shadow owner of Domain truth.
- Binding submitted, approved, issued, accepted, or signed versions are immutable.
- Corrections append governed facts and preserve the original record.
- External authority and provider outcomes retain their source and qualification.
- Tenant and represented-principal context come from trusted authentication and mandate evaluation.
- Retried writes are idempotent and concurrency guarded.
- Outcome Unknown is explicit and reconciled before risky duplicate execution.

2.2 Required standards

- stable identifiers
- typed contracts
- immutable binding versions
- transactional outbox
- idempotent Commands
- explicit Outcome Unknown
- correction lineage
- machine-verifiable conformance

3. Actors, Roles, and Authority

A conformant design distinguishes:

- the natural person acting;
- the service or AI identity invoking a Tool;
- the organization or principal represented;
- the role or relationship granting context;
- the mandate or delegation granting action authority;
- the Policy and jurisdiction limiting the action;
- the reviewer or approver responsible for a binding Decision.

Authority is evaluated at action time. Authentication alone does not grant business authority. Cached permissions and browser-supplied tenant headers are insufficient for high-impact actions.

4. Lifecycle and State Model

State	Semantics
PROPOSED	Distinct governed condition with entry, exit, timeout, correction and authority semantics
VALIDATION_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
APPROVAL_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
ACTIVE	Distinct governed condition with entry, exit, timeout, correction and authority semantics
SUSPENDED	Distinct governed condition with entry, exit, timeout, correction and authority semantics
CORRECTION_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
RETIRED	Distinct governed condition with entry, exit, timeout, correction and authority semantics

```
stateDiagram-v2
    [*] --> PROPOSED
    PROPOSED --> VALIDATION_PENDING
    VALIDATION_PENDING --> APPROVAL_PENDING
    APPROVAL_PENDING --> ACTIVE
    ACTIVE --> SUSPENDED
    SUSPENDED --> CORRECTION_PENDING
    CORRECTION_PENDING --> RETIRED
```

The diagram is a primary reading path. Real implementations may add parallel review, amendment, appeal, timeout, suspension, partial completion, reconciliation, correction, and terminal failure without collapsing their meaning.

4.1 Transition rules

- transition through a typed, authorized Command;
 - require exact expected version where concurrent change matters;
 - record actor, represented principal, mandate, Policy, reason, and Evidence;
 - use an idempotency key for retried writes;
 - publish a completed-fact Event after durable state change;
 - expose Outcome Unknown for unresolved external completion;
 - preserve original and corrected states in historical reconstruction.
-

5. Decision, Evidence, and Knowledge Cutoff

Reference Decision:

```
Decision {
    decisionId
    decisionType
    subjectRef
}
```

```

sourceVersionRefs[]
knowledgeCutoff
evidenceSetRef
policyVersionRefs[]
recommendationRefs[]
principalId
representedOrganizationId
mandateRef
rationale
outcome
decidedAt
effectiveAt
correctionOf?
}

```

Evidence records identify issuer or observer, subject, version, observed time, received time, effective interval, integrity, confidence, qualification, contradiction, correction, retention, residency, and access class. A recommendation may assist the Decision but cannot promote itself to the Decision.

6. Commands, Events, Queries, and Contracts

Reference Commands:

- CreateDatabaseMigration
- ValidateDatabaseMigration
- ApproveDatabaseMigration
- ActivateDatabaseMigration
- SuspendDatabaseMigration
- CorrectDatabaseMigration
- RetireDatabaseMigration

Reference Events:

- DatabaseMigrationCreated
- DatabaseMigrationValidated
- DatabaseMigrationApproved
- DatabaseMigrationActivated
- DatabaseMigrationSuspended
- DatabaseMigrationCorrected
- DatabaseMigrationRetired

Reference queries:

- GetDatabaseMigration
- ListDatabaseMigrationRecords
- GetDatabaseMigrationTimeline
- GetDatabaseMigrationEvidence
- GetDatabaseMigrationDecisionPackage
- GetDatabaseMigrationExceptions
- GetDatabaseMigrationAuditTrail

Reference API surface:

```

POST /api/v1/databasemigration/commands
GET  /api/v1/databasemigration/{id}
GET  /api/v1/databasemigration/{id}/timeline
GET  /api/v1/databasemigration/{id}/evidence
GET  /api/v1/databasemigration/{id}/decisions
POST /api/v1/databasemigration/{id}/exceptions
POST /api/v1/databasemigration/{id}/corrections

```

These endpoints are reference contracts, not claims that exact routes exist.

7. Workflow, Failure, and Compensation

flowchart TD

```

A[Validate identity version and authority] --> B[Collect required evidence]
B --> C{Policy and evidence gates pass?}
C -- No --> D[Open exception or request correction]
D --> B
C -- Yes --> E[Record Decision or execution Intent]
E --> F[Execute Domain command or external request]

```

```

F --> G{Outcome known?}
G -- Success --> H[Record fact and publish Event]
G -- Failure --> I[Record failure and compensation options]
G -- Unknown --> J[Enter Outcome Unknown]
J --> K[Query source or wait for verified callback]
K --> G
H --> L[Update projections and downstream workflow]

```

Representative failure modes:

- stale or wrong source version
- duplicate submission or replay
- external timeout after possible success
- contradictory source observations
- lost Event after committed state
- approval or mandate revoked during execution
- downstream action before a mandatory gate
- partial completion across multiple subjects
- evidence later retracted or corrected
- tenant, facility, channel or jurisdiction mismatch

Recovery shall query the authoritative source before duplicate execution, preserve partial success, use compensating Intent rather than destructive rewrite, publish corrections, quarantine high-impact ambiguity, and record affected downstream subjects.

8. Data and Persistence Architecture

Reference table families:

Table family	Purpose
databasemigration_aggregate	authoritative subject
databasemigration_version	immutable submitted, approved, issued, or signed versions
databasemigration_relationship	governed relationships
databasemigration_decision	binding Decisions and rationale
databasemigration_evidence_link	provenance and evidence relationships
databasemigration_event_outbox	transactional Event publication
databasemigration_idempotency	duplicate-write protection
databasemigration_exception	exception, claim, dispute, or hold
databasemigration_correction	correction and supersession lineage
databasemigration_audit	privileged action evidence

The owning service controls schema and migration. Cross-Domain references use stable identifiers or contracts. Analytics, reports, caches, search, and vector indexes remain projections. Backup, restore, retention, legal hold, privacy, residency, and decommissioning requirements are explicit.

9. Portal and Experience Requirements

- Primary participant Portal: initiate, review exact versions, supply evidence, and respond to required actions.
- Operator Portal: inspect exceptions, correlation, source responses, retries, compensation, and downstream impact.
- Executive or oversight view: consume governed metrics without becoming a source of Domain truth.

Every Portal displays exact version, owner, state, freshness, required action, and uncertainty. It supports loading, empty, partial, stale, denied, error, timeout, and Outcome Unknown states; accessibility; locale, currency, unit, date, and RTL/LTR correctness; evidence audit; safe retry; and correction history.

10. AI Participation and Tool Governance

Permitted AI tasks:

- assist schema and lineage analysis
- detect data-quality anomalies
- summarize migration and reconciliation risk
- never mutate authoritative data outside typed Domain Tools

AI shall not own Domain records, grant itself authority, suppress contradictions, silently edit approved versions, fabricate external outcomes, or execute high-impact actions outside typed Tools, policy, approval, idempotency, sandbox, and outcome observation.

Every Agent Run records source context, prompt/model/tool versions, structured output, confidence and limitations, Policy result, approval, Tool calls, provider responses, final outcome, and recovery actions.

11. Security, Privacy, and Trust Boundaries

- least privilege and separation of duties
- tenant, organization, facility, channel and jurisdiction isolation
- integrity protection for binding evidence
- verified external callbacks and anti-replay controls

Additional controls include secret isolation, callback signature verification, audit of privileged action, emergency-access review, sensitive-data minimization, purpose limitation, and threat models for impersonation, tampering, replay, confused deputy, fraud, prompt injection, and exfiltration.

12. Reliability, SLOs, and Operations

SLI	Reference objective
authoritative read availability	99.9% monthly unless a stricter tier applies
acknowledged Command durability	no acknowledged write may be silently lost
Event publication	99.9% within five minutes after committed fact
duplicate prevention	100% for same tenant, Command class, and idempotency key
binding Decision audit completeness	100%
state-state detection	bounded by business deadline

Dashboards and runbooks cover backlog, error budget, dependency failure, retry budget, DLQ, workflow stall, provider outage, reconciliation, capacity, backup/restore, release/rollback, incident command, and post-incident correction.

13. Testing and Continuous Conformance

- aggregate invariant and transition tests
- authorization, delegation, separation-of-duties and tenant-isolation tests
- idempotency and optimistic-concurrency tests
- database migration, key, constraint and rollback tests
- contract, outbox/inbox, duplicate, replay and DLQ tests
- workflow restart, timeout and compensation tests
- Portal E2E tests for success, partial, stale, denied, error and Outcome Unknown states
- accessibility, locale, currency, unit and RTL/LTR tests
- security adversarial, fraud and data-exfiltration tests
- AI evaluation, prompt-injection, policy-bypass and unauthorized-tool tests
- historical reconstruction, correction and audit-completeness tests
- performance, capacity, recovery and operational exercise tests

Release evidence links each invariant and control to the test, environment, input data, result, owner, and artifact proving it. Generated test counts never replace semantic coverage.

14. Repository Review Boundary

- This chapter is a repository-review specification. Exact executable implementation has not been inspected for every target behavior.
- Registered modules, catalog entries, README claims, frontend contracts, or plans are not proof of complete implementation.

Area	Classification
target aggregate, API, tables, state and workflow in this chapter	REFERENCE_ARCHITECTURE
registered module or catalog claim	DOCUMENTATION_ONLY_CLAIM until source inspection
uninspected integration	REQUIRES_REPOSITORY_REVIEW
behavior proven in another repository-grounded chapter	retain that chapter's evidence and classification
contradictory ownership or path claims	CONFLICT_REQUIRES_CANONICALIZATION

No statement upgrades implementation status merely because a file, directory, module name, frontend contract, database entry, plan, or report exists.

15. Worked Conformance Scenario

An authorized principal initiates a Command against an exact subject version. The service validates identity, tenant, mandate, Policy, Evidence, and idempotency. It records the state transition and outbox entry atomically. If an external dependency participates, timeout produces Outcome Unknown; reconciliation queries the source or waits for a verified callback instead of repeating the action blindly.

Portals show source, freshness, version, required action, and uncertainty. AI may extract, compare, summarize, detect anomalies, and recommend through typed Tools, but cannot own the subject or fabricate success. Later correction appends new facts, identifies affected projections and workflows, and preserves the historical Decision context.

16. Conformance Checklist

- ☐ stable identity and proposition ownership are explicit;
- ☐ authority and mandate are evaluated at action time;
- ☐ state, Recommendation, Decision, Intent, execution, and outcome remain distinct;
- ☐ Commands are typed, authorized, idempotent, and version guarded;
- ☐ Events describe completed facts;
- ☐ Evidence includes provenance, time, integrity, qualification, and correction;
- ☐ external outcomes retain their authority source;
- ☐ Portals show freshness and uncertainty;
- ☐ AI is bounded by typed Tools, Policy, approval, and outcome observation;
- ☐ security, tenant, facility, channel, and jurisdiction boundaries are enforced;
- ☐ Outcome Unknown and reconciliation exist where needed;
- ☐ corrections preserve history;
- ☐ tests prove invariants and recovery;
- ☐ implementation status is not upgraded without executable evidence.

17. Status Vocabulary

Status	Meaning
VERIFIED_IMPLEMENTED	Executable source and relevant behavior were inspected.
IMPLEMENTED_WITH_GAP	Executable behavior exists with a material governance or completeness gap.
PARTIAL_IMPLEMENTATION	Only part of the target capability is evidenced.
REFERENCE_ARCHITECTURE	Normative target behavior; no claim that it exists today.
PLANNED_NOT_IMPLEMENTED	Explicitly planned but not evidenced as executable.
DOCUMENTATION_ONLY_CLAIM	Documentation or client contract claims behavior without verified backend evidence.
CONFLICT_REQUIRES_CANONICALIZATION	Sources disagree and no final owner has been established.
REQUIRES_REPOSITORY_REVIEW	Necessary executable evidence has not yet been inspected.

Data Contract Evolution

Metadata	Value
Volume	10
Chapter ID	V10-23
Subject	DataContractEvolution
Status	Generated First-Draft Architecture Skeleton - Domain-Specific Expansion Required
Content maturity	TEMPLATE_DERIVED_REFERENCE_SKELETON unless repository citations prove otherwise
Default implementation classification	REFERENCE_ARCHITECTURE / REQUIRES_REPOSITORY_REVIEW
Accountable owner	Data Architecture Council
Evidence rule	No implementation claim without inspected executable evidence
Repository	chinair88-prog/allinb2c

1. Executive Orientation

This chapter defines how GTOS shall **define enterprise and Domain data ownership, semantics, persistence, quality, lineage, correction, protection, operations, and verification for Data Contract Evolution.**

The specification separates reality, observations, knowledge, Decisions, Intent, execution, and outcomes. A component name, database row, UI label, AI answer, or workflow state does not acquire authority merely because it is visible or technically convenient.

Reality or external outcome

- ≠ Observation
- ≠ Claim
- ≠ derived Perspective
- ≠ Prediction
- ≠ Recommendation
- ≠ authorized Decision
- ≠ execution Intent

1.1 Accountable ownership

The accountable owner is **Data Architecture Council**. Supporting applications, shared infrastructure, integration adapters, analytics, and AI coordinate or present the capability but shall not silently own its authoritative propositions.

1.2 Scope

- create and identify the subject
- validate evidence and policy
- make authorized Decisions
- execute typed Commands
- publish completed-fact Events
- query current and historical perspectives
- correct and reconcile outcomes

1.3 Non-goals

- treating a registered Gradle module, catalog record, README, frontend route, migration plan, or generated report as proof of end-to-end implementation;
- direct cross-Domain database writes;
- generic status values that combine lifecycle, approval, provider, document, payment, dispute, and correction state;

- silent replacement of history;
- AI-created authority or fabricated external outcomes.

2. Ubiquitous Language and Subject Model

Subject	Required interpretation
DataContractEvolution	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
DataContractEvolutionVersion	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
DataContractEvolutionDecision	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
DataContractEvolutionEvidenceSet	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
DataContractEvolutionException	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit

Reference aggregate:

```
DataContractEvolution {
  subjectId
  tenantId
  organizationContext
  sourceIdentityRefs[]
  sourceVersionRefs[]
  lifecycleState
  relationshipRefs[]
  decisionRefs[]
  intentRefs[]
  evidenceSetRef
  knowledgeCutoff
  policyRefs[]
  authorityRef
  jurisdictionRefs[]
  createdAt
  effectiveAt
  updatedAt
  version
  correctionOf?
  correlationId
}
```

2.1 Core invariants

- Stable identity precedes relationship, state, and cross-system correlation.
- Effective time and recorded time remain distinct where business truth changes over time.
- Recommendation, Decision, Intent, execution, and outcome are separate concepts.
- No Portal, report, search index, cache, workflow, or AI Agent becomes a shadow owner of Domain truth.
- Binding submitted, approved, issued, accepted, or signed versions are immutable.
- Corrections append governed facts and preserve the original record.
- External authority and provider outcomes retain their source and qualification.
- Tenant and represented-principal context come from trusted authentication and mandate evaluation.
- Retried writes are idempotent and concurrency guarded.
- Outcome Unknown is explicit and reconciled before risky duplicate execution.

2.2 Required standards

- stable identifiers
- typed contracts
- immutable binding versions
- transactional outbox
- idempotent Commands
- explicit Outcome Unknown
- correction lineage
- machine-verifiable conformance

3. Actors, Roles, and Authority

A conformant design distinguishes:

- the natural person acting;
- the service or AI identity invoking a Tool;
- the organization or principal represented;
- the role or relationship granting context;
- the mandate or delegation granting action authority;
- the Policy and jurisdiction limiting the action;
- the reviewer or approver responsible for a binding Decision.

Authority is evaluated at action time. Authentication alone does not grant business authority. Cached permissions and browser-supplied tenant headers are insufficient for high-impact actions.

4. Lifecycle and State Model

State	Semantics
PROPOSED	Distinct governed condition with entry, exit, timeout, correction and authority semantics
VALIDATION_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
APPROVAL_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
ACTIVE	Distinct governed condition with entry, exit, timeout, correction and authority semantics
SUSPENDED	Distinct governed condition with entry, exit, timeout, correction and authority semantics
CORRECTION_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
RETIRED	Distinct governed condition with entry, exit, timeout, correction and authority semantics

```
stateDiagram-v2
    [*] --> PROPOSED
    PROPOSED --> VALIDATION_PENDING
    VALIDATION_PENDING --> APPROVAL_PENDING
    APPROVAL_PENDING --> ACTIVE
    ACTIVE --> SUSPENDED
    SUSPENDED --> CORRECTION_PENDING
    CORRECTION_PENDING --> RETIRED
```

The diagram is a primary reading path. Real implementations may add parallel review, amendment, appeal, timeout, suspension, partial completion, reconciliation, correction, and terminal failure without collapsing their meaning.

4.1 Transition rules

- transition through a typed, authorized Command;
- require exact expected version where concurrent change matters;
- record actor, represented principal, mandate, Policy, reason, and Evidence;
- use an idempotency key for retried writes;
- publish a completed-fact Event after durable state change;
- expose Outcome Unknown for unresolved external completion;
- preserve original and corrected states in historical reconstruction.

5. Decision, Evidence, and Knowledge Cutoff

Reference Decision:

```
Decision {
    decisionId
    decisionType
    subjectRef
    sourceVersionRefs[]
    knowledgeCutoff
    evidenceSetRef
}
```

```

policyVersionRefs[]
recommendationRefs[]
principalId
representedOrganizationId
mandateRef
rationale
outcome
decidedAt
effectiveAt
correctionOf?
}

```

Evidence records identify issuer or observer, subject, version, observed time, received time, effective interval, integrity, confidence, qualification, contradiction, correction, retention, residency, and access class. A recommendation may assist the Decision but cannot promote itself to the Decision.

6. Commands, Events, Queries, and Contracts

Reference Commands:

- CreateDataContractEvolution
- ValidateDataContractEvolution
- ApproveDataContractEvolution
- ActivateDataContractEvolution
- SuspendDataContractEvolution
- CorrectDataContractEvolution
- RetireDataContractEvolution

Reference Events:

- DataContractEvolutionCreated
- DataContractEvolutionValidated
- DataContractEvolutionApproved
- DataContractEvolutionActivated
- DataContractEvolutionSuspended
- DataContractEvolutionCorrected
- DataContractEvolutionRetired

Reference queries:

- GetDataContractEvolution
- ListDataContractEvolutionRecords
- GetDataContractEvolutionTimeline
- GetDataContractEvolutionEvidence
- GetDataContractEvolutionDecisionPackage
- GetDataContractEvolutionExceptions
- GetDataContractEvolutionAuditTrail

Reference API surface:

```

POST /api/v1/datacontractevolution/commands
GET  /api/v1/datacontractevolution/{id}
GET  /api/v1/datacontractevolution/{id}/timeline
GET  /api/v1/datacontractevolution/{id}/evidence
GET  /api/v1/datacontractevolution/{id}/decisions
POST /api/v1/datacontractevolution/{id}/exceptions
POST /api/v1/datacontractevolution/{id}/corrections

```

These endpoints are reference contracts, not claims that exact routes exist.

7. Workflow, Failure, and Compensation

flowchart TD

```

A[Validate identity version and authority] --> B[Collect required evidence]
B --> C{Policy and evidence gates pass?}
C -- No --> D[Open exception or request correction]
D --> B
C -- Yes --> E[Record Decision or execution Intent]
E --> F[Execute Domain command or external request]
F --> G{Outcome known?}
G -- Success --> H[Record fact and publish Event]
G -- Failure --> I[Record failure and compensation options]

```

```

G -- Unknown --> J[Enter Outcome Unknown]
J --> K[Query source or wait for verified callback]
K --> G
H --> L[Update projections and downstream workflow]

```

Representative failure modes:

- stale or wrong source version
- duplicate submission or replay
- external timeout after possible success
- contradictory source observations
- lost Event after committed state
- approval or mandate revoked during execution
- downstream action before a mandatory gate
- partial completion across multiple subjects
- evidence later retracted or corrected
- tenant, facility, channel or jurisdiction mismatch

Recovery shall query the authoritative source before duplicate execution, preserve partial success, use compensating Intent rather than destructive rewrite, publish corrections, quarantine high-impact ambiguity, and record affected downstream subjects.

8. Data and Persistence Architecture

Reference table families:

Table family	Purpose
datacontractevolution_aggregate	authoritative subject
datacontractevolution_version	immutable submitted, approved, issued, or signed versions
datacontractevolution_relationship	governed relationships
datacontractevolution_decision	binding Decisions and rationale
datacontractevolution_evidence_link	provenance and evidence relationships
datacontractevolution_event_outbox	transactional Event publication
datacontractevolution_idempotency	duplicate-write protection
datacontractevolution_exception	exception, claim, dispute, or hold
datacontractevolution_correction	correction and supersession lineage
datacontractevolution_audit	privileged action evidence

The owning service controls schema and migration. Cross-Domain references use stable identifiers or contracts. Analytics, reports, caches, search, and vector indexes remain projections. Backup, restore, retention, legal hold, privacy, residency, and decommissioning requirements are explicit.

9. Portal and Experience Requirements

- Primary participant Portal: initiate, review exact versions, supply evidence, and respond to required actions.
- Operator Portal: inspect exceptions, correlation, source responses, retries, compensation, and downstream impact.
- Executive or oversight view: consume governed metrics without becoming a source of Domain truth.

Every Portal displays exact version, owner, state, freshness, required action, and uncertainty. It supports loading, empty, partial, stale, denied, error, timeout, and Outcome Unknown states; accessibility; locale, currency, unit, date, and RTL/LTR correctness; evidence audit; safe retry; and correction history.

10. AI Participation and Tool Governance

Permitted AI tasks:

- assist schema and lineage analysis
- detect data-quality anomalies
- summarize migration and reconciliation risk
- never mutate authoritative data outside typed Domain Tools

AI shall not own Domain records, grant itself authority, suppress contradictions, silently edit approved versions, fabricate external outcomes, or execute high-impact actions outside typed Tools, policy, approval, idempotency, sandbox, and outcome observation.

Every Agent Run records source context, prompt/model/tool versions, structured output, confidence and limitations, Policy result, approval, Tool calls, provider responses, final outcome, and recovery actions.

11. Security, Privacy, and Trust Boundaries

- least privilege and separation of duties
- tenant, organization, facility, channel and jurisdiction isolation
- integrity protection for binding evidence
- verified external callbacks and anti-replay controls

Additional controls include secret isolation, callback signature verification, audit of privileged action, emergency-access review, sensitive-data minimization, purpose limitation, and threat models for impersonation, tampering, replay, confused deputy, fraud, prompt injection, and exfiltration.

12. Reliability, SLOs, and Operations

SLI	Reference objective
authoritative read availability	99.9% monthly unless a stricter tier applies
acknowledged Command durability	no acknowledged write may be silently lost
Event publication	99.9% within five minutes after committed fact
duplicate prevention	100% for same tenant, Command class, and idempotency key
binding Decision audit completeness	100%
stale-state detection	bounded by business deadline

Dashboards and runbooks cover backlog, error budget, dependency failure, retry budget, DLQ, workflow stall, provider outage, reconciliation, capacity, backup/restore, release/rollback, incident command, and post-incident correction.

13. Testing and Continuous Conformance

- aggregate invariant and transition tests
- authorization, delegation, separation-of-duties and tenant-isolation tests
- idempotency and optimistic-concurrency tests
- database migration, key, constraint and rollback tests
- contract, outbox/inbox, duplicate, replay and DLQ tests
- workflow restart, timeout and compensation tests
- Portal E2E tests for success, partial, stale, denied, error and Outcome Unknown states
- accessibility, locale, currency, unit and RTL/LTR tests
- security adversarial, fraud and data-exfiltration tests
- AI evaluation, prompt-injection, policy-bypass and unauthorized-tool tests
- historical reconstruction, correction and audit-completeness tests
- performance, capacity, recovery and operational exercise tests

Release evidence links each invariant and control to the test, environment, input data, result, owner, and artifact proving it. Generated test counts never replace semantic coverage.

14. Repository Review Boundary

- This chapter is a repository-review specification. Exact executable implementation has not been inspected for every target behavior.
- Registered modules, catalog entries, README claims, frontend contracts, or plans are not proof of complete implementation.

Area	Classification
target aggregate, API, tables, state and workflow in this chapter	REFERENCE_ARCHITECTURE
registered module or catalog claim	DOCUMENTATION_ONLY_CLAIM until source inspection
uninspected integration	REQUIRES_REPOSITORY_REVIEW

Area	Classification
behavior proven in another repository-grounded chapter contradictory ownership or path claims	retain that chapter's evidence and classification CONFLICT_REQUIRES_CANONICALIZATION

No statement upgrades implementation status merely because a file, directory, module name, frontend contract, database entry, plan, or report exists.

15. Worked Conformance Scenario

An authorized principal initiates a Command against an exact subject version. The service validates identity, tenant, mandate, Policy, Evidence, and idempotency. It records the state transition and outbox entry atomically. If an external dependency participates, timeout produces Outcome Unknown; reconciliation queries the source or waits for a verified callback instead of repeating the action blindly.

Portals show source, freshness, version, required action, and uncertainty. AI may extract, compare, summarize, detect anomalies, and recommend through typed Tools, but cannot own the subject or fabricate success. Later correction appends new facts, identifies affected projections and workflows, and preserves the historical Decision context.

16. Conformance Checklist

- ☐ stable identity and proposition ownership are explicit;
- ☐ authority and mandate are evaluated at action time;
- ☐ state, Recommendation, Decision, Intent, execution, and outcome remain distinct;
- ☐ Commands are typed, authorized, idempotent, and version guarded;
- ☐ Events describe completed facts;
- ☐ Evidence includes provenance, time, integrity, qualification, and correction;
- ☐ external outcomes retain their authority source;
- ☐ Portals show freshness and uncertainty;
- ☐ AI is bounded by typed Tools, Policy, approval, and outcome observation;
- ☐ security, tenant, facility, channel, and jurisdiction boundaries are enforced;
- ☐ Outcome Unknown and reconciliation exist where needed;
- ☐ corrections preserve history;
- ☐ tests prove invariants and recovery;
- ☐ implementation status is not upgraded without executable evidence.

17. Status Vocabulary

Status	Meaning
VERIFIED_IMPLEMENTED	Executable source and relevant behavior were inspected.
IMPLEMENTED_WITH_GAP	Executable behavior exists with a material governance or completeness gap.
PARTIAL_IMPLEMENTATION	Only part of the target capability is evidenced.
REFERENCE_ARCHITECTURE	Normative target behavior; no claim that it exists today.
PLANNED_NOT_IMPLEMENTED	Explicitly planned but not evidenced as executable.
DOCUMENTATION_ONLY_CLAIM	Documentation or client contract claims behavior without verified backend evidence.
CONFLICT_REQUIRES_CANONICALIZATION	Sources disagree and no final owner has been established.
REQUIRES_REPOSITORY_REVIEW	Necessary executable evidence has not yet been inspected.

Decommissioned Databases and Tables

Metadata	Value
Volume	10
Chapter ID	V10-24
Subject	DecommissionedDatabasesandTables
Status	Generated First-Draft Architecture Skeleton - Domain-Specific Expansion Required
Content maturity	TEMPLATE_DERIVED_REFERENCE_SKELETON unless repository citations prove otherwise
Default implementation classification	REFERENCE_ARCHITECTURE / REQUIRES_REPOSITORY_REVIEW
Accountable owner	Data Architecture Council
Evidence rule	No implementation claim without inspected executable evidence
Repository	chinair88-prog/allinb2c

1. Executive Orientation

This chapter defines how GTOS shall **define enterprise and Domain data ownership, semantics, persistence, quality, lineage, correction, protection, operations, and verification for Decommissioned Databases and Tables.**

The specification separates reality, observations, knowledge, Decisions, Intent, execution, and outcomes. A component name, database row, UI label, AI answer, or workflow state does not acquire authority merely because it is visible or technically convenient.

Reality or external outcome
≠ Observation
≠ Claim
≠ derived Perspective
≠ Prediction
≠ Recommendation
≠ authorized Decision
≠ execution Intent

1.1 Accountable ownership

The accountable owner is **Data Architecture Council**. Supporting applications, shared infrastructure, integration adapters, analytics, and AI coordinate or present the capability but shall not silently own its authoritative propositions.

1.2 Scope

- create and identify the subject
- validate evidence and policy
- make authorized Decisions
- execute typed Commands
- publish completed-fact Events
- query current and historical perspectives
- correct and reconcile outcomes

1.3 Non-goals

- treating a registered Gradle module, catalog record, README, frontend route, migration plan, or generated report as proof of end-to-end implementation;

- direct cross-Domain database writes;
- generic status values that combine lifecycle, approval, provider, document, payment, dispute, and correction state;
- silent replacement of history;
- AI-created authority or fabricated external outcomes.

2. Ubiquitous Language and Subject Model

Subject	Required interpretation
DecommissionedDatabasesandTables	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
DecommissionedDatabasesandTablesVersion	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
DecommissionedDatabasesandTablesDecision	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
DecommissionedDatabasesandTablesEvidenceSet	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
DecommissionedDatabasesandTablesException	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit

Reference aggregate:

```
DecommissionedDatabasesandTables {
  subjectId
  tenantId
  organizationContext
  sourceIdentityRefs[]
  sourceVersionRefs[]
  lifecycleState
  relationshipRefs[]
  decisionRefs[]
  intentRefs[]
  evidenceSetRef
  knowledgeCutoff
  policyRefs[]
  authorityRef
  jurisdictionRefs[]
  createdAt
  effectiveAt
  updatedAt
  version
  correctionOf?
  correlationId
}
```

2.1 Core invariants

- Stable identity precedes relationship, state, and cross-system correlation.
- Effective time and recorded time remain distinct where business truth changes over time.
- Recommendation, Decision, Intent, execution, and outcome are separate concepts.
- No Portal, report, search index, cache, workflow, or AI Agent becomes a shadow owner of Domain truth.
- Binding submitted, approved, issued, accepted, or signed versions are immutable.
- Corrections append governed facts and preserve the original record.
- External authority and provider outcomes retain their source and qualification.
- Tenant and represented-principal context come from trusted authentication and mandate evaluation.
- Retried writes are idempotent and concurrency guarded.
- Outcome Unknown is explicit and reconciled before risky duplicate execution.

2.2 Required standards

- stable identifiers
- typed contracts
- immutable binding versions
- transactional outbox
- idempotent Commands
- explicit Outcome Unknown
- correction lineage

- machine-verifiable conformance

3. Actors, Roles, and Authority

A conformant design distinguishes:

- the natural person acting;
- the service or AI identity invoking a Tool;
- the organization or principal represented;
- the role or relationship granting context;
- the mandate or delegation granting action authority;
- the Policy and jurisdiction limiting the action;
- the reviewer or approver responsible for a binding Decision.

Authority is evaluated at action time. Authentication alone does not grant business authority. Cached permissions and browser-supplied tenant headers are insufficient for high-impact actions.

4. Lifecycle and State Model

State	Semantics
PROPOSED	Distinct governed condition with entry, exit, timeout, correction and authority semantics
VALIDATION_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
APPROVAL_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
ACTIVE	Distinct governed condition with entry, exit, timeout, correction and authority semantics
SUSPENDED	Distinct governed condition with entry, exit, timeout, correction and authority semantics
CORRECTION_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
RETIRED	Distinct governed condition with entry, exit, timeout, correction and authority semantics

```
stateDiagram-v2
    [*] --> PROPOSED
    PROPOSED --> VALIDATION_PENDING
    VALIDATION_PENDING --> APPROVAL_PENDING
    APPROVAL_PENDING --> ACTIVE
    ACTIVE --> SUSPENDED
    SUSPENDED --> CORRECTION_PENDING
    CORRECTION_PENDING --> RETIRED
```

The diagram is a primary reading path. Real implementations may add parallel review, amendment, appeal, timeout, suspension, partial completion, reconciliation, correction, and terminal failure without collapsing their meaning.

4.1 Transition rules

- transition through a typed, authorized Command;
- require exact expected version where concurrent change matters;
- record actor, represented principal, mandate, Policy, reason, and Evidence;
- use an idempotency key for retried writes;
- publish a completed-fact Event after durable state change;
- expose Outcome Unknown for unresolved external completion;
- preserve original and corrected states in historical reconstruction.

5. Decision, Evidence, and Knowledge Cutoff

Reference Decision:

```
Decision {
    decisionId
```

```

decisionType
subjectRef
sourceVersionRefs[]
knowledgeCutoff
evidenceSetRef
policyVersionRefs[]
recommendationRefs[]
principalId
representedOrganizationId
mandateRef
rationale
outcome
decidedAt
effectiveAt
correctionOf?
}

```

Evidence records identify issuer or observer, subject, version, observed time, received time, effective interval, integrity, confidence, qualification, contradiction, correction, retention, residency, and access class. A recommendation may assist the Decision but cannot promote itself to the Decision.

6. Commands, Events, Queries, and Contracts

Reference Commands:

- CreateDecommissionedDatabasesandTables
- ValidateDecommissionedDatabasesandTables
- ApproveDecommissionedDatabasesandTables
- ActivateDecommissionedDatabasesandTables
- SuspendDecommissionedDatabasesandTables
- CorrectDecommissionedDatabasesandTables
- RetireDecommissionedDatabasesandTables

Reference Events:

- DecommissionedDatabasesandTablesCreated
- DecommissionedDatabasesandTablesValidated
- DecommissionedDatabasesandTablesApproved
- DecommissionedDatabasesandTablesActivated
- DecommissionedDatabasesandTablesSuspended
- DecommissionedDatabasesandTablesCorrected
- DecommissionedDatabasesandTablesRetired

Reference queries:

- GetDecommissionedDatabasesandTables
- ListDecommissionedDatabasesandTablesRecords
- GetDecommissionedDatabasesandTablesTimeline
- GetDecommissionedDatabasesandTablesEvidence
- GetDecommissionedDatabasesandTablesDecisionPackage
- GetDecommissionedDatabasesandTablesExceptions
- GetDecommissionedDatabasesandTablesAuditTrail

Reference API surface:

```

POST /api/v1/decommissioneddatabasesandtables/commands
GET  /api/v1/decommissioneddatabasesandtables/{id}
GET  /api/v1/decommissioneddatabasesandtables/{id}/timeline
GET  /api/v1/decommissioneddatabasesandtables/{id}/evidence
GET  /api/v1/decommissioneddatabasesandtables/{id}/decisions
POST /api/v1/decommissioneddatabasesandtables/{id}/exceptions
POST /api/v1/decommissioneddatabasesandtables/{id}/corrections

```

These endpoints are reference contracts, not claims that exact routes exist.

7. Workflow, Failure, and Compensation

flowchart TD

```

A[Validate identity version and authority] --> B[Collect required evidence]
B --> C{Policy and evidence gates pass?}
C -- No --> D[Open exception or request correction]
D --> B

```

```

C -- Yes --> E[Record Decision or execution Intent]
E --> F[Execute Domain command or external request]
F --> G{Outcome known?}
G -- Success --> H[Record fact and publish Event]
G -- Failure --> I[Record failure and compensation options]
G -- Unknown --> J[Enter Outcome Unknown]
J --> K[Query source or wait for verified callback]
K --> G
H --> L[Update projections and downstream workflow]

```

Representative failure modes:

- stale or wrong source version
- duplicate submission or replay
- external timeout after possible success
- contradictory source observations
- lost Event after committed state
- approval or mandate revoked during execution
- downstream action before a mandatory gate
- partial completion across multiple subjects
- evidence later retracted or corrected
- tenant, facility, channel or jurisdiction mismatch

Recovery shall query the authoritative source before duplicate execution, preserve partial success, use compensating Intent rather than destructive rewrite, publish corrections, quarantine high-impact ambiguity, and record affected downstream subjects.

8. Data and Persistence Architecture

Reference table families:

Table family	Purpose
decommissioneddatabasesandtables_aggregate	authoritative subject
decommissioneddatabasesandtables_version	immutable submitted, approved, issued, or signed versions
decommissioneddatabasesandtables_relationship	governed relationships
decommissioneddatabasesandtables_decision	binding Decisions and rationale
decommissioneddatabasesandtables_evidence_link	provenance and evidence relationships
decommissioneddatabasesandtables_event_outbox	transactional Event publication
decommissioneddatabasesandtables_idempotency	duplicate-write protection
decommissioneddatabasesandtables_exception	exception, claim, dispute, or hold
decommissioneddatabasesandtables_correction	correction and supersession lineage
decommissioneddatabasesandtables_audit	privileged action evidence

The owning service controls schema and migration. Cross-Domain references use stable identifiers or contracts. Analytics, reports, caches, search, and vector indexes remain projections. Backup, restore, retention, legal hold, privacy, residency, and decommissioning requirements are explicit.

9. Portal and Experience Requirements

- Primary participant Portal: initiate, review exact versions, supply evidence, and respond to required actions.
- Operator Portal: inspect exceptions, correlation, source responses, retries, compensation, and downstream impact.
- Executive or oversight view: consume governed metrics without becoming a source of Domain truth.

Every Portal displays exact version, owner, state, freshness, required action, and uncertainty. It supports loading, empty, partial, stale, denied, error, timeout, and Outcome Unknown states; accessibility; locale, currency, unit, date, and RTL/LTR correctness; evidence audit; safe retry; and correction history.

10. AI Participation and Tool Governance

Permitted AI tasks:

- assist schema and lineage analysis
- detect data-quality anomalies
- summarize migration and reconciliation risk

- never mutate authoritative data outside typed Domain Tools

AI shall not own Domain records, grant itself authority, suppress contradictions, silently edit approved versions, fabricate external outcomes, or execute high-impact actions outside typed Tools, policy, approval, idempotency, sandbox, and outcome observation.

Every Agent Run records source context, prompt/model/tool versions, structured output, confidence and limitations, Policy result, approval, Tool calls, provider responses, final outcome, and recovery actions.

11. Security, Privacy, and Trust Boundaries

- least privilege and separation of duties
- tenant, organization, facility, channel and jurisdiction isolation
- integrity protection for binding evidence
- verified external callbacks and anti-replay controls

Additional controls include secret isolation, callback signature verification, audit of privileged action, emergency-access review, sensitive-data minimization, purpose limitation, and threat models for impersonation, tampering, replay, confused deputy, fraud, prompt injection, and exfiltration.

12. Reliability, SLOs, and Operations

SLI	Reference objective
authoritative read availability	99.9% monthly unless a stricter tier applies
acknowledged Command durability	no acknowledged write may be silently lost
Event publication	99.9% within five minutes after committed fact
duplicate prevention	100% for same tenant, Command class, and idempotency key
binding Decision audit completeness	100%
state-state detection	bounded by business deadline

Dashboards and runbooks cover backlog, error budget, dependency failure, retry budget, DLQ, workflow stall, provider outage, reconciliation, capacity, backup/restore, release/rollback, incident command, and post-incident correction.

13. Testing and Continuous Conformance

- aggregate invariant and transition tests
- authorization, delegation, separation-of-duties and tenant-isolation tests
- idempotency and optimistic-concurrency tests
- database migration, key, constraint and rollback tests
- contract, outbox/inbox, duplicate, replay and DLQ tests
- workflow restart, timeout and compensation tests
- Portal E2E tests for success, partial, stale, denied, error and Outcome Unknown states
- accessibility, locale, currency, unit and RTL/LTR tests
- security adversarial, fraud and data-exfiltration tests
- AI evaluation, prompt-injection, policy-bypass and unauthorized-tool tests
- historical reconstruction, correction and audit-completeness tests
- performance, capacity, recovery and operational exercise tests

Release evidence links each invariant and control to the test, environment, input data, result, owner, and artifact proving it. Generated test counts never replace semantic coverage.

14. Repository Review Boundary

- This chapter is a repository-review specification. Exact executable implementation has not been inspected for every target behavior.
- Registered modules, catalog entries, README claims, frontend contracts, or plans are not proof of complete implementation.

Area	Classification
target aggregate, API, tables, state and workflow in this chapter	REFERENCE_ARCHITECTURE
registered module or catalog claim	DOCUMENTATION_ONLY_CLAIM until source inspection
uninspected integration	REQUIRES_REPOSITORY_REVIEW
behavior proven in another repository-grounded chapter	retain that chapter's evidence and classification
contradictory ownership or path claims	CONFLICT_REQUIRES_CANONICALIZATION

No statement upgrades implementation status merely because a file, directory, module name, frontend contract, database entry, plan, or report exists.

15. Worked Conformance Scenario

An authorized principal initiates a Command against an exact subject version. The service validates identity, tenant, mandate, Policy, Evidence, and idempotency. It records the state transition and outbox entry atomically. If an external dependency participates, timeout produces Outcome Unknown; reconciliation queries the source or waits for a verified callback instead of repeating the action blindly.

Portals show source, freshness, version, required action, and uncertainty. AI may extract, compare, summarize, detect anomalies, and recommend through typed Tools, but cannot own the subject or fabricate success. Later correction appends new facts, identifies affected projections and workflows, and preserves the historical Decision context.

16. Conformance Checklist

- ☐ stable identity and proposition ownership are explicit;
- ☐ authority and mandate are evaluated at action time;
- ☐ state, Recommendation, Decision, Intent, execution, and outcome remain distinct;
- ☐ Commands are typed, authorized, idempotent, and version guarded;
- ☐ Events describe completed facts;
- ☐ Evidence includes provenance, time, integrity, qualification, and correction;
- ☐ external outcomes retain their authority source;
- ☐ Portals show freshness and uncertainty;
- ☐ AI is bounded by typed Tools, Policy, approval, and outcome observation;
- ☐ security, tenant, facility, channel, and jurisdiction boundaries are enforced;
- ☐ Outcome Unknown and reconciliation exist where needed;
- ☐ corrections preserve history;
- ☐ tests prove invariants and recovery;
- ☐ implementation status is not upgraded without executable evidence.

17. Status Vocabulary

Status	Meaning
VERIFIED_IMPLEMENTED	Executable source and relevant behavior were inspected.
IMPLEMENTED_WITH_GAP	Executable behavior exists with a material governance or completeness gap.
PARTIAL_IMPLEMENTATION	Only part of the target capability is evidenced.
REFERENCE_ARCHITECTURE	Normative target behavior; no claim that it exists today.
PLANNED_NOT_IMPLEMENTED	Explicitly planned but not evidenced as executable.
DOCUMENTATION_ONLY_CLAIM	Documentation or client contract claims behavior without verified backend evidence.
CONFLICT_REQUIRES_CANONICALIZATION	Sources disagree and no final owner has been established.
REQUIRES_REPOSITORY_REVIEW	Necessary executable evidence has not yet been inspected.

Database Users and Least Privilege

Metadata	Value
Volume	10
Chapter ID	V10-25
Subject	DatabaseUsersandLeastPrivilege
Status	Generated First-Draft Architecture Skeleton - Domain-Specific Expansion Required
Content maturity	TEMPLATE_DERIVED_REFERENCE_SKELETON unless repository citations prove otherwise
Default implementation classification	REFERENCE_ARCHITECTURE / REQUIRES_REPOSITORY_REVIEW
Accountable owner	Data Architecture Council
Evidence rule	No implementation claim without inspected executable evidence
Repository	chinair88-prog/allinb2c

1. Executive Orientation

This chapter defines how GTOS shall **define enterprise and Domain data ownership, semantics, persistence, quality, lineage, correction, protection, operations, and verification for Database Users and Least Privilege.**

The specification separates reality, observations, knowledge, Decisions, Intent, execution, and outcomes. A component name, database row, UI label, AI answer, or workflow state does not acquire authority merely because it is visible or technically convenient.

Reality or external outcome

- ≠ Observation
- ≠ Claim
- ≠ derived Perspective
- ≠ Prediction
- ≠ Recommendation
- ≠ authorized Decision
- ≠ execution Intent

1.1 Accountable ownership

The accountable owner is **Data Architecture Council**. Supporting applications, shared infrastructure, integration adapters, analytics, and AI coordinate or present the capability but shall not silently own its authoritative propositions.

1.2 Scope

- create and identify the subject
- validate evidence and policy
- make authorized Decisions
- execute typed Commands
- publish completed-fact Events
- query current and historical perspectives
- correct and reconcile outcomes

1.3 Non-goals

- treating a registered Gradle module, catalog record, README, frontend route, migration plan, or generated report as proof of end-to-end implementation;
- direct cross-Domain database writes;

- generic status values that combine lifecycle, approval, provider, document, payment, dispute, and correction state;
- silent replacement of history;
- AI-created authority or fabricated external outcomes.

2. Ubiquitous Language and Subject Model

Subject	Required interpretation
DatabaseUsersandLeastPrivilege	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
DatabaseUsersandLeastPrivilegeVersion	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
DatabaseUsersandLeastPrivilegeDecision	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
DatabaseUsersandLeastPrivilegeEvidenceSet	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
DatabaseUsersandLeastPrivilegeException	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit

Reference aggregate:

```
DatabaseUsersandLeastPrivilege {
  subjectId
  tenantId
  organizationContext
  sourceIdentityRefs[]
  sourceVersionRefs[]
  lifecycleState
  relationshipRefs[]
  decisionRefs[]
  intentRefs[]
  evidenceSetRef
  knowledgeCutoff
  policyRefs[]
  authorityRef
  jurisdictionRefs[]
  createdAt
  effectiveAt
  updatedAt
  version
  correctionOf?
  correlationId
}
```

2.1 Core invariants

- Stable identity precedes relationship, state, and cross-system correlation.
- Effective time and recorded time remain distinct where business truth changes over time.
- Recommendation, Decision, Intent, execution, and outcome are separate concepts.
- No Portal, report, search index, cache, workflow, or AI Agent becomes a shadow owner of Domain truth.
- Binding submitted, approved, issued, accepted, or signed versions are immutable.
- Corrections append governed facts and preserve the original record.
- External authority and provider outcomes retain their source and qualification.
- Tenant and represented-principal context come from trusted authentication and mandate evaluation.
- Retried writes are idempotent and concurrency guarded.
- Outcome Unknown is explicit and reconciled before risky duplicate execution.

2.2 Required standards

- stable identifiers
- typed contracts
- immutable binding versions
- transactional outbox
- idempotent Commands
- explicit Outcome Unknown
- correction lineage
- machine-verifiable conformance

3. Actors, Roles, and Authority

A conformant design distinguishes:

- the natural person acting;
- the service or AI identity invoking a Tool;
- the organization or principal represented;
- the role or relationship granting context;
- the mandate or delegation granting action authority;
- the Policy and jurisdiction limiting the action;
- the reviewer or approver responsible for a binding Decision.

Authority is evaluated at action time. Authentication alone does not grant business authority. Cached permissions and browser-supplied tenant headers are insufficient for high-impact actions.

4. Lifecycle and State Model

State	Semantics
PROPOSED	Distinct governed condition with entry, exit, timeout, correction and authority semantics
VALIDATION_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
APPROVAL_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
ACTIVE	Distinct governed condition with entry, exit, timeout, correction and authority semantics
SUSPENDED	Distinct governed condition with entry, exit, timeout, correction and authority semantics
CORRECTION_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
RETIRED	Distinct governed condition with entry, exit, timeout, correction and authority semantics

```
stateDiagram-v2
    [*] --> PROPOSED
    PROPOSED --> VALIDATION_PENDING
    VALIDATION_PENDING --> APPROVAL_PENDING
    APPROVAL_PENDING --> ACTIVE
    ACTIVE --> SUSPENDED
    SUSPENDED --> CORRECTION_PENDING
    CORRECTION_PENDING --> RETIRED
```

The diagram is a primary reading path. Real implementations may add parallel review, amendment, appeal, timeout, suspension, partial completion, reconciliation, correction, and terminal failure without collapsing their meaning.

4.1 Transition rules

- transition through a typed, authorized Command;
 - require exact expected version where concurrent change matters;
 - record actor, represented principal, mandate, Policy, reason, and Evidence;
 - use an idempotency key for retried writes;
 - publish a completed-fact Event after durable state change;
 - expose Outcome Unknown for unresolved external completion;
 - preserve original and corrected states in historical reconstruction.
-

5. Decision, Evidence, and Knowledge Cutoff

Reference Decision:

```
Decision {
    decisionId
    decisionType
    subjectRef
}
```

```

sourceVersionRefs[]
knowledgeCutoff
evidenceSetRef
policyVersionRefs[]
recommendationRefs[]
principalId
representedOrganizationId
mandateRef
rationale
outcome
decidedAt
effectiveAt
correctionOf?
}

```

Evidence records identify issuer or observer, subject, version, observed time, received time, effective interval, integrity, confidence, qualification, contradiction, correction, retention, residency, and access class. A recommendation may assist the Decision but cannot promote itself to the Decision.

6. Commands, Events, Queries, and Contracts

Reference Commands:

- CreateDatabaseUsersandLeastPrivilege
- ValidateDatabaseUsersandLeastPrivilege
- ApproveDatabaseUsersandLeastPrivilege
- ActivateDatabaseUsersandLeastPrivilege
- SuspendDatabaseUsersandLeastPrivilege
- CorrectDatabaseUsersandLeastPrivilege
- RetireDatabaseUsersandLeastPrivilege

Reference Events:

- DatabaseUsersandLeastPrivilegeCreated
- DatabaseUsersandLeastPrivilegeValidated
- DatabaseUsersandLeastPrivilegeApproved
- DatabaseUsersandLeastPrivilegeActivated
- DatabaseUsersandLeastPrivilegeSuspended
- DatabaseUsersandLeastPrivilegeCorrected
- DatabaseUsersandLeastPrivilegeRetired

Reference queries:

- GetDatabaseUsersandLeastPrivilege
- ListDatabaseUsersandLeastPrivilegeRecords
- GetDatabaseUsersandLeastPrivilegeTimeline
- GetDatabaseUsersandLeastPrivilegeEvidence
- GetDatabaseUsersandLeastPrivilegeDecisionPackage
- GetDatabaseUsersandLeastPrivilegeExceptions
- GetDatabaseUsersandLeastPrivilegeAuditTrail

Reference API surface:

```

POST /api/v1/databaseusersandleastprivilege/commands
GET  /api/v1/databaseusersandleastprivilege/{id}
GET  /api/v1/databaseusersandleastprivilege/{id}/timeline
GET  /api/v1/databaseusersandleastprivilege/{id}/evidence
GET  /api/v1/databaseusersandleastprivilege/{id}/decisions
POST /api/v1/databaseusersandleastprivilege/{id}/exceptions
POST /api/v1/databaseusersandleastprivilege/{id}/corrections

```

These endpoints are reference contracts, not claims that exact routes exist.

7. Workflow, Failure, and Compensation

flowchart TD

```

A[Validate identity version and authority] --> B[Collect required evidence]
B --> C{Policy and evidence gates pass?}
C -- No --> D[Open exception or request correction]
D --> B
C -- Yes --> E[Record Decision or execution Intent]
E --> F[Execute Domain command or external request]

```

```

F --> G{Outcome known?}
G -- Success --> H[Record fact and publish Event]
G -- Failure --> I[Record failure and compensation options]
G -- Unknown --> J[Enter Outcome Unknown]
J --> K[Query source or wait for verified callback]
K --> G
H --> L[Update projections and downstream workflow]

```

Representative failure modes:

- stale or wrong source version
- duplicate submission or replay
- external timeout after possible success
- contradictory source observations
- lost Event after committed state
- approval or mandate revoked during execution
- downstream action before a mandatory gate
- partial completion across multiple subjects
- evidence later retracted or corrected
- tenant, facility, channel or jurisdiction mismatch

Recovery shall query the authoritative source before duplicate execution, preserve partial success, use compensating Intent rather than destructive rewrite, publish corrections, quarantine high-impact ambiguity, and record affected downstream subjects.

8. Data and Persistence Architecture

Reference table families:

Table family	Purpose
databaseusersandleastprivilege_aggregate	authoritative subject
databaseusersandleastprivilege_version	immutable submitted, approved, issued, or signed versions
databaseusersandleastprivilege_relationship	governed relationships
databaseusersandleastprivilege_decision	binding Decisions and rationale
databaseusersandleastprivilege_evidence_link	provenance and evidence relationships
databaseusersandleastprivilege_event_outbox	transactional Event publication
databaseusersandleastprivilege_idempotency	duplicate-write protection
databaseusersandleastprivilege_exception	exception, claim, dispute, or hold
databaseusersandleastprivilege_correction	correction and supersession lineage
databaseusersandleastprivilege_audit	privileged action evidence

The owning service controls schema and migration. Cross-Domain references use stable identifiers or contracts. Analytics, reports, caches, search, and vector indexes remain projections. Backup, restore, retention, legal hold, privacy, residency, and decommissioning requirements are explicit.

9. Portal and Experience Requirements

- Primary participant Portal: initiate, review exact versions, supply evidence, and respond to required actions.
- Operator Portal: inspect exceptions, correlation, source responses, retries, compensation, and downstream impact.
- Executive or oversight view: consume governed metrics without becoming a source of Domain truth.

Every Portal displays exact version, owner, state, freshness, required action, and uncertainty. It supports loading, empty, partial, stale, denied, error, timeout, and Outcome Unknown states; accessibility; locale, currency, unit, date, and RTL/LTR correctness; evidence audit; safe retry; and correction history.

10. AI Participation and Tool Governance

Permitted AI tasks:

- assist schema and lineage analysis
- detect data-quality anomalies
- summarize migration and reconciliation risk
- never mutate authoritative data outside typed Domain Tools

AI shall not own Domain records, grant itself authority, suppress contradictions, silently edit approved versions, fabricate external outcomes, or execute high-impact actions outside typed Tools, policy, approval, idempotency, sandbox, and outcome observation.

Every Agent Run records source context, prompt/model/tool versions, structured output, confidence and limitations, Policy result, approval, Tool calls, provider responses, final outcome, and recovery actions.

11. Security, Privacy, and Trust Boundaries

- least privilege and separation of duties
- tenant, organization, facility, channel and jurisdiction isolation
- integrity protection for binding evidence
- verified external callbacks and anti-replay controls

Additional controls include secret isolation, callback signature verification, audit of privileged action, emergency-access review, sensitive-data minimization, purpose limitation, and threat models for impersonation, tampering, replay, confused deputy, fraud, prompt injection, and exfiltration.

12. Reliability, SLOs, and Operations

SLI	Reference objective
authoritative read availability	99.9% monthly unless a stricter tier applies
acknowledged Command durability	no acknowledged write may be silently lost
Event publication	99.9% within five minutes after committed fact
duplicate prevention	100% for same tenant, Command class, and idempotency key
binding Decision audit completeness	100%
state-state detection	bounded by business deadline

Dashboards and runbooks cover backlog, error budget, dependency failure, retry budget, DLQ, workflow stall, provider outage, reconciliation, capacity, backup/restore, release/rollback, incident command, and post-incident correction.

13. Testing and Continuous Conformance

- aggregate invariant and transition tests
- authorization, delegation, separation-of-duties and tenant-isolation tests
- idempotency and optimistic-concurrency tests
- database migration, key, constraint and rollback tests
- contract, outbox/inbox, duplicate, replay and DLQ tests
- workflow restart, timeout and compensation tests
- Portal E2E tests for success, partial, stale, denied, error and Outcome Unknown states
- accessibility, locale, currency, unit and RTL/LTR tests
- security adversarial, fraud and data-exfiltration tests
- AI evaluation, prompt-injection, policy-bypass and unauthorized-tool tests
- historical reconstruction, correction and audit-completeness tests
- performance, capacity, recovery and operational exercise tests

Release evidence links each invariant and control to the test, environment, input data, result, owner, and artifact proving it. Generated test counts never replace semantic coverage.

14. Repository Review Boundary

- This chapter is a repository-review specification. Exact executable implementation has not been inspected for every target behavior.
- Registered modules, catalog entries, README claims, frontend contracts, or plans are not proof of complete implementation.

Area	Classification
target aggregate, API, tables, state and workflow in this chapter	REFERENCE_ARCHITECTURE
registered module or catalog claim	DOCUMENTATION_ONLY_CLAIM until source inspection
uninspected integration	REQUIRES_REPOSITORY_REVIEW
behavior proven in another repository-grounded chapter	retain that chapter's evidence and classification
contradictory ownership or path claims	CONFLICT_REQUIRES_CANONICALIZATION

No statement upgrades implementation status merely because a file, directory, module name, frontend contract, database entry, plan, or report exists.

15. Worked Conformance Scenario

An authorized principal initiates a Command against an exact subject version. The service validates identity, tenant, mandate, Policy, Evidence, and idempotency. It records the state transition and outbox entry atomically. If an external dependency participates, timeout produces Outcome Unknown; reconciliation queries the source or waits for a verified callback instead of repeating the action blindly.

Portals show source, freshness, version, required action, and uncertainty. AI may extract, compare, summarize, detect anomalies, and recommend through typed Tools, but cannot own the subject or fabricate success. Later correction appends new facts, identifies affected projections and workflows, and preserves the historical Decision context.

16. Conformance Checklist

- ☐ stable identity and proposition ownership are explicit;
- ☐ authority and mandate are evaluated at action time;
- ☐ state, Recommendation, Decision, Intent, execution, and outcome remain distinct;
- ☐ Commands are typed, authorized, idempotent, and version guarded;
- ☐ Events describe completed facts;
- ☐ Evidence includes provenance, time, integrity, qualification, and correction;
- ☐ external outcomes retain their authority source;
- ☐ Portals show freshness and uncertainty;
- ☐ AI is bounded by typed Tools, Policy, approval, and outcome observation;
- ☐ security, tenant, facility, channel, and jurisdiction boundaries are enforced;
- ☐ Outcome Unknown and reconciliation exist where needed;
- ☐ corrections preserve history;
- ☐ tests prove invariants and recovery;
- ☐ implementation status is not upgraded without executable evidence.

17. Status Vocabulary

Status	Meaning
VERIFIED_IMPLEMENTED	Executable source and relevant behavior were inspected.
IMPLEMENTED_WITH_GAP	Executable behavior exists with a material governance or completeness gap.
PARTIAL_IMPLEMENTATION	Only part of the target capability is evidenced.
REFERENCE_ARCHITECTURE	Normative target behavior; no claim that it exists today.
PLANNED_NOT_IMPLEMENTED	Explicitly planned but not evidenced as executable.
DOCUMENTATION_ONLY_CLAIM	Documentation or client contract claims behavior without verified backend evidence.
CONFLICT_REQUIRES_CANONICALIZATION	Sources disagree and no final owner has been established.
REQUIRES_REPOSITORY_REVIEW	Necessary executable evidence has not yet been inspected.

Row-Level Security

Metadata	Value
Volume	10
Chapter ID	V10-26
Subject	Row-LevelSecurity
Status	Generated First-Draft Architecture Skeleton - Domain-Specific Expansion Required
Content maturity	TEMPLATE_DERIVED_REFERENCE_SKELETON unless repository citations prove otherwise
Default implementation classification	REFERENCE_ARCHITECTURE / REQUIRES_REPOSITORY_REVIEW
Accountable owner	Data Architecture Council
Evidence rule	No implementation claim without inspected executable evidence
Repository	chinair88-prog/allinb2c

1. Executive Orientation

This chapter defines how GTOS shall **define enterprise and Domain data ownership, semantics, persistence, quality, lineage, correction, protection, operations, and verification for Row-Level Security**.

The specification separates reality, observations, knowledge, Decisions, Intent, execution, and outcomes. A component name, database row, UI label, AI answer, or workflow state does not acquire authority merely because it is visible or technically convenient.

Reality or external outcome

- ≠ Observation
- ≠ Claim
- ≠ derived Perspective
- ≠ Prediction
- ≠ Recommendation
- ≠ authorized Decision
- ≠ execution Intent

1.1 Accountable ownership

The accountable owner is **Data Architecture Council**. Supporting applications, shared infrastructure, integration adapters, analytics, and AI coordinate or present the capability but shall not silently own its authoritative propositions.

1.2 Scope

- create and identify the subject
- validate evidence and policy
- make authorized Decisions
- execute typed Commands
- publish completed-fact Events
- query current and historical perspectives
- correct and reconcile outcomes

1.3 Non-goals

- treating a registered Gradle module, catalog record, README, frontend route, migration plan, or generated report as proof of end-to-end implementation;
- direct cross-Domain database writes;

- generic status values that combine lifecycle, approval, provider, document, payment, dispute, and correction state;
- silent replacement of history;
- AI-created authority or fabricated external outcomes.

2. Ubiquitous Language and Subject Model

Subject	Required interpretation
Row-LevelSecurity	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
Row-LevelSecurityVersion	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
Row-LevelSecurityDecision	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
Row-LevelSecurityEvidenceSet	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
Row-LevelSecurityException	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit

Reference aggregate:

```
Row-LevelSecurity {
  subjectId
  tenantId
  organizationContext
  sourceIdentityRefs[]
  sourceVersionRefs[]
  lifecycleState
  relationshipRefs[]
  decisionRefs[]
  intentRefs[]
  evidenceSetRef
  knowledgeCutoff
  policyRefs[]
  authorityRef
  jurisdictionRefs[]
  createdAt
  effectiveAt
  updatedAt
  version
  correctionOf?
  correlationId
}
```

2.1 Core invariants

- Stable identity precedes relationship, state, and cross-system correlation.
- Effective time and recorded time remain distinct where business truth changes over time.
- Recommendation, Decision, Intent, execution, and outcome are separate concepts.
- No Portal, report, search index, cache, workflow, or AI Agent becomes a shadow owner of Domain truth.
- Binding submitted, approved, issued, accepted, or signed versions are immutable.
- Corrections append governed facts and preserve the original record.
- External authority and provider outcomes retain their source and qualification.
- Tenant and represented-principal context come from trusted authentication and mandate evaluation.
- Retried writes are idempotent and concurrency guarded.
- Outcome Unknown is explicit and reconciled before risky duplicate execution.

2.2 Required standards

- stable identifiers
- typed contracts
- immutable binding versions
- transactional outbox
- idempotent Commands
- explicit Outcome Unknown
- correction lineage
- machine-verifiable conformance

3. Actors, Roles, and Authority

A conformant design distinguishes:

- the natural person acting;
- the service or AI identity invoking a Tool;
- the organization or principal represented;
- the role or relationship granting context;
- the mandate or delegation granting action authority;
- the Policy and jurisdiction limiting the action;
- the reviewer or approver responsible for a binding Decision.

Authority is evaluated at action time. Authentication alone does not grant business authority. Cached permissions and browser-supplied tenant headers are insufficient for high-impact actions.

4. Lifecycle and State Model

State	Semantics
PROPOSED	Distinct governed condition with entry, exit, timeout, correction and authority semantics
VALIDATION_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
APPROVAL_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
ACTIVE	Distinct governed condition with entry, exit, timeout, correction and authority semantics
SUSPENDED	Distinct governed condition with entry, exit, timeout, correction and authority semantics
CORRECTION_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
RETIRED	Distinct governed condition with entry, exit, timeout, correction and authority semantics

```
stateDiagram-v2
    [*] --> PROPOSED
    PROPOSED --> VALIDATION_PENDING
    VALIDATION_PENDING --> APPROVAL_PENDING
    APPROVAL_PENDING --> ACTIVE
    ACTIVE --> SUSPENDED
    SUSPENDED --> CORRECTION_PENDING
    CORRECTION_PENDING --> RETIRED
```

The diagram is a primary reading path. Real implementations may add parallel review, amendment, appeal, timeout, suspension, partial completion, reconciliation, correction, and terminal failure without collapsing their meaning.

4.1 Transition rules

- transition through a typed, authorized Command;
 - require exact expected version where concurrent change matters;
 - record actor, represented principal, mandate, Policy, reason, and Evidence;
 - use an idempotency key for retried writes;
 - publish a completed-fact Event after durable state change;
 - expose Outcome Unknown for unresolved external completion;
 - preserve original and corrected states in historical reconstruction.
-

5. Decision, Evidence, and Knowledge Cutoff

Reference Decision:

```
Decision {
    decisionId
    decisionType
    subjectRef
}
```

```

sourceVersionRefs[]
knowledgeCutoff
evidenceSetRef
policyVersionRefs[]
recommendationRefs[]
principalId
representedOrganizationId
mandateRef
rationale
outcome
decidedAt
effectiveAt
correctionOf?
}

```

Evidence records identify issuer or observer, subject, version, observed time, received time, effective interval, integrity, confidence, qualification, contradiction, correction, retention, residency, and access class. A recommendation may assist the Decision but cannot promote itself to the Decision.

6. Commands, Events, Queries, and Contracts

Reference Commands:

- CreateRow-LevelSecurity
- ValidateRow-LevelSecurity
- ApproveRow-LevelSecurity
- ActivateRow-LevelSecurity
- SuspendRow-LevelSecurity
- CorrectRow-LevelSecurity
- RetireRow-LevelSecurity

Reference Events:

- Row-LevelSecurityCreated
- Row-LevelSecurityValidated
- Row-LevelSecurityApproved
- Row-LevelSecurityActivated
- Row-LevelSecuritySuspended
- Row-LevelSecurityCorrected
- Row-LevelSecurityRetired

Reference queries:

- GetRow-LevelSecurity
- ListRow-LevelSecurityRecords
- GetRow-LevelSecurityTimeline
- GetRow-LevelSecurityEvidence
- GetRow-LevelSecurityDecisionPackage
- GetRow-LevelSecurityExceptions
- GetRow-LevelSecurityAuditTrail

Reference API surface:

```

POST /api/v1/row_levelsecurity/commands
GET  /api/v1/row_levelsecurity/{id}
GET  /api/v1/row_levelsecurity/{id}/timeline
GET  /api/v1/row_levelsecurity/{id}/evidence
GET  /api/v1/row_levelsecurity/{id}/decisions
POST /api/v1/row_levelsecurity/{id}/exceptions
POST /api/v1/row_levelsecurity/{id}/corrections

```

These endpoints are reference contracts, not claims that exact routes exist.

7. Workflow, Failure, and Compensation

flowchart TD

```

A[Validate identity version and authority] --> B[Collect required evidence]
B --> C{Policy and evidence gates pass?}
C -- No --> D[Open exception or request correction]
D --> B
C -- Yes --> E[Record Decision or execution Intent]
E --> F[Execute Domain command or external request]

```

```

F --> G{Outcome known?}
G -- Success --> H[Record fact and publish Event]
G -- Failure --> I[Record failure and compensation options]
G -- Unknown --> J[Enter Outcome Unknown]
J --> K[Query source or wait for verified callback]
K --> G
H --> L[Update projections and downstream workflow]

```

Representative failure modes:

- stale or wrong source version
- duplicate submission or replay
- external timeout after possible success
- contradictory source observations
- lost Event after committed state
- approval or mandate revoked during execution
- downstream action before a mandatory gate
- partial completion across multiple subjects
- evidence later retracted or corrected
- tenant, facility, channel or jurisdiction mismatch

Recovery shall query the authoritative source before duplicate execution, preserve partial success, use compensating Intent rather than destructive rewrite, publish corrections, quarantine high-impact ambiguity, and record affected downstream subjects.

8. Data and Persistence Architecture

Reference table families:

Table family	Purpose
row_levelsecurity_aggregate	authoritative subject
row_levelsecurity_version	immutable submitted, approved, issued, or signed versions
row_levelsecurity_relationship	governed relationships
row_levelsecurity_decision	binding Decisions and rationale
row_levelsecurity_evidence_link	provenance and evidence relationships
row_levelsecurity_event_outbox	transactional Event publication
row_levelsecurity_idempotency	duplicate-write protection
row_levelsecurity_exception	exception, claim, dispute, or hold
row_levelsecurity_correction	correction and supersession lineage
row_levelsecurity_audit	privileged action evidence

The owning service controls schema and migration. Cross-Domain references use stable identifiers or contracts. Analytics, reports, caches, search, and vector indexes remain projections. Backup, restore, retention, legal hold, privacy, residency, and decommissioning requirements are explicit.

9. Portal and Experience Requirements

- Primary participant Portal: initiate, review exact versions, supply evidence, and respond to required actions.
- Operator Portal: inspect exceptions, correlation, source responses, retries, compensation, and downstream impact.
- Executive or oversight view: consume governed metrics without becoming a source of Domain truth.

Every Portal displays exact version, owner, state, freshness, required action, and uncertainty. It supports loading, empty, partial, stale, denied, error, timeout, and Outcome Unknown states; accessibility; locale, currency, unit, date, and RTL/LTR correctness; evidence audit; safe retry; and correction history.

10. AI Participation and Tool Governance

Permitted AI tasks:

- assist schema and lineage analysis
- detect data-quality anomalies
- summarize migration and reconciliation risk
- never mutate authoritative data outside typed Domain Tools

AI shall not own Domain records, grant itself authority, suppress contradictions, silently edit approved versions, fabricate external outcomes, or execute high-impact actions outside typed Tools, policy, approval, idempotency, sandbox, and outcome observation.

Every Agent Run records source context, prompt/model/tool versions, structured output, confidence and limitations, Policy result, approval, Tool calls, provider responses, final outcome, and recovery actions.

11. Security, Privacy, and Trust Boundaries

- least privilege and separation of duties
- tenant, organization, facility, channel and jurisdiction isolation
- integrity protection for binding evidence
- verified external callbacks and anti-replay controls

Additional controls include secret isolation, callback signature verification, audit of privileged action, emergency-access review, sensitive-data minimization, purpose limitation, and threat models for impersonation, tampering, replay, confused deputy, fraud, prompt injection, and exfiltration.

12. Reliability, SLOs, and Operations

SLI	Reference objective
authoritative read availability	99.9% monthly unless a stricter tier applies
acknowledged Command durability	no acknowledged write may be silently lost
Event publication	99.9% within five minutes after committed fact
duplicate prevention	100% for same tenant, Command class, and idempotency key
binding Decision audit completeness	100%
stale-state detection	bounded by business deadline

Dashboards and runbooks cover backlog, error budget, dependency failure, retry budget, DLQ, workflow stall, provider outage, reconciliation, capacity, backup/restore, release/rollback, incident command, and post-incident correction.

13. Testing and Continuous Conformance

- aggregate invariant and transition tests
- authorization, delegation, separation-of-duties and tenant-isolation tests
- idempotency and optimistic-concurrency tests
- database migration, key, constraint and rollback tests
- contract, outbox/inbox, duplicate, replay and DLQ tests
- workflow restart, timeout and compensation tests
- Portal E2E tests for success, partial, stale, denied, error and Outcome Unknown states
- accessibility, locale, currency, unit and RTL/LTR tests
- security adversarial, fraud and data-exfiltration tests
- AI evaluation, prompt-injection, policy-bypass and unauthorized-tool tests
- historical reconstruction, correction and audit-completeness tests
- performance, capacity, recovery and operational exercise tests

Release evidence links each invariant and control to the test, environment, input data, result, owner, and artifact proving it. Generated test counts never replace semantic coverage.

14. Repository Review Boundary

- This chapter is a repository-review specification. Exact executable implementation has not been inspected for every target behavior.
- Registered modules, catalog entries, README claims, frontend contracts, or plans are not proof of complete implementation.

Area	Classification
target aggregate, API, tables, state and workflow in this chapter	REFERENCE_ARCHITECTURE
registered module or catalog claim	DOCUMENTATION_ONLY_CLAIM until source inspection
uninspected integration	REQUIRES_REPOSITORY_REVIEW
behavior proven in another repository-grounded chapter	retain that chapter's evidence and classification
contradictory ownership or path claims	CONFLICT_REQUIRES_CANONICALIZATION

No statement upgrades implementation status merely because a file, directory, module name, frontend contract, database entry, plan, or report exists.

15. Worked Conformance Scenario

An authorized principal initiates a Command against an exact subject version. The service validates identity, tenant, mandate, Policy, Evidence, and idempotency. It records the state transition and outbox entry atomically. If an external dependency participates, timeout produces Outcome Unknown; reconciliation queries the source or waits for a verified callback instead of repeating the action blindly.

Portals show source, freshness, version, required action, and uncertainty. AI may extract, compare, summarize, detect anomalies, and recommend through typed Tools, but cannot own the subject or fabricate success. Later correction appends new facts, identifies affected projections and workflows, and preserves the historical Decision context.

16. Conformance Checklist

- ☐ stable identity and proposition ownership are explicit;
- ☐ authority and mandate are evaluated at action time;
- ☐ state, Recommendation, Decision, Intent, execution, and outcome remain distinct;
- ☐ Commands are typed, authorized, idempotent, and version guarded;
- ☐ Events describe completed facts;
- ☐ Evidence includes provenance, time, integrity, qualification, and correction;
- ☐ external outcomes retain their authority source;
- ☐ Portals show freshness and uncertainty;
- ☐ AI is bounded by typed Tools, Policy, approval, and outcome observation;
- ☐ security, tenant, facility, channel, and jurisdiction boundaries are enforced;
- ☐ Outcome Unknown and reconciliation exist where needed;
- ☐ corrections preserve history;
- ☐ tests prove invariants and recovery;
- ☐ implementation status is not upgraded without executable evidence.

17. Status Vocabulary

Status	Meaning
VERIFIED_IMPLEMENTED	Executable source and relevant behavior were inspected.
IMPLEMENTED_WITH_GAP	Executable behavior exists with a material governance or completeness gap.
PARTIAL_IMPLEMENTATION	Only part of the target capability is evidenced.
REFERENCE_ARCHITECTURE	Normative target behavior; no claim that it exists today.
PLANNED_NOT_IMPLEMENTED	Explicitly planned but not evidenced as executable.
DOCUMENTATION_ONLY_CLAIM	Documentation or client contract claims behavior without verified backend evidence.
CONFLICT_REQUIRES_CANONICALIZATION	Sources disagree and no final owner has been established.
REQUIRES_REPOSITORY_REVIEW	Necessary executable evidence has not yet been inspected.

Encryption and Key Management

Metadata	Value
Volume	10
Chapter ID	V10-27
Subject	EncryptionandKeyManagement
Status	Generated First-Draft Architecture Skeleton - Domain-Specific Expansion Required
Content maturity	TEMPLATE_DERIVED_REFERENCE_SKELETON unless repository citations prove otherwise
Default implementation classification	REFERENCE_ARCHITECTURE / REQUIRES_REPOSITORY_REVIEW
Accountable owner	Data Architecture Council
Evidence rule	No implementation claim without inspected executable evidence
Repository	chinair88-prog/allinb2c

1. Executive Orientation

This chapter defines how GTOS shall **define enterprise and Domain data ownership, semantics, persistence, quality, lineage, correction, protection, operations, and verification for Encryption and Key Management**.

The specification separates reality, observations, knowledge, Decisions, Intent, execution, and outcomes. A component name, database row, UI label, AI answer, or workflow state does not acquire authority merely because it is visible or technically convenient.

Reality or external outcome

- ≠ Observation
- ≠ Claim
- ≠ derived Perspective
- ≠ Prediction
- ≠ Recommendation
- ≠ authorized Decision
- ≠ execution Intent

1.1 Accountable ownership

The accountable owner is **Data Architecture Council**. Supporting applications, shared infrastructure, integration adapters, analytics, and AI coordinate or present the capability but shall not silently own its authoritative propositions.

1.2 Scope

- create and identify the subject
- validate evidence and policy
- make authorized Decisions
- execute typed Commands
- publish completed-fact Events
- query current and historical perspectives
- correct and reconcile outcomes

1.3 Non-goals

- treating a registered Gradle module, catalog record, README, frontend route, migration plan, or generated report as proof of end-to-end implementation;
- direct cross-Domain database writes;

- generic status values that combine lifecycle, approval, provider, document, payment, dispute, and correction state;
- silent replacement of history;
- AI-created authority or fabricated external outcomes.

2. Ubiquitous Language and Subject Model

Subject	Required interpretation
EncryptionandKeyManagement	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
EncryptionandKeyManagementVersion	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
EncryptionandKeyManagementDecision	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
EncryptionandKeyManagementEvidenceSet	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
EncryptionandKeyManagementException	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit

Reference aggregate:

```
EncryptionandKeyManagement {
  subjectId
  tenantId
  organizationContext
  sourceIdentityRefs[]
  sourceVersionRefs[]
  lifecycleState
  relationshipRefs[]
  decisionRefs[]
  intentRefs[]
  evidenceSetRef
  knowledgeCutoff
  policyRefs[]
  authorityRef
  jurisdictionRefs[]
  createdAt
  effectiveAt
  updatedAt
  version
  correctionOf?
  correlationId
}
```

2.1 Core invariants

- Stable identity precedes relationship, state, and cross-system correlation.
- Effective time and recorded time remain distinct where business truth changes over time.
- Recommendation, Decision, Intent, execution, and outcome are separate concepts.
- No Portal, report, search index, cache, workflow, or AI Agent becomes a shadow owner of Domain truth.
- Binding submitted, approved, issued, accepted, or signed versions are immutable.
- Corrections append governed facts and preserve the original record.
- External authority and provider outcomes retain their source and qualification.
- Tenant and represented-principal context come from trusted authentication and mandate evaluation.
- Retried writes are idempotent and concurrency guarded.
- Outcome Unknown is explicit and reconciled before risky duplicate execution.

2.2 Required standards

- stable identifiers
- typed contracts
- immutable binding versions
- transactional outbox
- idempotent Commands
- explicit Outcome Unknown
- correction lineage
- machine-verifiable conformance

3. Actors, Roles, and Authority

A conformant design distinguishes:

- the natural person acting;
- the service or AI identity invoking a Tool;
- the organization or principal represented;
- the role or relationship granting context;
- the mandate or delegation granting action authority;
- the Policy and jurisdiction limiting the action;
- the reviewer or approver responsible for a binding Decision.

Authority is evaluated at action time. Authentication alone does not grant business authority. Cached permissions and browser-supplied tenant headers are insufficient for high-impact actions.

4. Lifecycle and State Model

State	Semantics
PROPOSED	Distinct governed condition with entry, exit, timeout, correction and authority semantics
VALIDATION_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
APPROVAL_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
ACTIVE	Distinct governed condition with entry, exit, timeout, correction and authority semantics
SUSPENDED	Distinct governed condition with entry, exit, timeout, correction and authority semantics
CORRECTION_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
RETIRED	Distinct governed condition with entry, exit, timeout, correction and authority semantics

```
stateDiagram-v2
    [*] --> PROPOSED
    PROPOSED --> VALIDATION_PENDING
    VALIDATION_PENDING --> APPROVAL_PENDING
    APPROVAL_PENDING --> ACTIVE
    ACTIVE --> SUSPENDED
    SUSPENDED --> CORRECTION_PENDING
    CORRECTION_PENDING --> RETIRED
```

The diagram is a primary reading path. Real implementations may add parallel review, amendment, appeal, timeout, suspension, partial completion, reconciliation, correction, and terminal failure without collapsing their meaning.

4.1 Transition rules

- transition through a typed, authorized Command;
 - require exact expected version where concurrent change matters;
 - record actor, represented principal, mandate, Policy, reason, and Evidence;
 - use an idempotency key for retried writes;
 - publish a completed-fact Event after durable state change;
 - expose Outcome Unknown for unresolved external completion;
 - preserve original and corrected states in historical reconstruction.
-

5. Decision, Evidence, and Knowledge Cutoff

Reference Decision:

```
Decision {
    decisionId
    decisionType
    subjectRef
}
```

```

sourceVersionRefs[]
knowledgeCutoff
evidenceSetRef
policyVersionRefs[]
recommendationRefs[]
principalId
representedOrganizationId
mandateRef
rationale
outcome
decidedAt
effectiveAt
correctionOf?
}

```

Evidence records identify issuer or observer, subject, version, observed time, received time, effective interval, integrity, confidence, qualification, contradiction, correction, retention, residency, and access class. A recommendation may assist the Decision but cannot promote itself to the Decision.

6. Commands, Events, Queries, and Contracts

Reference Commands:

- CreateEncryptionandKeyManagement
- ValidateEncryptionandKeyManagement
- ApproveEncryptionandKeyManagement
- ActivateEncryptionandKeyManagement
- SuspendEncryptionandKeyManagement
- CorrectEncryptionandKeyManagement
- RetireEncryptionandKeyManagement

Reference Events:

- EncryptionandKeyManagementCreated
- EncryptionandKeyManagementValidated
- EncryptionandKeyManagementApproved
- EncryptionandKeyManagementActivated
- EncryptionandKeyManagementSuspended
- EncryptionandKeyManagementCorrected
- EncryptionandKeyManagementRetired

Reference queries:

- GetEncryptionandKeyManagement
- ListEncryptionandKeyManagementRecords
- GetEncryptionandKeyManagementTimeline
- GetEncryptionandKeyManagementEvidence
- GetEncryptionandKeyManagementDecisionPackage
- GetEncryptionandKeyManagementExceptions
- GetEncryptionandKeyManagementAuditTrail

Reference API surface:

```

POST /api/v1/encryptionandkeymanagement/commands
GET  /api/v1/encryptionandkeymanagement/{id}
GET  /api/v1/encryptionandkeymanagement/{id}/timeline
GET  /api/v1/encryptionandkeymanagement/{id}/evidence
GET  /api/v1/encryptionandkeymanagement/{id}/decisions
POST /api/v1/encryptionandkeymanagement/{id}/exceptions
POST /api/v1/encryptionandkeymanagement/{id}/corrections

```

These endpoints are reference contracts, not claims that exact routes exist.

7. Workflow, Failure, and Compensation

flowchart TD

```

A[Validate identity version and authority] --> B[Collect required evidence]
B --> C{Policy and evidence gates pass?}
C -- No --> D[Open exception or request correction]
D --> B
C -- Yes --> E[Record Decision or execution Intent]
E --> F[Execute Domain command or external request]

```

```

F --> G{Outcome known?}
G -- Success --> H[Record fact and publish Event]
G -- Failure --> I[Record failure and compensation options]
G -- Unknown --> J[Enter Outcome Unknown]
J --> K[Query source or wait for verified callback]
K --> G
H --> L[Update projections and downstream workflow]

```

Representative failure modes:

- stale or wrong source version
- duplicate submission or replay
- external timeout after possible success
- contradictory source observations
- lost Event after committed state
- approval or mandate revoked during execution
- downstream action before a mandatory gate
- partial completion across multiple subjects
- evidence later retracted or corrected
- tenant, facility, channel or jurisdiction mismatch

Recovery shall query the authoritative source before duplicate execution, preserve partial success, use compensating Intent rather than destructive rewrite, publish corrections, quarantine high-impact ambiguity, and record affected downstream subjects.

8. Data and Persistence Architecture

Reference table families:

Table family	Purpose
encryptionandkeymanagement_aggregate	authoritative subject
encryptionandkeymanagement_version	immutable submitted, approved, issued, or signed versions
encryptionandkeymanagement_relationship	governed relationships
encryptionandkeymanagement_decision	binding Decisions and rationale
encryptionandkeymanagement_evidence_link	provenance and evidence relationships
encryptionandkeymanagement_event_outbox	transactional Event publication
encryptionandkeymanagement_idempotency	duplicate-write protection
encryptionandkeymanagement_exception	exception, claim, dispute, or hold
encryptionandkeymanagement_correction	correction and supersession lineage
encryptionandkeymanagement_audit	privileged action evidence

The owning service controls schema and migration. Cross-Domain references use stable identifiers or contracts. Analytics, reports, caches, search, and vector indexes remain projections. Backup, restore, retention, legal hold, privacy, residency, and decommissioning requirements are explicit.

9. Portal and Experience Requirements

- Primary participant Portal: initiate, review exact versions, supply evidence, and respond to required actions.
- Operator Portal: inspect exceptions, correlation, source responses, retries, compensation, and downstream impact.
- Executive or oversight view: consume governed metrics without becoming a source of Domain truth.

Every Portal displays exact version, owner, state, freshness, required action, and uncertainty. It supports loading, empty, partial, stale, denied, error, timeout, and Outcome Unknown states; accessibility; locale, currency, unit, date, and RTL/LTR correctness; evidence audit; safe retry; and correction history.

10. AI Participation and Tool Governance

Permitted AI tasks:

- assist schema and lineage analysis
- detect data-quality anomalies
- summarize migration and reconciliation risk
- never mutate authoritative data outside typed Domain Tools

AI shall not own Domain records, grant itself authority, suppress contradictions, silently edit approved versions, fabricate external outcomes, or execute high-impact actions outside typed Tools, policy, approval, idempotency, sandbox, and outcome observation.

Every Agent Run records source context, prompt/model/tool versions, structured output, confidence and limitations, Policy result, approval, Tool calls, provider responses, final outcome, and recovery actions.

11. Security, Privacy, and Trust Boundaries

- least privilege and separation of duties
- tenant, organization, facility, channel and jurisdiction isolation
- integrity protection for binding evidence
- verified external callbacks and anti-replay controls

Additional controls include secret isolation, callback signature verification, audit of privileged action, emergency-access review, sensitive-data minimization, purpose limitation, and threat models for impersonation, tampering, replay, confused deputy, fraud, prompt injection, and exfiltration.

12. Reliability, SLOs, and Operations

SLI	Reference objective
authoritative read availability	99.9% monthly unless a stricter tier applies
acknowledged Command durability	no acknowledged write may be silently lost
Event publication	99.9% within five minutes after committed fact
duplicate prevention	100% for same tenant, Command class, and idempotency key
binding Decision audit completeness	100%
stale-state detection	bounded by business deadline

Dashboards and runbooks cover backlog, error budget, dependency failure, retry budget, DLQ, workflow stall, provider outage, reconciliation, capacity, backup/restore, release/rollback, incident command, and post-incident correction.

13. Testing and Continuous Conformance

- aggregate invariant and transition tests
- authorization, delegation, separation-of-duties and tenant-isolation tests
- idempotency and optimistic-concurrency tests
- database migration, key, constraint and rollback tests
- contract, outbox/inbox, duplicate, replay and DLQ tests
- workflow restart, timeout and compensation tests
- Portal E2E tests for success, partial, stale, denied, error and Outcome Unknown states
- accessibility, locale, currency, unit and RTL/LTR tests
- security adversarial, fraud and data-exfiltration tests
- AI evaluation, prompt-injection, policy-bypass and unauthorized-tool tests
- historical reconstruction, correction and audit-completeness tests
- performance, capacity, recovery and operational exercise tests

Release evidence links each invariant and control to the test, environment, input data, result, owner, and artifact proving it. Generated test counts never replace semantic coverage.

14. Repository Review Boundary

- This chapter is a repository-review specification. Exact executable implementation has not been inspected for every target behavior.
- Registered modules, catalog entries, README claims, frontend contracts, or plans are not proof of complete implementation.

Area	Classification
target aggregate, API, tables, state and workflow in this chapter	REFERENCE_ARCHITECTURE
registered module or catalog claim	DOCUMENTATION_ONLY_CLAIM until source inspection
uninspected integration	REQUIRES_REPOSITORY_REVIEW
behavior proven in another repository-grounded chapter	retain that chapter's evidence and classification
contradictory ownership or path claims	CONFLICT_REQUIRES_CANONICALIZATION

No statement upgrades implementation status merely because a file, directory, module name, frontend contract, database entry, plan, or report exists.

15. Worked Conformance Scenario

An authorized principal initiates a Command against an exact subject version. The service validates identity, tenant, mandate, Policy, Evidence, and idempotency. It records the state transition and outbox entry atomically. If an external dependency participates, timeout produces Outcome Unknown; reconciliation queries the source or waits for a verified callback instead of repeating the action blindly.

Portals show source, freshness, version, required action, and uncertainty. AI may extract, compare, summarize, detect anomalies, and recommend through typed Tools, but cannot own the subject or fabricate success. Later correction appends new facts, identifies affected projections and workflows, and preserves the historical Decision context.

16. Conformance Checklist

- ☐ stable identity and proposition ownership are explicit;
- ☐ authority and mandate are evaluated at action time;
- ☐ state, Recommendation, Decision, Intent, execution, and outcome remain distinct;
- ☐ Commands are typed, authorized, idempotent, and version guarded;
- ☐ Events describe completed facts;
- ☐ Evidence includes provenance, time, integrity, qualification, and correction;
- ☐ external outcomes retain their authority source;
- ☐ Portals show freshness and uncertainty;
- ☐ AI is bounded by typed Tools, Policy, approval, and outcome observation;
- ☐ security, tenant, facility, channel, and jurisdiction boundaries are enforced;
- ☐ Outcome Unknown and reconciliation exist where needed;
- ☐ corrections preserve history;
- ☐ tests prove invariants and recovery;
- ☐ implementation status is not upgraded without executable evidence.

17. Status Vocabulary

Status	Meaning
VERIFIED_IMPLEMENTED	Executable source and relevant behavior were inspected.
IMPLEMENTED_WITH_GAP	Executable behavior exists with a material governance or completeness gap.
PARTIAL_IMPLEMENTATION	Only part of the target capability is evidenced.
REFERENCE_ARCHITECTURE	Normative target behavior; no claim that it exists today.
PLANNED_NOT_IMPLEMENTED	Explicitly planned but not evidenced as executable.
DOCUMENTATION_ONLY_CLAIM	Documentation or client contract claims behavior without verified backend evidence.
CONFLICT_REQUIRES_CANONICALIZATION	Sources disagree and no final owner has been established.
REQUIRES_REPOSITORY_REVIEW	Necessary executable evidence has not yet been inspected.

Analytical Stores and Warehouse

Metadata	Value
Volume	10
Chapter ID	V10-28
Subject	AnalyticalStoresandWarehouse
Status	Generated First-Draft Architecture Skeleton - Domain-Specific Expansion Required
Content maturity	TEMPLATE_DERIVED_REFERENCE_SKELETON unless repository citations prove otherwise
Default implementation classification	REFERENCE_ARCHITECTURE / REQUIRES_REPOSITORY_REVIEW
Accountable owner	Data Architecture Council
Evidence rule	No implementation claim without inspected executable evidence
Repository	chinair88-prog/allinb2c

1. Executive Orientation

This chapter defines how GTOS shall **define enterprise and Domain data ownership, semantics, persistence, quality, lineage, correction, protection, operations, and verification for Analytical Stores and Warehouse.**

The specification separates reality, observations, knowledge, Decisions, Intent, execution, and outcomes. A component name, database row, UI label, AI answer, or workflow state does not acquire authority merely because it is visible or technically convenient.

Reality or external outcome

- ≠ Observation
- ≠ Claim
- ≠ derived Perspective
- ≠ Prediction
- ≠ Recommendation
- ≠ authorized Decision
- ≠ execution Intent

1.1 Accountable ownership

The accountable owner is **Data Architecture Council**. Supporting applications, shared infrastructure, integration adapters, analytics, and AI coordinate or present the capability but shall not silently own its authoritative propositions.

1.2 Scope

- create and identify the subject
- validate evidence and policy
- make authorized Decisions
- execute typed Commands
- publish completed-fact Events
- query current and historical perspectives
- correct and reconcile outcomes

1.3 Non-goals

- treating a registered Gradle module, catalog record, README, frontend route, migration plan, or generated report as proof of end-to-end implementation;
- direct cross-Domain database writes;

- generic status values that combine lifecycle, approval, provider, document, payment, dispute, and correction state;
- silent replacement of history;
- AI-created authority or fabricated external outcomes.

2. Ubiquitous Language and Subject Model

Subject	Required interpretation
AnalyticalStoresandWarehouse	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
AnalyticalStoresandWarehouseVersion	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
AnalyticalStoresandWarehouseDecision	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
AnalyticalStoresandWarehouseEvidenceSet	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
AnalyticalStoresandWarehouseException	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit

Reference aggregate:

```

AnalyticalStoresandWarehouse {
  subjectId
  tenantId
  organizationContext
  sourceIdentityRefs[]
  sourceVersionRefs[]
  lifecycleState
  relationshipRefs[]
  decisionRefs[]
  intentRefs[]
  evidenceSetRef
  knowledgeCutoff
  policyRefs[]
  authorityRef
  jurisdictionRefs[]
  createdAt
  effectiveAt
  updatedAt
  version
  correctionOf?
  correlationId
}

```

2.1 Core invariants

- Stable identity precedes relationship, state, and cross-system correlation.
- Effective time and recorded time remain distinct where business truth changes over time.
- Recommendation, Decision, Intent, execution, and outcome are separate concepts.
- No Portal, report, search index, cache, workflow, or AI Agent becomes a shadow owner of Domain truth.
- Binding submitted, approved, issued, accepted, or signed versions are immutable.
- Corrections append governed facts and preserve the original record.
- External authority and provider outcomes retain their source and qualification.
- Tenant and represented-principal context come from trusted authentication and mandate evaluation.
- Retried writes are idempotent and concurrency guarded.
- Outcome Unknown is explicit and reconciled before risky duplicate execution.

2.2 Required standards

- stable identifiers
- typed contracts
- immutable binding versions
- transactional outbox
- idempotent Commands
- explicit Outcome Unknown
- correction lineage
- machine-verifiable conformance

3. Actors, Roles, and Authority

A conformant design distinguishes:

- the natural person acting;
- the service or AI identity invoking a Tool;
- the organization or principal represented;
- the role or relationship granting context;
- the mandate or delegation granting action authority;
- the Policy and jurisdiction limiting the action;
- the reviewer or approver responsible for a binding Decision.

Authority is evaluated at action time. Authentication alone does not grant business authority. Cached permissions and browser-supplied tenant headers are insufficient for high-impact actions.

4. Lifecycle and State Model

State	Semantics
PROPOSED	Distinct governed condition with entry, exit, timeout, correction and authority semantics
VALIDATION_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
APPROVAL_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
ACTIVE	Distinct governed condition with entry, exit, timeout, correction and authority semantics
SUSPENDED	Distinct governed condition with entry, exit, timeout, correction and authority semantics
CORRECTION_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
RETIRED	Distinct governed condition with entry, exit, timeout, correction and authority semantics

```
stateDiagram-v2
    [*] --> PROPOSED
    PROPOSED --> VALIDATION_PENDING
    VALIDATION_PENDING --> APPROVAL_PENDING
    APPROVAL_PENDING --> ACTIVE
    ACTIVE --> SUSPENDED
    SUSPENDED --> CORRECTION_PENDING
    CORRECTION_PENDING --> RETIRED
```

The diagram is a primary reading path. Real implementations may add parallel review, amendment, appeal, timeout, suspension, partial completion, reconciliation, correction, and terminal failure without collapsing their meaning.

4.1 Transition rules

- transition through a typed, authorized Command;
 - require exact expected version where concurrent change matters;
 - record actor, represented principal, mandate, Policy, reason, and Evidence;
 - use an idempotency key for retried writes;
 - publish a completed-fact Event after durable state change;
 - expose Outcome Unknown for unresolved external completion;
 - preserve original and corrected states in historical reconstruction.
-

5. Decision, Evidence, and Knowledge Cutoff

Reference Decision:

```
Decision {
    decisionId
    decisionType
    subjectRef
}
```

```

sourceVersionRefs[]
knowledgeCutoff
evidenceSetRef
policyVersionRefs[]
recommendationRefs[]
principalId
representedOrganizationId
mandateRef
rationale
outcome
decidedAt
effectiveAt
correctionOf?
}

```

Evidence records identify issuer or observer, subject, version, observed time, received time, effective interval, integrity, confidence, qualification, contradiction, correction, retention, residency, and access class. A recommendation may assist the Decision but cannot promote itself to the Decision.

6. Commands, Events, Queries, and Contracts

Reference Commands:

- CreateAnalyticalStoresandWarehouse
- ValidateAnalyticalStoresandWarehouse
- ApproveAnalyticalStoresandWarehouse
- ActivateAnalyticalStoresandWarehouse
- SuspendAnalyticalStoresandWarehouse
- CorrectAnalyticalStoresandWarehouse
- RetireAnalyticalStoresandWarehouse

Reference Events:

- AnalyticalStoresandWarehouseCreated
- AnalyticalStoresandWarehouseValidated
- AnalyticalStoresandWarehouseApproved
- AnalyticalStoresandWarehouseActivated
- AnalyticalStoresandWarehouseSuspended
- AnalyticalStoresandWarehouseCorrected
- AnalyticalStoresandWarehouseRetired

Reference queries:

- GetAnalyticalStoresandWarehouse
- ListAnalyticalStoresandWarehouseRecords
- GetAnalyticalStoresandWarehouseTimeline
- GetAnalyticalStoresandWarehouseEvidence
- GetAnalyticalStoresandWarehouseDecisionPackage
- GetAnalyticalStoresandWarehouseExceptions
- GetAnalyticalStoresandWarehouseAuditTrail

Reference API surface:

```

POST /api/v1/analyticalstoresandwarehouse/commands
GET  /api/v1/analyticalstoresandwarehouse/{id}
GET  /api/v1/analyticalstoresandwarehouse/{id}/timeline
GET  /api/v1/analyticalstoresandwarehouse/{id}/evidence
GET  /api/v1/analyticalstoresandwarehouse/{id}/decisions
POST /api/v1/analyticalstoresandwarehouse/{id}/exceptions
POST /api/v1/analyticalstoresandwarehouse/{id}/corrections

```

These endpoints are reference contracts, not claims that exact routes exist.

7. Workflow, Failure, and Compensation

flowchart TD

```

A[Validate identity version and authority] --> B[Collect required evidence]
B --> C{Policy and evidence gates pass?}
C -- No --> D[Open exception or request correction]
D --> B
C -- Yes --> E[Record Decision or execution Intent]
E --> F[Execute Domain command or external request]

```

```

F --> G{Outcome known?}
G -- Success --> H[Record fact and publish Event]
G -- Failure --> I[Record failure and compensation options]
G -- Unknown --> J[Enter Outcome Unknown]
J --> K[Query source or wait for verified callback]
K --> G
H --> L[Update projections and downstream workflow]

```

Representative failure modes:

- stale or wrong source version
- duplicate submission or replay
- external timeout after possible success
- contradictory source observations
- lost Event after committed state
- approval or mandate revoked during execution
- downstream action before a mandatory gate
- partial completion across multiple subjects
- evidence later retracted or corrected
- tenant, facility, channel or jurisdiction mismatch

Recovery shall query the authoritative source before duplicate execution, preserve partial success, use compensating Intent rather than destructive rewrite, publish corrections, quarantine high-impact ambiguity, and record affected downstream subjects.

8. Data and Persistence Architecture

Reference table families:

Table family	Purpose
analyticalstoresandwarehouse_aggregate	authoritative subject
analyticalstoresandwarehouse_version	immutable submitted, approved, issued, or signed versions
analyticalstoresandwarehouse_relationship	governed relationships
analyticalstoresandwarehouse_decision	binding Decisions and rationale
analyticalstoresandwarehouse_evidence_link	provenance and evidence relationships
analyticalstoresandwarehouse_event_outbox	transactional Event publication
analyticalstoresandwarehouse_idempotency	duplicate-write protection
analyticalstoresandwarehouse_exception	exception, claim, dispute, or hold
analyticalstoresandwarehouse_correction	correction and supersession lineage
analyticalstoresandwarehouse_audit	privileged action evidence

The owning service controls schema and migration. Cross-Domain references use stable identifiers or contracts. Analytics, reports, caches, search, and vector indexes remain projections. Backup, restore, retention, legal hold, privacy, residency, and decommissioning requirements are explicit.

9. Portal and Experience Requirements

- Primary participant Portal: initiate, review exact versions, supply evidence, and respond to required actions.
- Operator Portal: inspect exceptions, correlation, source responses, retries, compensation, and downstream impact.
- Executive or oversight view: consume governed metrics without becoming a source of Domain truth.

Every Portal displays exact version, owner, state, freshness, required action, and uncertainty. It supports loading, empty, partial, stale, denied, error, timeout, and Outcome Unknown states; accessibility; locale, currency, unit, date, and RTL/LTR correctness; evidence audit; safe retry; and correction history.

10. AI Participation and Tool Governance

Permitted AI tasks:

- assist schema and lineage analysis
- detect data-quality anomalies
- summarize migration and reconciliation risk
- never mutate authoritative data outside typed Domain Tools

AI shall not own Domain records, grant itself authority, suppress contradictions, silently edit approved versions, fabricate external outcomes, or execute high-impact actions outside typed Tools, policy, approval, idempotency, sandbox, and outcome observation.

Every Agent Run records source context, prompt/model/tool versions, structured output, confidence and limitations, Policy result, approval, Tool calls, provider responses, final outcome, and recovery actions.

11. Security, Privacy, and Trust Boundaries

- least privilege and separation of duties
- tenant, organization, facility, channel and jurisdiction isolation
- integrity protection for binding evidence
- verified external callbacks and anti-replay controls

Additional controls include secret isolation, callback signature verification, audit of privileged action, emergency-access review, sensitive-data minimization, purpose limitation, and threat models for impersonation, tampering, replay, confused deputy, fraud, prompt injection, and exfiltration.

12. Reliability, SLOs, and Operations

SLI	Reference objective
authoritative read availability	99.9% monthly unless a stricter tier applies
acknowledged Command durability	no acknowledged write may be silently lost
Event publication	99.9% within five minutes after committed fact
duplicate prevention	100% for same tenant, Command class, and idempotency key
binding Decision audit completeness	100%
stale-state detection	bounded by business deadline

Dashboards and runbooks cover backlog, error budget, dependency failure, retry budget, DLQ, workflow stall, provider outage, reconciliation, capacity, backup/restore, release/rollback, incident command, and post-incident correction.

13. Testing and Continuous Conformance

- aggregate invariant and transition tests
- authorization, delegation, separation-of-duties and tenant-isolation tests
- idempotency and optimistic-concurrency tests
- database migration, key, constraint and rollback tests
- contract, outbox/inbox, duplicate, replay and DLQ tests
- workflow restart, timeout and compensation tests
- Portal E2E tests for success, partial, stale, denied, error and Outcome Unknown states
- accessibility, locale, currency, unit and RTL/LTR tests
- security adversarial, fraud and data-exfiltration tests
- AI evaluation, prompt-injection, policy-bypass and unauthorized-tool tests
- historical reconstruction, correction and audit-completeness tests
- performance, capacity, recovery and operational exercise tests

Release evidence links each invariant and control to the test, environment, input data, result, owner, and artifact proving it. Generated test counts never replace semantic coverage.

14. Repository Review Boundary

- This chapter is a repository-review specification. Exact executable implementation has not been inspected for every target behavior.
- Registered modules, catalog entries, README claims, frontend contracts, or plans are not proof of complete implementation.

Area	Classification
target aggregate, API, tables, state and workflow in this chapter	REFERENCE_ARCHITECTURE
registered module or catalog claim	DOCUMENTATION_ONLY_CLAIM until source inspection
uninspected integration	REQUIRES_REPOSITORY_REVIEW
behavior proven in another repository-grounded chapter	retain that chapter's evidence and classification
contradictory ownership or path claims	CONFLICT_REQUIRES_CANONICALIZATION

No statement upgrades implementation status merely because a file, directory, module name, frontend contract, database entry, plan, or report exists.

15. Worked Conformance Scenario

An authorized principal initiates a Command against an exact subject version. The service validates identity, tenant, mandate, Policy, Evidence, and idempotency. It records the state transition and outbox entry atomically. If an external dependency participates, timeout produces Outcome Unknown; reconciliation queries the source or waits for a verified callback instead of repeating the action blindly.

Portals show source, freshness, version, required action, and uncertainty. AI may extract, compare, summarize, detect anomalies, and recommend through typed Tools, but cannot own the subject or fabricate success. Later correction appends new facts, identifies affected projections and workflows, and preserves the historical Decision context.

16. Conformance Checklist

- ☐ stable identity and proposition ownership are explicit;
- ☐ authority and mandate are evaluated at action time;
- ☐ state, Recommendation, Decision, Intent, execution, and outcome remain distinct;
- ☐ Commands are typed, authorized, idempotent, and version guarded;
- ☐ Events describe completed facts;
- ☐ Evidence includes provenance, time, integrity, qualification, and correction;
- ☐ external outcomes retain their authority source;
- ☐ Portals show freshness and uncertainty;
- ☐ AI is bounded by typed Tools, Policy, approval, and outcome observation;
- ☐ security, tenant, facility, channel, and jurisdiction boundaries are enforced;
- ☐ Outcome Unknown and reconciliation exist where needed;
- ☐ corrections preserve history;
- ☐ tests prove invariants and recovery;
- ☐ implementation status is not upgraded without executable evidence.

17. Status Vocabulary

Status	Meaning
VERIFIED_IMPLEMENTED	Executable source and relevant behavior were inspected.
IMPLEMENTED_WITH_GAP	Executable behavior exists with a material governance or completeness gap.
PARTIAL_IMPLEMENTATION	Only part of the target capability is evidenced.
REFERENCE_ARCHITECTURE	Normative target behavior; no claim that it exists today.
PLANNED_NOT_IMPLEMENTED	Explicitly planned but not evidenced as executable.
DOCUMENTATION_ONLY_CLAIM	Documentation or client contract claims behavior without verified backend evidence.
CONFLICT_REQUIRES_CANONICALIZATION	Sources disagree and no final owner has been established.
REQUIRES_REPOSITORY_REVIEW	Necessary executable evidence has not yet been inspected.

Vector Stores

Metadata	Value
Volume	10
Chapter ID	V10-29
Subject	VectorStores
Status	Generated First-Draft Architecture Skeleton - Domain-Specific Expansion Required
Content maturity	TEMPLATE_DERIVED_REFERENCE_SKELETON unless repository citations prove otherwise
Default implementation classification	REFERENCE_ARCHITECTURE / REQUIRES_REPOSITORY_REVIEW
Accountable owner	Data Architecture Council
Evidence rule	No implementation claim without inspected executable evidence
Repository	chinair88-prog/allinb2c

1. Executive Orientation

This chapter defines how GTOS shall **define enterprise and Domain data ownership, semantics, persistence, quality, lineage, correction, protection, operations, and verification for Vector Stores**.

The specification separates reality, observations, knowledge, Decisions, Intent, execution, and outcomes. A component name, database row, UI label, AI answer, or workflow state does not acquire authority merely because it is visible or technically convenient.

Reality or external outcome

- ≠ Observation
- ≠ Claim
- ≠ derived Perspective
- ≠ Prediction
- ≠ Recommendation
- ≠ authorized Decision
- ≠ execution Intent

1.1 Accountable ownership

The accountable owner is **Data Architecture Council**. Supporting applications, shared infrastructure, integration adapters, analytics, and AI coordinate or present the capability but shall not silently own its authoritative propositions.

1.2 Scope

- create and identify the subject
- validate evidence and policy
- make authorized Decisions
- execute typed Commands
- publish completed-fact Events
- query current and historical perspectives
- correct and reconcile outcomes

1.3 Non-goals

- treating a registered Gradle module, catalog record, README, frontend route, migration plan, or generated report as proof of end-to-end implementation;
- direct cross-Domain database writes;
- generic status values that combine lifecycle, approval, provider, document, payment, dispute, and correction state;

- silent replacement of history;
- AI-created authority or fabricated external outcomes.

2. Ubiquitous Language and Subject Model

Subject	Required interpretation
VectorStores	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
VectorStoresVersion	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
VectorStoresDecision	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
VectorStoresEvidenceSet	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
VectorStoresException	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit

Reference aggregate:

```
VectorStores {
  subjectId
  tenantId
  organizationContext
  sourceIdentityRefs[]
  sourceVersionRefs[]
  lifecycleState
  relationshipRefs[]
  decisionRefs[]
  intentRefs[]
  evidenceSetRef
  knowledgeCutoff
  policyRefs[]
  authorityRef
  jurisdictionRefs[]
  createdAt
  effectiveAt
  updatedAt
  version
  correctionOf?
  correlationId
}
```

2.1 Core invariants

- Stable identity precedes relationship, state, and cross-system correlation.
- Effective time and recorded time remain distinct where business truth changes over time.
- Recommendation, Decision, Intent, execution, and outcome are separate concepts.
- No Portal, report, search index, cache, workflow, or AI Agent becomes a shadow owner of Domain truth.
- Binding submitted, approved, issued, accepted, or signed versions are immutable.
- Corrections append governed facts and preserve the original record.
- External authority and provider outcomes retain their source and qualification.
- Tenant and represented-principal context come from trusted authentication and mandate evaluation.
- Retried writes are idempotent and concurrency guarded.
- Outcome Unknown is explicit and reconciled before risky duplicate execution.

2.2 Required standards

- stable identifiers
- typed contracts
- immutable binding versions
- transactional outbox
- idempotent Commands
- explicit Outcome Unknown
- correction lineage
- machine-verifiable conformance

3. Actors, Roles, and Authority

A conformant design distinguishes:

- the natural person acting;
- the service or AI identity invoking a Tool;
- the organization or principal represented;
- the role or relationship granting context;
- the mandate or delegation granting action authority;
- the Policy and jurisdiction limiting the action;
- the reviewer or approver responsible for a binding Decision.

Authority is evaluated at action time. Authentication alone does not grant business authority. Cached permissions and browser-supplied tenant headers are insufficient for high-impact actions.

4. Lifecycle and State Model

State	Semantics
PROPOSED	Distinct governed condition with entry, exit, timeout, correction and authority semantics
VALIDATION_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
APPROVAL_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
ACTIVE	Distinct governed condition with entry, exit, timeout, correction and authority semantics
SUSPENDED	Distinct governed condition with entry, exit, timeout, correction and authority semantics
CORRECTION_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
RETIRED	Distinct governed condition with entry, exit, timeout, correction and authority semantics

```
stateDiagram-v2
    [*] --> PROPOSED
    PROPOSED --> VALIDATION_PENDING
    VALIDATION_PENDING --> APPROVAL_PENDING
    APPROVAL_PENDING --> ACTIVE
    ACTIVE --> SUSPENDED
    SUSPENDED --> CORRECTION_PENDING
    CORRECTION_PENDING --> RETIRED
```

The diagram is a primary reading path. Real implementations may add parallel review, amendment, appeal, timeout, suspension, partial completion, reconciliation, correction, and terminal failure without collapsing their meaning.

4.1 Transition rules

- transition through a typed, authorized Command;
- require exact expected version where concurrent change matters;
- record actor, represented principal, mandate, Policy, reason, and Evidence;
- use an idempotency key for retried writes;
- publish a completed-fact Event after durable state change;
- expose Outcome Unknown for unresolved external completion;
- preserve original and corrected states in historical reconstruction.

5. Decision, Evidence, and Knowledge Cutoff

Reference Decision:

```
Decision {
    decisionId
    decisionType
    subjectRef
    sourceVersionRefs[]
    knowledgeCutoff
    evidenceSetRef
}
```

```

policyVersionRefs[]
recommendationRefs[]
principalId
representedOrganizationId
mandateRef
rationale
outcome
decidedAt
effectiveAt
correctionOf?
}

```

Evidence records identify issuer or observer, subject, version, observed time, received time, effective interval, integrity, confidence, qualification, contradiction, correction, retention, residency, and access class. A recommendation may assist the Decision but cannot promote itself to the Decision.

6. Commands, Events, Queries, and Contracts

Reference Commands:

- CreateVectorStores
- ValidateVectorStores
- ApproveVectorStores
- ActivateVectorStores
- SuspendVectorStores
- CorrectVectorStores
- RetireVectorStores

Reference Events:

- VectorStoresCreated
- VectorStoresValidated
- VectorStoresApproved
- VectorStoresActivated
- VectorStoresSuspended
- VectorStoresCorrected
- VectorStoresRetired

Reference queries:

- GetVectorStores
- ListVectorStoresRecords
- GetVectorStoresTimeline
- GetVectorStoresEvidence
- GetVectorStoresDecisionPackage
- GetVectorStoresExceptions
- GetVectorStoresAuditTrail

Reference API surface:

```

POST /api/v1/vectorstores/commands
GET  /api/v1/vectorstores/{id}
GET  /api/v1/vectorstores/{id}/timeline
GET  /api/v1/vectorstores/{id}/evidence
GET  /api/v1/vectorstores/{id}/decisions
POST /api/v1/vectorstores/{id}/exceptions
POST /api/v1/vectorstores/{id}/corrections

```

These endpoints are reference contracts, not claims that exact routes exist.

7. Workflow, Failure, and Compensation

flowchart TD

```

A[Validate identity version and authority] --> B[Collect required evidence]
B --> C{Policy and evidence gates pass?}
C -- No --> D[Open exception or request correction]
D --> B
C -- Yes --> E[Record Decision or execution Intent]
E --> F[Execute Domain command or external request]
F --> G{Outcome known?}
G -- Success --> H[Record fact and publish Event]
G -- Failure --> I[Record failure and compensation options]

```

```

G -- Unknown --> J[Enter Outcome Unknown]
J --> K[Query source or wait for verified callback]
K --> G
H --> L[Update projections and downstream workflow]

```

Representative failure modes:

- stale or wrong source version
- duplicate submission or replay
- external timeout after possible success
- contradictory source observations
- lost Event after committed state
- approval or mandate revoked during execution
- downstream action before a mandatory gate
- partial completion across multiple subjects
- evidence later retracted or corrected
- tenant, facility, channel or jurisdiction mismatch

Recovery shall query the authoritative source before duplicate execution, preserve partial success, use compensating Intent rather than destructive rewrite, publish corrections, quarantine high-impact ambiguity, and record affected downstream subjects.

8. Data and Persistence Architecture

Reference table families:

Table family	Purpose
vectorstores_aggregate	authoritative subject
vectorstores_version	immutable submitted, approved, issued, or signed versions
vectorstores_relationship	governed relationships
vectorstores_decision	binding Decisions and rationale
vectorstores_evidence_link	provenance and evidence relationships
vectorstores_event_outbox	transactional Event publication
vectorstores_idempotency	duplicate-write protection
vectorstores_exception	exception, claim, dispute, or hold
vectorstores_correction	correction and supersession lineage
vectorstores_audit	privileged action evidence

The owning service controls schema and migration. Cross-Domain references use stable identifiers or contracts. Analytics, reports, caches, search, and vector indexes remain projections. Backup, restore, retention, legal hold, privacy, residency, and decommissioning requirements are explicit.

9. Portal and Experience Requirements

- Primary participant Portal: initiate, review exact versions, supply evidence, and respond to required actions.
- Operator Portal: inspect exceptions, correlation, source responses, retries, compensation, and downstream impact.
- Executive or oversight view: consume governed metrics without becoming a source of Domain truth.

Every Portal displays exact version, owner, state, freshness, required action, and uncertainty. It supports loading, empty, partial, stale, denied, error, timeout, and Outcome Unknown states; accessibility; locale, currency, unit, date, and RTL/LTR correctness; evidence audit; safe retry; and correction history.

10. AI Participation and Tool Governance

Permitted AI tasks:

- assist schema and lineage analysis
- detect data-quality anomalies
- summarize migration and reconciliation risk
- never mutate authoritative data outside typed Domain Tools

AI shall not own Domain records, grant itself authority, suppress contradictions, silently edit approved versions, fabricate external outcomes, or execute high-impact actions outside typed Tools, policy, approval, idempotency, sandbox, and outcome observation.

Every Agent Run records source context, prompt/model/tool versions, structured output, confidence and limitations, Policy result, approval, Tool calls, provider responses, final outcome, and recovery actions.

11. Security, Privacy, and Trust Boundaries

- least privilege and separation of duties
- tenant, organization, facility, channel and jurisdiction isolation
- integrity protection for binding evidence
- verified external callbacks and anti-replay controls

Additional controls include secret isolation, callback signature verification, audit of privileged action, emergency-access review, sensitive-data minimization, purpose limitation, and threat models for impersonation, tampering, replay, confused deputy, fraud, prompt injection, and exfiltration.

12. Reliability, SLOs, and Operations

SLI	Reference objective
authoritative read availability	99.9% monthly unless a stricter tier applies
acknowledged Command durability	no acknowledged write may be silently lost
Event publication	99.9% within five minutes after committed fact
duplicate prevention	100% for same tenant, Command class, and idempotency key
binding Decision audit completeness	100%
stale-state detection	bounded by business deadline

Dashboards and runbooks cover backlog, error budget, dependency failure, retry budget, DLQ, workflow stall, provider outage, reconciliation, capacity, backup/restore, release/rollback, incident command, and post-incident correction.

13. Testing and Continuous Conformance

- aggregate invariant and transition tests
- authorization, delegation, separation-of-duties and tenant-isolation tests
- idempotency and optimistic-concurrency tests
- database migration, key, constraint and rollback tests
- contract, outbox/inbox, duplicate, replay and DLQ tests
- workflow restart, timeout and compensation tests
- Portal E2E tests for success, partial, stale, denied, error and Outcome Unknown states
- accessibility, locale, currency, unit and RTL/LTR tests
- security adversarial, fraud and data-exfiltration tests
- AI evaluation, prompt-injection, policy-bypass and unauthorized-tool tests
- historical reconstruction, correction and audit-completeness tests
- performance, capacity, recovery and operational exercise tests

Release evidence links each invariant and control to the test, environment, input data, result, owner, and artifact proving it. Generated test counts never replace semantic coverage.

14. Repository Review Boundary

- This chapter is a repository-review specification. Exact executable implementation has not been inspected for every target behavior.
- Registered modules, catalog entries, README claims, frontend contracts, or plans are not proof of complete implementation.

Area	Classification
target aggregate, API, tables, state and workflow in this chapter	REFERENCE_ARCHITECTURE
registered module or catalog claim	DOCUMENTATION_ONLY_CLAIM until source inspection
uninspected integration	REQUIRES_REPOSITORY_REVIEW

Area	Classification
behavior proven in another repository-grounded chapter contradictory ownership or path claims	retain that chapter's evidence and classification CONFLICT_REQUIRES_CANONICALIZATION

No statement upgrades implementation status merely because a file, directory, module name, frontend contract, database entry, plan, or report exists.

15. Worked Conformance Scenario

An authorized principal initiates a Command against an exact subject version. The service validates identity, tenant, mandate, Policy, Evidence, and idempotency. It records the state transition and outbox entry atomically. If an external dependency participates, timeout produces Outcome Unknown; reconciliation queries the source or waits for a verified callback instead of repeating the action blindly.

Portals show source, freshness, version, required action, and uncertainty. AI may extract, compare, summarize, detect anomalies, and recommend through typed Tools, but cannot own the subject or fabricate success. Later correction appends new facts, identifies affected projections and workflows, and preserves the historical Decision context.

16. Conformance Checklist

- ☐ stable identity and proposition ownership are explicit;
- ☐ authority and mandate are evaluated at action time;
- ☐ state, Recommendation, Decision, Intent, execution, and outcome remain distinct;
- ☐ Commands are typed, authorized, idempotent, and version guarded;
- ☐ Events describe completed facts;
- ☐ Evidence includes provenance, time, integrity, qualification, and correction;
- ☐ external outcomes retain their authority source;
- ☐ Portals show freshness and uncertainty;
- ☐ AI is bounded by typed Tools, Policy, approval, and outcome observation;
- ☐ security, tenant, facility, channel, and jurisdiction boundaries are enforced;
- ☐ Outcome Unknown and reconciliation exist where needed;
- ☐ corrections preserve history;
- ☐ tests prove invariants and recovery;
- ☐ implementation status is not upgraded without executable evidence.

17. Status Vocabulary

Status	Meaning
VERIFIED_IMPLEMENTED	Executable source and relevant behavior were inspected.
IMPLEMENTED_WITH_GAP	Executable behavior exists with a material governance or completeness gap.
PARTIAL_IMPLEMENTATION	Only part of the target capability is evidenced.
REFERENCE_ARCHITECTURE	Normative target behavior; no claim that it exists today.
PLANNED_NOT_IMPLEMENTED	Explicitly planned but not evidenced as executable.
DOCUMENTATION_ONLY_CLAIM	Documentation or client contract claims behavior without verified backend evidence.
CONFLICT_REQUIRES_CANONICALIZATION	Sources disagree and no final owner has been established.
REQUIRES_REPOSITORY_REVIEW	Necessary executable evidence has not yet been inspected.

Search Indexes

Metadata	Value
Volume	10
Chapter ID	V10-30
Subject	SearchIndexes
Status	Generated First-Draft Architecture Skeleton - Domain-Specific Expansion Required
Content maturity	TEMPLATE_DERIVED_REFERENCE_SKELETON unless repository citations prove otherwise
Default implementation classification	REFERENCE_ARCHITECTURE / REQUIRES_REPOSITORY_REVIEW
Accountable owner	Data Architecture Council
Evidence rule	No implementation claim without inspected executable evidence
Repository	chinair88-prog/allinb2c

1. Executive Orientation

This chapter defines how GTOS shall **define enterprise and Domain data ownership, semantics, persistence, quality, lineage, correction, protection, operations, and verification for Search Indexes.**

The specification separates reality, observations, knowledge, Decisions, Intent, execution, and outcomes. A component name, database row, UI label, AI answer, or workflow state does not acquire authority merely because it is visible or technically convenient.

Reality or external outcome

- ≠ Observation
- ≠ Claim
- ≠ derived Perspective
- ≠ Prediction
- ≠ Recommendation
- ≠ authorized Decision
- ≠ execution Intent

1.1 Accountable ownership

The accountable owner is **Data Architecture Council**. Supporting applications, shared infrastructure, integration adapters, analytics, and AI coordinate or present the capability but shall not silently own its authoritative propositions.

1.2 Scope

- create and identify the subject
- validate evidence and policy
- make authorized Decisions
- execute typed Commands
- publish completed-fact Events
- query current and historical perspectives
- correct and reconcile outcomes

1.3 Non-goals

- treating a registered Gradle module, catalog record, README, frontend route, migration plan, or generated report as proof of end-to-end implementation;
- direct cross-Domain database writes;
- generic status values that combine lifecycle, approval, provider, document, payment, dispute, and correction state;

- silent replacement of history;
- AI-created authority or fabricated external outcomes.

2. Ubiquitous Language and Subject Model

Subject	Required interpretation
SearchIndexes	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
SearchIndexesVersion	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
SearchIndexesDecision	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
SearchIndexesEvidenceSet	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
SearchIndexesException	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit

Reference aggregate:

```
SearchIndexes {
  subjectId
  tenantId
  organizationContext
  sourceIdentityRefs[]
  sourceVersionRefs[]
  lifecycleState
  relationshipRefs[]
  decisionRefs[]
  intentRefs[]
  evidenceSetRef
  knowledgeCutoff
  policyRefs[]
  authorityRef
  jurisdictionRefs[]
  createdAt
  effectiveAt
  updatedAt
  version
  correctionOf?
  correlationId
}
```

2.1 Core invariants

- Stable identity precedes relationship, state, and cross-system correlation.
- Effective time and recorded time remain distinct where business truth changes over time.
- Recommendation, Decision, Intent, execution, and outcome are separate concepts.
- No Portal, report, search index, cache, workflow, or AI Agent becomes a shadow owner of Domain truth.
- Binding submitted, approved, issued, accepted, or signed versions are immutable.
- Corrections append governed facts and preserve the original record.
- External authority and provider outcomes retain their source and qualification.
- Tenant and represented-principal context come from trusted authentication and mandate evaluation.
- Retried writes are idempotent and concurrency guarded.
- Outcome Unknown is explicit and reconciled before risky duplicate execution.

2.2 Required standards

- stable identifiers
- typed contracts
- immutable binding versions
- transactional outbox
- idempotent Commands
- explicit Outcome Unknown
- correction lineage
- machine-verifiable conformance

3. Actors, Roles, and Authority

A conformant design distinguishes:

- the natural person acting;
- the service or AI identity invoking a Tool;
- the organization or principal represented;
- the role or relationship granting context;
- the mandate or delegation granting action authority;
- the Policy and jurisdiction limiting the action;
- the reviewer or approver responsible for a binding Decision.

Authority is evaluated at action time. Authentication alone does not grant business authority. Cached permissions and browser-supplied tenant headers are insufficient for high-impact actions.

4. Lifecycle and State Model

State	Semantics
PROPOSED	Distinct governed condition with entry, exit, timeout, correction and authority semantics
VALIDATION_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
APPROVAL_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
ACTIVE	Distinct governed condition with entry, exit, timeout, correction and authority semantics
SUSPENDED	Distinct governed condition with entry, exit, timeout, correction and authority semantics
CORRECTION_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
RETIRED	Distinct governed condition with entry, exit, timeout, correction and authority semantics

```
stateDiagram-v2
    [*] --> PROPOSED
    PROPOSED --> VALIDATION_PENDING
    VALIDATION_PENDING --> APPROVAL_PENDING
    APPROVAL_PENDING --> ACTIVE
    ACTIVE --> SUSPENDED
    SUSPENDED --> CORRECTION_PENDING
    CORRECTION_PENDING --> RETIRED
```

The diagram is a primary reading path. Real implementations may add parallel review, amendment, appeal, timeout, suspension, partial completion, reconciliation, correction, and terminal failure without collapsing their meaning.

4.1 Transition rules

- transition through a typed, authorized Command;
- require exact expected version where concurrent change matters;
- record actor, represented principal, mandate, Policy, reason, and Evidence;
- use an idempotency key for retried writes;
- publish a completed-fact Event after durable state change;
- expose Outcome Unknown for unresolved external completion;
- preserve original and corrected states in historical reconstruction.

5. Decision, Evidence, and Knowledge Cutoff

Reference Decision:

```
Decision {
    decisionId
    decisionType
    subjectRef
    sourceVersionRefs[]
    knowledgeCutoff
    evidenceSetRef
}
```

```

policyVersionRefs[]
recommendationRefs[]
principalId
representedOrganizationId
mandateRef
rationale
outcome
decidedAt
effectiveAt
correctionOf?
}

```

Evidence records identify issuer or observer, subject, version, observed time, received time, effective interval, integrity, confidence, qualification, contradiction, correction, retention, residency, and access class. A recommendation may assist the Decision but cannot promote itself to the Decision.

6. Commands, Events, Queries, and Contracts

Reference Commands:

- CreateSearchIndexes
- ValidateSearchIndexes
- ApproveSearchIndexes
- ActivateSearchIndexes
- SuspendSearchIndexes
- CorrectSearchIndexes
- RetireSearchIndexes

Reference Events:

- SearchIndexesCreated
- SearchIndexesValidated
- SearchIndexesApproved
- SearchIndexesActivated
- SearchIndexesSuspended
- SearchIndexesCorrected
- SearchIndexesRetired

Reference queries:

- GetSearchIndexes
- ListSearchIndexesRecords
- GetSearchIndexesTimeline
- GetSearchIndexesEvidence
- GetSearchIndexesDecisionPackage
- GetSearchIndexesExceptions
- GetSearchIndexesAuditTrail

Reference API surface:

```

POST /api/v1/searchindexes/commands
GET  /api/v1/searchindexes/{id}
GET  /api/v1/searchindexes/{id}/timeline
GET  /api/v1/searchindexes/{id}/evidence
GET  /api/v1/searchindexes/{id}/decisions
POST /api/v1/searchindexes/{id}/exceptions
POST /api/v1/searchindexes/{id}/corrections

```

These endpoints are reference contracts, not claims that exact routes exist.

7. Workflow, Failure, and Compensation

flowchart TD

```

A[Validate identity version and authority] --> B[Collect required evidence]
B --> C{Policy and evidence gates pass?}
C -- No --> D[Open exception or request correction]
D --> B
C -- Yes --> E[Record Decision or execution Intent]
E --> F[Execute Domain command or external request]
F --> G{Outcome known?}
G -- Success --> H[Record fact and publish Event]
G -- Failure --> I[Record failure and compensation options]

```

```

G -- Unknown --> J[Enter Outcome Unknown]
J --> K[Query source or wait for verified callback]
K --> G
H --> L[Update projections and downstream workflow]

```

Representative failure modes:

- stale or wrong source version
- duplicate submission or replay
- external timeout after possible success
- contradictory source observations
- lost Event after committed state
- approval or mandate revoked during execution
- downstream action before a mandatory gate
- partial completion across multiple subjects
- evidence later retracted or corrected
- tenant, facility, channel or jurisdiction mismatch

Recovery shall query the authoritative source before duplicate execution, preserve partial success, use compensating Intent rather than destructive rewrite, publish corrections, quarantine high-impact ambiguity, and record affected downstream subjects.

8. Data and Persistence Architecture

Reference table families:

Table family	Purpose
searchindexes_aggregate	authoritative subject
searchindexes_version	immutable submitted, approved, issued, or signed versions
searchindexes_relationship	governed relationships
searchindexes_decision	binding Decisions and rationale
searchindexes_evidence_link	provenance and evidence relationships
searchindexes_event_outbox	transactional Event publication
searchindexes_idempotency	duplicate-write protection
searchindexes_exception	exception, claim, dispute, or hold
searchindexes_correction	correction and supersession lineage
searchindexes_audit	privileged action evidence

The owning service controls schema and migration. Cross-Domain references use stable identifiers or contracts. Analytics, reports, caches, search, and vector indexes remain projections. Backup, restore, retention, legal hold, privacy, residency, and decommissioning requirements are explicit.

9. Portal and Experience Requirements

- Primary participant Portal: initiate, review exact versions, supply evidence, and respond to required actions.
- Operator Portal: inspect exceptions, correlation, source responses, retries, compensation, and downstream impact.
- Executive or oversight view: consume governed metrics without becoming a source of Domain truth.

Every Portal displays exact version, owner, state, freshness, required action, and uncertainty. It supports loading, empty, partial, stale, denied, error, timeout, and Outcome Unknown states; accessibility; locale, currency, unit, date, and RTL/LTR correctness; evidence audit; safe retry; and correction history.

10. AI Participation and Tool Governance

Permitted AI tasks:

- assist schema and lineage analysis
- detect data-quality anomalies
- summarize migration and reconciliation risk
- never mutate authoritative data outside typed Domain Tools

AI shall not own Domain records, grant itself authority, suppress contradictions, silently edit approved versions, fabricate external outcomes, or execute high-impact actions outside typed Tools, policy, approval, idempotency, sandbox, and outcome observation.

Every Agent Run records source context, prompt/model/tool versions, structured output, confidence and limitations, Policy result, approval, Tool calls, provider responses, final outcome, and recovery actions.

11. Security, Privacy, and Trust Boundaries

- least privilege and separation of duties
- tenant, organization, facility, channel and jurisdiction isolation
- integrity protection for binding evidence
- verified external callbacks and anti-replay controls

Additional controls include secret isolation, callback signature verification, audit of privileged action, emergency-access review, sensitive-data minimization, purpose limitation, and threat models for impersonation, tampering, replay, confused deputy, fraud, prompt injection, and exfiltration.

12. Reliability, SLOs, and Operations

SLI	Reference objective
authoritative read availability	99.9% monthly unless a stricter tier applies
acknowledged Command durability	no acknowledged write may be silently lost
Event publication	99.9% within five minutes after committed fact
duplicate prevention	100% for same tenant, Command class, and idempotency key
binding Decision audit completeness	100%
stale-state detection	bounded by business deadline

Dashboards and runbooks cover backlog, error budget, dependency failure, retry budget, DLQ, workflow stall, provider outage, reconciliation, capacity, backup/restore, release/rollback, incident command, and post-incident correction.

13. Testing and Continuous Conformance

- aggregate invariant and transition tests
- authorization, delegation, separation-of-duties and tenant-isolation tests
- idempotency and optimistic-concurrency tests
- database migration, key, constraint and rollback tests
- contract, outbox/inbox, duplicate, replay and DLQ tests
- workflow restart, timeout and compensation tests
- Portal E2E tests for success, partial, stale, denied, error and Outcome Unknown states
- accessibility, locale, currency, unit and RTL/LTR tests
- security adversarial, fraud and data-exfiltration tests
- AI evaluation, prompt-injection, policy-bypass and unauthorized-tool tests
- historical reconstruction, correction and audit-completeness tests
- performance, capacity, recovery and operational exercise tests

Release evidence links each invariant and control to the test, environment, input data, result, owner, and artifact proving it. Generated test counts never replace semantic coverage.

14. Repository Review Boundary

- This chapter is a repository-review specification. Exact executable implementation has not been inspected for every target behavior.
- Registered modules, catalog entries, README claims, frontend contracts, or plans are not proof of complete implementation.

Area	Classification
target aggregate, API, tables, state and workflow in this chapter	REFERENCE_ARCHITECTURE
registered module or catalog claim	DOCUMENTATION_ONLY_CLAIM until source inspection
uninspected integration	REQUIRES_REPOSITORY_REVIEW

Area	Classification
behavior proven in another repository-grounded chapter contradictory ownership or path claims	retain that chapter's evidence and classification CONFLICT_REQUIRES_CANONICALIZATION

No statement upgrades implementation status merely because a file, directory, module name, frontend contract, database entry, plan, or report exists.

15. Worked Conformance Scenario

An authorized principal initiates a Command against an exact subject version. The service validates identity, tenant, mandate, Policy, Evidence, and idempotency. It records the state transition and outbox entry atomically. If an external dependency participates, timeout produces Outcome Unknown; reconciliation queries the source or waits for a verified callback instead of repeating the action blindly.

Portals show source, freshness, version, required action, and uncertainty. AI may extract, compare, summarize, detect anomalies, and recommend through typed Tools, but cannot own the subject or fabricate success. Later correction appends new facts, identifies affected projections and workflows, and preserves the historical Decision context.

16. Conformance Checklist

- ☐ stable identity and proposition ownership are explicit;
- ☐ authority and mandate are evaluated at action time;
- ☐ state, Recommendation, Decision, Intent, execution, and outcome remain distinct;
- ☐ Commands are typed, authorized, idempotent, and version guarded;
- ☐ Events describe completed facts;
- ☐ Evidence includes provenance, time, integrity, qualification, and correction;
- ☐ external outcomes retain their authority source;
- ☐ Portals show freshness and uncertainty;
- ☐ AI is bounded by typed Tools, Policy, approval, and outcome observation;
- ☐ security, tenant, facility, channel, and jurisdiction boundaries are enforced;
- ☐ Outcome Unknown and reconciliation exist where needed;
- ☐ corrections preserve history;
- ☐ tests prove invariants and recovery;
- ☐ implementation status is not upgraded without executable evidence.

17. Status Vocabulary

Status	Meaning
VERIFIED_IMPLEMENTED IMPLEMENTED_WITH_GAP	Executable source and relevant behavior were inspected. Executable behavior exists with a material governance or completeness gap.
PARTIAL_IMPLEMENTATION REFERENCE_ARCHITECTURE PLANNED_NOT_IMPLEMENTED DOCUMENTATION_ONLY_CLAIM	Only part of the target capability is evidenced. Normative target behavior; no claim that it exists today. Explicitly planned but not evidenced as executable. Documentation or client contract claims behavior without verified backend evidence.
CONFLICT_REQUIRES_CANONICALIZATION REQUIRES_REPOSITORY_REVIEW	Sources disagree and no final owner has been established. Necessary executable evidence has not yet been inspected.

Caching and Invalidation

Metadata	Value
Volume	10
Chapter ID	V10-31
Subject	CachingandInvalidation
Status	Generated First-Draft Architecture Skeleton - Domain-Specific Expansion Required
Content maturity	TEMPLATE_DERIVED_REFERENCE_SKELETON unless repository citations prove otherwise
Default implementation classification	REFERENCE_ARCHITECTURE / REQUIRES_REPOSITORY_REVIEW
Accountable owner	Data Architecture Council
Evidence rule	No implementation claim without inspected executable evidence
Repository	chinair88-prog/allinb2c

1. Executive Orientation

This chapter defines how GTOS shall **define enterprise and Domain data ownership, semantics, persistence, quality, lineage, correction, protection, operations, and verification for Caching and Invalidation.**

The specification separates reality, observations, knowledge, Decisions, Intent, execution, and outcomes. A component name, database row, UI label, AI answer, or workflow state does not acquire authority merely because it is visible or technically convenient.

Reality or external outcome

- ≠ Observation
- ≠ Claim
- ≠ derived Perspective
- ≠ Prediction
- ≠ Recommendation
- ≠ authorized Decision
- ≠ execution Intent

1.1 Accountable ownership

The accountable owner is **Data Architecture Council**. Supporting applications, shared infrastructure, integration adapters, analytics, and AI coordinate or present the capability but shall not silently own its authoritative propositions.

1.2 Scope

- create and identify the subject
- validate evidence and policy
- make authorized Decisions
- execute typed Commands
- publish completed-fact Events
- query current and historical perspectives
- correct and reconcile outcomes

1.3 Non-goals

- treating a registered Gradle module, catalog record, README, frontend route, migration plan, or generated report as proof of end-to-end implementation;
- direct cross-Domain database writes;

- generic status values that combine lifecycle, approval, provider, document, payment, dispute, and correction state;
- silent replacement of history;
- AI-created authority or fabricated external outcomes.

2. Ubiquitous Language and Subject Model

Subject	Required interpretation
CachingandInvalidation	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
CachingandInvalidationVersion	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
CachingandInvalidationDecision	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
CachingandInvalidationEvidenceSet	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
CachingandInvalidationException	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit

Reference aggregate:

```
CachingandInvalidation {
  subjectId
  tenantId
  organizationContext
  sourceIdentityRefs[]
  sourceVersionRefs[]
  lifecycleState
  relationshipRefs[]
  decisionRefs[]
  intentRefs[]
  evidenceSetRef
  knowledgeCutoff
  policyRefs[]
  authorityRef
  jurisdictionRefs[]
  createdAt
  effectiveAt
  updatedAt
  version
  correctionOf?
  correlationId
}
```

2.1 Core invariants

- Stable identity precedes relationship, state, and cross-system correlation.
- Effective time and recorded time remain distinct where business truth changes over time.
- Recommendation, Decision, Intent, execution, and outcome are separate concepts.
- No Portal, report, search index, cache, workflow, or AI Agent becomes a shadow owner of Domain truth.
- Binding submitted, approved, issued, accepted, or signed versions are immutable.
- Corrections append governed facts and preserve the original record.
- External authority and provider outcomes retain their source and qualification.
- Tenant and represented-principal context come from trusted authentication and mandate evaluation.
- Retried writes are idempotent and concurrency guarded.
- Outcome Unknown is explicit and reconciled before risky duplicate execution.

2.2 Required standards

- stable identifiers
- typed contracts
- immutable binding versions
- transactional outbox
- idempotent Commands
- explicit Outcome Unknown
- correction lineage
- machine-verifiable conformance

3. Actors, Roles, and Authority

A conformant design distinguishes:

- the natural person acting;
- the service or AI identity invoking a Tool;
- the organization or principal represented;
- the role or relationship granting context;
- the mandate or delegation granting action authority;
- the Policy and jurisdiction limiting the action;
- the reviewer or approver responsible for a binding Decision.

Authority is evaluated at action time. Authentication alone does not grant business authority. Cached permissions and browser-supplied tenant headers are insufficient for high-impact actions.

4. Lifecycle and State Model

State	Semantics
PROPOSED	Distinct governed condition with entry, exit, timeout, correction and authority semantics
VALIDATION_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
APPROVAL_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
ACTIVE	Distinct governed condition with entry, exit, timeout, correction and authority semantics
SUSPENDED	Distinct governed condition with entry, exit, timeout, correction and authority semantics
CORRECTION_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
RETIRED	Distinct governed condition with entry, exit, timeout, correction and authority semantics

```
stateDiagram-v2
    [*] --> PROPOSED
    PROPOSED --> VALIDATION_PENDING
    VALIDATION_PENDING --> APPROVAL_PENDING
    APPROVAL_PENDING --> ACTIVE
    ACTIVE --> SUSPENDED
    SUSPENDED --> CORRECTION_PENDING
    CORRECTION_PENDING --> RETIRED
```

The diagram is a primary reading path. Real implementations may add parallel review, amendment, appeal, timeout, suspension, partial completion, reconciliation, correction, and terminal failure without collapsing their meaning.

4.1 Transition rules

- transition through a typed, authorized Command;
 - require exact expected version where concurrent change matters;
 - record actor, represented principal, mandate, Policy, reason, and Evidence;
 - use an idempotency key for retried writes;
 - publish a completed-fact Event after durable state change;
 - expose Outcome Unknown for unresolved external completion;
 - preserve original and corrected states in historical reconstruction.
-

5. Decision, Evidence, and Knowledge Cutoff

Reference Decision:

```
Decision {
    decisionId
    decisionType
    subjectRef
}
```

```

sourceVersionRefs[]
knowledgeCutoff
evidenceSetRef
policyVersionRefs[]
recommendationRefs[]
principalId
representedOrganizationId
mandateRef
rationale
outcome
decidedAt
effectiveAt
correctionOf?
}

```

Evidence records identify issuer or observer, subject, version, observed time, received time, effective interval, integrity, confidence, qualification, contradiction, correction, retention, residency, and access class. A recommendation may assist the Decision but cannot promote itself to the Decision.

6. Commands, Events, Queries, and Contracts

Reference Commands:

- CreateCachingandInvalidation
- ValidateCachingandInvalidation
- ApproveCachingandInvalidation
- ActivateCachingandInvalidation
- SuspendCachingandInvalidation
- CorrectCachingandInvalidation
- RetireCachingandInvalidation

Reference Events:

- CachingandInvalidationCreated
- CachingandInvalidationValidated
- CachingandInvalidationApproved
- CachingandInvalidationActivated
- CachingandInvalidationSuspended
- CachingandInvalidationCorrected
- CachingandInvalidationRetired

Reference queries:

- GetCachingandInvalidation
- ListCachingandInvalidationRecords
- GetCachingandInvalidationTimeline
- GetCachingandInvalidationEvidence
- GetCachingandInvalidationDecisionPackage
- GetCachingandInvalidationExceptions
- GetCachingandInvalidationAuditTrail

Reference API surface:

```

POST /api/v1/cachingandinvalidation/commands
GET  /api/v1/cachingandinvalidation/{id}
GET  /api/v1/cachingandinvalidation/{id}/timeline
GET  /api/v1/cachingandinvalidation/{id}/evidence
GET  /api/v1/cachingandinvalidation/{id}/decisions
POST /api/v1/cachingandinvalidation/{id}/exceptions
POST /api/v1/cachingandinvalidation/{id}/corrections

```

These endpoints are reference contracts, not claims that exact routes exist.

7. Workflow, Failure, and Compensation

flowchart TD

```

A[Validate identity version and authority] --> B[Collect required evidence]
B --> C{Policy and evidence gates pass?}
C -- No --> D[Open exception or request correction]
D --> B
C -- Yes --> E[Record Decision or execution Intent]
E --> F[Execute Domain command or external request]

```

```

F --> G{Outcome known?}
G -- Success --> H[Record fact and publish Event]
G -- Failure --> I[Record failure and compensation options]
G -- Unknown --> J[Enter Outcome Unknown]
J --> K[Query source or wait for verified callback]
K --> G
H --> L[Update projections and downstream workflow]

```

Representative failure modes:

- stale or wrong source version
- duplicate submission or replay
- external timeout after possible success
- contradictory source observations
- lost Event after committed state
- approval or mandate revoked during execution
- downstream action before a mandatory gate
- partial completion across multiple subjects
- evidence later retracted or corrected
- tenant, facility, channel or jurisdiction mismatch

Recovery shall query the authoritative source before duplicate execution, preserve partial success, use compensating Intent rather than destructive rewrite, publish corrections, quarantine high-impact ambiguity, and record affected downstream subjects.

8. Data and Persistence Architecture

Reference table families:

Table family	Purpose
cachingandinvalidation_aggregate	authoritative subject
cachingandinvalidation_version	immutable submitted, approved, issued, or signed versions
cachingandinvalidation_relationship	governed relationships
cachingandinvalidation_decision	binding Decisions and rationale
cachingandinvalidation_evidence_link	provenance and evidence relationships
cachingandinvalidation_event_outbox	transactional Event publication
cachingandinvalidation_idempotency	duplicate-write protection
cachingandinvalidation_exception	exception, claim, dispute, or hold
cachingandinvalidation_correction	correction and supersession lineage
cachingandinvalidation_audit	privileged action evidence

The owning service controls schema and migration. Cross-Domain references use stable identifiers or contracts. Analytics, reports, caches, search, and vector indexes remain projections. Backup, restore, retention, legal hold, privacy, residency, and decommissioning requirements are explicit.

9. Portal and Experience Requirements

- Primary participant Portal: initiate, review exact versions, supply evidence, and respond to required actions.
- Operator Portal: inspect exceptions, correlation, source responses, retries, compensation, and downstream impact.
- Executive or oversight view: consume governed metrics without becoming a source of Domain truth.

Every Portal displays exact version, owner, state, freshness, required action, and uncertainty. It supports loading, empty, partial, stale, denied, error, timeout, and Outcome Unknown states; accessibility; locale, currency, unit, date, and RTL/LTR correctness; evidence audit; safe retry; and correction history.

10. AI Participation and Tool Governance

Permitted AI tasks:

- assist schema and lineage analysis
- detect data-quality anomalies
- summarize migration and reconciliation risk
- never mutate authoritative data outside typed Domain Tools

AI shall not own Domain records, grant itself authority, suppress contradictions, silently edit approved versions, fabricate external outcomes, or execute high-impact actions outside typed Tools, policy, approval, idempotency, sandbox, and outcome observation.

Every Agent Run records source context, prompt/model/tool versions, structured output, confidence and limitations, Policy result, approval, Tool calls, provider responses, final outcome, and recovery actions.

11. Security, Privacy, and Trust Boundaries

- least privilege and separation of duties
- tenant, organization, facility, channel and jurisdiction isolation
- integrity protection for binding evidence
- verified external callbacks and anti-replay controls

Additional controls include secret isolation, callback signature verification, audit of privileged action, emergency-access review, sensitive-data minimization, purpose limitation, and threat models for impersonation, tampering, replay, confused deputy, fraud, prompt injection, and exfiltration.

12. Reliability, SLOs, and Operations

SLI	Reference objective
authoritative read availability	99.9% monthly unless a stricter tier applies
acknowledged Command durability	no acknowledged write may be silently lost
Event publication	99.9% within five minutes after committed fact
duplicate prevention	100% for same tenant, Command class, and idempotency key
binding Decision audit completeness	100%
stale-state detection	bounded by business deadline

Dashboards and runbooks cover backlog, error budget, dependency failure, retry budget, DLQ, workflow stall, provider outage, reconciliation, capacity, backup/restore, release/rollback, incident command, and post-incident correction.

13. Testing and Continuous Conformance

- aggregate invariant and transition tests
- authorization, delegation, separation-of-duties and tenant-isolation tests
- idempotency and optimistic-concurrency tests
- database migration, key, constraint and rollback tests
- contract, outbox/inbox, duplicate, replay and DLQ tests
- workflow restart, timeout and compensation tests
- Portal E2E tests for success, partial, stale, denied, error and Outcome Unknown states
- accessibility, locale, currency, unit and RTL/LTR tests
- security adversarial, fraud and data-exfiltration tests
- AI evaluation, prompt-injection, policy-bypass and unauthorized-tool tests
- historical reconstruction, correction and audit-completeness tests
- performance, capacity, recovery and operational exercise tests

Release evidence links each invariant and control to the test, environment, input data, result, owner, and artifact proving it. Generated test counts never replace semantic coverage.

14. Repository Review Boundary

- This chapter is a repository-review specification. Exact executable implementation has not been inspected for every target behavior.
- Registered modules, catalog entries, README claims, frontend contracts, or plans are not proof of complete implementation.

Area	Classification
target aggregate, API, tables, state and workflow in this chapter	REFERENCE_ARCHITECTURE
registered module or catalog claim	DOCUMENTATION_ONLY_CLAIM until source inspection
uninspected integration	REQUIRES_REPOSITORY_REVIEW
behavior proven in another repository-grounded chapter	retain that chapter's evidence and classification
contradictory ownership or path claims	CONFLICT_REQUIRES_CANONICALIZATION

No statement upgrades implementation status merely because a file, directory, module name, frontend contract, database entry, plan, or report exists.

15. Worked Conformance Scenario

An authorized principal initiates a Command against an exact subject version. The service validates identity, tenant, mandate, Policy, Evidence, and idempotency. It records the state transition and outbox entry atomically. If an external dependency participates, timeout produces Outcome Unknown; reconciliation queries the source or waits for a verified callback instead of repeating the action blindly.

Portals show source, freshness, version, required action, and uncertainty. AI may extract, compare, summarize, detect anomalies, and recommend through typed Tools, but cannot own the subject or fabricate success. Later correction appends new facts, identifies affected projections and workflows, and preserves the historical Decision context.

16. Conformance Checklist

- ☐ stable identity and proposition ownership are explicit;
- ☐ authority and mandate are evaluated at action time;
- ☐ state, Recommendation, Decision, Intent, execution, and outcome remain distinct;
- ☐ Commands are typed, authorized, idempotent, and version guarded;
- ☐ Events describe completed facts;
- ☐ Evidence includes provenance, time, integrity, qualification, and correction;
- ☐ external outcomes retain their authority source;
- ☐ Portals show freshness and uncertainty;
- ☐ AI is bounded by typed Tools, Policy, approval, and outcome observation;
- ☐ security, tenant, facility, channel, and jurisdiction boundaries are enforced;
- ☐ Outcome Unknown and reconciliation exist where needed;
- ☐ corrections preserve history;
- ☐ tests prove invariants and recovery;
- ☐ implementation status is not upgraded without executable evidence.

17. Status Vocabulary

Status	Meaning
VERIFIED_IMPLEMENTED	Executable source and relevant behavior were inspected.
IMPLEMENTED_WITH_GAP	Executable behavior exists with a material governance or completeness gap.
PARTIAL_IMPLEMENTATION	Only part of the target capability is evidenced.
REFERENCE_ARCHITECTURE	Normative target behavior; no claim that it exists today.
PLANNED_NOT_IMPLEMENTED	Explicitly planned but not evidenced as executable.
DOCUMENTATION_ONLY_CLAIM	Documentation or client contract claims behavior without verified backend evidence.
CONFLICT_REQUIRES_CANONICALIZATION	Sources disagree and no final owner has been established.
REQUIRES_REPOSITORY_REVIEW	Necessary executable evidence has not yet been inspected.

CDC, Outbox and Replication

Metadata	Value
Volume	10
Chapter ID	V10-32
Subject	CDC, Outbox and Replication
Status	Generated First-Draft Architecture Skeleton - Domain-Specific Expansion Required
Content maturity	TEMPLATE_DERIVED_REFERENCE_SKELETON unless repository citations prove otherwise
Default implementation classification	REFERENCE_ARCHITECTURE / REQUIRES_REPOSITORY_REVIEW
Accountable owner	Data Architecture Council
Evidence rule	No implementation claim without inspected executable evidence
Repository	chinair88-prog/allinb2c

1. Executive Orientation

This chapter defines how GTOS shall **define enterprise and Domain data ownership, semantics, persistence, quality, lineage, correction, protection, operations, and verification for CDC, Outbox and Replication.**

The specification separates reality, observations, knowledge, Decisions, Intent, execution, and outcomes. A component name, database row, UI label, AI answer, or workflow state does not acquire authority merely because it is visible or technically convenient.

Reality or external outcome

- ≠ Observation
- ≠ Claim
- ≠ derived Perspective
- ≠ Prediction
- ≠ Recommendation
- ≠ authorized Decision
- ≠ execution Intent

1.1 Accountable ownership

The accountable owner is **Data Architecture Council**. Supporting applications, shared infrastructure, integration adapters, analytics, and AI coordinate or present the capability but shall not silently own its authoritative propositions.

1.2 Scope

- create and identify the subject
- validate evidence and policy
- make authorized Decisions
- execute typed Commands
- publish completed-fact Events
- query current and historical perspectives
- correct and reconcile outcomes

1.3 Non-goals

- treating a registered Gradle module, catalog record, README, frontend route, migration plan, or generated report as proof of end-to-end implementation;
- direct cross-Domain database writes;

- generic status values that combine lifecycle, approval, provider, document, payment, dispute, and correction state;
- silent replacement of history;
- AI-created authority or fabricated external outcomes.

2. Ubiquitous Language and Subject Model

Subject	Required interpretation
CDC,OutboxandReplication	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
CDC,OutboxandReplicationVersion	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
CDC,OutboxandReplicationDecision	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
CDC,OutboxandReplicationEvidenceSet	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
CDC,OutboxandReplicationException	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit

Reference aggregate:

```
CDC,OutboxandReplication {
  subjectId
  tenantId
  organizationContext
  sourceIdentityRefs[]
  sourceVersionRefs[]
  lifecycleState
  relationshipRefs[]
  decisionRefs[]
  intentRefs[]
  evidenceSetRef
  knowledgeCutoff
  policyRefs[]
  authorityRef
  jurisdictionRefs[]
  createdAt
  effectiveAt
  updatedAt
  version
  correctionOf?
  correlationId
}
```

2.1 Core invariants

- Stable identity precedes relationship, state, and cross-system correlation.
- Effective time and recorded time remain distinct where business truth changes over time.
- Recommendation, Decision, Intent, execution, and outcome are separate concepts.
- No Portal, report, search index, cache, workflow, or AI Agent becomes a shadow owner of Domain truth.
- Binding submitted, approved, issued, accepted, or signed versions are immutable.
- Corrections append governed facts and preserve the original record.
- External authority and provider outcomes retain their source and qualification.
- Tenant and represented-principal context come from trusted authentication and mandate evaluation.
- Retried writes are idempotent and concurrency guarded.
- Outcome Unknown is explicit and reconciled before risky duplicate execution.

2.2 Required standards

- stable identifiers
- typed contracts
- immutable binding versions
- transactional outbox
- idempotent Commands
- explicit Outcome Unknown
- correction lineage
- machine-verifiable conformance

3. Actors, Roles, and Authority

A conformant design distinguishes:

- the natural person acting;
- the service or AI identity invoking a Tool;
- the organization or principal represented;
- the role or relationship granting context;
- the mandate or delegation granting action authority;
- the Policy and jurisdiction limiting the action;
- the reviewer or approver responsible for a binding Decision.

Authority is evaluated at action time. Authentication alone does not grant business authority. Cached permissions and browser-supplied tenant headers are insufficient for high-impact actions.

4. Lifecycle and State Model

State	Semantics
PROPOSED	Distinct governed condition with entry, exit, timeout, correction and authority semantics
VALIDATION_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
APPROVAL_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
ACTIVE	Distinct governed condition with entry, exit, timeout, correction and authority semantics
SUSPENDED	Distinct governed condition with entry, exit, timeout, correction and authority semantics
CORRECTION_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
RETIRED	Distinct governed condition with entry, exit, timeout, correction and authority semantics

```
stateDiagram-v2
    [*] --> PROPOSED
    PROPOSED --> VALIDATION_PENDING
    VALIDATION_PENDING --> APPROVAL_PENDING
    APPROVAL_PENDING --> ACTIVE
    ACTIVE --> SUSPENDED
    SUSPENDED --> CORRECTION_PENDING
    CORRECTION_PENDING --> RETIRED
```

The diagram is a primary reading path. Real implementations may add parallel review, amendment, appeal, timeout, suspension, partial completion, reconciliation, correction, and terminal failure without collapsing their meaning.

4.1 Transition rules

- transition through a typed, authorized Command;
 - require exact expected version where concurrent change matters;
 - record actor, represented principal, mandate, Policy, reason, and Evidence;
 - use an idempotency key for retried writes;
 - publish a completed-fact Event after durable state change;
 - expose Outcome Unknown for unresolved external completion;
 - preserve original and corrected states in historical reconstruction.
-

5. Decision, Evidence, and Knowledge Cutoff

Reference Decision:

```
Decision {
    decisionId
    decisionType
    subjectRef
}
```

```

sourceVersionRefs[]
knowledgeCutoff
evidenceSetRef
policyVersionRefs[]
recommendationRefs[]
principalId
representedOrganizationId
mandateRef
rationale
outcome
decidedAt
effectiveAt
correctionOf?
}

```

Evidence records identify issuer or observer, subject, version, observed time, received time, effective interval, integrity, confidence, qualification, contradiction, correction, retention, residency, and access class. A recommendation may assist the Decision but cannot promote itself to the Decision.

6. Commands, Events, Queries, and Contracts

Reference Commands:

- CreateCDC,OutboxandReplication
- ValidateCDC,OutboxandReplication
- ApproveCDC,OutboxandReplication
- ActivateCDC,OutboxandReplication
- SuspendCDC,OutboxandReplication
- CorrectCDC,OutboxandReplication
- RetireCDC,OutboxandReplication

Reference Events:

- CDC,OutboxandReplicationCreated
- CDC,OutboxandReplicationValidated
- CDC,OutboxandReplicationApproved
- CDC,OutboxandReplicationActivated
- CDC,OutboxandReplicationSuspended
- CDC,OutboxandReplicationCorrected
- CDC,OutboxandReplicationRetired

Reference queries:

- GetCDC,OutboxandReplication
- ListCDC,OutboxandReplicationRecords
- GetCDC,OutboxandReplicationTimeline
- GetCDC,OutboxandReplicationEvidence
- GetCDC,OutboxandReplicationDecisionPackage
- GetCDC,OutboxandReplicationExceptions
- GetCDC,OutboxandReplicationAuditTrail

Reference API surface:

```

POST /api/v1/cdc_outboxandreplication/commands
GET  /api/v1/cdc_outboxandreplication/{id}
GET  /api/v1/cdc_outboxandreplication/{id}/timeline
GET  /api/v1/cdc_outboxandreplication/{id}/evidence
GET  /api/v1/cdc_outboxandreplication/{id}/decisions
POST /api/v1/cdc_outboxandreplication/{id}/exceptions
POST /api/v1/cdc_outboxandreplication/{id}/corrections

```

These endpoints are reference contracts, not claims that exact routes exist.

7. Workflow, Failure, and Compensation

flowchart TD

```

A[Validate identity version and authority] --> B[Collect required evidence]
B --> C{Policy and evidence gates pass?}
C -- No --> D[Open exception or request correction]
D --> B
C -- Yes --> E[Record Decision or execution Intent]
E --> F[Execute Domain command or external request]

```

```

F --> G{Outcome known?}
G -- Success --> H[Record fact and publish Event]
G -- Failure --> I[Record failure and compensation options]
G -- Unknown --> J[Enter Outcome Unknown]
J --> K[Query source or wait for verified callback]
K --> G
H --> L[Update projections and downstream workflow]

```

Representative failure modes:

- stale or wrong source version
- duplicate submission or replay
- external timeout after possible success
- contradictory source observations
- lost Event after committed state
- approval or mandate revoked during execution
- downstream action before a mandatory gate
- partial completion across multiple subjects
- evidence later retracted or corrected
- tenant, facility, channel or jurisdiction mismatch

Recovery shall query the authoritative source before duplicate execution, preserve partial success, use compensating Intent rather than destructive rewrite, publish corrections, quarantine high-impact ambiguity, and record affected downstream subjects.

8. Data and Persistence Architecture

Reference table families:

Table family	Purpose
cdc_outboxandreplication_aggregate	authoritative subject
cdc_outboxandreplication_version	immutable submitted, approved, issued, or signed versions
cdc_outboxandreplication_relationship	governed relationships
cdc_outboxandreplication_decision	binding Decisions and rationale
cdc_outboxandreplication_evidence_link	provenance and evidence relationships
cdc_outboxandreplication_event_outbox	transactional Event publication
cdc_outboxandreplication_idempotency	duplicate-write protection
cdc_outboxandreplication_exception	exception, claim, dispute, or hold
cdc_outboxandreplication_correction	correction and supersession lineage
cdc_outboxandreplication_audit	privileged action evidence

The owning service controls schema and migration. Cross-Domain references use stable identifiers or contracts. Analytics, reports, caches, search, and vector indexes remain projections. Backup, restore, retention, legal hold, privacy, residency, and decommissioning requirements are explicit.

9. Portal and Experience Requirements

- Primary participant Portal: initiate, review exact versions, supply evidence, and respond to required actions.
- Operator Portal: inspect exceptions, correlation, source responses, retries, compensation, and downstream impact.
- Executive or oversight view: consume governed metrics without becoming a source of Domain truth.

Every Portal displays exact version, owner, state, freshness, required action, and uncertainty. It supports loading, empty, partial, stale, denied, error, timeout, and Outcome Unknown states; accessibility; locale, currency, unit, date, and RTL/LTR correctness; evidence audit; safe retry; and correction history.

10. AI Participation and Tool Governance

Permitted AI tasks:

- assist schema and lineage analysis
- detect data-quality anomalies
- summarize migration and reconciliation risk
- never mutate authoritative data outside typed Domain Tools

AI shall not own Domain records, grant itself authority, suppress contradictions, silently edit approved versions, fabricate external outcomes, or execute high-impact actions outside typed Tools, policy, approval, idempotency, sandbox, and outcome observation.

Every Agent Run records source context, prompt/model/tool versions, structured output, confidence and limitations, Policy result, approval, Tool calls, provider responses, final outcome, and recovery actions.

11. Security, Privacy, and Trust Boundaries

- least privilege and separation of duties
- tenant, organization, facility, channel and jurisdiction isolation
- integrity protection for binding evidence
- verified external callbacks and anti-replay controls

Additional controls include secret isolation, callback signature verification, audit of privileged action, emergency-access review, sensitive-data minimization, purpose limitation, and threat models for impersonation, tampering, replay, confused deputy, fraud, prompt injection, and exfiltration.

12. Reliability, SLOs, and Operations

SLI	Reference objective
authoritative read availability	99.9% monthly unless a stricter tier applies
acknowledged Command durability	no acknowledged write may be silently lost
Event publication	99.9% within five minutes after committed fact
duplicate prevention	100% for same tenant, Command class, and idempotency key
binding Decision audit completeness	100%
state-state detection	bounded by business deadline

Dashboards and runbooks cover backlog, error budget, dependency failure, retry budget, DLQ, workflow stall, provider outage, reconciliation, capacity, backup/restore, release/rollback, incident command, and post-incident correction.

13. Testing and Continuous Conformance

- aggregate invariant and transition tests
- authorization, delegation, separation-of-duties and tenant-isolation tests
- idempotency and optimistic-concurrency tests
- database migration, key, constraint and rollback tests
- contract, outbox/inbox, duplicate, replay and DLQ tests
- workflow restart, timeout and compensation tests
- Portal E2E tests for success, partial, stale, denied, error and Outcome Unknown states
- accessibility, locale, currency, unit and RTL/LTR tests
- security adversarial, fraud and data-exfiltration tests
- AI evaluation, prompt-injection, policy-bypass and unauthorized-tool tests
- historical reconstruction, correction and audit-completeness tests
- performance, capacity, recovery and operational exercise tests

Release evidence links each invariant and control to the test, environment, input data, result, owner, and artifact proving it. Generated test counts never replace semantic coverage.

14. Repository Review Boundary

- This chapter is a repository-review specification. Exact executable implementation has not been inspected for every target behavior.
- Registered modules, catalog entries, README claims, frontend contracts, or plans are not proof of complete implementation.

Area	Classification
target aggregate, API, tables, state and workflow in this chapter	REFERENCE_ARCHITECTURE
registered module or catalog claim	DOCUMENTATION_ONLY_CLAIM until source inspection
uninspected integration	REQUIRES_REPOSITORY_REVIEW
behavior proven in another repository-grounded chapter	retain that chapter's evidence and classification
contradictory ownership or path claims	CONFLICT_REQUIRES_CANONICALIZATION

No statement upgrades implementation status merely because a file, directory, module name, frontend contract, database entry, plan, or report exists.

15. Worked Conformance Scenario

An authorized principal initiates a Command against an exact subject version. The service validates identity, tenant, mandate, Policy, Evidence, and idempotency. It records the state transition and outbox entry atomically. If an external dependency participates, timeout produces Outcome Unknown; reconciliation queries the source or waits for a verified callback instead of repeating the action blindly.

Portals show source, freshness, version, required action, and uncertainty. AI may extract, compare, summarize, detect anomalies, and recommend through typed Tools, but cannot own the subject or fabricate success. Later correction appends new facts, identifies affected projections and workflows, and preserves the historical Decision context.

16. Conformance Checklist

- ☐ stable identity and proposition ownership are explicit;
- ☐ authority and mandate are evaluated at action time;
- ☐ state, Recommendation, Decision, Intent, execution, and outcome remain distinct;
- ☐ Commands are typed, authorized, idempotent, and version guarded;
- ☐ Events describe completed facts;
- ☐ Evidence includes provenance, time, integrity, qualification, and correction;
- ☐ external outcomes retain their authority source;
- ☐ Portals show freshness and uncertainty;
- ☐ AI is bounded by typed Tools, Policy, approval, and outcome observation;
- ☐ security, tenant, facility, channel, and jurisdiction boundaries are enforced;
- ☐ Outcome Unknown and reconciliation exist where needed;
- ☐ corrections preserve history;
- ☐ tests prove invariants and recovery;
- ☐ implementation status is not upgraded without executable evidence.

17. Status Vocabulary

Status	Meaning
VERIFIED_IMPLEMENTED	Executable source and relevant behavior were inspected.
IMPLEMENTED_WITH_GAP	Executable behavior exists with a material governance or completeness gap.
PARTIAL_IMPLEMENTATION	Only part of the target capability is evidenced.
REFERENCE_ARCHITECTURE	Normative target behavior; no claim that it exists today.
PLANNED_NOT_IMPLEMENTED	Explicitly planned but not evidenced as executable.
DOCUMENTATION_ONLY_CLAIM	Documentation or client contract claims behavior without verified backend evidence.
CONFLICT_REQUIRES_CANONICALIZATION	Sources disagree and no final owner has been established.
REQUIRES_REPOSITORY_REVIEW	Necessary executable evidence has not yet been inspected.

Master and Reference Data

Metadata	Value
Volume	10
Chapter ID	V10-33
Subject	MasterandReferenceData
Status	Generated First-Draft Architecture Skeleton - Domain-Specific Expansion Required
Content maturity	TEMPLATE_DERIVED_REFERENCE_SKELETON unless repository citations prove otherwise
Default implementation classification	REFERENCE_ARCHITECTURE / REQUIRES_REPOSITORY_REVIEW
Accountable owner	Data Architecture Council
Evidence rule	No implementation claim without inspected executable evidence
Repository	chinair88-prog/allinb2c

1. Executive Orientation

This chapter defines how GTOS shall **define enterprise and Domain data ownership, semantics, persistence, quality, lineage, correction, protection, operations, and verification for Master and Reference Data.**

The specification separates reality, observations, knowledge, Decisions, Intent, execution, and outcomes. A component name, database row, UI label, AI answer, or workflow state does not acquire authority merely because it is visible or technically convenient.

Reality or external outcome

- ≠ Observation
- ≠ Claim
- ≠ derived Perspective
- ≠ Prediction
- ≠ Recommendation
- ≠ authorized Decision
- ≠ execution Intent

1.1 Accountable ownership

The accountable owner is **Data Architecture Council**. Supporting applications, shared infrastructure, integration adapters, analytics, and AI coordinate or present the capability but shall not silently own its authoritative propositions.

1.2 Scope

- create and identify the subject
- validate evidence and policy
- make authorized Decisions
- execute typed Commands
- publish completed-fact Events
- query current and historical perspectives
- correct and reconcile outcomes

1.3 Non-goals

- treating a registered Gradle module, catalog record, README, frontend route, migration plan, or generated report as proof of end-to-end implementation;
- direct cross-Domain database writes;
- generic status values that combine lifecycle, approval, provider, document, payment, dispute, and correction state;

- silent replacement of history;
- AI-created authority or fabricated external outcomes.

2. Ubiquitous Language and Subject Model

Subject	Required interpretation
MasterandReferenceData	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
MasterandReferenceDataVersion	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
MasterandReferenceDataDecision	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
MasterandReferenceDataEvidenceSet	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
MasterandReferenceDataException	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit

Reference aggregate:

```
MasterandReferenceData {
  subjectId
  tenantId
  organizationContext
  sourceIdentityRefs[]
  sourceVersionRefs[]
  lifecycleState
  relationshipRefs[]
  decisionRefs[]
  intentRefs[]
  evidenceSetRef
  knowledgeCutoff
  policyRefs[]
  authorityRef
  jurisdictionRefs[]
  createdAt
  effectiveAt
  updatedAt
  version
  correctionOf?
  correlationId
}
```

2.1 Core invariants

- Stable identity precedes relationship, state, and cross-system correlation.
- Effective time and recorded time remain distinct where business truth changes over time.
- Recommendation, Decision, Intent, execution, and outcome are separate concepts.
- No Portal, report, search index, cache, workflow, or AI Agent becomes a shadow owner of Domain truth.
- Binding submitted, approved, issued, accepted, or signed versions are immutable.
- Corrections append governed facts and preserve the original record.
- External authority and provider outcomes retain their source and qualification.
- Tenant and represented-principal context come from trusted authentication and mandate evaluation.
- Retried writes are idempotent and concurrency guarded.
- Outcome Unknown is explicit and reconciled before risky duplicate execution.

2.2 Required standards

- stable identifiers
- typed contracts
- immutable binding versions
- transactional outbox
- idempotent Commands
- explicit Outcome Unknown
- correction lineage
- machine-verifiable conformance

3. Actors, Roles, and Authority

A conformant design distinguishes:

- the natural person acting;
- the service or AI identity invoking a Tool;
- the organization or principal represented;
- the role or relationship granting context;
- the mandate or delegation granting action authority;
- the Policy and jurisdiction limiting the action;
- the reviewer or approver responsible for a binding Decision.

Authority is evaluated at action time. Authentication alone does not grant business authority. Cached permissions and browser-supplied tenant headers are insufficient for high-impact actions.

4. Lifecycle and State Model

State	Semantics
PROPOSED	Distinct governed condition with entry, exit, timeout, correction and authority semantics
VALIDATION_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
APPROVAL_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
ACTIVE	Distinct governed condition with entry, exit, timeout, correction and authority semantics
SUSPENDED	Distinct governed condition with entry, exit, timeout, correction and authority semantics
CORRECTION_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
RETIRED	Distinct governed condition with entry, exit, timeout, correction and authority semantics

```
stateDiagram-v2
    [*] --> PROPOSED
    PROPOSED --> VALIDATION_PENDING
    VALIDATION_PENDING --> APPROVAL_PENDING
    APPROVAL_PENDING --> ACTIVE
    ACTIVE --> SUSPENDED
    SUSPENDED --> CORRECTION_PENDING
    CORRECTION_PENDING --> RETIRED
```

The diagram is a primary reading path. Real implementations may add parallel review, amendment, appeal, timeout, suspension, partial completion, reconciliation, correction, and terminal failure without collapsing their meaning.

4.1 Transition rules

- transition through a typed, authorized Command;
- require exact expected version where concurrent change matters;
- record actor, represented principal, mandate, Policy, reason, and Evidence;
- use an idempotency key for retried writes;
- publish a completed-fact Event after durable state change;
- expose Outcome Unknown for unresolved external completion;
- preserve original and corrected states in historical reconstruction.

5. Decision, Evidence, and Knowledge Cutoff

Reference Decision:

```
Decision {
    decisionId
    decisionType
    subjectRef
    sourceVersionRefs[]
    knowledgeCutoff
    evidenceSetRef
}
```

```

policyVersionRefs[]
recommendationRefs[]
principalId
representedOrganizationId
mandateRef
rationale
outcome
decidedAt
effectiveAt
correctionOf?
}

```

Evidence records identify issuer or observer, subject, version, observed time, received time, effective interval, integrity, confidence, qualification, contradiction, correction, retention, residency, and access class. A recommendation may assist the Decision but cannot promote itself to the Decision.

6. Commands, Events, Queries, and Contracts

Reference Commands:

- CreateMasterandReferenceData
- ValidateMasterandReferenceData
- ApproveMasterandReferenceData
- ActivateMasterandReferenceData
- SuspendMasterandReferenceData
- CorrectMasterandReferenceData
- RetireMasterandReferenceData

Reference Events:

- MasterandReferenceDataCreated
- MasterandReferenceDataValidated
- MasterandReferenceDataApproved
- MasterandReferenceDataActivated
- MasterandReferenceDataSuspended
- MasterandReferenceDataCorrected
- MasterandReferenceDataRetired

Reference queries:

- GetMasterandReferenceData
- ListMasterandReferenceDataRecords
- GetMasterandReferenceDataTimeline
- GetMasterandReferenceDataEvidence
- GetMasterandReferenceDataDecisionPackage
- GetMasterandReferenceDataExceptions
- GetMasterandReferenceDataAuditTrail

Reference API surface:

```

POST /api/v1/masterandreferencedata/commands
GET  /api/v1/masterandreferencedata/{id}
GET  /api/v1/masterandreferencedata/{id}/timeline
GET  /api/v1/masterandreferencedata/{id}/evidence
GET  /api/v1/masterandreferencedata/{id}/decisions
POST /api/v1/masterandreferencedata/{id}/exceptions
POST /api/v1/masterandreferencedata/{id}/corrections

```

These endpoints are reference contracts, not claims that exact routes exist.

7. Workflow, Failure, and Compensation

flowchart TD

```

A[Validate identity version and authority] --> B[Collect required evidence]
B --> C{Policy and evidence gates pass?}
C -- No --> D[Open exception or request correction]
D --> B
C -- Yes --> E[Record Decision or execution Intent]
E --> F[Execute Domain command or external request]
F --> G{Outcome known?}
G -- Success --> H[Record fact and publish Event]
G -- Failure --> I[Record failure and compensation options]

```

```

G -- Unknown --> J[Enter Outcome Unknown]
J --> K[Query source or wait for verified callback]
K --> G
H --> L[Update projections and downstream workflow]

```

Representative failure modes:

- stale or wrong source version
- duplicate submission or replay
- external timeout after possible success
- contradictory source observations
- lost Event after committed state
- approval or mandate revoked during execution
- downstream action before a mandatory gate
- partial completion across multiple subjects
- evidence later retracted or corrected
- tenant, facility, channel or jurisdiction mismatch

Recovery shall query the authoritative source before duplicate execution, preserve partial success, use compensating Intent rather than destructive rewrite, publish corrections, quarantine high-impact ambiguity, and record affected downstream subjects.

8. Data and Persistence Architecture

Reference table families:

Table family	Purpose
masterandreferencedata_aggregate	authoritative subject
masterandreferencedata_version	immutable submitted, approved, issued, or signed versions
masterandreferencedata_relationship	governed relationships
masterandreferencedata_decision	binding Decisions and rationale
masterandreferencedata_evidence_link	provenance and evidence relationships
masterandreferencedata_event_outbox	transactional Event publication
masterandreferencedata_idempotency	duplicate-write protection
masterandreferencedata_exception	exception, claim, dispute, or hold
masterandreferencedata_correction	correction and supersession lineage
masterandreferencedata_audit	privileged action evidence

The owning service controls schema and migration. Cross-Domain references use stable identifiers or contracts. Analytics, reports, caches, search, and vector indexes remain projections. Backup, restore, retention, legal hold, privacy, residency, and decommissioning requirements are explicit.

9. Portal and Experience Requirements

- Primary participant Portal: initiate, review exact versions, supply evidence, and respond to required actions.
- Operator Portal: inspect exceptions, correlation, source responses, retries, compensation, and downstream impact.
- Executive or oversight view: consume governed metrics without becoming a source of Domain truth.

Every Portal displays exact version, owner, state, freshness, required action, and uncertainty. It supports loading, empty, partial, stale, denied, error, timeout, and Outcome Unknown states; accessibility; locale, currency, unit, date, and RTL/LTR correctness; evidence audit; safe retry; and correction history.

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Permitted AI tasks:

- assist schema and lineage analysis
- detect data-quality anomalies
- summarize migration and reconciliation risk
- never mutate authoritative data outside typed Domain Tools

AI shall not own Domain records, grant itself authority, suppress contradictions, silently edit approved versions, fabricate external outcomes, or execute high-impact actions outside typed Tools, policy, approval, idempotency, sandbox, and outcome observation.

Every Agent Run records source context, prompt/model/tool versions, structured output, confidence and limitations, Policy result, approval, Tool calls, provider responses, final outcome, and recovery actions.

11. Security, Privacy, and Trust Boundaries

- least privilege and separation of duties
- tenant, organization, facility, channel and jurisdiction isolation
- integrity protection for binding evidence
- verified external callbacks and anti-replay controls

Additional controls include secret isolation, callback signature verification, audit of privileged action, emergency-access review, sensitive-data minimization, purpose limitation, and threat models for impersonation, tampering, replay, confused deputy, fraud, prompt injection, and exfiltration.

12. Reliability, SLOs, and Operations

SLI	Reference objective
authoritative read availability	99.9% monthly unless a stricter tier applies
acknowledged Command durability	no acknowledged write may be silently lost
Event publication	99.9% within five minutes after committed fact
duplicate prevention	100% for same tenant, Command class, and idempotency key
binding Decision audit completeness	100%
stale-state detection	bounded by business deadline

Dashboards and runbooks cover backlog, error budget, dependency failure, retry budget, DLQ, workflow stall, provider outage, reconciliation, capacity, backup/restore, release/rollback, incident command, and post-incident correction.

13. Testing and Continuous Conformance

- aggregate invariant and transition tests
- authorization, delegation, separation-of-duties and tenant-isolation tests
- idempotency and optimistic-concurrency tests
- database migration, key, constraint and rollback tests
- contract, outbox/inbox, duplicate, replay and DLQ tests
- workflow restart, timeout and compensation tests
- Portal E2E tests for success, partial, stale, denied, error and Outcome Unknown states
- accessibility, locale, currency, unit and RTL/LTR tests
- security adversarial, fraud and data-exfiltration tests
- AI evaluation, prompt-injection, policy-bypass and unauthorized-tool tests
- historical reconstruction, correction and audit-completeness tests
- performance, capacity, recovery and operational exercise tests

Release evidence links each invariant and control to the test, environment, input data, result, owner, and artifact proving it. Generated test counts never replace semantic coverage.

14. Repository Review Boundary

- This chapter is a repository-review specification. Exact executable implementation has not been inspected for every target behavior.
- Registered modules, catalog entries, README claims, frontend contracts, or plans are not proof of complete implementation.

Area	Classification
target aggregate, API, tables, state and workflow in this chapter	REFERENCE_ARCHITECTURE
registered module or catalog claim	DOCUMENTATION_ONLY_CLAIM until source inspection
uninspected integration	REQUIRES_REPOSITORY_REVIEW

Area	Classification
behavior proven in another repository-grounded chapter contradictory ownership or path claims	retain that chapter's evidence and classification CONFLICT_REQUIRES_CANONICALIZATION

No statement upgrades implementation status merely because a file, directory, module name, frontend contract, database entry, plan, or report exists.

15. Worked Conformance Scenario

An authorized principal initiates a Command against an exact subject version. The service validates identity, tenant, mandate, Policy, Evidence, and idempotency. It records the state transition and outbox entry atomically. If an external dependency participates, timeout produces Outcome Unknown; reconciliation queries the source or waits for a verified callback instead of repeating the action blindly.

Portals show source, freshness, version, required action, and uncertainty. AI may extract, compare, summarize, detect anomalies, and recommend through typed Tools, but cannot own the subject or fabricate success. Later correction appends new facts, identifies affected projections and workflows, and preserves the historical Decision context.

16. Conformance Checklist

- ☐ stable identity and proposition ownership are explicit;
- ☐ authority and mandate are evaluated at action time;
- ☐ state, Recommendation, Decision, Intent, execution, and outcome remain distinct;
- ☐ Commands are typed, authorized, idempotent, and version guarded;
- ☐ Events describe completed facts;
- ☐ Evidence includes provenance, time, integrity, qualification, and correction;
- ☐ external outcomes retain their authority source;
- ☐ Portals show freshness and uncertainty;
- ☐ AI is bounded by typed Tools, Policy, approval, and outcome observation;
- ☐ security, tenant, facility, channel, and jurisdiction boundaries are enforced;
- ☐ Outcome Unknown and reconciliation exist where needed;
- ☐ corrections preserve history;
- ☐ tests prove invariants and recovery;
- ☐ implementation status is not upgraded without executable evidence.

17. Status Vocabulary

Status	Meaning
VERIFIED_IMPLEMENTED	Executable source and relevant behavior were inspected.
IMPLEMENTED_WITH_GAP	Executable behavior exists with a material governance or completeness gap.
PARTIAL_IMPLEMENTATION	Only part of the target capability is evidenced.
REFERENCE_ARCHITECTURE	Normative target behavior; no claim that it exists today.
PLANNED_NOT_IMPLEMENTED	Explicitly planned but not evidenced as executable.
DOCUMENTATION_ONLY_CLAIM	Documentation or client contract claims behavior without verified backend evidence.
CONFLICT_REQUIRES_CANONICALIZATION	Sources disagree and no final owner has been established.
REQUIRES_REPOSITORY_REVIEW	Necessary executable evidence has not yet been inspected.

Data Observability and Reconciliation

Metadata	Value
Volume	10
Chapter ID	V10-34
Subject	DataObservabilityandReconciliation
Status	Generated First-Draft Architecture Skeleton - Domain-Specific Expansion Required
Content maturity	TEMPLATE_DERIVED_REFERENCE_SKELETON unless repository citations prove otherwise
Default implementation classification	REFERENCE_ARCHITECTURE / REQUIRES_REPOSITORY_REVIEW
Accountable owner	Data Architecture Council
Evidence rule	No implementation claim without inspected executable evidence
Repository	chinair88-prog/allinb2c

1. Executive Orientation

This chapter defines how GTOS shall **define enterprise and Domain data ownership, semantics, persistence, quality, lineage, correction, protection, operations, and verification for Data Observability and Reconciliation**.

The specification separates reality, observations, knowledge, Decisions, Intent, execution, and outcomes. A component name, database row, UI label, AI answer, or workflow state does not acquire authority merely because it is visible or technically convenient.

Reality or external outcome

- ≠ Observation
- ≠ Claim
- ≠ derived Perspective
- ≠ Prediction
- ≠ Recommendation
- ≠ authorized Decision
- ≠ execution Intent

1.1 Accountable ownership

The accountable owner is **Data Architecture Council**. Supporting applications, shared infrastructure, integration adapters, analytics, and AI coordinate or present the capability but shall not silently own its authoritative propositions.

1.2 Scope

- create and identify the subject
- validate evidence and policy
- make authorized Decisions
- execute typed Commands
- publish completed-fact Events
- query current and historical perspectives
- correct and reconcile outcomes

1.3 Non-goals

- treating a registered Gradle module, catalog record, README, frontend route, migration plan, or generated report as proof of end-to-end implementation;
- direct cross-Domain database writes;

- generic status values that combine lifecycle, approval, provider, document, payment, dispute, and correction state;
- silent replacement of history;
- AI-created authority or fabricated external outcomes.

2. Ubiquitous Language and Subject Model

Subject	Required interpretation
DataObservabilityandReconciliation	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
DataObservabilityandReconciliationVersion	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
DataObservabilityandReconciliationDecision	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
DataObservabilityandReconciliationEvidenceSet	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
DataObservabilityandReconciliationException	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit

Reference aggregate:

```
DataObservabilityandReconciliation {
  subjectId
  tenantId
  organizationContext
  sourceIdentityRefs[]
  sourceVersionRefs[]
  lifecycleState
  relationshipRefs[]
  decisionRefs[]
  intentRefs[]
  evidenceSetRef
  knowledgeCutoff
  policyRefs[]
  authorityRef
  jurisdictionRefs[]
  createdAt
  effectiveAt
  updatedAt
  version
  correctionOf?
  correlationId
}
```

2.1 Core invariants

- Stable identity precedes relationship, state, and cross-system correlation.
- Effective time and recorded time remain distinct where business truth changes over time.
- Recommendation, Decision, Intent, execution, and outcome are separate concepts.
- No Portal, report, search index, cache, workflow, or AI Agent becomes a shadow owner of Domain truth.
- Binding submitted, approved, issued, accepted, or signed versions are immutable.
- Corrections append governed facts and preserve the original record.
- External authority and provider outcomes retain their source and qualification.
- Tenant and represented-principal context come from trusted authentication and mandate evaluation.
- Retried writes are idempotent and concurrency guarded.
- Outcome Unknown is explicit and reconciled before risky duplicate execution.

2.2 Required standards

- stable identifiers
- typed contracts
- immutable binding versions
- transactional outbox
- idempotent Commands
- explicit Outcome Unknown
- correction lineage
- machine-verifiable conformance

3. Actors, Roles, and Authority

A conformant design distinguishes:

- the natural person acting;
- the service or AI identity invoking a Tool;
- the organization or principal represented;
- the role or relationship granting context;
- the mandate or delegation granting action authority;
- the Policy and jurisdiction limiting the action;
- the reviewer or approver responsible for a binding Decision.

Authority is evaluated at action time. Authentication alone does not grant business authority. Cached permissions and browser-supplied tenant headers are insufficient for high-impact actions.

4. Lifecycle and State Model

State	Semantics
PROPOSED	Distinct governed condition with entry, exit, timeout, correction and authority semantics
VALIDATION_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
APPROVAL_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
ACTIVE	Distinct governed condition with entry, exit, timeout, correction and authority semantics
SUSPENDED	Distinct governed condition with entry, exit, timeout, correction and authority semantics
CORRECTION_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
RETIRED	Distinct governed condition with entry, exit, timeout, correction and authority semantics

```
stateDiagram-v2
    [*] --> PROPOSED
    PROPOSED --> VALIDATION_PENDING
    VALIDATION_PENDING --> APPROVAL_PENDING
    APPROVAL_PENDING --> ACTIVE
    ACTIVE --> SUSPENDED
    SUSPENDED --> CORRECTION_PENDING
    CORRECTION_PENDING --> RETIRED
```

The diagram is a primary reading path. Real implementations may add parallel review, amendment, appeal, timeout, suspension, partial completion, reconciliation, correction, and terminal failure without collapsing their meaning.

4.1 Transition rules

- transition through a typed, authorized Command;
 - require exact expected version where concurrent change matters;
 - record actor, represented principal, mandate, Policy, reason, and Evidence;
 - use an idempotency key for retried writes;
 - publish a completed-fact Event after durable state change;
 - expose Outcome Unknown for unresolved external completion;
 - preserve original and corrected states in historical reconstruction.
-

5. Decision, Evidence, and Knowledge Cutoff

Reference Decision:

```
Decision {
    decisionId
    decisionType
    subjectRef
}
```

```

sourceVersionRefs[]
knowledgeCutoff
evidenceSetRef
policyVersionRefs[]
recommendationRefs[]
principalId
representedOrganizationId
mandateRef
rationale
outcome
decidedAt
effectiveAt
correctionOf?
}

```

Evidence records identify issuer or observer, subject, version, observed time, received time, effective interval, integrity, confidence, qualification, contradiction, correction, retention, residency, and access class. A recommendation may assist the Decision but cannot promote itself to the Decision.

6. Commands, Events, Queries, and Contracts

Reference Commands:

- CreateDataObservabilityandReconciliation
- ValidateDataObservabilityandReconciliation
- ApproveDataObservabilityandReconciliation
- ActivateDataObservabilityandReconciliation
- SuspendDataObservabilityandReconciliation
- CorrectDataObservabilityandReconciliation
- RetireDataObservabilityandReconciliation

Reference Events:

- DataObservabilityandReconciliationCreated
- DataObservabilityandReconciliationValidated
- DataObservabilityandReconciliationApproved
- DataObservabilityandReconciliationActivated
- DataObservabilityandReconciliationSuspended
- DataObservabilityandReconciliationCorrected
- DataObservabilityandReconciliationRetired

Reference queries:

- GetDataObservabilityandReconciliation
- ListDataObservabilityandReconciliationRecords
- GetDataObservabilityandReconciliationTimeline
- GetDataObservabilityandReconciliationEvidence
- GetDataObservabilityandReconciliationDecisionPackage
- GetDataObservabilityandReconciliationExceptions
- GetDataObservabilityandReconciliationAuditTrail

Reference API surface:

```

POST /api/v1/dataobservabilityandreconciliation/commands
GET  /api/v1/dataobservabilityandreconciliation/{id}
GET  /api/v1/dataobservabilityandreconciliation/{id}/timeline
GET  /api/v1/dataobservabilityandreconciliation/{id}/evidence
GET  /api/v1/dataobservabilityandreconciliation/{id}/decisions
POST /api/v1/dataobservabilityandreconciliation/{id}/exceptions
POST /api/v1/dataobservabilityandreconciliation/{id}/corrections

```

These endpoints are reference contracts, not claims that exact routes exist.

7. Workflow, Failure, and Compensation

flowchart TD

```

A[Validate identity version and authority] --> B[Collect required evidence]
B --> C{Policy and evidence gates pass?}
C -- No --> D[Open exception or request correction]
D --> B
C -- Yes --> E[Record Decision or execution Intent]
E --> F[Execute Domain command or external request]

```

```

F --> G{Outcome known?}
G -- Success --> H[Record fact and publish Event]
G -- Failure --> I[Record failure and compensation options]
G -- Unknown --> J[Enter Outcome Unknown]
J --> K[Query source or wait for verified callback]
K --> G
H --> L[Update projections and downstream workflow]

```

Representative failure modes:

- stale or wrong source version
- duplicate submission or replay
- external timeout after possible success
- contradictory source observations
- lost Event after committed state
- approval or mandate revoked during execution
- downstream action before a mandatory gate
- partial completion across multiple subjects
- evidence later retracted or corrected
- tenant, facility, channel or jurisdiction mismatch

Recovery shall query the authoritative source before duplicate execution, preserve partial success, use compensating Intent rather than destructive rewrite, publish corrections, quarantine high-impact ambiguity, and record affected downstream subjects.

8. Data and Persistence Architecture

Reference table families:

Table family	Purpose
dataobservabilityandreconciliation_aggregate	authoritative subject
dataobservabilityandreconciliation_version	immutable submitted, approved, issued, or signed versions
dataobservabilityandreconciliation_relationship	governed relationships
dataobservabilityandreconciliation_decision	binding Decisions and rationale
dataobservabilityandreconciliation_evidence_link	provenance and evidence relationships
dataobservabilityandreconciliation_event_outbox	transactional Event publication
dataobservabilityandreconciliation_idempotency	duplicate-write protection
dataobservabilityandreconciliation_exception	exception, claim, dispute, or hold
dataobservabilityandreconciliation_correction	correction and supersession lineage
dataobservabilityandreconciliation_audit	privileged action evidence

The owning service controls schema and migration. Cross-Domain references use stable identifiers or contracts. Analytics, reports, caches, search, and vector indexes remain projections. Backup, restore, retention, legal hold, privacy, residency, and decommissioning requirements are explicit.

9. Portal and Experience Requirements

- Primary participant Portal: initiate, review exact versions, supply evidence, and respond to required actions.
- Operator Portal: inspect exceptions, correlation, source responses, retries, compensation, and downstream impact.
- Executive or oversight view: consume governed metrics without becoming a source of Domain truth.

Every Portal displays exact version, owner, state, freshness, required action, and uncertainty. It supports loading, empty, partial, stale, denied, error, timeout, and Outcome Unknown states; accessibility; locale, currency, unit, date, and RTL/LTR correctness; evidence audit; safe retry; and correction history.

10. AI Participation and Tool Governance

Permitted AI tasks:

- assist schema and lineage analysis
- detect data-quality anomalies
- summarize migration and reconciliation risk
- never mutate authoritative data outside typed Domain Tools

AI shall not own Domain records, grant itself authority, suppress contradictions, silently edit approved versions, fabricate external outcomes, or execute high-impact actions outside typed Tools, policy, approval, idempotency, sandbox, and outcome observation.

Every Agent Run records source context, prompt/model/tool versions, structured output, confidence and limitations, Policy result, approval, Tool calls, provider responses, final outcome, and recovery actions.

11. Security, Privacy, and Trust Boundaries

- least privilege and separation of duties
- tenant, organization, facility, channel and jurisdiction isolation
- integrity protection for binding evidence
- verified external callbacks and anti-replay controls

Additional controls include secret isolation, callback signature verification, audit of privileged action, emergency-access review, sensitive-data minimization, purpose limitation, and threat models for impersonation, tampering, replay, confused deputy, fraud, prompt injection, and exfiltration.

12. Reliability, SLOs, and Operations

SLI	Reference objective
authoritative read availability	99.9% monthly unless a stricter tier applies
acknowledged Command durability	no acknowledged write may be silently lost
Event publication	99.9% within five minutes after committed fact
duplicate prevention	100% for same tenant, Command class, and idempotency key
binding Decision audit completeness	100%
stale-state detection	bounded by business deadline

Dashboards and runbooks cover backlog, error budget, dependency failure, retry budget, DLQ, workflow stall, provider outage, reconciliation, capacity, backup/restore, release/rollback, incident command, and post-incident correction.

13. Testing and Continuous Conformance

- aggregate invariant and transition tests
- authorization, delegation, separation-of-duties and tenant-isolation tests
- idempotency and optimistic-concurrency tests
- database migration, key, constraint and rollback tests
- contract, outbox/inbox, duplicate, replay and DLQ tests
- workflow restart, timeout and compensation tests
- Portal E2E tests for success, partial, stale, denied, error and Outcome Unknown states
- accessibility, locale, currency, unit and RTL/LTR tests
- security adversarial, fraud and data-exfiltration tests
- AI evaluation, prompt-injection, policy-bypass and unauthorized-tool tests
- historical reconstruction, correction and audit-completeness tests
- performance, capacity, recovery and operational exercise tests

Release evidence links each invariant and control to the test, environment, input data, result, owner, and artifact proving it. Generated test counts never replace semantic coverage.

14. Repository Review Boundary

- This chapter is a repository-review specification. Exact executable implementation has not been inspected for every target behavior.
- Registered modules, catalog entries, README claims, frontend contracts, or plans are not proof of complete implementation.

Area	Classification
target aggregate, API, tables, state and workflow in this chapter	REFERENCE_ARCHITECTURE
registered module or catalog claim	DOCUMENTATION_ONLY_CLAIM until source inspection
uninspected integration	REQUIRES_REPOSITORY_REVIEW
behavior proven in another repository-grounded chapter	retain that chapter's evidence and classification
contradictory ownership or path claims	CONFLICT_REQUIRES_CANONICALIZATION

No statement upgrades implementation status merely because a file, directory, module name, frontend contract, database entry, plan, or report exists.

15. Worked Conformance Scenario

An authorized principal initiates a Command against an exact subject version. The service validates identity, tenant, mandate, Policy, Evidence, and idempotency. It records the state transition and outbox entry atomically. If an external dependency participates, timeout produces Outcome Unknown; reconciliation queries the source or waits for a verified callback instead of repeating the action blindly.

Portals show source, freshness, version, required action, and uncertainty. AI may extract, compare, summarize, detect anomalies, and recommend through typed Tools, but cannot own the subject or fabricate success. Later correction appends new facts, identifies affected projections and workflows, and preserves the historical Decision context.

16. Conformance Checklist

- ☐ stable identity and proposition ownership are explicit;
- ☐ authority and mandate are evaluated at action time;
- ☐ state, Recommendation, Decision, Intent, execution, and outcome remain distinct;
- ☐ Commands are typed, authorized, idempotent, and version guarded;
- ☐ Events describe completed facts;
- ☐ Evidence includes provenance, time, integrity, qualification, and correction;
- ☐ external outcomes retain their authority source;
- ☐ Portals show freshness and uncertainty;
- ☐ AI is bounded by typed Tools, Policy, approval, and outcome observation;
- ☐ security, tenant, facility, channel, and jurisdiction boundaries are enforced;
- ☐ Outcome Unknown and reconciliation exist where needed;
- ☐ corrections preserve history;
- ☐ tests prove invariants and recovery;
- ☐ implementation status is not upgraded without executable evidence.

17. Status Vocabulary

Status	Meaning
VERIFIED_IMPLEMENTED	Executable source and relevant behavior were inspected.
IMPLEMENTED_WITH_GAP	Executable behavior exists with a material governance or completeness gap.
PARTIAL_IMPLEMENTATION	Only part of the target capability is evidenced.
REFERENCE_ARCHITECTURE	Normative target behavior; no claim that it exists today.
PLANNED_NOT_IMPLEMENTED	Explicitly planned but not evidenced as executable.
DOCUMENTATION_ONLY_CLAIM	Documentation or client contract claims behavior without verified backend evidence.
CONFLICT_REQUIRES_CANONICALIZATION	Sources disagree and no final owner has been established.
REQUIRES_REPOSITORY_REVIEW	Necessary executable evidence has not yet been inspected.

Database Ownership and Data Lineage Matrix

Metadata	Value
Volume	10
Chapter ID	V10-SYNTH
Subject	DatabaseOwnershipMatrix
Status	Generated First-Draft Architecture Skeleton - Domain-Specific Expansion Required
Content maturity	TEMPLATE_DERIVED_REFERENCE_SKELETON unless repository citations prove otherwise
Default implementation classification	REFERENCE_ARCHITECTURE / REQUIRES_REPOSITORY_REVIEW
Accountable owner	Data Architecture Council
Evidence rule	No implementation claim without inspected executable evidence
Repository	chinair88-prog/allinb2c

1. Executive Orientation

This chapter defines how GTOS shall **trace every database, schema, table family, service owner, migration, Event, projection, report, AI store, backup, retention, and decommissioning state.**

The specification separates reality, observations, knowledge, Decisions, Intent, execution, and outcomes. A component name, database row, UI label, AI answer, or workflow state does not acquire authority merely because it is visible or technically convenient.

- Reality or external outcome
- ≠ Observation
 - ≠ Claim
 - ≠ derived Perspective
 - ≠ Prediction
 - ≠ Recommendation
 - ≠ authorized Decision
 - ≠ execution Intent

1.1 Accountable ownership

The accountable owner is **Data Architecture Council**. Supporting applications, shared infrastructure, integration adapters, analytics, and AI coordinate or present the capability but shall not silently own its authoritative propositions.

1.2 Scope

- create and identify the subject
- validate evidence and policy
- make authorized Decisions
- execute typed Commands
- publish completed-fact Events
- query current and historical perspectives
- correct and reconcile outcomes

1.3 Non-goals

- treating a registered Gradle module, catalog record, README, frontend route, migration plan, or generated report as proof of end-to-end implementation;

- direct cross-Domain database writes;
- generic status values that combine lifecycle, approval, provider, document, payment, dispute, and correction state;
- silent replacement of history;
- AI-created authority or fabricated external outcomes.

2. Ubiquitous Language and Subject Model

Subject	Required interpretation
DatabaseOwnershipMatrix	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
DatabaseOwnershipMatrixVersion	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
DatabaseOwnershipMatrixDecision	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
DatabaseOwnershipMatrixEvidenceSet	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit
DatabaseOwnershipMatrixException	Independent identified subject, governed subordinate entity, or qualified perspective; ownership must be explicit

Reference aggregate:

```
DatabaseOwnershipMatrix {
  subjectId
  tenantId
  organizationContext
  sourceIdentityRefs[]
  sourceVersionRefs[]
  lifecycleState
  relationshipRefs[]
  decisionRefs[]
  intentRefs[]
  evidenceSetRef
  knowledgeCutoff
  policyRefs[]
  authorityRef
  jurisdictionRefs[]
  createdAt
  effectiveAt
  updatedAt
  version
  correctionOf?
  correlationId
}
```

2.1 Core invariants

- Stable identity precedes relationship, state, and cross-system correlation.
- Effective time and recorded time remain distinct where business truth changes over time.
- Recommendation, Decision, Intent, execution, and outcome are separate concepts.
- No Portal, report, search index, cache, workflow, or AI Agent becomes a shadow owner of Domain truth.
- Binding submitted, approved, issued, accepted, or signed versions are immutable.
- Corrections append governed facts and preserve the original record.
- External authority and provider outcomes retain their source and qualification.
- Tenant and represented-principal context come from trusted authentication and mandate evaluation.
- Retried writes are idempotent and concurrency guarded.
- Outcome Unknown is explicit and reconciled before risky duplicate execution.

2.2 Required standards

- stable identifiers
- typed contracts
- immutable binding versions
- transactional outbox
- idempotent Commands
- explicit Outcome Unknown
- correction lineage

- machine-verifiable conformance

3. Actors, Roles, and Authority

A conformant design distinguishes:

- the natural person acting;
- the service or AI identity invoking a Tool;
- the organization or principal represented;
- the role or relationship granting context;
- the mandate or delegation granting action authority;
- the Policy and jurisdiction limiting the action;
- the reviewer or approver responsible for a binding Decision.

Authority is evaluated at action time. Authentication alone does not grant business authority. Cached permissions and browser-supplied tenant headers are insufficient for high-impact actions.

4. Lifecycle and State Model

State	Semantics
PROPOSED	Distinct governed condition with entry, exit, timeout, correction and authority semantics
VALIDATION_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
APPROVAL_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
ACTIVE	Distinct governed condition with entry, exit, timeout, correction and authority semantics
SUSPENDED	Distinct governed condition with entry, exit, timeout, correction and authority semantics
CORRECTION_PENDING	Distinct governed condition with entry, exit, timeout, correction and authority semantics
RETIRED	Distinct governed condition with entry, exit, timeout, correction and authority semantics

```
stateDiagram-v2
    [*] --> PROPOSED
    PROPOSED --> VALIDATION_PENDING
    VALIDATION_PENDING --> APPROVAL_PENDING
    APPROVAL_PENDING --> ACTIVE
    ACTIVE --> SUSPENDED
    SUSPENDED --> CORRECTION_PENDING
    CORRECTION_PENDING --> RETIRED
```

The diagram is a primary reading path. Real implementations may add parallel review, amendment, appeal, timeout, suspension, partial completion, reconciliation, correction, and terminal failure without collapsing their meaning.

4.1 Transition rules

- transition through a typed, authorized Command;
- require exact expected version where concurrent change matters;
- record actor, represented principal, mandate, Policy, reason, and Evidence;
- use an idempotency key for retried writes;
- publish a completed-fact Event after durable state change;
- expose Outcome Unknown for unresolved external completion;
- preserve original and corrected states in historical reconstruction.

5. Decision, Evidence, and Knowledge Cutoff

Reference Decision:

```
Decision {
    decisionId
```

```

decisionType
subjectRef
sourceVersionRefs[]
knowledgeCutoff
evidenceSetRef
policyVersionRefs[]
recommendationRefs[]
principalId
representedOrganizationId
mandateRef
rationale
outcome
decidedAt
effectiveAt
correctionOf?
}

```

Evidence records identify issuer or observer, subject, version, observed time, received time, effective interval, integrity, confidence, qualification, contradiction, correction, retention, residency, and access class. A recommendation may assist the Decision but cannot promote itself to the Decision.

6. Commands, Events, Queries, and Contracts

Reference Commands:

- CreateDatabaseOwnershipMatrix
- ValidateDatabaseOwnershipMatrix
- ApproveDatabaseOwnershipMatrix
- ActivateDatabaseOwnershipMatrix
- SuspendDatabaseOwnershipMatrix
- CorrectDatabaseOwnershipMatrix
- RetireDatabaseOwnershipMatrix

Reference Events:

- DatabaseOwnershipMatrixCreated
- DatabaseOwnershipMatrixValidated
- DatabaseOwnershipMatrixApproved
- DatabaseOwnershipMatrixActivated
- DatabaseOwnershipMatrixSuspended
- DatabaseOwnershipMatrixCorrected
- DatabaseOwnershipMatrixRetired

Reference queries:

- GetDatabaseOwnershipMatrix
- ListDatabaseOwnershipMatrixRecords
- GetDatabaseOwnershipMatrixTimeline
- GetDatabaseOwnershipMatrixEvidence
- GetDatabaseOwnershipMatrixDecisionPackage
- GetDatabaseOwnershipMatrixExceptions
- GetDatabaseOwnershipMatrixAuditTrail

Reference API surface:

```

POST /api/v1/databaseownershipmatrix/commands
GET  /api/v1/databaseownershipmatrix/{id}
GET  /api/v1/databaseownershipmatrix/{id}/timeline
GET  /api/v1/databaseownershipmatrix/{id}/evidence
GET  /api/v1/databaseownershipmatrix/{id}/decisions
POST /api/v1/databaseownershipmatrix/{id}/exceptions
POST /api/v1/databaseownershipmatrix/{id}/corrections

```

These endpoints are reference contracts, not claims that exact routes exist.

7. Workflow, Failure, and Compensation

flowchart TD

```

A[Validate identity version and authority] --> B[Collect required evidence]
B --> C[Policy and evidence gates pass?]
C -- No --> D[Open exception or request correction]
D --> B

```

```

C -- Yes --> E[Record Decision or execution Intent]
E --> F[Execute Domain command or external request]
F --> G{Outcome known?}
G -- Success --> H[Record fact and publish Event]
G -- Failure --> I[Record failure and compensation options]
G -- Unknown --> J[Enter Outcome Unknown]
J --> K[Query source or wait for verified callback]
K --> G
H --> L[Update projections and downstream workflow]

```

Representative failure modes:

- stale or wrong source version
- duplicate submission or replay
- external timeout after possible success
- contradictory source observations
- lost Event after committed state
- approval or mandate revoked during execution
- downstream action before a mandatory gate
- partial completion across multiple subjects
- evidence later retracted or corrected
- tenant, facility, channel or jurisdiction mismatch

Recovery shall query the authoritative source before duplicate execution, preserve partial success, use compensating Intent rather than destructive rewrite, publish corrections, quarantine high-impact ambiguity, and record affected downstream subjects.

8. Data and Persistence Architecture

Reference table families:

Table family	Purpose
databaseownershipmatrix_aggregate	authoritative subject
databaseownershipmatrix_version	immutable submitted, approved, issued, or signed versions
databaseownershipmatrix_relationship	governed relationships
databaseownershipmatrix_decision	binding Decisions and rationale
databaseownershipmatrix_evidence_link	provenance and evidence relationships
databaseownershipmatrix_event_outbox	transactional Event publication
databaseownershipmatrix_idempotency	duplicate-write protection
databaseownershipmatrix_exception	exception, claim, dispute, or hold
databaseownershipmatrix_correction	correction and supersession lineage
databaseownershipmatrix_audit	privileged action evidence

The owning service controls schema and migration. Cross-Domain references use stable identifiers or contracts. Analytics, reports, caches, search, and vector indexes remain projections. Backup, restore, retention, legal hold, privacy, residency, and decommissioning requirements are explicit.

9. Portal and Experience Requirements

- Primary participant Portal: initiate, review exact versions, supply evidence, and respond to required actions.
- Operator Portal: inspect exceptions, correlation, source responses, retries, compensation, and downstream impact.
- Executive or oversight view: consume governed metrics without becoming a source of Domain truth.

Every Portal displays exact version, owner, state, freshness, required action, and uncertainty. It supports loading, empty, partial, stale, denied, error, timeout, and Outcome Unknown states; accessibility; locale, currency, unit, date, and RTL/LTR correctness; evidence audit; safe retry; and correction history.

10. AI Participation and Tool Governance

Permitted AI tasks:

- extract and classify evidence
- detect anomalies and missing fields
- summarize timelines and contradictions

- recommend next actions with sources and uncertainty

AI shall not own Domain records, grant itself authority, suppress contradictions, silently edit approved versions, fabricate external outcomes, or execute high-impact actions outside typed Tools, policy, approval, idempotency, sandbox, and outcome observation.

Every Agent Run records source context, prompt/model/tool versions, structured output, confidence and limitations, Policy result, approval, Tool calls, provider responses, final outcome, and recovery actions.

11. Security, Privacy, and Trust Boundaries

- least privilege and separation of duties
- tenant, organization, facility, channel and jurisdiction isolation
- integrity protection for binding evidence
- verified external callbacks and anti-replay controls

Additional controls include secret isolation, callback signature verification, audit of privileged action, emergency-access review, sensitive-data minimization, purpose limitation, and threat models for impersonation, tampering, replay, confused deputy, fraud, prompt injection, and exfiltration.

12. Reliability, SLOs, and Operations

SLI	Reference objective
authoritative read availability	99.9% monthly unless a stricter tier applies
acknowledged Command durability	no acknowledged write may be silently lost
Event publication	99.9% within five minutes after committed fact
duplicate prevention	100% for same tenant, Command class, and idempotency key
binding Decision audit completeness	100%
state-state detection	bounded by business deadline

Dashboards and runbooks cover backlog, error budget, dependency failure, retry budget, DLQ, workflow stall, provider outage, reconciliation, capacity, backup/restore, release/rollback, incident command, and post-incident correction.

13. Testing and Continuous Conformance

- aggregate invariant and transition tests
- authorization, delegation, separation-of-duties and tenant-isolation tests
- idempotency and optimistic-concurrency tests
- database migration, key, constraint and rollback tests
- contract, outbox/inbox, duplicate, replay and DLQ tests
- workflow restart, timeout and compensation tests
- Portal E2E tests for success, partial, stale, denied, error and Outcome Unknown states
- accessibility, locale, currency, unit and RTL/LTR tests
- security adversarial, fraud and data-exfiltration tests
- AI evaluation, prompt-injection, policy-bypass and unauthorized-tool tests
- historical reconstruction, correction and audit-completeness tests
- performance, capacity, recovery and operational exercise tests

Release evidence links each invariant and control to the test, environment, input data, result, owner, and artifact proving it. Generated test counts never replace semantic coverage.

14. Repository Review Boundary

- This chapter is a repository-review specification. Exact executable implementation has not been inspected for every target behavior.
- Registered modules, catalog entries, README claims, frontend contracts, or plans are not proof of complete implementation.

Area	Classification
target aggregate, API, tables, state and workflow in this chapter	REFERENCE_ARCHITECTURE
registered module or catalog claim	DOCUMENTATION_ONLY_CLAIM until source inspection
uninspected integration	REQUIRES_REPOSITORY_REVIEW
behavior proven in another repository-grounded chapter	retain that chapter's evidence and classification
contradictory ownership or path claims	CONFLICT_REQUIRES_CANONICALIZATION

No statement upgrades implementation status merely because a file, directory, module name, frontend contract, database entry, plan, or report exists.

15. Worked Conformance Scenario

An authorized principal initiates a Command against an exact subject version. The service validates identity, tenant, mandate, Policy, Evidence, and idempotency. It records the state transition and outbox entry atomically. If an external dependency participates, timeout produces Outcome Unknown; reconciliation queries the source or waits for a verified callback instead of repeating the action blindly.

Portals show source, freshness, version, required action, and uncertainty. AI may extract, compare, summarize, detect anomalies, and recommend through typed Tools, but cannot own the subject or fabricate success. Later correction appends new facts, identifies affected projections and workflows, and preserves the historical Decision context.

16. Conformance Checklist

- ☐ stable identity and proposition ownership are explicit;
- ☐ authority and mandate are evaluated at action time;
- ☐ state, Recommendation, Decision, Intent, execution, and outcome remain distinct;
- ☐ Commands are typed, authorized, idempotent, and version guarded;
- ☐ Events describe completed facts;
- ☐ Evidence includes provenance, time, integrity, qualification, and correction;
- ☐ external outcomes retain their authority source;
- ☐ Portals show freshness and uncertainty;
- ☐ AI is bounded by typed Tools, Policy, approval, and outcome observation;
- ☐ security, tenant, facility, channel, and jurisdiction boundaries are enforced;
- ☐ Outcome Unknown and reconciliation exist where needed;
- ☐ corrections preserve history;
- ☐ tests prove invariants and recovery;
- ☐ implementation status is not upgraded without executable evidence.

17. Status Vocabulary

Status	Meaning
VERIFIED_IMPLEMENTED	Executable source and relevant behavior were inspected.
IMPLEMENTED_WITH_GAP	Executable behavior exists with a material governance or completeness gap.
PARTIAL_IMPLEMENTATION	Only part of the target capability is evidenced.
REFERENCE_ARCHITECTURE	Normative target behavior; no claim that it exists today.
PLANNED_NOT_IMPLEMENTED	Explicitly planned but not evidenced as executable.
DOCUMENTATION_ONLY_CLAIM	Documentation or client contract claims behavior without verified backend evidence.
CONFLICT_REQUIRES_CANONICALIZATION	Sources disagree and no final owner has been established.
REQUIRES_REPOSITORY_REVIEW	Necessary executable evidence has not yet been inspected.